

RTL Design Sherpa

RTL Common Library Module Reference — Rev 0.90

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1 rtl-common Library Overview

The rtl-common library is a collection of fundamental digital design building blocks — parameterizable, synthesis-friendly RTL modules built to a consistent quality bar, ready to drop into a wide range of digital systems. If you've hand-rolled a counter or a CRC one too many times (and if you've been doing this long enough, you have), this library is where that stops.

1.1 Library Philosophy

1.1.1 Design Principles

Five principles run through everything in here:

- **Modularity**: Each module does one specific function and composes easily with the others
- **Parameterizability**: Configurable width and functionality parameters for maximum reusability
- **Synthesis Friendly**: Optimized for modern synthesis tools with predictable area/timing results
- **Technology Independence**: Clean RTL that works across ASIC and FPGA technologies
- **Broad Coverage**: counters, FIFOs, arbiters, data integrity, and clock/reset control (arithmetic is its own library — rtl-math; CDC is its own too — rtl-cdc)

1.1.2 Quality Standards

Every module in the tree is held to the same bar:

- **Lint Clean**: All modules pass rigorous RTL linting checks
- **Simulation Verified**: Comprehensive testbenches validate functionality
- **Synthesizable**: Written in standard SystemVerilog with no vendor primitives
- **Documentation Complete**: Detailed documentation with usage examples

1.2 Architecture Overview

1.2.1 Module Hierarchy

Per-category counts live in one place — the count table in index.md — so this tree lists categories only and cannot drift from it.

```
rtl-common Library
 (Arithmetic & Math moved to rtl/math/ — see rtl-math book)
 Data Integrity
    Error Detection (CRC, Checksum, Parity)
    Error Correction (Hamming ECC)
 Control & Arbitration
    Round-Robin Arbiters
    Priority Arbiters
 Clock & Reset
    Clock Management (ICG, Dividers)
    Reset Synchronization
 Counters & Sequences
    Binary Counters (Gray/Johnson counters moved to rtl/cdc/)
    Specialized Counting
 Data Conversion
    Code Converters (Binary→BCD; Gray/Johnson moved to rtl/cdc/)
    Display Decoders
 Bit Operations
    Bit Searching & Counting
    Vector Manipulation
 Shift & LFSR
    Barrel Shifters
    Pseudorandom Generation
 Memory & Storage
    FIFO Implementations
    Content Addressable Memory
 Utilities
     PWM Generation
     Sorting Algorithms
```

1.3 Functional Categories

1.3.1 1. Arithmetic and Mathematical Operations

Moved. The `math_*` arithmetic library (adders, subtractors, multipliers, compressors, prefix cells, and the BF16 / FP8 / FP16 / FP32 / IEEE-754 floating-point modules) is now its own subsystem and its own book: **rtl-math** — RTL in `rtl/math/`, tests in `val/math/`.

1.3.2 2. Data Integrity and Error Management

If data has to survive storage or transmission, these are the blocks that make that happen.

1.3.2.1 Cyclic Redundancy Check (CRC)

- **Generic CRC Engine**: Configurable polynomial, width, and initialization
- **Cascade Implementation**: High-throughput parallel CRC processing
- **XOR-Shift Variants**: Optimized for specific polynomial types

1.3.2.2 Error Correction Coding (ECC)

- **Hamming SECDED**: Single Error Correction, Double Error Detection
- **Configurable Width**: Support for various data widths
- **Encoder/Decoder Pairs**: Complete ECC implementation

1.3.2.3 Simple Integrity Checks

- **Checksum**: Software-compatible checksum algorithms
- **Parity**: Even/odd parity generation and checking

Key Features: - Standards-compliant implementations - Parameterizable for various data widths - Optimized for both latency and throughput scenarios

1.3.3 3. Control Logic and Arbitration

Shared resources need a referee. These blocks are it — the essential components for managing shared resources and coordinating the system.

1.3.3.1 Round-Robin Arbitration

- **Simple Round-Robin**: Basic fair arbitration with minimal overhead
- **Weighted Round-Robin**: Priority-based fair arbitration
- **Advanced Features**: Configurable agent counts and weighting

1.3.3.2 Priority-Based Control

- **Priority Encoders**: Fixed priority arbitration schemes
- **Configurable Priority**: Runtime priority adjustment capability

Applications: - Bus arbitration in multi-master systems - Resource allocation in processors - Quality of Service (QoS) implementations

1.3.4 4. Clock and Reset Management

Power and timing management — the unglamorous foundation of a dependable digital system.

1.3.4.1 Clock Control

- **Integrated Clock Gating (ICG)**: Power optimization through controlled clock disabling
- **Clock Dividers**: Frequency synthesis and clock domain generation
- **Pulse Generators**: Controlled clock pulse generation

1.3.4.2 Reset and Synchronization

- **Reset Synchronizers**: Safe reset assertion/deassertion
- **Debouncing**: Input signal stabilization

Power Management Benefits: - Clock gating (ICG) removes switching on idle functional blocks; the actual dynamic-power saving depends on activity factor and is design-specific - Proper reset sequencing for reliable startup - Signal integrity maintenance

1.3.5 5. Counting and Sequence Generation

Fundamental building blocks for timing, addressing, and control.

1.3.5.1 Binary Counters

- **Basic Counters**: Up/down counting with enable/reset
- **Gray and Johnson Counters**: `counter_bingray`, `counter_johnson` — these moved to `rtl/cdc/`; see the rtl-cdc book

1.3.5.2 Specialized Counting

- **Frequency-Invariant**: Clock-independent counting
- **Load/Clear**: Programmable counter initialization
- **Ring Counters**: One-hot sequence generation

Design Features: - Parameterizable width and functionality - Optional synchronous/asynchronous controls - Overflow/underflow detection

1.3.6 6. Data Conversion and Encoding

Code conversion and display interfacing utilities.

1.3.6.1 Number System Conversion

- **Binary → BCD**: Decimal display conversion

- **Binary ↔️ Gray, Johnson decode**: `bin2gray`, `gray2bin`, `johnson2bin` — these moved to `rtl/cdc/`; see the rtl-cdc book

1.3.6.2 Encoding/Decoding

- **Priority Encoders**: Multiple input to binary conversion
- **Binary Decoders**: Address decoding and control
- **7-Segment Display**: Hexadecimal character display

Applications: - Display interfaces - Address decoding - Safe async signal crossing

1.3.7 7. Bit Manipulation and Searching

Efficient bit-level operations for high-performance computing.

1.3.7.1 Bit Detection

- **Leading Zero Count**: Normalization and floating-point support
- **Find First/Last Set**: Efficient bit searching
- **Priority Detection**: Leading one and trailing one detection

1.3.7.2 Vector Operations

- **Bit Reversal**: Endianness conversion and FFT applications
- **Vector Manipulation**: Bit-level data reorganization

Performance Characteristics: - Logarithmic complexity algorithms where possible - Optimized for synthesis tool recognition - Support for various data widths

1.3.8 8. Shift Operations and LFSRs

Shift-based operations and pseudorandom sequence generation.

1.3.8.1 Shifters

- **Barrel Shifters**: Arbitrary shift/rotate in single cycle
- **Universal Shifters**: Combined left/right, logical/arithmetic shifting

1.3.8.2 Linear Feedback Shift Registers (LFSRs)

- **Fibonacci LFSRs**: External feedback topology
- **Galois LFSRs**: Internal feedback topology
- **Configurable Polynomials**: Various sequence lengths and properties

Applications: - Pseudorandom number generation - Built-in self-test (BIST) patterns - Encryption and scrambling

1.3.9 9. Memory and Storage

Buffering and storage solutions for data flow management.

1.3.9.1 FIFO Implementations

- **Synchronous FIFO**: Single clock domain buffering (`fifo_sync`, `fifo_control`)
- **Asynchronous FIFO**: `fifo_async` moved to `rtl/cdc/`; production FIFOs live in `rtl/amba/gaxi/`

1.3.9.2 Associative Memory

- **Content Addressable Memory (CAM)**: Tag matching and lookup

Features: - Configurable depth and width - Full/empty flag generation - Error handling and overflow protection

1.3.10 10. Utility and Specialized Functions

Specialized functions for specific application domains.

1.3.10.1 Signal Generation

- **PWM**: Pulse width modulation for analog control
- **Timing Control**: Precise pulse and timing generation

1.3.10.2 Data Processing

- **Hardware Sorting**: Parallel sorting implementations
- **Algorithmic Processing**: Specialized computational functions

1.4 Performance and Optimization

Picking the right block is half the job — how you configure it decides whether you close timing and meet the power budget.

1.4.1 Synthesis Guidelines

1.4.1.1 Area Optimization

- Use ripple-carry arithmetic for low-area requirements
- Select appropriate FIFO depths based on data flow analysis

- Consider shared resources for infrequently used functions

1.4.1.2 Speed Optimization

- Use parallel prefix adders (Brent-Kung, Han-Carlson) for critical paths
- Implement pipelining in high-throughput applications
- Use barrel shifters for single-cycle shift operations

1.4.1.3 Power Optimization

- Apply clock gating (ICG) to inactive functional blocks
- Use Gray code counters for reduced switching activity
- Implement frequency-invariant designs where applicable

1.4.2 Technology-Specific Considerations

1.4.2.1 ASIC Implementation

- Use dedicated adder and multiplier cells
- Use technology-specific clock gating cells
- Optimize for specific voltage and temperature ranges

1.4.2.2 FPGA Implementation

- Use DSP blocks for arithmetic operations
- Use BRAM for deep FIFOs
- Use the dedicated clock management resources

1.5 Verification and Validation

1.5.1 Testbench Coverage

- **Functional Coverage**: All operational modes and edge cases
- **Directed Tests**: Specific functionality verification
- **Random Testing**: Stress testing with random inputs
- **Corner Cases**: Boundary conditions and error scenarios

1.5.2 Compliance Testing

- **Timing Analysis**: Setup/hold requirements validation
- **Power Analysis**: Dynamic and static power characterization
- **Area Analysis**: Resource utilization across technologies

1.6 Usage Guidelines

1.6.1 Module Selection

1. **Identify Requirements**: Performance, area, and power constraints
2. **Select Architecture**: Choose appropriate implementation (ripple vs. parallel, etc.)
3. **Configure Parameters**: Set widths and options for specific application
4. **Verify Integration**: Ensure proper interfacing and timing

1.6.2 Integration Best Practices

The example below instantiates `math_adder_full_nbit`, which now lives in `rtl/math/` — modules compose across the two libraries.

```
// Example: Building a complete arithmetic unit (math_* modules are in
rtl/math/)
module arithmetic_unit #(
    parameter int DATA_WIDTH = 32
) (
    input  logic                    clk,
    input  logic                    rst_n,
    input  logic [DATA_WIDTH-1:0]   operand_a,
    input  logic [DATA_WIDTH-1:0]   operand_b,
    input  logic [1:0]              operation,
    output logic [DATA_WIDTH-1:0]   result,
    output logic                    overflow
);

    // Use appropriate math modules based on requirements
    logic [DATA_WIDTH-1:0] add_result, sub_result, mult_result;
    logic add_carry, sub_borrow;

    // High-speed addition using parallel prefix
    math_adder_full_nbit #(
        .N(DATA_WIDTH)
```

```
    ) u_adder (
        .i_a(operand_a), .i_b(operand_b), .i_c(1'b0),
        .ow_sum(add_result), .ow_carry(add_carry)
    );

    // Subtraction using complement method
    math_subtractor_full_nbit #(
        .N(DATA_WIDTH)
    ) u_subtractor (
        .i_a(operand_a), .i_b(operand_b), .i_b_in(1'b0),
        .ow_d(sub_result), .ow_b(sub_borrow)
    );

    // Result selection based on operation
    always_comb begin
        case (operation)
            2'b00: begin result = add_result; overflow = add_carry;
end
            2'b01: begin result = sub_result; overflow = sub_borrow;
end
            default: begin result = '0; overflow = 1'b0; end
        endcase
    end

endmodule
```

1.7 Future Enhancements

1.7.1 Planned Additions

- **DSP Functions**: FFT, filtering, and signal processing primitives
- **Security Primitives**: Hardware security modules and cryptographic functions
- **AI/ML Accelerators**: Matrix operations and neural network primitives

1.7.2 Technology Evolution

- **Emerging Technologies**: Support for new process nodes and device types
- **Tool Integration**: Enhanced synthesis and verification tool compatibility
- **Standards Compliance**: Updates for evolving industry standards

That's the library. The blocks are verified, documented, and built to a consistent bar — so you can spend your design effort on the parts of the system that are actually new.

1.8 References

- Seminal papers, defining documents and textbook chapters for every technique in this area — arbitration (round-robin, weighted, deficit round robin, matrix, token bucket), CAM and sorting networks, FIFO wrap-bit pointers, clock gating, CRC/checksum/SECDED, LFSRs, barrel shifters, and leading-zero detection — are indexed in `rtl/common/references/README.md`, with open-access PDFs archived beside it. Per-module pages carry the citations that apply to them.

2 Deficit Round Robin Arbiter

Module: `rtl/common/arbiter_deficit_round_robin.sv`

2.1 Overview

Deficit round-robin (DRR) arbiter — the sibling wrapper to arbiter_round_robin_weighted around the same arbiter_round_robin core. The two disciplines answer different questions:

- **WRR:** shares proportional to **grant count**. Weight 4 means four grants per replenish round, whatever each grant costs.
- **DRR:** shares proportional to **cost served**. Quantum 4 means four cost-units (bytes, beats, bus cycles) per round, however many grants that takes.

With equal-cost requests the two give identical long-run shares — use the cheaper WRR there. Reach for the DRR when a grant's resource usage varies per request: packet lengths, burst sizes, variable-beat DMA descriptors.

```
module arbiter_deficit_round_robin #(
    parameter int CLIENTS      = 4,
    parameter int MAX_QUANTUM  = 16,
    parameter int COST_WIDTH   = 4,
    parameter int WAIT_GNT_ACK = 0
) (
    input  logic                clk,
    input  logic                rst_n,
    input  logic                block_arb,
    input  logic [CXQW-1:0]     quantum,
    input  logic [CXCW-1:0]     req_cost,
    input  logic [C-1:0]        request,
    input  logic [C-1:0]        grant_ack,
    output logic                grant_valid,
    output logic [C-1:0]        grant,
    output logic [N-1:0]        grant_id
);
```

2.2 Parameters

2.2.1 User-Settable Parameters

Parameter	Type	Default	Description
CLIENTS	int	4	Number of requesting clients (range: 2-32)

Parameter	Type	Default	Description
MAX_QUANTUM	int	16	Exclusive bound on per-client quantum; field width = $clog2(MAX_QUANTUM) (range: 2-256)
COST_WIDTH	int	4	Width of each client's request-cost input (costs 1..2^COST_WIDTH-1; range: 1-16)
WAIT_GNT_ACK	int	0	0 = grant completes immediately, 1 = grant held until grant_ack

2.2.2 Derived Parameters

Declared in the parameter list so the port declarations can use them (strict front ends reject body localparams in port ranges); overridable in principle, and must not be — leave them to their derivations. Only DW is a true localparam.

Parameter	Default	Description
QW	$clog2(MAX_QUANTUM)	Quantum field width
DW	$clog2(2^COST_WIDTH + MAX_QUANTUM) + 1	Deficit counter width — sized so every legal cost is reachable by accumulation, which is the no-livelock guarantee
N	$clog2(CLIENTS)	grant_id width
CXQW / CXCW	CLIENTS x QW / CLIENTS x COST_WIDTH	Packed array widths

2.3 Ports

2.3.1 Inputs

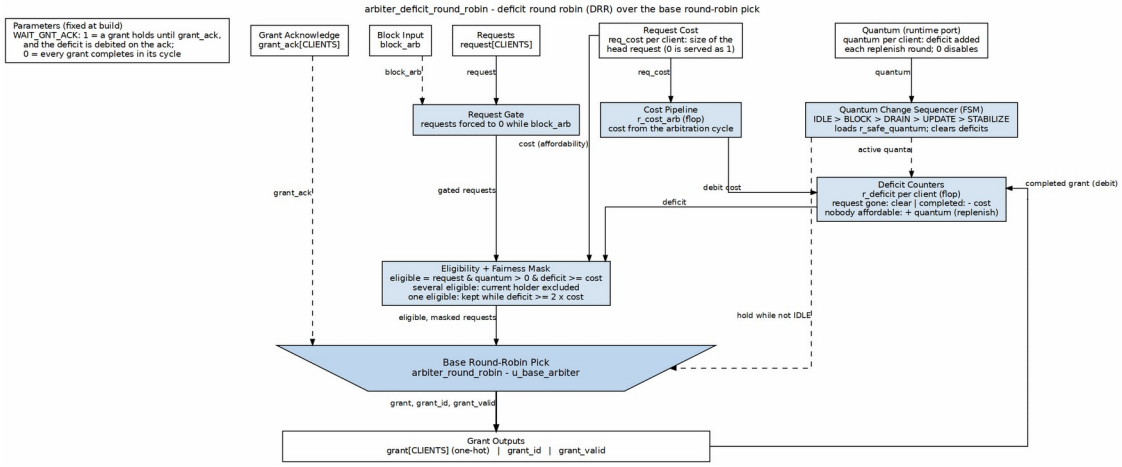
Port	Width	Description
clk	1	Clock
rst_n	1	Async active-low reset
block_arb	1	Block all arbitration while high
quantum	CXQW	Packed per-client quanta, client 0 in the low field. Zero disables a client. Changes go through the atomic-update FSM
req_cost	CXCW	Packed cost of each client's HEAD request. Stable while request held; the next frame's cost may be presented as soon as the current frame's grant is observed
request	CLIENTS	Request vector
grant_ack	CLIENTS	Grant acknowledgment (ACK mode only)

2.3.2 Outputs

Port	Width	Description
grant_valid	1	Grant output valid
grant	CLIENTS	One-hot grant vector
grant_id	N	Binary-encoded winner

2.4 Functional Description

2.4.1 Block diagram



arbiter_deficit_round_robin block diagram

Requests drop down the center through the `block_arb` gate into the eligibility and fairness mask, which passes only clients whose deficit covers the cost of their head request. The masked requests feed the base `arbiter_round_robin` (the trapezium), whose registered grant goes straight to the outputs. The deficit counters sit on the right. `req_cost` feeds affordability directly and, one cycle later through `r_cost_arb`, the debit; each completed grant climbs the right edge to debit its client, and when nobody can afford service every enabled client gains its quantum. A change on the `quantum` port goes through the change sequencer, which holds the pick until the new quanta are loaded.

2.4.2 The Deficit Discipline

Each client holds a deficit counter (reset 0). A requester is **eligible** when its deficit covers its head cost. When requests exist but nobody can afford service, a **replenish round** adds each requesting client's quantum to its deficit — repeating on consecutive cycles until someone becomes eligible, which is how a cost larger than one quantum is saved up for (multi-round accumulation; the deficit width guarantees termination). On grant completion the winner is debited the served cost and the **remainder carries** to its next request — the carry is what keeps long-run shares on the quantum ratio regardless of how costs divide into it.

Anti-hoarding: a client's deficit clears when its request deasserts (classic DRR empty-queue rule). Service share is earned while competing — an idle client cannot bank deficit and burst later. Hold request through frame gaps if the carry should survive.

2.4.3 The Cost Pipeline

The grant registers one cycle after the arbitration that won it. A consumer naturally pops its frame on grant and presents the next frame's cost immediately — so in the completion cycle `req_cost` may already belong to the *next* frame. The arbiter therefore debits the **arbitration-**

cycle cost, captured in an internal one-deep pipeline. Without it, a back-to-back client is debited the wrong frame's cost (this was caught during bring-up by the testbench's deficit mirror, not by inspection — see the test).

2.4.4 Share Example

Quanta [4,2,1,1], every request cost 2:

- Replenish gives deficits [4,2,1,1]: clients 0 and 1 afford immediately, clients 2 and 3 must accumulate a second round to reach 2.
- Long-run cost-units served settle at 4:2:1:1 — the quantum ratio — even though clients 2 and 3 are only served every other round.

2.4.5 Quantum Changes and Disable

Runtime quantum changes pass through the same shadow-register FSM as the WRR's weights (IDLE → BLOCK → DRAIN → UPDATE → STABILIZE; 10 cycles minimum, ~25 with the BLOCK timeout); deficits clear at STABILIZE so old carry cannot distort the new policy. Quantum 0 disables a client entirely. A cost of 0 is defensively served as cost 1.

2.5 Design Notes

Deficit round robin shapes by CREDIT, not by turn. A client with a larger quantum receives proportionally more service over a round rather than more turns, so short-burst and long-burst clients with the same quantum get the same bandwidth, not the same grant count. Size quanta in the unit the downstream consumes.

The compliance model lives in the DV framework. Round-robin order and fairness verdicts come from `CocoTBFramework`'s `ArbiterCompliance`; several historical "violations" in this family were defects in that model rather than in any arbiter (common BUG-006 (was COMMON-016) through common BUG-009 (was COMMON-019)).

2.6 Timing Characteristics

Property	Value
Latency	1 cycle steady-state (deficit compare, masking and RR decision combinational; grant registered in the core)
Throughput	1 grant/cycle max
Replenish	1 cycle per round, repeats until affordable

Property	Value
Quantum update	10-25 cycles (FSM; DRAIN occupies 3 cycles, STABILIZE 4)

Critical path: the deficit >= cost comparator into the request mask into the base arbiter — one comparator wider than the WRR's credit-nonzero check. At high client counts consider registering eligibility (costs a cycle of latency).

2.7 Usage Examples

Every parameter and port below is taken from the module declaration.

```
arbiter_deficit_round_robin #(
    .CLIENTS                (4),
    .MAX_QUANTUM            (16),
    .COST_WIDTH             (4),
    .WAIT_GNT_ACK           (0),
    .QW                     ($clog2(MAX_QUANTUM)),
    .N                      ($clog2(CLIENTS)),
    .C                      (CLIENTS),
    .CXQW                   (CLIENTS * QW),
    .CXCW                   (CLIENTS * COST_WIDTH),
) u_arbiter_deficit_round_robin (
    .clk                    (clk),
    .rst_n                  (rst_n),
    .block_arb              (block_arb),
    .quantum                (quantum),
    .req_cost               (req_cost),
    .request                (request),
    .grant_ack              (grant_ack),
    .grant_valid            (grant_valid),
    .grant                  (grant),
    .grant_id               (grant_id)
);
```

2.8 Related Modules

- arbiter_round_robin — the shared base core
- arbiter_round_robin_weighted — the grant-count sibling; use it when requests are equal-sized
- arbiter_round_robin_simple, arbiter_priority_encoder

2.9 Testing

Location: `val/common/test_arbiter_deficit_round_robin.py`

The property under test is cost-proportionality: served cost-units per client must follow the quantum ratio whatever the per-request costs are. Scenarios: equal-cost anchor, mixed random costs (the DRR-defining case), cost > quantum accumulation, anti-hoarding, zero-quantum disable, dynamic quantum change; both ACK modes, 4-16 clients. Every scenario also runs a cycle mirror of the deficit discipline that fails the test on any grant to a client whose deficit did not cover its arbitration-cycle cost.

```
pytest val/common/test_arbiter_deficit_round_robin.py -v
```

2.10 References

- Shreedhar, M., Varghese, G. “Efficient Fair Queuing Using Deficit Round-Robin.” *IEEE/ACM Transactions on Networking*, 4(3), June 1996. DOI: 10.1109/90.502236 — the original DRR scheme (quantum added per round, cost charged per service, remainder carried as deficit) that this module’s cost-proportional sharing implements. Conference version: *Proc. ACM SIGCOMM ’95*, pp. 231–242, DOI: 10.1145/217382.217453.
- Dally, W.J., Towles, B. *Principles and Practices of Interconnection Networks*, Morgan Kaufmann, 2004, ch. 18 “Arbitration”, §18.4.4 (weighted round-robin) — textbook context for the credit-based family this module extends (secondary source).
- Indexed in `rtl/common/references/README.md` (local PDF: `DeficitRoundRobin__Shreedhar1996.pdf`)

2.11 Navigation

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3 Matrix Arbiter

Fair arbitration with no rotating pointer. Every requester keeps its own scorecard of who it outranks, so there's no barrel shifter and no priority-encoder chain in the winner path — just a grid of pairwise priority bits settling the argument in one combinational pass.

3.1 Overview

I lifted the scheme straight from Dally & Towles, *Principles and Practices of Interconnection Networks*, chapter 17 §18.5. Priority lives in a `CLIENTS×CLIENTS` bit matrix `M`:

- `M[i][j] = 1` means requester `i` currently outranks requester `j`.
- `M[i][j] = 0` means requester `i` is outranked by requester `j`.
- The diagonal is a don't-care.

Half those bits are redundant — `M[j][i]` is just `~M[i][j]` — so the module stores only the `CLIENTS·(CLIENTS-1)/2` upper-triangle bits and derives the rest combinationally. Every grant rewrites the winner's row and column, shoving it to the back of the line. That's the entire fairness argument: keep requesting and you'll be served within `CLIENTS` consecutive grants. Guaranteed, no starvation corner cases hiding anywhere.

One naming note so the parameter tables don't trip you: `N` is the bit width of `grant_id` — `$clog2(CLIENTS)`, guarded to 1 when `CLIENTS==1`. It's not the client count. The client count is always `CLIENTS`.

3.2 Parameters

Parameter	Type	Default	Description
CLIENTS	int	4	Number of requesting clients (must be ≥ 1)
WAIT_GNT_ACK	int	0	When 1, holds the grant until the winner acknowledges it
N	int	$clog2(CLIENTS) (guarded)	**Derived** bit width of `grant_id`; do not override

3.3 Ports

3.3.1 Inputs

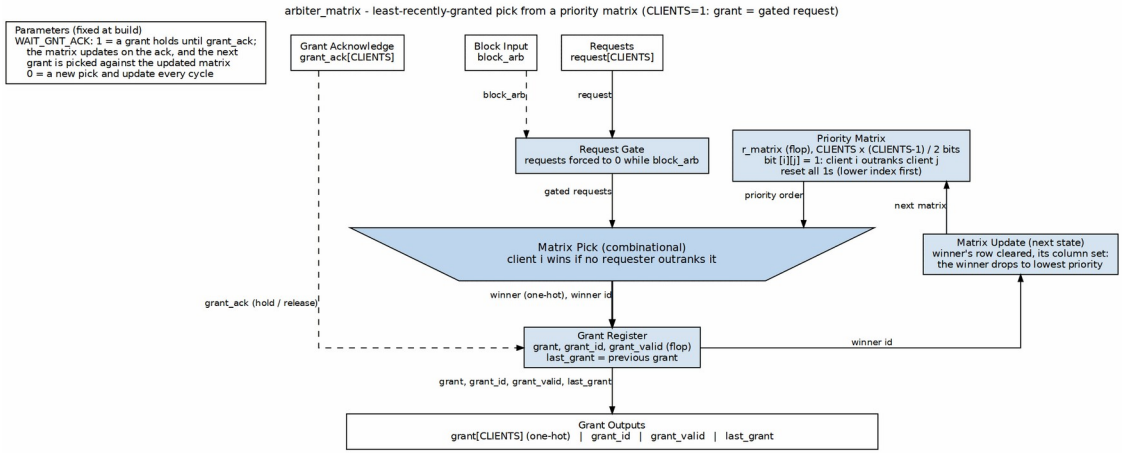
Port	Width	Description
clk	1	System clock
rst_n	1	Active-low asynchronous reset
block_arb	1	When asserted, masks all requests to zero and stops grants
request	CLIENTS	One-bit request per client
grant_ack	CLIENTS	Per-client grant acknowledgment (used when `WAIT_GNT_ACK = 1`)

3.3.2 Outputs

Port	Width	Description
grant_valid	1	High when a valid grant is being issued
grant	CLIENTS	One-hot grant vector (exactly one bit set when `grant_valid=1`)
grant_id	`N ($clog2(CLIENTS)`, guarded)	Binary-encoded ID of the granted client
last_grant	CLIENTS	Previous cycle's grant (for debugging/history)

3.4 Functional Description

3.4.1 Block diagram



arbiter_matrix block diagram

Requests drop down the center through the `block_arb` gate into the matrix pick (the trapezium): a client wins when no other requester outranks it in the priority matrix. The winner is registered as the grant. Its id runs out to the right into the matrix update, which moves the winner to lowest priority, and the next matrix is loaded back into `r_matrix`, so the least recently granted requester always wins. With `WAIT_GNT_ACK=1` the grant holds until `grant_ack`, and the update happens on the ack.

3.4.2 Matrix Representation

The stored state is the upper triangle of `M`, packed into `CLIENTS*(CLIENTS-1)/2` bits. Out of reset every stored bit is 1, so power-on priority is the plain total order:

```
0 > 1 > 2 > ... > CLIENTS-1
```

Client 0 wins everything until that first grant shuffles the deck.

3.4.3 Grant Logic

Each grant bit is a purely combinational vote:

```
g[i] = request[i] AND (for all j != i: NOT(request[j] AND M[j][i]))
```

Read it out loud: client `i` wins unless some other *requesting* client outranks it. The matrix is a tournament — antisymmetric and complete — so whenever anybody asks, exactly one winner steps forward. No tie-breaking. No encoder to consult.

3.4.4 Matrix Update

When client `k` wins, two things happen:

- **Row k cleared**: `M[k][j] = 0` for all `j > k` — k loses to all higher-index clients.
- **Column k set**: `M[i][k] = 1` for all `i < k` — all lower-index clients outrank k.

Net effect: k goes last. The new order becomes:

```
0 > 1 > ... > k-1 > k+1 > ... > CLIENTS-1 > k
```

3.4.5 No-ACK Mode (`WAIT_GNT_ACK = 0`)

Grant, rotate, repeat. A one-cycle grant goes out every cycle a winner exists, and the matrix updates right behind it.

3.4.6 ACK Mode (`WAIT_GNT_ACK = 1`)

Here the grant holds until the granted client asserts its own `grant_ack` bit, and the matrix doesn't move until that ACK arrives — priority state stays frozen mid-transaction. The part worth your attention: the post-ACK update and the next-winner calculation happen combinationally in the ACK cycle itself, and the new grant registers on the same edge. Feed it a steady stream of ACKs and grants come out back-to-back, no idle cycles.

One guard, and it's deliberate: only the granted client's ack bit can retire the grant. ACKs from other clients are ignored (foreign-ack immunity). That hazard is real — an errant ack retiring a grant it didn't owns you a debug session — so the immunity is cheap insurance.

3.4.7 Block Arbitration

`block_arb` squashes every request to zero, so nothing grants while it's high. Release it and arbitration picks up exactly where the matrix left off. Pausing costs you cycles, never state.

3.5 Timing Characteristics

Grants are registered — expect one full cycle between a request and the grant it earns. The matrix updates on the same clock, so in no-ACK mode each new winner is chosen from the freshly updated matrix.

ACK mode pulls the update forward into the ACK cycle: combinationally evaluated when the ACK arrives, registered on that same edge. Keep the ACKs coming and grants stream out with zero idle cycles.

No frequency or area numbers on this page. Nothing's been measured against a real device yet, and I'd rather admit that than invent a figure.

3.6 Usage Examples

```
arbiter_matrix #(
    .CLIENTS     (8),
    .WAIT_GNT_ACK(0)
) u_arbiter_matrix (
    .clk         (clk),
    .rst_n       (rst_n),
    .block_arb   (1'b0),
    .request     (request_vec),
    .grant_ack   ('0),                  // Unused in No-ACK mode
    .grant_valid (grant_valid),
    .grant       (grant_vec),
    .grant_id    (winner_id),
    .last_grant  (last_grant_vec)
);
```

3.7 Design Notes

3.7.1 Usage Notes

- Out of reset, client 0 is king — the first all-requesting grant goes to client 0.
- There's no central pointer, no mask register. Just the pairwise bits updating themselves, locally. Synthesis tools eat this structure up.
- The grant is registered, so a lone requester waits exactly one cycle after raising its request.
- Sizing honesty: lovely at small `CLIENTS`. At large counts the `O(CLIENTS²)` matrix storage and the cross-client comparison web get fat, and you'll feel it in both area and timing. Dally & Towles land on the same conclusion.

3.8 Related Modules

Instantiated by: (none yet)

See also: - `arbiter_round_robin` — pointer/mask-based round-robin arbiter - `arbiter_round_robin_simple` — minimal rotating-priority arbiter - `arbiter_priority_encoder` — fixed-priority encoder used by the round-robin variants

---

3.9 Testing

`val/common/test_arbiter_matrix.py` puts the module through its paces. It validates:

- One-hot or zero grant, never granted to an unrequested client
- Strong fairness over `4·CLIENTS` consecutive grants with all clients requesting
- Correct rotation from cold start and after partial-request idle gaps
- Immediate service for a single requester regardless of matrix state
- Reset restoring client 0 as the highest-priority requester
- In ACK mode: hold-until-owner-ack, foreign-ack immunity, back-to-back same-cycle handoff, and single-requester re-grant
- Parameterization across `CLIENTS = 2, 4, 5, 8` in no-ACK mode, plus ACK mode at `REG_LEVEL=GATE` and `FULL`

```
source env_python
pytest val/common/test_arbiter_matrix.py -v
```

The testbench (`bin/TBClasses/common/arbiter_matrix_tb.py`) sticks to the same `ArbiterMaster` / `ArbiterMonitor` methodology as the round-robin testbench:

- A `RoundRobinArbiterMonitor` is attached because the observable rotation of the matrix arbiter is equivalent to round-robin.
- Randomized scenario phases (grant-signals, basic-arbitration, walking-requests, block-arb, bursty-traffic, rapid-request-changes, single-client-saturation, fairness) drive stimulus through `ArbiterMaster` and assert one-hot, no-unrequested-grant, and rotation behavior through the monitor.
- Deterministic directed supplements cover corner cases the randomized master/monitor cannot express cleanly: one-hot/request-match, strong-fairness-continuous, fairness-cold-start-and-gaps, single-requester-immediate, and reset-order. ACK-mode equivalents are exercised separately (hold-until-owner-ack, foreign-ack immunity, back-to-back handoff, single-requester).
- `ArbiterCompliance` protocol checking is enabled and asserted at the end of the run.

- Directed tests shut the master down via `enter_directed_mode()` / `exit_directed_mode()` so the testbench pokes the DUT signals directly without interference.
- Verification passes at `REG_LEVEL=GATE`, `FUNC`, and `FULL` across `CLIENTS = 2, 4, 5, 8` (no-ACK), plus ACK mode at `GATE` and `FULL`.

3.10 References

- No peer paper — the matrix (least-recently-served) arbiter is a standard technique with no defensible single origin (marked "standard technique" in the references index).
- Dally, W.J., Towles, B. *Principles and Practices of Interconnection Networks*, Morgan Kaufmann, 2004, ch. 18 "Arbitration", §18.5 "Matrix Arbiter". Backs the grant/update scheme and the observation that matrix arbiters are well suited to small arbiters.
- Indexed in `rtl/common/references/README.md`

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4 Weighted Matrix Arbiter

Weighted fairness, matrix style. Every client gets a credit counter, and the priority matrix only rotates after the current winner has burned its full weight in grants — so a client with weight `W` can string together up to `W` consecutive wins before it heads to the back of the line.

4.1 Overview

This module takes the matrix arbiter from Dally & Towles, *Principles and Practices of Interconnection Networks*, chapter 18 §18.5 and bolts on the weighted-arbiter semantics from §18.4.4. Priority still lives in a `CLIENTS×CLIENTS` bit matrix `M`:

- `M[i][j] = 1` means requester `i` currently outranks requester `j`.
- `M[i][j] = 0` means requester `i` is outranked by requester `j`.
- The diagonal is a don't-care.

As in the unweighted module, only the `CLIENTS·(CLIENTS-1)/2` upper-triangle bits are stored; the lower triangle derives combinationally. The difference: the update that drops the winner to last place waits until the winner's credit counter hits 0.

One naming note, same as the unweighted page: `N` is the bit width of `grant_id` — `$clog2(CLIENTS)`, guarded to 1 when `CLIENTS==1`. It's not the client count. The client count is always `CLIENTS`.

4.2 Parameters

Parameter	Type	Default	Description
MAX_LEVELS	int	16	Maximum weight value per client (must be ≥ 1)
CLIENTS	int	4	Number of requesting clients (must be ≥ 1)
WAIT_GNT_ACK	int	0	When 1, holds the grant until the winner acknowledges it
MAX_LEVELS_WIDTH	int	`$clog2(MAX_LEVELS)`	**Derived** credit counter width
N	int	`$clog2(CLIENTS)` (guarded)	**Derived** bit width of `grant_id`; do not override

4.3 Ports

4.3.1 Inputs

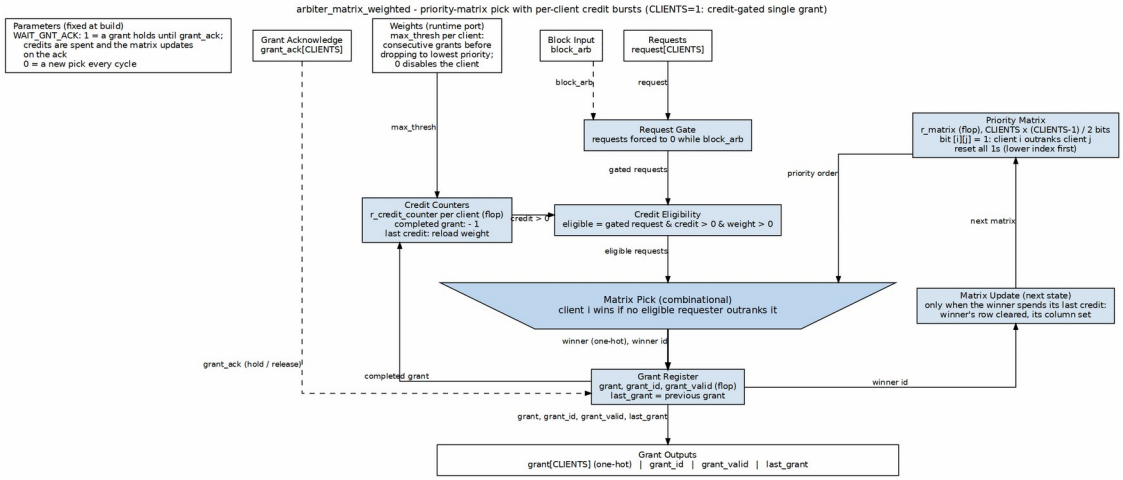
Port	Width	Description
clk	1	System clock
rst_n	1	Active-low asynchronous reset
block_arb	1	When asserted, masks all requests to zero and stops new grants
max_thresh	`CLIENTS * MAX_LEVELS_WIDTH`	Packed per-client weights `{weight[N-1], ..., weight[0]}`
request	CLIENTS	One-bit request per client
grant_ack	CLIENTS	Per-client grant acknowledgment (used when `WAIT_GNT_ACK = 1`)

4.3.2 Outputs

Port	Width	Description
grant_valid	1	High when a valid grant is being issued
grant	CLIENTS	One-hot grant vector (exactly one bit set when `grant_valid=1`)
grant_id	`N`	Binary-encoded ID of the granted client
last_grant	CLIENTS	Previous cycle's grant (for debugging/history)

4.4 Functional Description

4.4.1 Block diagram



arbiter_matrix_weighted block diagram

Requests drop down the center through the `block_arb` gate and the credit eligibility check into the matrix pick (the trapezium). The credit loop is on the left: `max_thresh` sets each client's weight, and each completed grant returns up the left edge to spend one credit. The priority loop is on the right: the winner id goes to the matrix update, which moves the winner to lowest priority only when it spends its last credit. A client therefore keeps top priority for `weight` consecutive grants before the others get their turn.

4.4.2 Matrix Representation

The stored state is the upper triangle of `M`, packed into `CLIENTS*(CLIENTS-1)/2` bits. Out of reset every stored bit is 1, so power-on priority is the plain total order:

```
0 > 1 > 2 > ... > CLIENTS-1
```

Client 0 wins everything until that first grant shuffles the deck.

4.4.3 Credit Counters and Weight Semantics

Each client owns a down-counter loaded from its `max_thresh` field. Eligibility is strict — weight non-zero **and** counter greater than 0:

- **Grant completion** decrements the winner's counter.
    - No-ACK mode: completion is the grant cycle itself.
    - ACK mode: completion is the cycle in which the winner's `grant_ack` bit is asserted.

- **Reload and rotation**: When a decrement drives a counter to 0, the counter is reloaded from the current `max_thresh` field and the matrix is updated for that winner.
- **Weight = 0**: The client is immediately disabled; its counter is forced to 0 and it is never eligible.
- **Dynamic weight changes**: A new weight takes effect at the next reload for that client (i.e., the next time its counter reaches 0). If the weight is increased from 0 while the counter is 0, the counter loads the new weight on the next cycle.

4.4.4 Grant Logic

Same vote as the unweighted module, just over the eligible set:

```
g[i] = eligible[i] AND (for all j != i: NOT(eligible[j] AND M[j][i]))
```

where `eligible[i] = request[i] AND block_arb==0 AND weight[i]>0 AND credit[i]>0`.

The matrix is still a tournament, so whenever at least one client is eligible, exactly one wins.

4.4.5 Matrix Update

When client `k` wins **and** that grant drives its counter to 0:

- **Row `k` cleared**: `M[k][j] = 0` for all `j > k` — `k` loses to all higher-index clients.
- **Column `k` set**: `M[i][k] = 1` for all `i < k` — all lower-index clients outrank `k`.

Client `k` drops to the lowest priority, and the new order becomes:

```
0 > 1 > ... > k-1 > k+1 > ... > CLIENTS-1 > k
```

Between rotations, the same winner keeps re-winning as long as it has credits and keeps asking. That's the entire trick — the matrix holds still through a client's burst.

4.4.6 Weighted Sequence Example

Concretely: `CLIENTS = 3`, weights `[3, 1, 2]`, everybody requesting forever. The grant order repeats:

```
0, 0, 0, 1, 2, 2
```

Three in a row for client 0, one for client 1, two for client 2. Notice the burstiness — that's what this scheme buys you. If you'd rather see shares interleave cycle by cycle, that's the round-robin weighted module's party trick, not this one.

4.4.7 No-ACK Mode (`WAIT_GNT_ACK = 0`)

One-cycle grants go out every cycle an eligible winner exists, and the matrix only moves when the winner's credits run dry. Request-to-grant latency is one cycle, plus a single credit-load bubble right after reset.

4.4.8 ACK Mode (`WAIT_GNT_ACK = 1`)

The grant holds until the granted client asserts its own `grant_ack` bit, and the matrix rotates only when that ACK exhausts the winner's credits — otherwise the same winner re-wins on the same-cycle handoff. The handoff works exactly like the unweighted module: next winner computed combinationally in the ACK cycle, registered on the same edge.

Foreign-ack immunity applies here too. Only the granted client's ack bit can retire the grant; everyone else's ACKs are noise.

One subtlety worth spelling out: a held grant only *completes* when it's ACK'd, and the handoff itself doesn't spend a credit. Credits track completed grants, not ACK transactions.

4.4.9 Block Arbitration

`block_arb` squashes all requests to zero, so no new grants issue while it's high. A held ACK-mode grant stays held until its ACK shows up — that's deliberate. Release the block and arbitration resumes from the current matrix and credit state.

4.5 Timing Characteristics

Registered outputs, same as the unweighted module: one cycle from request to grant, with the matrix and credit updates registered alongside.

ACK mode pulls the update into the ACK cycle, combinationally, so a continuous ACK stream produces back-to-back grants — though if an ACK doesn't exhaust the winner's credits you'll watch the same winner re-grant instead of the priority rotating. Still no idle cycles.

Same disclaimer as the unweighted page: no measured MHz or LUT numbers yet. Nothing's been run against a target device.

4.6 Usage Examples

```
localparam int NUM_CLIENTS = 4;
localparam int MAX_WEIGHT  = 16;
localparam int WEIGHT_W    = $clog2(MAX_WEIGHT);
```

```
logic [NUM_CLIENTS-1:0] req, gnt;
logic [WEIGHT_W-1:0] weights [NUM_CLIENTS];
logic [NUM_CLIENTS*WEIGHT_W-1:0] packed_weights;

// Pack weights: client 0 occupies the least-significant field.
assign packed_weights = {weights[3], weights[2], weights[1],
weights[0]};

arbiter_matrix_weighted #(
    .CLIENTS     (NUM_CLIENTS),
    .MAX_LEVELS  (MAX_WEIGHT),
    .WAIT_GNT_ACK(0)
) u_weighted_arbiter (
    .clk         (clk),
    .rst_n       (rst_n),
    .block_arb   (1'b0),
    .max_thresh  (packed_weights),
    .request     (req),
    .grant_ack   ('0),
    .grant_valid (grant_valid),
    .grant       (gnt),
    .grant_id    (winner_id),
    .last_grant  (last_grant_vec)
);
```

4.7 Design Notes

4.7.1 Usage Notes

- Out of reset, client 0 is king — the first all-requesting grant goes to client 0.
- Weight 0 disables a client immediately, leftover credits or not. The deviation note below explains why that's safe here.
- Weight changes aren't instant. They take effect at the client's next reload — the next time its counter reaches 0.
- No central pointer, no mask register. Just the pairwise bits, updating locally.
- The grant is registered, so a lone requester waits exactly one cycle after raising its request.

- Same sizing advice as the unweighted module: great at small `CLIENTS`. As the port count climbs, the `O(CLIENTS²)` matrix and comparison web get expensive, and you'll feel it in area and timing.

4.7.2 Deviation from `arbiter_round_robin_weighted`

So why doesn't weight-0 disable route through the multi-cycle `WEIGHT_BLOCK → DRAIN → UPDATE → STABILIZE` FSM that `arbiter_round_robin_weighted` uses? Because that FSM exists to protect the RR module's rotating pointer and request mask — mask a client mid-grant over there and you can leave stale state behind.

This datapath has no pointer and no mask. Eligibility is just `request[i] && weight[i] > 0 && credit[i] > 0`, and the matrix only rotates on a completed grant. So when a weight drops to 0 I force the credit counter to 0 and the client simply stops being eligible next cycle. Nothing stale, nothing to drain. The deviation is deliberate, and it's safe for exactly the reason the FSM exists on the other side.

4.8 Related Modules

Instantiated by: (none yet)

See also: - `arbiter_matrix` — unweighted matrix (distributed-state) fair arbiter - `arbiter_round_robin` — pointer/mask-based round-robin arbiter - `arbiter_round_robin_weighted` — weighted round-robin arbiter (credit-interleaved shares) - `arbiter_deficit_round_robin` — deficit round-robin arbiter (cost-proportional shares)

4.9 Testing

`val/common/test_arbiter_matrix_weighted.py` puts the module through its paces. It validates:

- Documented weighted grant sequences (e.g., `CLIENTS=3`, weights `[3,1,2]` → `[0,0,0,1,2,2]`)
- One-hot or zero grant, never granted to an unrequested or credit-ineligible client
- Weight = 0 disables the client
- Dynamic weight changes take effect at the next credit reload
- Immediate service for a single requester regardless of matrix state
- Reset restoring client 0 as the highest-priority requester

- `block_arb` pauses new grants
- In ACK mode: hold-until-owner-ack, foreign-ack immunity, back-to-back same-cycle handoff, and single-requester re-grant
- Parameterization across `CLIENTS = 1, 2, 4, 5, 8` in no-ACK mode, plus ACK mode at `REG_LEVEL=GATE` and `FULL`
- Dedicated single-client (`CLIENTS==1`) directed tests for both ACK and no-ACK modes

```
source env_python
pytest val/common/test_arbiter_matrix_weighted.py -v
```

The testbench (`bin/TBClasses/common/arbiter_matrix_weighted_tb.py`) sticks to the same `ArbiterMaster` / `ArbiterMonitor` methodology as the round-robin testbench:

- A `WeightedRoundRobinArbiterMonitor` is attached with `is_weighted=True` so it tracks `max_thresh` and runs WRR compliance checks.
- Randomized scenario phases (grant-signals, basic-arbitration, walking-requests, block-arb, bursty-traffic, rapid-request-changes, single-client-saturation) drive stimulus through the weighted `ArbiterMaster` and assert one-hot, no-unrequested-grant, and weighted behavior through the monitor.
- The weighted-distribution phase resets the DUT and uses `ArbiterMaster` manual control to hold all requests continuously (with auto-ACK in ACK mode). This produces a clean saturation stimulus so the monitor's weight-compliance window sees the expected share ratio.
- Deterministic directed supplements cover weighted sequences, zero-weight, dynamic weight changes, idle gaps, mid-credit idle, single-requester immediacy, single-client no-ACK/ACK behavior, reset-order, and ACK-mode protocol.
- `ArbiterCompliance` protocol checking is enabled and asserted at the end of the run.
- Directed tests shut the master down via `enter_directed_mode()` / `exit_directed_mode()` so the testbench pokes the DUT signals directly without interference.

- Verification passes at `REG_LEVEL=GATE`, `FUNC`, and `FULL` across `CLIENTS = 1, 2, 4, 5, 8` in no-ACK mode, plus ACK mode at `GATE` and `FULL`.

4.10 References

- No peer paper — the weighted matrix arbiter is a standard technique with no defensible single origin (marked “standard technique” in the references index).
- Katevenis, M., Sidiropoulos, S., Courcoubetis, C. “Weighted Round-Robin Cell Multiplexing in a General-Purpose ATM Switch Chip.” *IEEE Journal on Selected Areas in Communications*, 9(8), Oct. 1991. DOI: 10.1109/49.105173 — the seminal hardware weighted round-robin with interleaved (not bursted) per-weight service, the behavior this module’s credit scheme produces.
- Dally, W.J., Towles, B. *Principles and Practices of Interconnection Networks*, Morgan Kaufmann, 2004, ch. 18 “Arbitration”, §18.5 “Matrix Arbiter”. Backs the grant/update scheme and the observation that matrix arbiters are well suited to small arbiters.
- Dally, W.J., Towles, B. *Principles and Practices of Interconnection Networks*, Morgan Kaufmann, 2004, ch. 18 “Arbitration”, §18.4.4 “Weighted Round-Robin Arbiter”. Backs the credit-based weighting semantics and the trade-off between controlled long-term shares and short-term burstiness.
- Indexed in `rtl/common/references/README.md`

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5 Priority Encoder Arbiter

A high-speed combinational priority encoder built for arbitration. Priority is fixed: lower-indexed clients always have higher priority.

5.1 Overview

The `arbiter_priority_encoder` module gives you single-cycle priority encoding, with optimized implementations for the common client counts (4, 8, 16, 32) built on fully unrolled casez logic. For other client counts, it falls back to a synthesizable loop-based approach. You also get dual request inputs (masked and unmasked) for use in more elaborate arbitration schemes — the round-robin arbiter in this library is built on exactly that trick.

```
module arbiter_priority_encoder #(
    parameter int CLIENTS = 4,          // Number of requesting
clients
    parameter int N = $clog2(CLIENTS)   // Winner ID width (derived)
) (
    input  logic [CLIENTS-1:0]  requests_masked,      // Masked
request vector
    input  logic [CLIENTS-1:0]  requests_unmasked,    // Unmasked
request vector
    input  logic                any_masked_requests,  // Select masked
vs unmasked
    output logic [N-1:0]        winner,               // Binary-
encoded winner ID
    output logic                winner_valid          // Winner output
is valid
);
```

5.2 Parameters

5.2.1 User-Settable Parameters

Parameter	Type	Default	Description
CLIENTS	int	4	Number of requesting clients (range: 2-1024)

5.2.2 Derived Parameters

Parameter	Default	Description
N	$clog2(CLIENTS)	Winner ID width in bits

Note: `N` defaults to `$clog2(CLIENTS)` and should be left alone – but it is an ordinary `parameter`, not a `localparam`, so it *can* be overridden at instantiation. It is declared that way so it can size the `winner` port; `arbiter_round_robin` relies on this and passes `.N(N)` explicitly.

5.3 Ports

5.3.1 Inputs

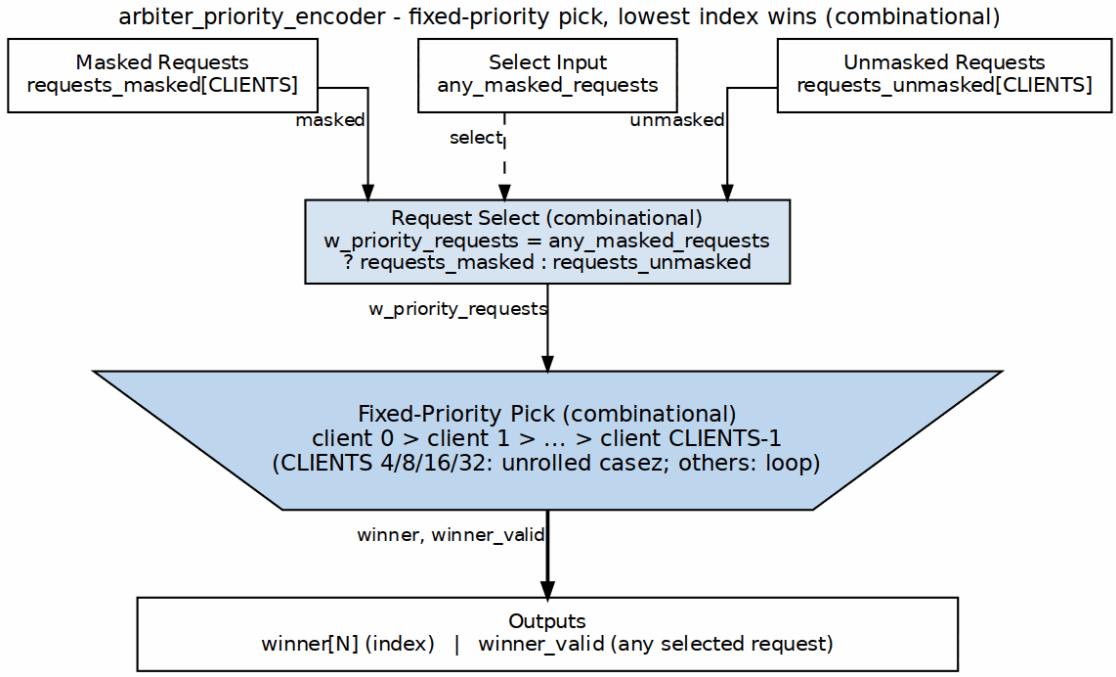
Port	Width	Description
requests_masked	CLIENTS	Masked request vector (higher priority source)
requests_unmasked	CLIENTS	Unmasked request vector (fallback source)
any_masked_requests	1	Select masked (1) or unmasked (0) requests

5.3.2 Outputs

Port	Width	Description
winner	N	Binary-encoded winner client ID
winner_valid	1	Asserted when at least one request is active

5.4 Functional Description

5.4.1 Block diagram



arbiter_priority_encoder block diagram

The block is purely combinational. Two request vectors enter at the top, and the caller's `any_masked_requests` selects one of them; because the caller drives the select, it is a plain box, not a pick. The fixed-priority pick is the trapezium: the lowest-numbered requesting client wins. `winner` and `winner_valid` leave its narrow edge into the outputs. `CLIENTS` changes only the implementation (unrolled `casez` for 4, 8, 16 and 32, a loop otherwise), not the behavior.

5.4.2 Priority Order

The module implements **fixed priority** arbitration: - **Client 0**: Highest priority (always wins if requesting) - **Client 1**: Second highest priority - **...** - **Client CLIENTS-1**: Lowest priority

5.4.3 Request Selection

The module evaluates requests in priority order:

1. **If `any_masked_requests = 1`**: Use `requests_masked` input
2. **If `any_masked_requests = 0`**: Use `requests_unmasked` input
3. **Find lowest-indexed asserted bit** in selected request vector
4. **If no requests**: `winner = 0` (don't care), `winner_valid = 0`

5.4.4 Combinational Operation

- **Zero cycles latency**: Purely combinational logic
- **No clock or reset**: Outputs update immediately with input changes
- **Single-cycle response**: Winner determined in one combinational path

5.4.5 Fixed Priority vs Round-Robin

Worth knowing what you're signing up for when you pick fixed priority:

Aspect	Fixed Priority	Round-Robin
Fairness	No (low-indexed clients favored)	Yes (all clients equal)
Latency	Deterministic for high-priority	Variable
Starvation	Possible for low-priority	Prevented
Simplicity	Very simple (this module)	More complex
Best Use	Interrupt handling, critical tasks	Data paths, fair scheduling

5.4.6 Priority Inversion

Issue: High-priority client blocked by low-priority client holding resource

Mitigation Strategies: 1. **Priority Inheritance**: Temporarily boost blocked client's priority 2. **Resource Preemption**: Force low-priority client to release resource 3. **Priority Ceiling**: Limit maximum priority of resource holders

5.4.7 Optimized Implementations

The module picks its implementation based on client count:

Power-of-2 Counts (4, 8, 16, 32):

```
// Fully unrolled casez logic for maximum performance
always_comb begin
    casez (w_priority_requests)
        4'b???1: begin winner = 2'd0; winner_valid = 1'b1; end  //
Client 0
        4'b??10: begin winner = 2'd1; winner_valid = 1'b1; end  //
Client 1
        4'b?100: begin winner = 2'd2; winner_valid = 1'b1; end  //
Client 2
        4'b1000: begin winner = 2'd3; winner_valid = 1'b1; end  //
Client 3
        default: begin winner = 2'd0; winner_valid = 1'b0; end
```

```
    endcase
end
```

Other Client Counts:

```
// Generic loop-based implementation
always_comb begin
    winner = '0;
    winner_valid = 1'b0;

    // Find lowest-indexed requesting client
    for (int i = 0; i < CLIENTS; i++) begin
        if (w_priority_requests[i] && !winner_valid) begin
            winner = i[N-1:0];
            winner_valid = 1'b1;
        end
    end
end
```

5.4.8 Priority Selection Logic

```
// Internal priority request selection
logic [CLIENTS-1:0] w_priority_requests;
assign w_priority_requests = any_masked_requests ?
                             requests_masked :
                             requests_unmasked;
```

5.5 Timing Characteristics

5.5.1 Latency and Propagation

Metric	Optimized (4,8,16,32)	Generic (<64)	Generic (>64)
Latency	0 cycles (combinational)	0 cycles	0 cycles
Propagation Delay	1-2 LUT levels	2-3 LUT levels	3-4 LUT levels
Max Frequency	>500 MHz (FPGA)	>400 MHz	>300 MHz

5.5.2 Resource Utilization

Client Count	Implementation	LUTs (Est.)	Area Complexity
4	Optimized	~8	O(CLIENTS)
8	Optimized	~16	O(CLIENTS)

Client Count	Implementation	LUTs (Est.)	Area Complexity
16	Optimized	~32	O(CLIENTS)
32	Optimized	~64	O(CLIENTS)
Other	Generic	~CLIENTS×log(CLIENTS)	O(CLIENTS×log(CLIENTS))

5.6 Usage Examples

5.6.1 Standalone Priority Encoder

```
// Simple fixed-priority arbiter (single request input)
arbiter_priority_encoder #(
    .CLIENTS(8)
) u_pri_enc (
    .requests_masked   (8'b0),          // Unused
    .requests_unmasked (client_reqs),    // Primary request vector
    .any_masked_requests(1'b0),         // Always use unmasked
    .winner            (winner_id),
    .winner_valid      (grant_valid)
);

// Convert winner ID to one-hot grant signal
logic [7:0] grant_one_hot;
always_comb begin
    grant_one_hot = '0;
    if (grant_valid) begin
        grant_one_hot[winner_id] = 1'b1;
    end
end
```

5.6.2 Dual-Input Priority Encoder

```
// Masked + unmasked request inputs
arbiter_priority_encoder #(
    .CLIENTS(16)
) u_pri_enc_dual (
    .requests_masked    (masked_reqs),   // Higher priority requests
    .requests_unmasked  (all_reqs),      // All requests
    .any_masked_requests(|masked_reqs),  // Select masked if any
present
    .winner             (winner_id),
    .winner_valid       (grant_valid)
);
```

5.6.3 Integration in Round-Robin Arbiter

```
// Used within arbiter_round_robin.sv
// Masked requests = clients above last winner
// Unmasked requests = all clients
arbiter_priority_encoder #(
    .CLIENTS(CLIENTS)
) u_priority_encoder (
    .requests_masked    (w_requests_masked),
    .requests_unmasked  (w_requests_unmasked),
    .any_masked_requests(w_any_masked_requests),
    .winner             (w_winner),
    .winner_valid       (w_winner_valid)
);
```

5.6.4 Multi-Level Priority Groups

```
// Implement priority groups: High (C0-C3), Med (C4-C7), Low (C8-C11)
logic [3:0] high_reqs, med_reqs, low_reqs;
logic [1:0] high_winner, med_winner, low_winner;
logic high_valid, med_valid, low_valid;

// Separate encoders for each priority group
arbiter_priority_encoder #(.CLIENTS(4)) u_high_pri (
    .requests_masked(4'b0),
    .requests_unmasked(high_reqs),
    .any_masked_requests(1'b0),
    .winner(high_winner),
    .winner_valid(high_valid)
);

arbiter_priority_encoder #(.CLIENTS(4)) u_med_pri (
    .requests_masked(4'b0),
    .requests_unmasked(med_reqs),
    .any_masked_requests(1'b0),
    .winner(med_winner),
    .winner_valid(med_valid)
);

arbiter_priority_encoder #(.CLIENTS(4)) u_low_pri (
    .requests_masked(4'b0),
    .requests_unmasked(low_reqs),
    .any_masked_requests(1'b0),
    .winner(low_winner),
    .winner_valid(low_valid)
);

// Select winner from highest priority group with requests
```

```
logic [3:0] final_winner;
always_comb begin
    if (high_valid)      final_winner = {2'b00, high_winner};
    else if (med_valid)  final_winner = {2'b01, med_winner};
    else if (low_valid)  final_winner = {2'b10, low_winner};
    else                 final_winner = 4'b0;
end
```

5.6.5 Example Priority Scenarios

Four cases worth tracing by hand before you trust the thing in a bigger design.

5.6.5.1 Scenario 1: Single Requester (Client 2)

```
requests_unmasked = 8'b0000_0100
any_masked_requests = 0

→ winner = 3'd2
→ winner_valid = 1
```

5.6.5.2 Scenario 2: Multiple Requesters (Clients 0, 2, 5)

```
requests_unmasked = 8'b0010_0101
any_masked_requests = 0

→ winner = 3'd0  (lowest index wins)
→ winner_valid = 1
```

5.6.5.3 Scenario 3: Masked Priority (Clients 3, 6 masked)

```
requests_masked   = 8'b0100_1000
requests_unmasked = 8'b0110_1101
any_masked_requests = 1

→ winner = 3'd3  (from masked, index 3 < 6)
→ winner_valid = 1
```

5.6.5.4 Scenario 4: No Requests

```
requests_unmasked = 8'b0000_0000
any_masked_requests = 0

→ winner = 3'd0 (don't care)
→ winner_valid = 0
```

5.7 Design Notes

5.7.1 When to Use

Appropriate Use Cases: - Fixed-priority arbitration systems - Interrupt controllers - Bus arbitration with priority levels - Resource allocation with service levels - Component within round-robin or weighted arbiters

5.7.2 Performance Optimization

For High-Speed Designs: - Use power-of-2 client counts (4, 8, 16, 32) for optimized implementation - Pipeline outputs if critical path is too long - Consider hierarchical encoding for >32 clients

For Area Optimization: - Non-power-of-2 counts use smaller generic implementation - Share encoder between multiple arbitration paths - Use smallest practical CLIENTS parameter

5.7.3 Synthesis Considerations

Timing Optimization:

```
// Pipeline option for high-speed designs
logic [N-1:0] r_winner;
logic r_winner_valid;

always_ff @(posedge clk) begin
    r_winner <= winner;
    r_winner_valid <= winner_valid;
end
```

Resource Optimization: - Optimized implementations use casez (efficient mux tree) - Generic implementation uses priority loop (more logic levels) - For CLIENTS > 32, consider hierarchical structure

5.7.4 Common Pitfalls

Here's the part that bites people:

Anti-Pattern 1: Assuming fairness

```
// WRONG: Fixed priority is NOT fair
// Client 0 will always win if requesting
```

Anti-Pattern 2: Ignoring winner_valid

```
// WRONG: Using winner when no requests
if (winner == 2'd0) ...  // Ambiguous: valid client 0 or no requests?

// RIGHT: Check winner_valid first
if (winner_valid && winner == 2'd0) ...
```

Anti-Pattern 3: Mixing priority schemes

```
// WRONG: Trying to implement round-robin with priority encoder alone
// Use arbiter_round_robin.sv instead
```

5.8 Related Modules

- **arbiter_round_robin.sv** - Uses this module internally for winner selection
- **arbiter_round_robin_simple.sv** - Alternative simplified arbiter
- **arbiter_round_robin_weighted.sv** - Weighted arbitration with QoS
- **encoder.sv** - General-purpose binary encoder
- **priority_encoder.sv** - Alternative naming (may be same module)

5.9 Testing

From the test suite (`val/common/test_arbiter_priority_encoder.py`):

- **Coverage**: 95%
- **Key Test Scenarios**:
  - Priority order verification (all bit patterns)
  - Masked vs unmasked selection
  - No requests (winner_valid = 0)
  - All clients requesting simultaneously
  - Single client at a time
  - Power-of-2 sizes (4, 8, 16, 32)
  - Non-power-of-2 sizes (5, 12)

Test Command: `pytest val/common/test_arbiter_priority_encoder.py -v`

5.10 References

- No peer paper — fixed-priority (lowest index wins) encoding is a standard technique with no defensible single origin (marked “standard technique” in the references index).

- Gupta, P., McKeown, N. “Designing and Implementing a Fast Crossbar Scheduler.” *IEEE Micro*, 19(1), Jan.–Feb. 1999. DOI: 10.1109/40.748793 — the programmable priority encoder built from two fixed-priority encoders, one fed thermometer-masked requests and one fed unmasked requests; the same masked/unmasked dual-encoder structure in which this encoder serves as the masked and unmasked halves.
- Dally, W.J., Towles, B. *Principles and Practices of Interconnection Networks*, Morgan Kaufmann, 2004, ch. 18 “Arbitration”, §18.3 “Fixed Priority Arbiter” — textbook treatment (secondary source).
- Indexed in `rtl/common/references/README.md`

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6 Round Robin Arbiter

Fair round-robin arbitration for multiple clients — every requesting client gets equal access to the shared resource, and no client starves.

6.1 Overview

The `arbiter_round_robin` module cycles through requesting clients in order, so access to the shared resource stays fair over time. It is built around a pre-computed mask lookup table (no logic cost at runtime) and delegates winner selection to the `arbiter_priority_encoder` submodule.

6.2 Parameters

Parameter	Type	Default	Description
CLIENTS	int	4	Number of clients
WAIT_GNT_ACK	int	0	When set to 1, waits for grant acknowledgment before moving to next client
N	int	$clog2(CLIENTS)	Internal parameter for address width calculation

6.3 Ports

6.3.1 Inputs

Port	Width	Description
clk	1	System clock
rst_n	1	Active-low asynchronous reset
block_arb	1	When asserted, blocks all arbitration (forces no grants)
request	CLIENTS	Request signals from each client
grant_ack	CLIENTS	Grant acknowledgment signals (used when `WAIT_GNT_ACK = 1`)

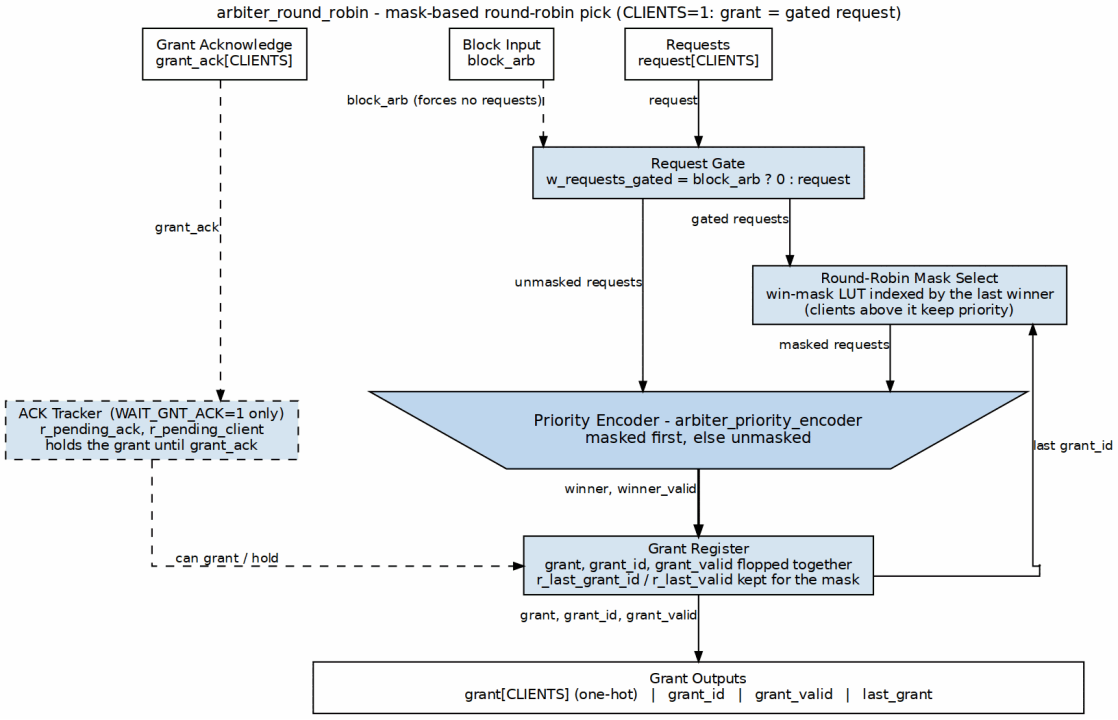
6.3.2 Outputs

Port	Width	Description
grant_valid	1	Indicates when a valid grant is being

Port	Width	Description
		issued
grant	CLIENTS	One-hot encoded grant signals
grant_id	N	Binary encoded ID of the granted client
last_grant	CLIENTS	Previous cycle's grant (for debugging/history)

6.4 Functional Description

6.4.1 Block diagram



arbiter_round_robin block diagram

Requests come down the center. `block_arb` gates them all to zero, and the gated set reaches the priority encoder twice: unmasked, and through the round-robin mask, which keeps only the clients above the last winner. The encoder (`arbiter_priority_encoder`) picks the lowest masked request, or the lowest unmasked one if no masked request is left. The grant register flops `grant`, `grant_id` and `grant_valid` together, and the last winner climbs back up the right edge to select the next mask. With `WAIT_GNT_ACK=1`, the dashed ACK tracker holds each grant until its `grant_ack` arrives.

6.4.2 Fair Round-Robin Operation

The rotation order is `0 → 1 → 2 → ... → (CLIENTS-1) → 0` (see the RTL header). The algorithm ensures fairness by: 1. After serving client `i`, giving priority to the **higher-indexed** clients (`i+1 .. CLIENTS-1`) via `w_win_mask_decode[i]` 2. Checking those masked (higher-index) requests first 3. When no higher-indexed clients are requesting, wrapping around to the **lowest** indices (client 0 upward)

6.4.3 Block Arbitration

Assert `block_arb` and all requests are masked to zero — arbitration is effectively disabled.

A blocked interval RESETS the rotation. This is the part that surprises people. `r_last_valid` follows `grant_valid` every cycle, and a block produces no grants, so `r_last_valid` falls to 0 and the mask select drops to its third branch:

```
assign w_curr_mask_decode = grant_valid  ? w_win_mask_decode[grant_id]
:
                            r_last_valid ?
w_win_mask_decode[r_last_grant_id] :
                                           CLIENTS'(1);
```

`CLIENTS'(1)` masks off everything except client 0, so **the first grant after `block_arb` releases goes to the lowest-numbered requester, not to the client that was next in line.** Traced on a 32-client instance: block released at 25060 ns, and the next two grants went to clients 0 and 3 while the pre-block rotation had reached the high teens.

If you need fairness to survive a blocked interval, `block_arb` is the wrong tool — gate the requests upstream instead, so the arbiter keeps seeing a valid grant history.

6.4.4 Grant Acknowledgment Support

When `WAIT_GNT_ACK = 1`, the arbiter waits for the granted client to acknowledge receipt before updating internal state and moving to the next client.

6.4.5 Key Internal Signals

- `r_last_grant_id`: Tracks the last winner's client ID (smaller than full mask)
- `r_last_valid`: Indicates if last winner should be used for mask generation
- `r_pending_ack`: ACK mode state (indicates ACK pending)
- `r_pending_client`: Which client has pending ACK (only in ACK mode)
- `w_requests_masked`: Requests with priority mask applied
- `w_requests_unmasked`: Raw gated requests without masking

6.4.6 Priority Mechanism

The arbiter is built around a pre-computed mask lookup table:

1. **Mask Lookup Tables**: Pre-computed at elaboration time (no logic cost)
   - `w_win_mask_decode[i] = ~((1 << (i+1)) - 1)`: selects clients **i+1 and above** (all higher indices). After client `i` wins, this is the mask that gives priority to the next-higher clients. This is the only LUT the arbiter actually reads (via `w_curr_mask_decode`).
   - `w_mask_decode[i] = (1 << i) - 1`: covers clients **0 through i-1**. Note: this table is generated but **never read** in the current RTL — it is dead logic left from an earlier scheme, not part of the active datapath.
2. **Fast Request Preprocessing**: Single LUT level for request gating
   - Block arbitration immediately gates all requests
   - Masked and unmasked request vectors computed in parallel
3. **Last Winner Tracking**: Uses client ID instead of full one-hot mask
   - `r_last_grant_id`: More efficient than storing full grant vector
   - `r_last_valid`: Indicates if mask should be applied

6.4.7 Winner Selection

Winner selection is delegated to the `arbiter_priority_encoder` submodule: - Takes both masked and unmasked request vectors - Returns binary-encoded winner ID - Outputs validity signal

```
// Priority encoder selects highest priority requester
arbiter_priority_encoder #(.CLIENTS(CLIENTS), .N(N))
u_priority_encoder (
    .requests_masked    (w_requests_masked),
    .requests_unmasked  (w_requests_unmasked),
    .any_masked_requests(w_any_masked_requests),
    .winner             (w_winner),
    .winner_valid       (w_winner_valid)
);

// Grant decision with ACK permission check
assign w_should_grant = w_winner_valid && w_any_requests &&
w_can_grant;
```

6.4.8 Mask Update Logic

- **No-ACK Mode**: Mask updates immediately when grant issued (1-cycle round-robin)
- **ACK Mode**: Mask updates only when ACK received (prevents premature rotation)

6.5 Timing Characteristics

This module is **sequential**: it contains clocked logic (via `always_ff` or the repository's `ALWAYS_FF_RST` macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

6.6 Usage Examples

Every parameter and port below is taken from the module declaration.

```
arbiter_round_robin #(
    .CLIENTS                (4),
    .WAIT_GNT_ACK           (0),
    .N                      ($clog2(CLIENTS))
) u_arbiter_round_robin (
    .clk                    (clk),
    .rst_n                  (rst_n),
    .block_arb              (block_arb),
    .request                (request),
    .grant_ack              (grant_ack),
    .grant_valid            (grant_valid),
    .grant                  (grant),
    .grant_id               (grant_id),
    .last_grant             (last_grant)
);
```

6.7 Design Notes

6.7.1 Usage Notes

- On the first grant after reset (no last winner), the mask defaults to client 0, so the lowest-indexed requester wins ties initially; thereafter priority rotates upward from the last winner
- The round-robin nature ensures long-term fairness across all clients
- Grant acknowledgment feature is useful in systems where the granted client needs time to process the grant

6.8 Related Modules

Read out of the RTL, not curated: these are the modules this one instantiates and the modules that instantiate it.

Instantiates: - `arbiter_priority_encoder`

Instantiated by: - `arbiter_deficit_round_robin` - `arbiter_round_robin_weighted` - `arbiter_rr_pwm_monbus` - `axi_read_engine` - `axi_read_engine_beats` - `axi_write_engine`

6.9 Testing

`val/common/test_arbiter_round_robin.py` exercises this module. It collects 6 parameter cases at the default `REG_LEVEL`.

```
source env_python
pytest val/common/test_arbiter_round_robin.py -v
```

6.10 References

- Gupta, P., McKeown, N. “Designing and Implementing a Fast Crossbar Scheduler.” *IEEE Micro*, 19(1), Jan.–Feb. 1999. DOI: 10.1109/40.748793 — the programmable priority encoder built from two fixed-priority encoders, one fed thermometer-masked requests and one fed unmasked requests; the same masked/unmasked dual-encoder structure this module uses for its masked/unmasked request paths.
- Dally, W.J., Towles, B. *Principles and Practices of Interconnection Networks*, Morgan Kaufmann, 2004, ch. 18 “Arbitration”, §18.4.2 “Round-Robin

Arbiter" and §18.4.3 "Grant-Hold Circuit" (the hold-until-ACK mode) — textbook treatment (secondary source).

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← Back to Main Index | **rtl-common Index** # Simple Round-Robin Arbiter with Grant/ACK

A rotating-priority arbiter whose grant is registered and held until the granted agent acknowledges it, with back-to-back handoff on the acknowledge cycle.

6.12 Overview

`arbiter_round_robin_simple_ack` uses the same priority datapath as arbiter_round_robin_simple. A win-mask keeps only the agents above the last winner eligible. If none of them is requesting, the scan falls back to the unmasked request vector, and a lowest-set-bit isolate picks the winner. The difference is the handshake. The grant is a flop, it stays asserted (one-hot) until the owner returns `grant_ack`, and only then does the arbiter move on. Use it when the granted agent needs the resource for a variable number of cycles, and the agent, not the arbiter, decides when the transfer is over. It is a simplified, educational module: the clearest place in the library to see a held-grant handshake in isolation.

```
module arbiter_round_robin_simple_ack #(
    parameter int unsigned N = 4,
    parameter int unsigned W = (N > 1) ? $clog2(N) : 1
) (
    input  logic          clk,
    input  logic          rst_n,         // active-low reset
    input  logic [N-1:0]  request,       // request bits [N-1:0]
    input  logic [N-1:0]  grant_ack,     // per-agent ack; only the
granted agent's bit counts
    output logic          grant_valid,   // a grant is outstanding
    output logic [N-1:0]  grant,         // one-hot grant, held until
acked
    output logic [W-1:0]  grant_id       // encoded grant (undef if
grant_valid==0)
);
```

6.12.1 Key Features

- **Grant held until ACK**: once issued, `grant` does not change until `grant_ack[grant_id]` is sampled high
- **Back-to-back handoff**: on the ack cycle the arbiter re-arbitrates among the other requesters, so a waiting agent is granted on the next cycle with no bubble
- **No re-grant in the ack cycle**: if only the acked agent (or nobody) is requesting, the grant clears for one cycle
- **Stray acks ignored**: `grant_ack` bits from agents that do not hold the grant have no effect
- **N=1 elaborates**: `W` is guarded so `$clog2(1) = 0` does not produce a `[-1:0]` vector

6.13 Parameters

Parameter	Type	Default	Description
N	int unsigned	4	Number of requesting agents (N >= 1)
W	int unsigned	`(N > 1) ? $clog2(N) : 1`	Width of `grant_id`. Derived from N – leave it at the default; it is a parameter rather than a localparam only so it can be referenced in the port list.

6.14 Ports

6.14.1 Inputs

Port	Width	Description
clk	1	Clock signal
rst_n	1	Active-low asynchronous reset
request	N	Request bits from N agents [N-1:0]
grant_ack	N	Per-agent acknowledge; only the bit of

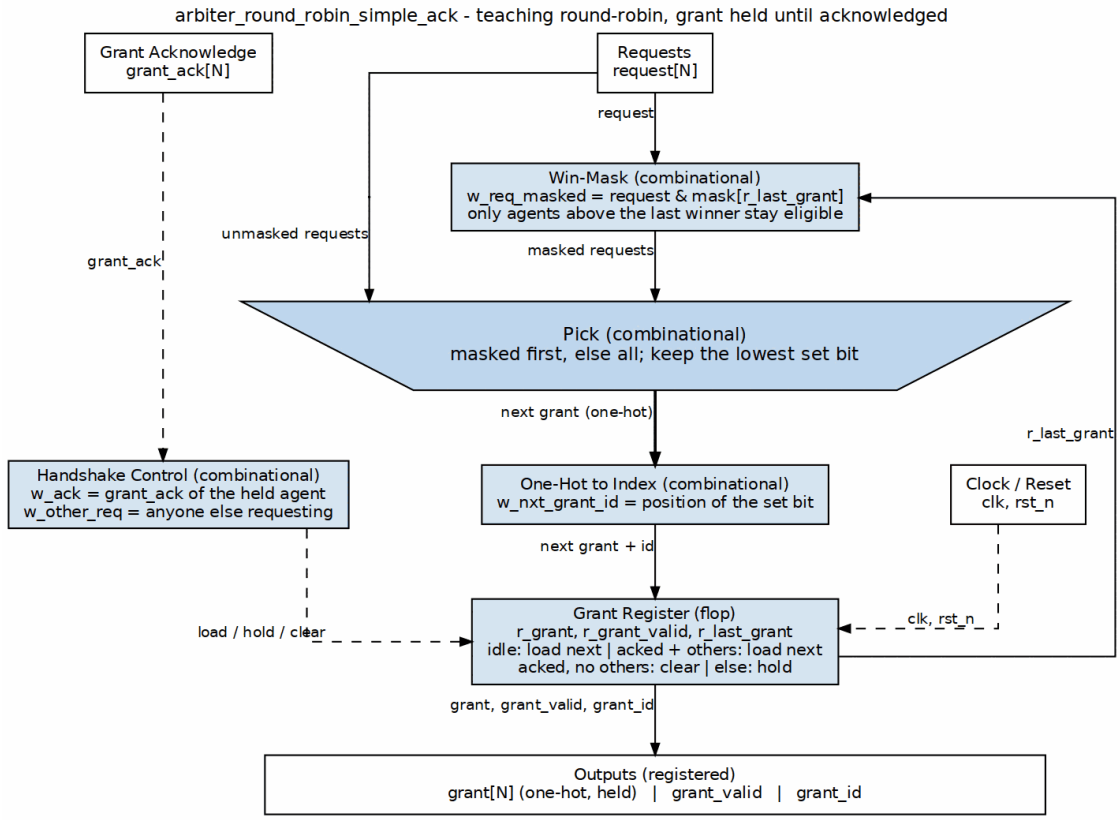
Port	Width	Description
		the currently granted agent is used

6.14.2 Outputs

Port	Width	Description
grant_valid	1	A grant is outstanding and waiting on its ack
grant	N	One-hot grant, registered, held until acked [N-1:0]
grant_id	W	Binary index of the held grant (undefined if grant_valid==0)

6.15 Functional Description

6.15.1 Block diagram



arbiter_round_robin_simple_ack block diagram

This is the teaching round-robin pick with the grant held in a register until the granted agent acknowledges it. The request path is the same as in `arbiter_round_robin_simple`: win-mask, then the pick (masked first, else all, lowest set bit), then the index encoder. The winner goes into the Grant Register instead of straight out, so the outputs are registered. `grant_ack` comes back as a dashed return into the Handshake Control, which tells the register to load the next winner, hold, or clear (the four rules are in the register's label). `r_last_grant` climbs the right edge back into the mask.

6.15.2 Priority Selection

The priority path is entirely combinational and is evaluated every cycle. It is only loaded into the grant flops when the arbiter is idle or on an ack:

```
// Agents above the last winner first; wrap to the full vector if none
of them requests.
assign w_req_masked = request & w_win_mask_decode[r_last_grant];
assign w_req_sel    = (|w_req_masked) ? w_req_masked : request;

// Isolate lowest set bit (one-hot).
assign w_nxt_grant  = w_req_sel & ((~w_req_sel) + {{(N-1){1'b0}},
1'b1});
```

`w_win_mask_decode[i]` is an elaboration-time constant, `~((1 << (i+1)) - 1)`: all agents strictly above `i`. After agent N-1 wins, the mask is all zero, so the scan restarts from agent 0.

6.15.3 Grant Lifecycle

The state is three registers: `r_grant` (one-hot), `r_grant_valid`, and `r_last_grant` (the index of the current or most recent winner, which also drives `grant_id`).

State	Condition	Next cycle
Idle (`grant_valid=0`)	any request	load the winner; `grant_valid=1`
Idle	no request	stay idle
Granted	owner's `grant_ack` low	hold `grant` unchanged
Granted	owner acks, another agent requests	load the next winner (back-to-back)
Granted	owner acks, no other agent requests	clear; `grant_valid=0` for at least one cycle

On the ack cycle the acked agent is the last winner, so it is masked out whenever an agent above it is requesting. When it is the highest requester, the fallback to the unmasked vector picks the lowest requester, which is below it. Either way the handoff never re-grants the agent that just acked.

6.15.4 Why the Grant Clears When Only the Acked Agent Requests

The ack completes the transfer. After it, the requester may legally drop its request, so re-granting it in the ack cycle would hand it a grant it may no longer want. This is the same contract as the `WAIT_GNT_ACK` mode of arbiter_round_robin. If the agent still has work, it is re-granted from the idle state one cycle later.

6.15.5 Request Rules

- An agent must hold `request` until it sees `grant`.
- The winner is captured in the cycle before the grant registers. A request dropped in that cycle is still granted and must still be acked; an un-acked grant is held forever.

6.15.6 Reset Behavior

- `r_last_grant` resets to N-1, so the first arbitration after reset starts from agent 0.
- `r_grant` and `r_grant_valid` reset to zero.

6.16 Timing Characteristics

This module is **sequential**: one `always_ff` block clocked on `clk` with active-low asynchronous reset `rst_n`. All three outputs come straight from flops.

Characteristic	Value
Request to grant (idle)	1 cycle
Ack to next grant (another agent waiting)	1 cycle, back-to-back
Ack to re-grant of the same agent (nobody else waiting)	2 cycles (one idle cycle)
Grant hold	until the owner's `grant_ack`

No synthesis frequency or area figures are quoted – none have been measured against a target device.

6.16.1 Handshake Sequence Example

N=3. All three agents request through cycle 4, then only agent 0; each owner acks in the cycle shown.

Cycle	request	grant_ack	grant_valid	grant	grant_id	Note
0	111	000	0	000	-	idle;

Cycle	request	grant_ack	grant_valid	grant	grant_id	Note
						winner agent 0 is loaded
1	111	000	1	001	0	agent 0 holds the grant
2	111	001	1	001	0	agent 0 acks; agent 1 wins the handoff
3	111	010	1	010	1	agent 1 acks in its first cycle
4	111	100	1	100	2	agent 2 acks; mask wraps to agent 0
5	001	001	1	001	0	agent 0 acks; nobody else requests
6	001	000	0	000	-	grant clears for one cycle

Cycle	request	grant_ack	grant_valid	grant	grant_id	Note
7	001	000	1	001	0	agent 0 re-granted from idle

6.17 Usage Examples

```
arbiter_round_robin_simple_ack #(
    .N(4)
) u_arb (
    .clk        (clk),
    .rst_n      (rst_n),
    .request    (req_bits),     // [3:0], held until granted
    .grant_ack  (ack_bits),     // [3:0], owner pulses its bit when
done
    .grant_valid(grant_valid),
    .grant      (grant_oh),     // [3:0] one-hot, held until acked
    .grant_id   (grant_bin)     // [1:0]
);
```

6.18 Design Notes

- **Choosing between the round-robin variants.** Use arbiter_round_robin_simple when a grant lasts exactly one cycle. Use this module when the owner holds the resource for a variable time and signals completion. Use arbiter_round_robin with `WAIT_GNT_ACK=1` when you also need its other features.
- **The idle cycle is deliberate.** It costs one cycle only when a single agent re-requests back-to-back with nobody else waiting. Under contention the handoff has no bubble.
- **`grant_id` is undefined while idle.** It holds the last winner's index; qualify it with `grant_valid`.
- **A teaching module.** This is a deliberately simplified example: it shows the grant/ack handshake on top of the smallest round-robin datapath, without the extra modes of `arbiter_round_robin`. It is meant to be read and experimented with, which is why no production RTL in the repository instantiates it.

6.19 Related Modules

- arbiter_round_robin_simple - Same priority datapath, single-cycle grant, no handshake
- arbiter_round_robin - Full round-robin arbiter; its `WAIT_GNT_ACK` mode defines the ack contract this module follows
- arbiter_single_client - Degenerate one-client arbiter with the same registered, ack-held grant
- arbiter_round_robin_weighted - Weighted / QoS round-robin variant
- arbiter_priority_encoder - Fixed-priority arbiter

Source: `rtl/common/arbiter_round_robin_simple_ack.sv` (filelist `rtl/common/filelists/arbiter_round_robin_simple_ack.f`).

6.20 Testing

From the test suite (`val/common/test_arbiter_round_robin_simple_ack.py`), parameterized over N:

- **GATE**: N = 4, 8
- **FUNC / FULL**: N = 2, 3, 4, 5, 6, 8, 16
- **Key Test Scenarios**:
  - ARB-01 to ARB-02: grant signal checks and basic arbitration
  - ARB-03 to ARB-05: request-set fairness, request-set changes, single-client saturation
  - ARB-06 to ARB-08: walking requests, bursty traffic, rapid request changes
  - ARB-09: grant held until the owner acks
  - ARB-10: back-to-back handoff on the ack cycle
  - ARB-11: dynamic arbitration liveness

Test Command: `make -C val/common clean-all && make -C val/common run-arbiter_round_robin_simple_ack-func`

6.21 References

- Gupta, P., McKeown, N. “Designing and Implementing a Fast Crossbar Scheduler.” *IEEE Micro*, 19(1), Jan.–Feb. 1999. DOI: 10.1109/40.748793 — the programmable priority encoder built from a masked and an unmasked fixed-priority encoder; the same dual-encoder structure as this module’s win-mask with fallback to the unmasked requests.
- Dally, W.J., Towles, B. *Principles and Practices of Interconnection Networks*, Morgan Kaufmann, 2004, ch. 18 “Arbitration”, §18.4.2 “Round-Robin Arbiter” and §18.4.3 “Grant-Hold Circuit” — textbook treatment of the rotating priority and of holding a grant until the owner releases it (secondary source).
- Indexed in `rtl/common/references/README.md`

6.22 Navigation

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7 Simple Round-Robin Arbiter

A generic rotating-priority arbiter that gives you fair arbitration among multiple requesting agents using a win-mask — no if/case ladders in the priority path.

7.1 Overview

The `arbiter_round_robin_simple` module implements round-robin arbitration with fair access to a shared resource among N requesting agents. Fairness comes from remembering the last granted agent and giving priority to the agents above it on the next arbitration. The grant is combinational and lasts one cycle; for a grant that is held until the owner acknowledges it, see arbiter_round_robin_simple_ack.

```
module arbiter_round_robin_simple #(
    parameter int unsigned N = 4,
    parameter int unsigned W = $clog2(N)
) (
    input  logic          clk,
    input  logic          rst_n,         // active-low reset
    input  logic [N-1:0]  request,       // request bits [N-1:0]
    output logic          grant_valid,   // any grant
    output logic [N-1:0]  grant,         // one-hot grant
    output logic [W-1:0]  grant_id       // encoded grant (undef if
grant_valid==0)
);
```

7.2 Parameters

Parameter	Type	Default	Description
N	int unsigned	4	Number of requesting agents (N >= 2; see note)
W	int unsigned	`$clog2(N)`	Width of `grant_id`. Derived from N – leave it at the default; it is a parameter rather than a localparam only so it can be referenced in the port list.

7.3 Ports

7.3.1 Inputs

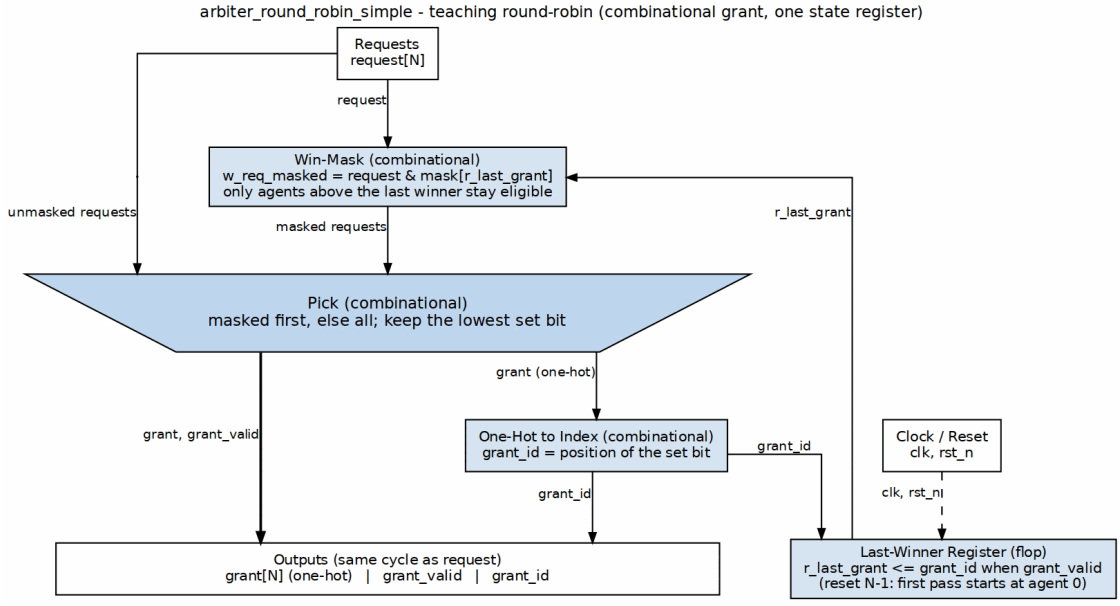
Port	Width	Description
clk	1	Clock signal
rst_n	1	Active-low asynchronous reset
request	N	Request bits from N agents [N-1:0]

7.3.2 Outputs

Port	Width	Description
grant_valid	1	Indicates if any grant is active
grant	N	One-hot grant output [N-1:0]
grant_id	$clog2(N)	Binary encoded grant ID (undefined if grant_valid==0)

7.4 Functional Description

7.4.1 Block diagram



arbiter_round_robin_simple block diagram

This is the teaching version of `arbiter_round_robin`, with nothing around the pick. Requests enter at the top and split two ways: the raw vector runs down the left as the unmasked requests, and the win-mask keeps only the agents above the last winner. The pick takes the masked set if it has any bit set, otherwise the whole vector, and keeps the lowest set bit. `grant` and `grant_valid` leave straight into the outputs, in the same cycle as the request, and the index encoder produces `grant_id`. The only state is `r_last_grant`, which loads the winner's index and climbs the right edge back into the mask.

7.4.2 Round-Robin Algorithm

The arbiter implements a rotating priority scheme with a mask rather than a rotation:

1. **State Memory**: `r_last_grant` holds the index of the last granted agent
2. **Win-Mask**: requests are ANDed with a mask that keeps only the agents strictly above the last winner
3. **Wrap**: if no masked request remains, the unmasked request vector is used instead
4. **Priority Selection**: lowest-set-bit isolation (`x & (~x + 1)`) picks the winner in one step

7.4.3 Key Features

- **Fair Arbitration**: Ensures all requesting agents get equal opportunity over time
- **Efficient Logic**: Uses bit manipulation instead of complex if/case structures
- **Single-Cycle Response**: Combinational grant generation with registered state update
- **Scalable**: Parameterizable for any number of agents (N >= 2). At N=1, `$clog2(1) = 0` collapses the shift/ID arithmetic into zero-size casts and zero replications that strict front ends reject (the same degeneracy the arbiter_single_client page documents for the base arbiter) — use `arbiter_single_client` for the one-agent case

7.4.4 Reset Behavior

- **Power-on Reset**: `r_last_grant` initialized to N-1, causing first arbitration to start from agent 0
- **Active Reset**: All state registers cleared, grant outputs become invalid

- **Reset Release**: Next arbitration cycle begins with agent 0 having highest priority

7.4.5 Win-Mask Logic

```
// After agent i wins, only agents above i stay eligible (elaboration-
time constants).
for (genvar i = 0; i < N; i++) begin : gen_mask_lut
    assign w_win_mask_decode[i] = ~((N'(1) << (i + 1)) - N'(1));
end

// Agents above the last winner first; if none of them is requesting,
wrap to
// the full request vector.
assign w_req_masked = request & w_win_mask_decode[r_last_grant];
assign w_any_masked = |w_req_masked;
assign w_req_sel    = w_any_masked ? w_req_masked : request;
```

After agent N-1 wins, its mask is all zero, so the fallback to the unmasked vector restarts the scan from agent 0. This is the same mask-plus-fallback structure as the programmable priority encoder of Gupta & McKeown (see References), and the same decode as `arbiter_round_robin`. Unlike a rotate-based round-robin, nothing is shifted, so the request vector never has to be rotated into place and the result rotated back.

7.4.6 Priority Selection

The module uses lowest-set-bit isolation for efficient one-hot selection:

```
// Isolate lowest set bit (one-hot)
w_nxt_grant = w_req_sel & ((~w_req_sel) + {{(N-1){1'b0}}, 1'b1});
```

7.4.7 Grant Encoding

A compact one-hot to binary encoder generates the grant ID:

```
always_comb begin
    w_grant_id = r_last_grant; // default to last grant
    for (int i = 0; i < N; i++) begin
        if (w_nxt_grant[i]) w_grant_id = i[W-1:0];
    end
end
```

7.5 Timing Characteristics

Characteristic	Value
Setup Time	1 clock cycle for state update
Response Time	Combinational (0 cycles)
Reset Recovery	1 clock cycle

7.6 Usage Examples

7.6.1 Basic 4-Agent Arbiter

```
arbiter_round_robin_simple #(
    .N(4)
) u_arbiter (
    .clk        (clk),
    .rst_n      (rst_n),
    .request    (req_bits),     // [3:0]
    .grant_valid(grant_valid),
    .grant      (grant_oh),     // [3:0] one-hot
    .grant_id   (grant_bin)     // [1:0] binary
);
```

7.6.2 8-Agent System

```
arbiter_round_robin_simple #(
    .N(8)
) u_8way_arbiter (
    .clk        (clk),
    .rst_n      (rst_n),
    .request    (agent_requests), // [7:0]
    .grant_valid(resource_granted),
    .grant      (agent_grants),   // [7:0]
    .grant_id   (winning_agent)   // [2:0]
);
```

7.6.3 Arbitration Sequence Example

For N=4 with requests [3:0]:

Cycle	Last Grant	Requests	Win Mask	Masked	Selected	Grant	Grant ID
0	3 (init)	1111	0000	0000	1111 (wrap)	0001	0
1	0	1110	1110	1110	1110	0010	1
2	1	1101	1100	1100	1100	0100	2
3	2	1011	1000	1000	1000	1000	3
4	3	1111	0000	0000	1111 (wrap)	0001	0

`Win Mask` keeps the agents above the last grant. `Selected` is the masked vector, or the full request vector when the masked one is empty (marked “wrap”). `Grant` is the lowest set bit of `Selected`.

7.7 Design Notes

7.7.1 Advantages

- **Simple Logic**: A constant mask, a 2:1 mux and a lowest-set-bit isolate; no shifter
- **Scalable**: Works for any number of agents with minimal logic overhead
- **Fair**: Guarantees round-robin fairness over time
- **Fast**: Single combinational path from request to grant

7.7.2 Limitations

- **Grant Undefined**: `grant_id` is undefined when `grant_valid` is low
- **Request Requirement**: At least one request must be active for meaningful arbitration
- **Single-Cycle Grant**: The grant is not held; an agent that needs the resource for several cycles should use arbiter_round_robin_simple_ack

7.7.3 Synthesis Considerations

- **Area**: N mask constants selected by `r_last_grant` (an N-way mux of N-bit constants), one N-bit 2:1 mux, an N-bit increment-and-AND for the isolate, and the one-hot to index encoder
- **Timing**: Critical path is `r_last_grant` -> mask select -> AND -> any-reduce -> fallback mux -> carry chain of the isolate -> `grant`
- **No barrel shifter**: the mask approach avoids the O(N log N) rotate-and-unrotate network of a rotation-based design

7.8 Related Modules

- `arbiter_round_robin`: Enhanced round-robin arbiter with additional features
- arbiter_round_robin_simple_ack: Same datapath with a registered grant held until the owner acks
- `arbiter_priority_encoder`: Fixed priority arbiter implementation

- `encoder_priority_enable`: Priority encoding utility module

For most round-robin arbitration needs in digital systems, `arbiter_round_robin_simple` is the efficient, fair answer.

7.9 Testing

From the test suite (`val/common/test_arbiter_round_robin_simple.py`), parameterized over N:

- **GATE**: N = 4, 8
- **FUNC / FULL**: N = 2, 3, 4, 5, 6, 8, 16
- **Key Test Scenarios**:
  - ARB-01 to ARB-02: grant signal checks and basic arbitration
  - ARB-03 to ARB-05: request-set fairness, request-set changes, single-client saturation
  - ARB-06 to ARB-08: walking requests, bursty traffic, rapid request changes
  - ARB-09: dynamic arbitration liveness

Test Command: `make -C val/common clean-all && make -C val/common run-arbiter_round_robin_simple-func`

7.10 References

- Gupta, P., McKeown, N. “Designing and Implementing a Fast Crossbar Scheduler.” *IEEE Micro*, 19(1), Jan.–Feb. 1999. DOI: 10.1109/40.748793 — the programmable priority encoder built from two fixed-priority encoders, one fed thermometer-masked requests and one fed unmasked requests; the same masked/unmasked dual-encoder structure (here the mask is the win-mask above the last winner, with fallback to the unmasked requests).
- Dally, W.J., Towles, B. *Principles and Practices of Interconnection Networks*, Morgan Kaufmann, 2004, ch. 18 “Arbitration”, §18.4.2 “Round-Robin Arbiter” — textbook treatment (secondary source).
- Indexed in `rtl/common/references/README.md`

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8 Weighted Round Robin Arbiter

A credit-based weighted round-robin arbiter with Quality of Service (QoS) support. Clients get bandwidth proportional to their weights, and no client can starve.

8.1 Overview

The `arbiter_round_robin_weighted` module layers configurable per-client weights on top of fair round-robin arbitration, giving you proportional bandwidth allocation. Each client receives grants proportional to its assigned weight while starvation is prevented. The module supports an optional acknowledgment protocol and dynamic weight changes with atomic update semantics.

```
module arbiter_round_robin_weighted #(
    parameter int MAX_LEVELS = 16,      // Maximum weight value per
client
    parameter int CLIENTS = 4,          // Number of requesting
clients
    parameter int WAIT_GNT_ACK = 0      // Enable ACK protocol (0=No-
ACK, 1=ACK)
) (
    // Derived parameter - in the list for port-range use; do not
override:
    // localparam int MAX_LEVELS_WIDTH = $clog2(MAX_LEVELS);  //
Credit counter width
    // localparam int N = $clog2(CLIENTS);                    //
Client ID width
    // localparam int C = CLIENTS;                            //
Convenience alias
    // localparam int MTW = MAX_LEVELS_WIDTH;                 //
Weight width alias
    // localparam int CXMTW = CLIENTS * MAX_LEVELS_WIDTH;     //
Packed weight array width

    input  logic              clk,              // Clock input
    input  logic              rst_n,            // Active-low
asynchronous reset
    input  logic              block_arb,        // Block all
arbitration
    input  logic [CXMTW-1:0]  max_thresh,       // Packed weight
values for all clients
    input  logic [C-1:0]      request,          // Request signals
from clients
    input  logic [C-1:0]      grant_ack,        // Grant
acknowledgment (ACK mode only)
    output logic              grant_valid,      // Grant output valid
    output logic [C-1:0]      grant,            // Grant vector (one-
hot)
    output logic [N-1:0]      grant_id          // Grant client ID
```

```
(binary encoded)
);
```

8.2 Parameters

8.2.1 User-Settable Parameters

Parameter	Type	Default	Description
MAX_LEVELS	int	16	Sizes the weight/credit fields at `$clog2(MAX_LEVELS)` bits. **The largest usable weight is `MAX_LEVELS-1`** (15 for the default 16), *not* MAX_LEVELS: a weight of `MAX_LEVELS` truncates to 0 in the field and silently **disables** that client (`w_valid_clients[j] = client_weight[j] > 0`). Usable weight range per client: `1 .. MAX_LEVELS-1` (0 = disabled).
CLIENTS	int	4	Number of requesting clients (range: 2-32)
WAIT_GNT_ACK	int	0	Enable ACK protocol (0=No-ACK, 1=ACK required)

8.2.2 Derived Parameters

Parameter	Default	Description
MAX_LEVELS_WIDTH	$clog2(MAX_LEVELS)	Credit counter width in bits
N	$clog2(CLIENTS)	Client ID width in bits
C	CLIENTS	Convenience alias for port widths

Parameter	Default	Description
MTW	MAX_LEVELS_WIDTH	Convenience alias for weight width
CXMTW	CLIENTS * MAX_LEVELS_WIDTH	Total packed weight array width

Note: MAX_LEVELS_WIDTH, N, C, MTW, and CXMTW are derived `parameters` — declared in the parameter list so the port declarations can use them (strict front ends reject body localparams in port ranges). They *can* be overridden at instantiation, and must not be; leave them to their derivations. (The arbiter_priority_encoder page describes the same pattern.)

8.3 Ports

8.3.1 Inputs

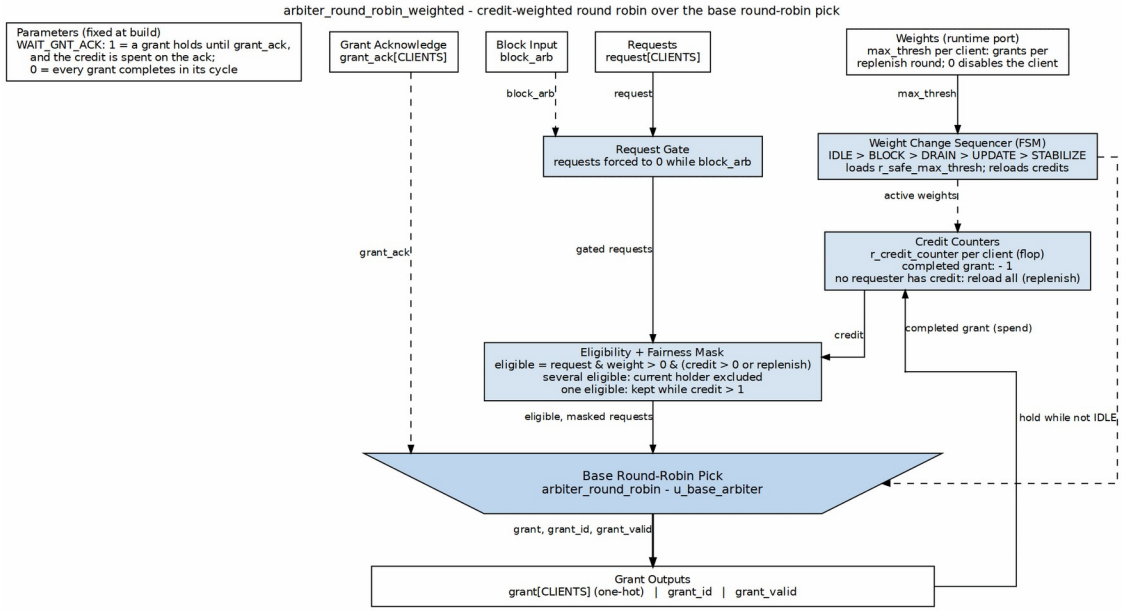
Port	Width	Description
clk	1	System clock
rst_n	1	Active-low asynchronous reset
block_arb	1	Block all arbitration (external gate)
max_thresh	CXMTW	Packed weight values: {weight[N-1], ..., weight[1], weight[0]}
request	C	Request signals from clients
grant_ack	C	Grant acknowledgment signals (ACK mode only)

8.3.2 Outputs

Port	Width	Description
grant_valid	1	Valid grant indicator
grant	C	Grant vector (one-hot encoded)
grant_id	N	Grant client ID (binary encoded)

8.4 Functional Description

8.4.1 Block diagram



arbiter_round_robin_weighted block diagram

Requests drop down the center through the `block_arb` gate into the eligibility and fairness mask, which passes clients that still hold credit (or that a replenish round is about to refill). The masked requests feed the base `arbiter_round_robin` (the trapezium), whose registered grant goes straight to the outputs. The credit counters sit on the right: each completed grant climbs the right edge to spend one credit, and when no requester has credit left all counters reload from their weights. A change on `max_thresh` goes through the change sequencer, which holds the pick until the new weights are loaded.

8.4.2 Credit-Based Weighting

Weighted bandwidth allocation here runs on a credit system:

1. **Credit Initialization**: reset loads **1 credit** per client (`r_credit_counter[i] <= MTW'(1)`) — but the shadow weight register resets to all-ones, so for any normal configuration the weight-change FSM runs immediately after reset (with the sub-arbiter blocked) and loads the CONFIGURED weights into the credits at WEIGHT_STABILIZE, before the first grant can issue. The first post-reset round IS weighted; only a config equal to the all-ones reset value skips the init pass

2. **Grant Completion**: When a grant completes, the client's credit counter decrements

3. **Credit Eligibility**: Only clients with non-zero credits are eligible for round-robin arbitration
4. **Global Replenishment**: When no requesting clients have credits, all credits are reloaded

8.4.3 Arbitration Stages

1. **Credit Eligibility**: Filter requests to only clients with credits > 0
2. **Round-Robin Selection**: Base arbiter (`arbiter_round_robin`) selects among eligible clients
3. **Grant Output**: Winning client receives one-hot grant signal
4. **Credit Update**: Winner's credit decrements on completion (immediate or ACK-based)

8.4.4 Weight Ratio Example

For weights [4, 2, 1, 1] with all clients requesting continuously: - **Client 0**: 4 credits per round - **Client 1**: 2 credits per round - **Client 2**: 1 credit per round - **Client 3**: 1 credit per round - **Pattern**: grants are **interleaved, not bursty**. The masking logic excludes the just-granted client whenever more than one client is still eligible (`w_mask_multi_req[j] = w_requesting_eligible[j] && !grant[j]`), so client 0 does **not** receive 4 back-to-back grants while others are eligible. A representative sequence is `C0, C1, C2, C3, C0, C1, C0, (bubble), C0,  [replenish], repeat` — client 0 still gets 4 grants per round, but spread out. - **Bandwidth** (the quantity that actually holds): C0=50%, C1=25%, C2=12.5%, C3=12.5% per round.

Consecutive per-client grants only occur once a client is the **sole** eligible requester (e.g., others have exhausted their credits), at which point the `w_mask_last_client` term lets it keep the resource.

8.4.5 Zero-Weight Client Handling

Clients with `max_thresh = 0` are effectively disabled: - Never accumulate credits - Never participate in arbitration - Credit counters remain at zero

8.4.6 ACK Protocol Support

When `WAIT_GNT_ACK = 1`: - Grant remains asserted until `grant_ack` received - Credit decrement waits for ACK - Weight change FSM waits for pending ACKs - Enables pipelined/posted transaction masters

When `WAIT_GNT_ACK = 0`: - Grant completes immediately (1 cycle) - Credit decrements same cycle as grant - Optimal for simple masters (no pipeline)

8.4.7 Starvation Prevention

The arbiter guarantees starvation-free operation: - All non-zero weight clients eventually served - Maximum starvation time: Sum of all other clients' weights - Global replenishment ensures fairness

8.4.8 Dynamic Weight Changes

8.4.8.1 Weight Management FSM

A finite state machine handles atomic weight updates:

State	Description
WEIGHT_IDLE	Normal operation
WEIGHT_BLOCK	Detect weight change, block new grants
WEIGHT_DRAIN	Wait for pending ACK completion (timer 2 → 3 cycles in state)
WEIGHT_UPDATE	Atomic weight update, reset credits
WEIGHT_STABILIZE	System stabilization (timer 3 → 4 cycles in state)

8.4.8.2 Weight Update Sequence

1. **Detection**: FSM detects `max_thresh` change from shadow register
2. **Block**: New grants blocked, pending grants drain
3. **Drain**: Wait for ACK completion (ACK mode) or fixed delay
4. **Update**: Atomic update of weights, credit counters reset
5. **Stabilize**: System stabilization period
6. **Resume**: Return to normal operation

Total Latency: 10 cycles minimum (DRAIN occupies 3 cycles, STABILIZE 4 — each timed state runs timer+1 cycles); up to ~25 with the 15-cycle BLOCK timeout on a pending ACK

8.4.8.3 Safety Features

- **Shadow Register**: Active weights stored in `r_safe_max_thresh` for race-free updates
- **Timeout Protection**: 15-cycle timeout prevents FSM lockup
- **Credit Reset**: Credits reset to new weights during WEIGHT_STABILIZE state

- **ACK Awareness**: FSM waits for pending `grant_ack` before updating weights

8.4.9 Credit Management

```
// Credit decrement on grant completion
if (w_grant_completed[i] && r_credit_counter[i] > 0) begin
    w_credit_counter[i] = r_credit_counter[i] - 1;
end
```

Grant Completion Modes: - **No-ACK Mode**: Completion when `grant[i] && grant_valid` (same cycle) - **ACK Mode**: Completion when `grant[i] && grant_valid && grant_ack[i]` (delayed)

8.4.10 Global Replenishment Logic

```
// Replenish when no requesting clients have credits.
// Note w_req_post, not request: w_req_post = block_arb ? '0 :
request, so while
// block_arb is high there are no requests to replenish for.
assign w_global_replenish = (w_normal_operation && !w_pending_grants
&&
                             (|w_req_post) && !
w_requesting_clients_with_credits);

// On replenish: reload all credit counters to their weights
if (w_global_replenish) begin
    w_credit_counter[i] = client_weight[i];
end
```

8.4.11 Request Masking for Fairness

```
// Only pass eligible clients to round-robin arbiter. Again
w_req_post, so a
// blocked arbiter makes every client ineligible rather than merely
ungranted.
assign w_requesting_eligible[i] = w_req_post[i] &&
                                  ((w_has_crd[i]) ||
                                   (w_global_replenish &&
w_valid_clients[i]));

// Apply masking for fairness among multiple eligible clients.
// The two terms are OR'd, NOT selected by a ternary: the "eligible &&
!grant"
// term is ALWAYS active, so a lone eligible client can never be
masked to 0.
assign w_mask_multi_req[i]   = w_requesting_eligible[i] && !grant[i];
assign w_mask_last_client[i] = !multiple_eligible &&
                               w_requesting_eligible[i] &&
```

```
                                (r_credit_counter[i] > 1);
assign w_mask_req[i] = w_mask_multi_req[i] || w_mask_last_client[i];
```

Do not re-implement this as a ternary (`multiple_eligible ? ... : ...`). Dropping the unconditional `eligible && !grant` term would leave a lone eligible client on its last credit (`r_credit_counter == 1`) with `w_mask_req = 0` forever: no grant, so no decrement, so no replenish — a deadlock. The OR form grants it every other cycle.

8.4.12 Sub-Module: arbiter_round_robin

The base `arbiter_round_robin` module handles fair selection among eligible clients:

```
arbiter_round_robin #(
    .CLIENTS      (CLIENTS),
    .WAIT_GNT_ACK (WAIT_GNT_ACK)
) u_base_arbiter (
    .clk        (clk),
    .rst_n      (rst_n),
    .block_arb  (w_sub_block_arb),
    .request    (w_mask_req),
    .grant_ack  (grant_ack),
    .grant_valid(grant_valid),
    .grant      (grant),
    .grant_id   (grant_id),
    .last_grant (r_last_grant)
);
```

8.5 Timing Characteristics

8.5.1 Latency and Throughput

- **Latency**: 1 cycle in steady state. Credit calculation, eligibility masking and the round-robin decision are all combinational (`request` -> `w_req_post` -> `w_requesting_eligible` -> `w_mask_req`); the only register stage is the base arbiter's `grant <= w_next_grant`. Measured on this RTL: a request sampled at edge N produces `grant_valid` at edge N+1. Two exceptions add cycles – the first grant after reset waits for the weight-init FSM, 10 cycles to return to IDLE traced from the FSM constants (detect 1 + BLOCK 1 + DRAIN 3 + UPDATE 1 + STABILIZE 4) plus one for the grant register, ~11 cycles total – and a global replenish inserts a one-cycle bubble.
- **Throughput**: 1 grant per cycle (maximum)
- **Grant Hold**: No-ACK=1 cycle, ACK=Until grant_ack asserted

- **Weight Update**: 10-25 cycles (FSM: BLOCK→DRAIN→UPDATE→STABILIZE)
- **Reset**: Asynchronous (all credits→1, weights→default)

8.5.2 Critical Paths

1. **Credit Comparison**: Credit counter → eligibility logic
2. **Request Filtering**: Eligible clients → masked requests
3. **Round-Robin**: Base arbiter selection logic

8.5.3 Resource Utilization

CLIENTS	MAX_LEVELS	FFs (Est.)	LUTs (Est.)
4	16	~40	~50
8	16	~70	~90
4	32	~45	~55
8	32	~80	~110

8.5.4 Maximum Frequency

- **Typical**: 300-500 MHz (modern FPGAs)
- **Limiting Factor**: Credit comparison and request filtering logic

8.6 Usage Examples

8.6.1 Basic QoS Arbiter (No-ACK Mode)

```
// 4-client arbiter with QoS weights: 8:4:2:1
localparam int NUM_CLIENTS = 4;
localparam int MAX_WEIGHT = 16;
localparam int WEIGHT_W = $clog2(MAX_WEIGHT);

logic [NUM_CLIENTS-1:0] client_req, client_grant;
logic grant_vld;
logic [$clog2(NUM_CLIENTS)-1:0] grant_idx;
logic [NUM_CLIENTS*WEIGHT_W-1:0] qos_weights;

// Assign QoS weights (high priority = higher weight)
assign qos_weights = {4'd1, 4'd2, 4'd4, 4'd8};  // {C3, C2, C1, C0}

arbiter_round_robin_weighted #(
    .CLIENTS(NUM_CLIENTS),
    .MAX_LEVELS(MAX_WEIGHT),
```

```
    .WAIT_GNT_ACK(0)           // No-ACK mode
) u_qos_arbiter (
    .clk        (clk),
    .rst_n      (rst_n),
    .block_arb  (1'b0),
    .max_thresh (qos_weights),
    .request    (client_req),
    .grant_ack  ('0),           // Unused in No-ACK mode
    .grant_valid(grant_vld),
    .grant      (client_grant),
    .grant_id   (grant_idx)
);
```

8.6.2 Dynamic Weight Adjustment

Slice exactly one client's field. Client `j` owns bits `[(j+1)*MAX_LEVELS_WIDTH-1 : j*MAX_LEVELS_WIDTH]`, so with `MAX_LEVELS=16` (4-bit weights) client 0 is `[3:0]` and client 1 is `[7:4]`. Writing a 4-bit value into a wider slice zero-extends across the neighbour: `r_qos_weights[7:0] <= 4'd15` sets client 1's weight to 0, and `w_valid_clients[j] = (client_weight[j] > 0)` means a zero-weight client is never granted again.

```
// Adjust weights based on priority events
logic [NUM_CLIENTS*WEIGHT_W-1:0] r_qos_weights;

always_ff @(posedge clk) begin
    if (high_priority_event) begin
        r_qos_weights[3:0] <= 4'd15;  // Boost Client 0 weight
    end else if (normal_priority) begin
        r_qos_weights[3:0] <= 4'd4;   // Reset Client 0 weight
    end
end

arbiter_round_robin_weighted #(
    .CLIENTS(NUM_CLIENTS),
    .MAX_LEVELS(MAX_WEIGHT),
    .WAIT_GNT_ACK(0)
) u_dynamic_arbiter (
    .clk        (clk),
    .rst_n      (rst_n),
    .block_arb  (1'b0),
    .max_thresh (r_qos_weights),  // Dynamic weights
    .request    (client_req),
    .grant_ack  ('0),
    .grant_valid(grant_vld),
    .grant      (client_grant),
    .grant_id   (grant_idx)
);
```

8.6.3 ACK Mode for Pipelined Masters

```
// ACK mode arbiter for pipelined masters
arbiter_round_robin_weighted #(
    .CLIENTS(2),
    .MAX_LEVELS(8),
    .WAIT_GNT_ACK(1)            // ACK required
) u_ack_arbiter (
    .clk        (clk),
    .rst_n      (rst_n),
    .block_arb  (bus_busy),
    .max_thresh ({3'd3, 3'd5}),  // client1=3, client0=5; fields are
$clog2(8)=3 bits each
    .request    (m_req),
    .grant_ack  (m_done),        // Master completion signal
    .grant_valid(m_grant_vld),
    .grant      (m_grant),
    .grant_id   (m_id)
);
```

8.7 Design Notes

8.7.1 When to Use Each Mode

No-ACK Mode (WAIT_GNT_ACK=0): - Simple masters without pipelining - Combinational logic masters - Maximum throughput (1 grant/cycle) - Caveat: Grant completes same cycle (no hold)

ACK Mode (WAIT_GNT_ACK=1): - Pipelined masters (e.g., AXI) - Posted transaction buses - Multi-cycle operations - Caveat: Reduced throughput (grant held until ACK)

8.7.2 Weight Selection Guidelines

- **Finer QoS Control**: Increase `MAX_LEVELS` (but more credit counter bits)
- **Typical Range**: MAX_LEVELS=8 to 32 for most applications
- **Weight Ratio**: Should reflect desired bandwidth ratios
- **Zero Weights**: Effectively disable clients (never granted)

8.7.3 Dynamic Weight Change Best Practices

- **DO NOT** change weights every cycle (causes FSM thrashing)
- **Recommended**: Change weights on policy updates (millisecond timescale)
- **Safe**: FSM ensures atomic updates without race conditions
- **Timeout**: 15-cycle timeout prevents lockup

8.8 Related Modules

- **arbiter_round_robin.sv** - Base round-robin arbiter (used internally)
- **arbiter_round_robin_simple.sv** - Lightweight RR arbiter (no weights)
- **arbiter_priority_encoder.sv** - Fixed priority arbiter

8.9 Testing

From the test suite (`val/common/test_arbiter_round_robin_weighted.py`):

- **Coverage**: 94%
- **Key Test Scenarios**:
  - Weight ratio verification ([4,2,1,1] patterns)
  - Global credit replenishment
  - Dynamic weight changes (atomic updates)
  - ACK mode timing and weight updates
  - Zero weight handling (client disable)
  - Single client optimization
  - FSM timeout protection
  - Starvation prevention

Test Command: `pytest val/common/test_arbiter_round_robin_weighted.py -v`

8.10 References

- Katevenis, M., Sidiropoulos, S., Courcoubetis, C. “Weighted Round-Robin Cell Multiplexing in a General-Purpose ATM Switch Chip.” *IEEE Journal on Selected Areas in Communications*, 9(8), Oct. 1991. DOI: 10.1109/49.105173 — the seminal hardware weighted round-robin with interleaved (not bursted) per-weight service, the behavior this module’s credit scheme produces. A preprint is on the author’s group site: archvlsi.ics.forth.gr (PostScript).
- Dally, W.J., Towles, B. *Principles and Practices of Interconnection Networks*, Morgan Kaufmann, 2004, ch. 18 “Arbitration”, §18.4.4 “Weighted Round-

Robin Arbiter" — textbook treatment of weighted shares (secondary source).

- Indexed in `rtl/common/references/README.md`

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9 Single-Client Arbiter

Module: `arbiter_single_client.sv` **Location:** `rtl/common/` **Status:** Production Ready

9.1 Overview

`arbiter_single_client` is the degenerate, one-requester member of the round-robin arbiter family. It exists to give a single-channel datapath the **exact same registered, ack-held grant lifecycle** as `arbiter_round_robin` running in `WAIT_GNT_ACK` mode, without the parameter degeneracy that breaks the multi-client arbiter when `CLIENTS == 1`.

`arbiter_round_robin` underflows at `CLIENTS == 1` because `$clog2(1) = 0` produces a `grant_id[-1:0]` slice. Historically single-channel builds worked around this by substituting a purely *combinational* passthrough (`grant = request`). That passthrough does **not** reproduce the arbiter's registered grant/ack handshake, so it injects a bubble beat at every burst boundary in the AXI read/write engines that depend on the "request, grant, hold-for-ack" timing. This module is the faithful single-client reduction of those rules.

9.1.1 Key Features

- **Timing-faithful:** Mirrors `arbiter_round_robin` `WAIT_GNT_ACK` `always_ff` rules exactly for one client
- **Registered grant:** Grant is a flip-flop output, held until acknowledged (no combinational passthrough bubble)
- **Ack-held lifecycle:** Grant persists until `grant_ack`, then clears and can re-grant the next cycle
- **Bubble-free bursts:** Preserves the grant timing the AXI read/write engines rely on for back-to-back burst beats
- **Optional ack mode:** `WAIT_GNT_ACK = 0` falls back to a simple registered request-to-grant with no ack hold
- **Trivial decode:** `grant` is a one-hot of width 1 (equal to `grant_valid`); `grant_id` is always 0

9.1.2 Use Cases

The module provides a drop-in single-client arbiter so that a design parameterized down to one channel behaves identically to the same design with two or more channels. Rather than special-casing the `NUM_CHANNELS == 1` build with a passthrough, the datapath instantiates `arbiter_single_client` and gets the identical grant/ack sequencing the general arbiter would have produced for one requester.

- Single-channel builds of a datapath that is otherwise parameterized for N channels
- AXI read/write engines that require the registered, ack-held grant handshake for correct burst sequencing
- Any place a `arbiter_round_robin` instance would be used, but with `CLIENTS == 1`
- Replacing a combinational `grant = request` passthrough that was injecting per-burst bubble beats

Key Benefit: A one-channel build behaves bit-for-bit like the multi-channel arbiter path, eliminating the burst-boundary bubble that a combinational passthrough introduced.

9.2 Parameters

Parameter	Type	Default	Description
WAIT_GNT_ACK	int	1	`1` = registered grant is held until `grant_ack` is received (round-robin ACK lifecycle). `0` = simple registered request-to-grant with no ack hold.

9.3 Ports

9.3.1 Inputs

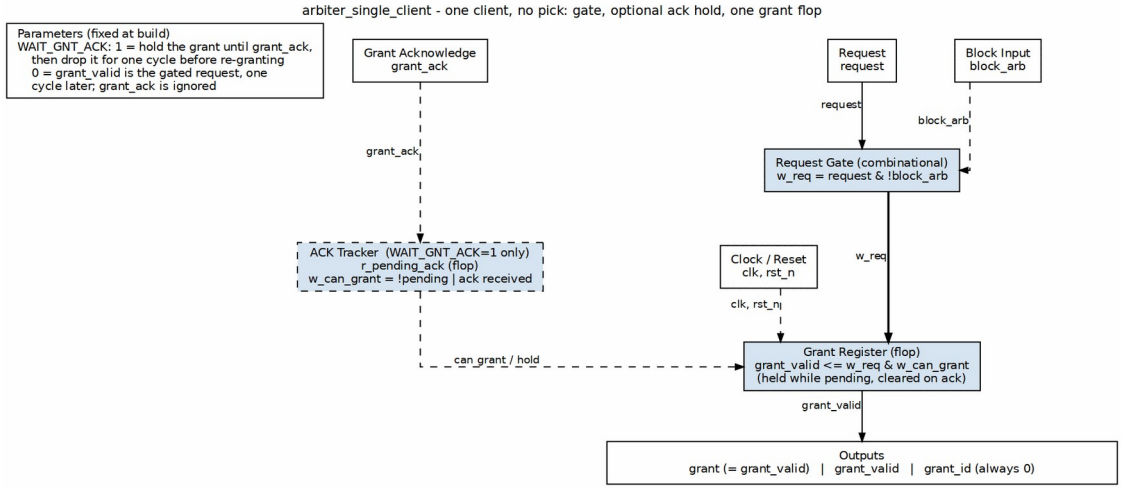
Port	Width	Description
clk	1	Clock
rst_n	1	Active-low asynchronous reset
block_arb	1	Block arbitration; when high the request is masked and no grant is issued
request	1	The single client's request
grant_ack	1	The single client's grant acknowledgment (used only when `WAIT_GNT_ACK = 1`)

9.3.2 Outputs

Port	Width	Description
grant_valid	1	Grant valid (registered)
grant	1	One-hot grant of width 1; equal to `grant_valid`
grant_id	1	Granted client index; always `0` for a single client

9.4 Functional Description

9.4.1 Block diagram



arbiter_single_client block diagram

With one client there is nothing to pick between, so the diagram has no trapezium. The request passes the `block_arb` gate and loads the single grant flop, one cycle later. When `WAIT_GNT_ACK=1`, the ACK tracker (dashed, because it exists only in that build) holds the grant until `grant_ack` arrives and then lets it drop for a cycle; the parameter legend gives both modes. `grant` is `grant_valid` and `grant_id` is always 0.

9.4.2 Request Qualification

The incoming request is masked by `block_arb` first:

```
assign w_req = request && !block_arb;
```

A blocked arbiter behaves as if the client were not requesting at all.

9.4.3 Grant-Eligibility Logic (ACK Mode)

When `WAIT_GNT_ACK = 1`, three combinational signals gate whether a new grant may be issued:

```
assign w_ack_received = r_pending_ack && grant_ack;          // ack
seen for the outstanding grant
assign w_can_grant    = !r_pending_ack || w_ack_received;    // free
to grant, or ack just closed the last grant
assign w_should_grant = w_req && w_can_grant;
```

`r_pending_ack` records that a grant has been issued and is awaiting its acknowledgment. A new grant can only be launched when there is no pending ack, or when the ack for the current grant arrives this cycle.

9.4.4 Grant Lifecycle (ACK Mode)

The registered state machine is the single-client reduction of the `arbiter_round_robin` `WAIT_GNT_ACK` rules. Because there is only one client, `w_other_requests` is always 0, so the arbiter's Rule 4 (switch to another requester) never applies and Rule 3 simply clears:

State	Condition	Action	`grant_valid` next	`r_pending_ack` next
Rule 1	`grant_valid == 0`	Not granted; grant if requesting	`w_should_grant`	`w_should_grant`
Rule 2	granted, `!w_ack_received`	Hold the grant	`1`	`1`
Rule 3	granted, `w_ack_received`	Ack received; clear (re-grant next cycle)	`0`	`0`

The grant is thus **registered and held** across cycles until the client acknowledges, then it drops for one cycle before it can be re-issued.

9.4.5 No-ACK Fallback Mode

When `WAIT_GNT_ACK = 0`, the ack machinery is bypassed. `grant_valid` simply follows a registered version of the qualified request and `r_pending_ack` stays 0:

```
grant_valid   <= w_should_grant;   // w_can_grant is constant 1 in
this mode
r_pending_ack <= 1'b0;
```

This gives a one-cycle registered request-to-grant with no hold.

9.4.6 Output Mapping

```
assign grant    = grant_valid;   // one-hot of width 1
assign grant_id = 1'b0;          // single client
```

Because there is exactly one client, `grant` and `grant_valid` are identical, and `grant_id` is a constant 0.

9.5 Timing Characteristics

This module is **sequential**: it contains 1 `always_ff` block(s), clocked on `clk` with active-low asynchronous reset `rst_n`. Outputs derived in those blocks are registered and therefore appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

9.6 Usage Examples

```
// Single-channel build: use arbiter_single_client where a >1-channel
// build would instantiate arbiter_round_robin in WAIT_GNT_ACK mode.
arbiter_single_client #(
    .WAIT_GNT_ACK(1)
) u_arb (
    .clk        (aclk),
    .rst_n      (aresetn),
    .block_arb  (arb_blocked),
    .request    (engine_request),
    .grant_ack  (engine_grant_ack),
    .grant_valid(engine_grant_valid),
    .grant      (engine_grant),      // width-1 one-hot
    .grant_id   ()                   // always 0, usually unconnected
);
```

9.7 Design Notes

- **Why not a combinational passthrough?** `grant = request` looks correct for one client but loses the registered, ack-held timing. The AXI read/write engines interpret the missing hold as an early grant drop and insert a

bubble beat at every burst boundary. This module preserves the exact cycle-by-cycle behavior.

- **`grant_id` degeneracy.** The general arbiter's `grant_id` is `$clog2(CLIENTS)` bits wide, which collapses to an illegal `[-1:0]` slice at `CLIENTS == 1`. Here `grant_id` is simply tied to `0`, sidestepping the elaboration problem.
- **Block behavior.** `block_arb` masks the request combinationally, so raising it prevents new grants; an already-pending ack-held grant continues to follow the state machine.
- **Reset.** Active-low asynchronous reset clears both `grant_valid` and `r_pending_ack`.

9.8 Related Modules

Used by single-channel builds of AXI read/write datapath engines that otherwise instantiate `arbiter_round_robin`. The module itself is self-contained — no submodule instantiations.

- arbiter_round_robin - The general N-client arbiter whose `WAIT_GNT_ACK` timing this module reproduces for one client
- arbiter_round_robin_simple - Minimal-area round-robin variant
- arbiter_round_robin_weighted - Weighted / QoS round-robin variant
- arbiter_priority_encoder - Fixed-priority arbiter

Source: `rtl/common/arbiter_single_client.sv`, with `rtl/common/arbiter_round_robin.sv` as the timing reference, and `docs/markdown/rtl-common/index.md`.

Last Updated: 2026-07-15

9.9 Testing

No dedicated testbench, by decision. A single-client arbiter grants its one requester unconditionally; the behaviour is verified in situ in STREAM rather than through a standalone bench. Recorded as exempt in the coverage baseline (common TASK-010, was COMMON-024).

Treat any behaviour described on this page as unverified by simulation.

9.10 References

- No peer paper — a one-client registered grant/ACK holder is a standard technique with no defensible single origin (marked “standard technique” in the references index).
- Dally, W.J., Towles, B. *Principles and Practices of Interconnection Networks*, Morgan Kaufmann, 2004, ch. 18 “Arbitration”, §18.4.3 “Grant-Hold Circuit” — textbook treatment of holding a grant until release (secondary source).
- Indexed in `rtl/common/references/README.md`

9.11 Navigation

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10 Token-Bucket Request Shaper

Module: `rtl/common/arbiter_token_bucket.sv`

10.1 Overview

Per-client token-bucket request shaper. It sits in front of any arbiter in the family and gates each client's request by its token balance: tokens accumulate at a configured per-client rate on an external refill tick (up to a burst cap), and one token is spent per completed grant. A client out of tokens simply stops requesting until the next refill.

This is **rate shaping, not fairness** — the arbiter behind it still decides who wins among the requests that pass; the shaper decides how *often* each client may compete. That separation is why it is a free-standing block rather than a mode of any one arbiter: it composes with arbiter_round_robin for rate-limited fair sharing, or with arbiter_round_robin_weighted / arbiter_deficit_round_robin for shaped rate *and* weighted share together.

```
module arbiter_token_bucket #(
    parameter int CLIENTS      = 4,
    parameter int MAX_TOKENS   = 64,
    parameter int RATE_WIDTH   = 4,
    parameter int WAIT_GNT_ACK = 0
) (
    input  logic              clk,
    input  logic              rst_n,
    input  logic              refill_tick,
    input  logic [CXRW-1:0]   rate,
    input  logic [CXTW-1:0]   bucket_cap,
    input  logic [C-1:0]      request_in,
    input  logic [C-1:0]      grant,
    input  logic              grant_valid,
    input  logic [C-1:0]      grant_ack,
    output logic [C-1:0]      request_out,
    output logic [CXTW-1:0]   tokens
);
```

10.2 Parameters

Parameter	Type	Default	Description
CLIENTS	int	4	Number of clients being shaped
MAX_TOKENS	int	64	Exclusive bound on buckets/caps; field width =

Parameter	Type	Default	Description
			$clog2(MAX_TOKENS)
RATE_WIDTH	int	4	Width of each per-client tokens-per-tick rate field
WAIT_GNT_ACK	int	0	Completion contract of the DOWNSTREAM arbiter (must match it)

10.2.1 Derived Parameters (do not override)

Declared as `parameter` so the elaborator can size the packed ports, not so callers can set them. Overriding one desynchronises it from the parameters above and the design mis-sizes or fails to elaborate.

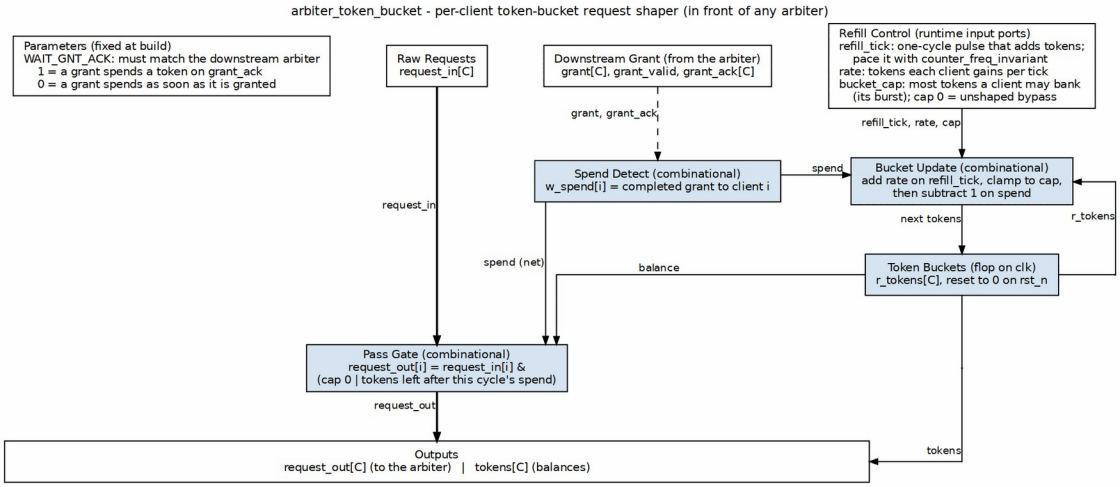
Derived parameter	Default expression
`TW`	`$clog2(MAX_TOKENS)`
`CXTW`	`CLIENTS * TW`

10.3 Ports

Port	Dir	Width	Description
`clk`	In	1	
`rst_n`	In	1	
`refill_tick`	In	1	
`rate`	In	`[CXRW-1:0]`	
`bucket_cap`	In	`[CXTW-1:0]`	
`request_in`	In	`[C-1:0]`	
`grant`	In	`[C-1:0]`	
`grant_valid`	In	1	
`grant_ack`	In	`[C-1:0]`	
`request_out`	Out	`[C-1:0]`	
`tokens`	Out	`[CXTW-1:0]`	

10.4 Functional Description

10.4.1 Block diagram



arbiter_token_bucket block diagram

The shaper never picks between clients; the arbiter behind it does, so there is no trapezium. Raw requests run down the left through the Pass Gate to `request_out`. A client passes only while it holds tokens (or its cap is 0, which bypasses shaping), counted net of a spend completing this cycle. The downstream arbiter's grant comes back as a dashed return into Spend Detect. Bucket Update adds `rate` on each `refill_tick`, clamps to `bucket_cap`, and subtracts one per spend; the balance climbs the right edge back into the update and feeds the gate. The two legends describe the build parameter and the runtime refill controls.

- **`refill_tick` is external.** Pair it with counter_freq_invariant's microsecond tick so rates carry real-time meaning across clock frequencies, or with any pulse of your choosing.
- **cap = 0 means UNSHAPED, not blocked** (fail-open). A zeroed config must degrade to "no shaping", never to "no service". To block a client, gate its request upstream or zero its weight/quantum in the downstream arbiter. Corollary: real caps live in 1..MAX_TOKENS-1 — a cap value equal to MAX_TOKENS wraps the field to 0 and turns shaping OFF.
- **The cap clamp is an invariant, applied every cycle** — not only at refill. A runtime cap decrease bites immediately; banked burst above the new cap is forfeited (with rate 0 a refill-time-only clamp would never fire at all).
- **Overspend-proof gate.** The downstream grant registers one cycle after arbitration saw `request_out`, so in the completion cycle the bucket register

still shows the pre-spend value. The pass gate subtracts the in-flight spend, so a one-token client cannot win twice on one token — the same registered-decision window the DRR's cost pipeline handles.

- **Burst semantics:** a bucket at cap C allows C back-to-back grants, then the client throttles to its refill rate. Reset starts every bucket empty (the conservative choice for a rate limiter).
- **No config-update FSM** — unlike the WRR/DRR there is no cross-client invariant to protect; rate/cap changes take effect per the clamp rules.

10.5 Design Notes

`TW` and `CXTW` are derived; do not override them. They are `$clog2(MAX_TOKENS)` and `CLIENTS * TW`, declared as `parameter` so the elaborator can size the packed ports. Setting one directly desynchronises it from `MAX_TOKENS` and the packed vectors mis-size.

`WAIT_GNT_ACK` must match the DOWNSTREAM arbiter. It states the completion contract this shaper is feeding, not a preference; a mismatch means the shaper and the arbiter disagree about when a grant is finished.

10.6 Timing Characteristics

Property	Value
Gate	Combinational: request_in to request_out through one compare
Buckets	Registered; refill-then-clamp-then-spend per cycle
Sustained rate	Exactly rate[i] tokens per tick interval under saturation

10.7 Usage Examples

Every parameter and port below is taken from the module declaration.

```
arbiter_token_bucket #(
    .CLIENTS                (4),
    .MAX_TOKENS             (64),
    .RATE_WIDTH             (4),
    .WAIT_GNT_ACK           (0),
    .TW                     ($clog2(MAX_TOKENS)),
    .C                      (CLIENTS),
```

```
    .CXTW                   (CLIENTS * TW),
    .CXRW                   (CLIENTS * RATE_WIDTH),
) u_arbiter_token_bucket (
    .clk                    (clk),
    .rst_n                  (rst_n),
    .refill_tick            (refill_tick),
    .rate                   (rate),
    .bucket_cap             (bucket_cap),
    .request_in             (request_in),
    .grant                  (grant),
    .grant_valid            (grant_valid),
    .grant_ack              (grant_ack),
    .request_out            (request_out),
    .tokens                 (tokens)
);
```

10.8 Related Modules

- arbiter_round_robin, arbiter_round_robin_weighted, arbiter_deficit_round_robin — the arbiters this shaper feeds (welded to none of them)
- counter_freq_invariant — the natural refill_tick source

10.9 Testing

Location: `val/common/test_arbiter_token_bucket.py`

The TB plays the downstream arbiter (registered grants, both ACK modes) and keeps a never-overspend ledger asserted at every completion: cumulative grants can never exceed cumulative banked refill. Scenarios: sustained rate under saturation, burst allowance exactly the cap, cap-0 bypass, and a runtime rate cut that drains then fully blocks.

```
pytest val/common/test_arbiter_token_bucket.py -v
```

10.10 References

- Turner, J. “New Directions in Communications (or Which Way to the Information Age?).” *IEEE Communications Magazine*, 24(10), Oct. 1986. DOI: 10.1109/MCOM.1986.1092946 — origin of the leaky-bucket rate regulator from which the token-bucket shaper derives.
- Shenker, S., Wroclawski, J. “General Characterization Parameters for Integrated Service Network Elements.” RFC 2215, IETF, Sept. 1997. rfc-

editor.org/rfc/rfc2215 — defines the token-bucket specification by a token rate *r* and bucket depth *b* (standards document, not a peer paper); this module's per-client rate and burst cap are exactly that (r, b) pair.

- Indexed in `rtl/common/references/README.md`

10.11 Navigation

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11 Binary to BCD Converter

Binary in, decimal out — the module you need the moment a number has to end up on a display.

11.1 Overview

The `bin_to_bcd` module converts binary numbers to Binary Coded Decimal (BCD) format using the Double-Dabble algorithm (also known as the Add-3-Then-Shift algorithm). It's a fundamental operation in digital systems for displaying numeric values on seven-segment displays, LCD panels, and other decimal-based output devices.

```
module bin_to_bcd #(
    parameter int WIDTH  = 8,
    parameter int DIGITS = 3
) (
    input  logic                clk,     // Clock signal
    input  logic                rst_n,   // Active-low reset signal
    input  logic                start,
    input  logic [   WIDTH-1:0] binary,
    output logic [DIGITS*4-1:0] bcd,
    output logic                done
);
```

11.2 Parameters

Parameter	Type	Default	Description
WIDTH	int	8	Bit width of the input binary number. Range: 1 to 32 (practical range). Determines conversion complexity and timing. Sizing rule: for n-bit binary input, maximum decimal value is 2^n - 1.
DIGITS	int	3	Number of BCD digits in the output. Must be sufficient for the maximum input value: `DIGITS ≥ ceil(log10(2^WIDTH))`. Output width is `DIGITS` × 4 bits (4 bits per BCD digit).

11.2.1 Parameter Relationship Table

WIDTH	Max Binary Value	Max Decimal	Min DIGITS Required	BCD Width
4	15	15	2	8 bits
8	255	255	3	12 bits
10	1023	1023	4	16 bits
12	4095	4095	4	16 bits
16	65535	65535	5	20 bits
20	1048575	1048575	7	28 bits
24	16777215	16777215	8	32 bits

11.3 Ports

11.3.1 Inputs

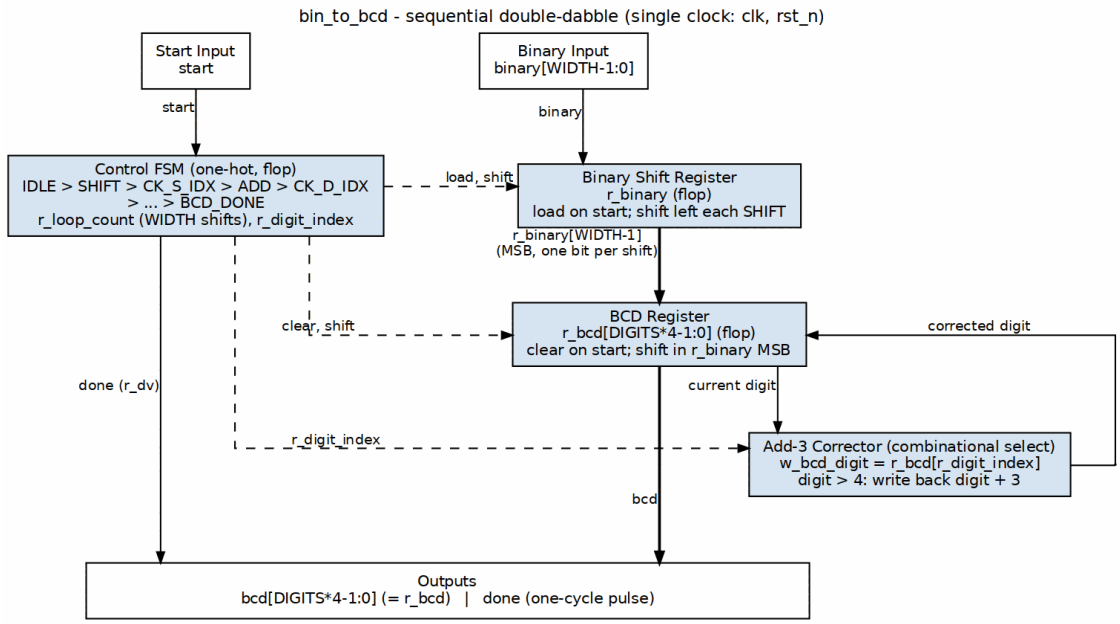
Port	Width	Description
clk	1	System clock
rst_n	1	Active-low asynchronous reset
start	1	Initiate conversion (pulse)
binary	WIDTH	Binary number to convert

11.3.2 Outputs

Port	Width	Description
bcd	DIGITS×4	BCD result (packed format)
done	1	Conversion complete flag

11.4 Functional Description

11.4.1 Block diagram



bin_to_bcd block diagram

The control FSM loads `binary` into the binary shift register on `start` and clears the BCD register. Each SHIFT moves the binary register's MSB into the bottom of the BCD register. Between shifts, the FSM steps `r_digit_index` across the digits, and the add-3 corrector writes `digit + 3` back into any digit above 4 (the loop on the right edge). After `WIDTH` shifts the FSM pulses `done` for one cycle, and `bcd` holds the result.

11.4.2 The Double-Dabble Algorithm

The Double-Dabble algorithm converts binary to BCD through iterative shifting and adjustment:

1. **Initialize**: BCD accumulator = 0, binary shift register = input value
2. **Check**: For each BCD digit, if value ≥ 5, add 3
3. **Shift**: Shift entire register (BCD + binary) left by 1 bit
4. **Repeat**: Continue until all binary bits are processed
5. **Result**: BCD accumulator contains the decimal equivalent

11.4.3 Why Add 3?

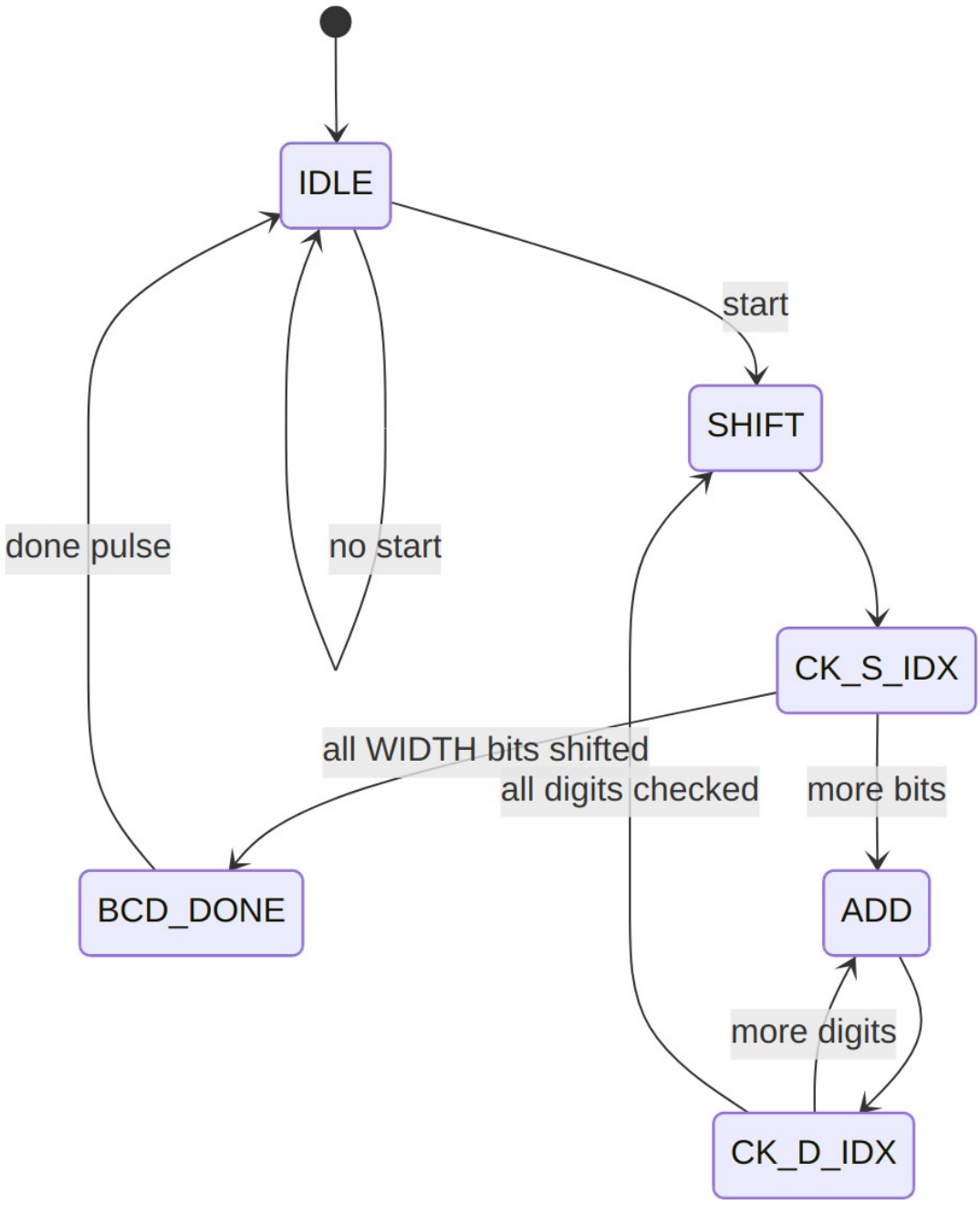
The “add 3” step exists to prevent BCD overflow during shifting: - **BCD Range**: Each digit must stay within 0-9 (4 bits) - **Problem**: Shifting a BCD digit ≥ 5 creates values > 9 - **Solution**: Adding 3 before shifting ensures proper BCD after shift

11.4.4 Mathematical Proof

For BCD digit D where D ≥ 5: - Before shift: D + 3 - After shift: 2(D + 3) = 2D + 6 - For D = 5: 2(5) + 6 = 16 = 0x10 (BCD: 1,6 - carry generated) - For D = 9: 2(9) + 6 = 24 = 0x18 (BCD: 2,4 - carry generated)

11.4.5 State Machine

```
typedef enum logic [5:0] {
    IDLE     = 6'b000001,  // Wait for start signal
    SHIFT    = 6'b000010,  // Shift binary bit into BCD
    CK_S_IDX = 6'b000100,  // Check if all bits shifted
    ADD      = 6'b001000,  // Add 3 to digits > 4
    CK_D_IDX = 6'b010000,  // Check if all digits processed
    BCD_DONE = 6'b100000   // Conversion complete
} fsm_state_t;
```



State Transition Diagram

11.4.6 State Descriptions

11.4.6.1 IDLE State

- **Function**: Wait for start signal

- **Duration**: Variable (until start asserted)
- **Actions**:
  - Monitor `start` input
  - Initialize counters and registers when start detected
- **Next State**: SHIFT when `start == 1'b1`

11.4.6.2 SHIFT State

- **Function**: Shift one bit from binary to BCD
- **Duration**: 1 clock cycle
- **Actions**:
  - Left-shift BCD accumulator by 1 bit
  - Move MSB of binary into LSB of BCD
  - Left-shift binary register by 1 bit
  - Increment loop counter
- **Next State**: Always CK_S_IDX

11.4.6.3 CK_S_IDX State

- **Function**: Check if all binary bits processed
- **Duration**: 1 clock cycle
- **Actions**:
  - Compare loop counter with WIDTH
  - Reset digit index for ADD phase
- **Next State**:
  - BCD_DONE if `loop_count == WIDTH-1`
  - ADD if more bits to process

11.4.6.4 ADD State

- **Function**: Add 3 to BCD digits ≥ 5
- **Duration**: 1 clock cycle
- **Actions**:
  - Extract current BCD digit
  - Add 3 if digit > 4
  - Update BCD accumulator
- **Next State**: Always CK_D_IDX

11.4.6.5 CK_D_IDX State

- **Function**: Check if all digits processed
- **Duration**: 1 clock cycle
- **Actions**:
  - Compare digit index with DIGITS
  - Increment digit index
- **Next State**:
  - SHIFT if `digit_index == DIGITS-1`
  - ADD if more digits to process

11.4.6.6 BCD_DONE State

- **Function**: Signal completion
- **Duration**: 1 clock cycle
- **Actions**:
  - Assert `done` flag
  - Prepare for next conversion
- **Next State**: Always IDLE

11.4.7 Register Definitions

```
// FSM state register
fsm_state_t r_fsm_main;

// BCD accumulator - stores the converted result
logic [DIGITS*4-1:0] r_bcd;

// Binary shift register - input value being processed
logic [WIDTH-1:0] r_binary;

// Loop counter - tracks processed bits
localparam int LOOP_COUNT_WIDTH = $clog2(WIDTH + 1);
logic [LOOP_COUNT_WIDTH-1:0] r_loop_count;

// Digit index - tracks current BCD digit being processed
localparam int DIGIT_INDEX_WIDTH = $clog2(DIGITS);
logic [DIGIT_INDEX_WIDTH-1:0] r_digit_index;

// Data valid flag
logic r_dv;
```

11.4.8 Core Processing Logic

```
// Extract current BCD digit being processed
logic [3:0] w_bcd_digit;
assign w_bcd_digit = r_bcd[r_digit_index*4+:4];

// Main FSM
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_fsm_main    <= IDLE;
        r_bcd         <= '0;
        r_binary      <= '0;
        r_digit_index <= '0;
        r_loop_count  <= '0;
        r_dv          <= '0;
    end else begin
        r_dv <= 1'b0;  // Default: done is low

        casez (r_fsm_main)
            IDLE: begin
                if (start == 1'b1) begin
                    r_binary      <= binary;
                    r_fsm_main    <= SHIFT;
                    r_bcd         <= '0;
                    r_loop_count  <= '0;
```

```
                r_digit_index <= '0;
            end else begin
                r_fsm_main <= IDLE;
            end
        end

        SHIFT: begin
            // Shift BCD left and insert MSB of binary
            r_bcd      <= r_bcd << 1;
            r_bcd[0]   <= r_binary[WIDTH-1];
            r_binary   <= r_binary << 1;
            r_fsm_main <= CK_S_IDX;
        end

        CK_S_IDX: begin
            if (r_loop_count == LOOP_COUNT_WIDTH'(WIDTH - 1))
begin
                r_loop_count <= '0;
                r_fsm_main   <= BCD_DONE;
            end else begin
                r_loop_count <= r_loop_count + 1'b1;
                r_digit_index <= '0;
                r_fsm_main   <= ADD;
            end
        end

        ADD: begin
            if (w_bcd_digit > 4'd4) begin
                r_bcd[(r_digit_index*4)+:4] <= w_bcd_digit + 4'd3;
            end
            r_fsm_main <= CK_D_IDX;
        end

        CK_D_IDX: begin
            if (r_digit_index == DIGIT_INDEX_WIDTH'(DIGITS - 1))
begin
                r_digit_index <= '0;
                r_fsm_main    <= SHIFT;
            end else begin
                r_digit_index <= r_digit_index + 1'b1;
                r_fsm_main    <= ADD;
            end
        end

        BCD_DONE: begin
            r_dv       <= 1'b1;
            r_fsm_main <= IDLE;
```

```
            end

            default: begin
                r_fsm_main <= IDLE;
            end
        endcase
    end
end
```

11.4.9 BCD Format and Encoding

Each BCD digit uses 4 bits to represent decimal values 0-9:

Decimal	BCD (4-bit)	Hex
0	0000	0x0
1	0001	0x1
2	0010	0x2
3	0011	0x3
4	0100	0x4
5	0101	0x5
6	0110	0x6
7	0111	0x7
8	1000	0x8
9	1001	0x9
Invalid	1010-1111	0xA-0xF

The output `bcd` uses packed BCD format where multiple digits are concatenated:

```
// For DIGITS = 3, bcd[11:0] contains:
// bcd[11:8]  = hundreds digit
// bcd[7:4]   = tens digit
// bcd[3:0]   = ones digit

// Example: decimal 123 → bcd = 12'h123
```

Extracting individual digits from the packed vector:

```
// Extract individual digits from packed BCD
logic [3:0] ones, tens, hundreds;

assign ones     = bcd[3:0];
assign tens     = bcd[7:4];
assign hundreds = bcd[11:8];
```

```
// Convert to ASCII for display
logic [7:0] ascii_ones, ascii_tens, ascii_hundreds;

assign ascii_ones     = 8'h30 + ones;     // Add ASCII '0'
assign ascii_tens     = 8'h30 + tens;
assign ascii_hundreds = 8'h30 + hundreds;
```

11.5 Related Modules

Nothing in the tree instantiates this module and it instantiates nothing: it is a leaf, used directly by whatever design needs it. Its nearest neighbours in `rtl/common/` are:

- (none in this category)

11.6 Timing Characteristics

11.6.1 Conversion Latency

The total conversion time depends on the input width and number of digits:

Formula: `Latency = (WIDTH-1) × (2×DIGITS + 2) + 4` clock cycles (measured from the cycle `start` is accepted to the cycle `done` asserts). The +4 tail is `SHIFT`, `CK_S_IDX`, `BCD_DONE`, and one more cycle for the registered `done`: `r_dv` is *set* in the BCD_DONE state and `done = r_dv` only reads 1 the following cycle, after the FSM is back in IDLE.

The per-bit cost is dominated by the add-3 phase: each shift is followed by a full `ADD → CK_D_IDX` pass over **every** digit, so each bit position costs `2×DIGITS` cycles there in addition to its `SHIFT` and `CK_S_IDX` cycles. The `CK_S_IDX` and `CK_D_IDX` states are distinct clock cycles, not free.

11.6.2 Latency Breakdown

- **SHIFT** + **CK_S_IDX**: 2 cycles per input bit → `2 × WIDTH` cycles.
- **ADD** + **CK_D_IDX**: `2 × DIGITS` cycles after each shift **except the last** (the final bit skips the add phase and goes straight to `BCD_DONE`) → `(WIDTH-1) × 2 × DIGITS` cycles.
- **Completion**: 1 cycle for BCD_DONE state.
- **Total**: `(WIDTH-1) × (2×DIGITS + 2) + 3` cycles.

11.6.3 Latency Examples

WIDTH	DIGITS	Total Cycles	@ 100MHz	@ 50MHz
4	2	3×6 + 4 = 22	220ns	440ns
8	3	7×8 + 4 = 60	600ns	1.20µs
12	4	11×10 + 4 = 114	1.14µs	2.28µs
16	5	15×12 + 4 = 184	1.84µs	3.68µs
20	7	19×16 + 4 = 308	3.08µs	6.16µs

11.6.4 Pipeline Considerations

The conversion is inherently sequential, making pipelining challenging: - **Data Dependency**: Each iteration depends on previous results - **State-Based**: FSM prevents overlapping conversions - **Optimization**: Consider parallel converters for high throughput

11.6.5 Synthesis Results (Typical FPGA)

Configuration	LUTs	FFs	BRAM	Max Freq	Notes
8-bit, 3-digit	85	45	0	250 MHz	Basic config
12-bit, 4-digit	145	65	0	220 MHz	Medium size
16-bit, 5-digit	220	85	0	200 MHz	Larger design
20-bit, 7-digit	350	125	0	180 MHz	Complex design

11.6.6 Power Consumption

- **Dynamic Power**: Proportional to conversion frequency
- **Static Power**: Minimal (standard CMOS)
- **Optimization**: Clock gating during IDLE state
- **Trade-off**: Performance vs. power efficiency

11.7 Usage Examples

Every parameter and port below is read from the module declaration.

```
bin_to_bcd #(
    .WIDTH                  (8),
    .DIGITS                 (3)
) u_bin_to_bcd (
    .clk                    (clk),
    .rst_n                  (rst_n),
    .start                  (start),
    .binary                 (binary),
    .bcd                    (bcd),
    .done                   (done)
);
```

11.8 Design Notes

11.8.1 Optimization Techniques

11.8.1.1 Early Termination Optimization

```
// Optimize for small numbers by detecting leading zeros
logic [WIDTH-1:0] leading_zero_mask;
logic [$clog2(WIDTH):0] significant_bits;

// Count significant bits
count_leading_zeros #(.WIDTH(WIDTH)) clz_inst (
    .data(binary),
    .clz(leading_zeros)
);

assign significant_bits = WIDTH - leading_zeros;

// Modify loop termination condition
always_ff @(posedge clk) begin
    // ... other logic ...

    CK_S_IDX: begin
        if (r_loop_count == significant_bits - 1) begin
            r_fsm_main <= BCD_DONE; // Early termination
        end else begin
            // Continue normal processing
        end
    end
end
```

11.8.1.2 Parallel Digit Processing

```
// Process all digits in parallel during ADD state
logic [3:0] digit_array [DIGITS];
logic [3:0] adjusted_digits [DIGITS];

// Extract all digits
genvar i;
generate
    for (i = 0; i < DIGITS; i++) begin : extract_digits
        assign digit_array[i] = r_bcd[i*4+:4];
        assign adjusted_digits[i] = (digit_array[i] > 4) ?
                                    (digit_array[i] + 3) :
digit_array[i];
    end
endgenerate

// Update all digits simultaneously
always_ff @(posedge clk) begin
    // ... other states ...

    ADD: begin
        for (int j = 0; j < DIGITS; j++) begin
            r_bcd[j*4+:4] <= adjusted_digits[j];
        end
        r_fsm_main <= CK_S_IDX; // Skip CK_D_IDX state
    end
end
```

11.8.1.3 Pipelined Implementation

```
// Multi-stage pipeline for high throughput
module bin_to_bcd_pipelined #(
    parameter int WIDTH = 8,
    parameter int DIGITS = 3,
    parameter int STAGES = 4  // Pipeline depth
) (
    input  logic                clk,
    input  logic                rst_n,
    input  logic                valid_in,
    input  logic [WIDTH-1:0]    binary_in,
    output logic                valid_out,
    output logic [DIGITS*4-1:0] bcd_out
);

    // Pipeline stages
    logic [STAGES-1:0] valid_pipe;
    logic [WIDTH-1:0] binary_pipe [STAGES];
    logic [DIGITS*4-1:0] bcd_pipe [STAGES];
```

```
    // Stage processing
    generate
        for (genvar s = 0; s < STAGES; s++) begin : pipeline_stages
            logic [7:0] bits_to_process;
            assign bits_to_process = WIDTH * s / STAGES;

            // Process subset of bits in each stage
            // Implementation depends on specific algorithm
partitioning
        end
    endgenerate

endmodule
```

11.8.2 Common Design Patterns

11.8.2.1 Pattern 1: Auto-Converting Display Interface

```
module auto_display_interface #(
    parameter int WIDTH = 10
) (
    input  logic [WIDTH-1:0] sensor_data,
    input  logic             data_valid,
    output logic [6:0]       display_segments[4],
    output logic             display_valid
);

    // Auto-trigger conversion on new data
    logic prev_data_valid;
    logic start_conversion;

    always_ff @(posedge clk) begin
        prev_data_valid <= data_valid;
    end

    assign start_conversion = data_valid && !prev_data_valid;

    // BCD converter and display driver
    // ... (implementation similar to previous examples)

endmodule
```

11.8.2.2 Pattern 2: Buffered Converter with FIFO

```
module buffered_bcd_converter #(
    parameter int WIDTH = 12,
    parameter int DIGITS = 4,
```

```
    parameter int BUFFER_DEPTH = 8
) (
    input  logic             clk,
    input  logic             rst_n,
    input  logic [WIDTH-1:0] binary_in,
    input  logic             write_enable,
    output logic [DIGITS*4-1:0] bcd_out,
    output logic             bcd_valid,
    output logic             buffer_full,
    output logic             buffer_empty
);

    // Input FIFO
    logic [WIDTH-1:0] fifo_data_out;
    logic fifo_read_enable, fifo_empty_int;

    fifo_sync #(
        .DATA_WIDTH(WIDTH),
        .DEPTH(BUFFER_DEPTH)
    ) input_buffer (
        .clk(clk),
        .rst_n(rst_n),
        .write(write_enable),
        .wr_data(binary_in),
        .wr_full(buffer_full),
        .wr_almost_full(),
        .read(fifo_read_enable),
        .rd_data(fifo_data_out),
        .rd_empty(fifo_empty_int),
        .rd_almost_empty()
    );

    // BCD converter control
    logic conversion_idle;

    assign fifo_read_enable = !fifo_empty_int && conversion_idle;
    assign buffer_empty = fifo_empty_int;

    // BCD converter instance
    bin_to_bcd #(
        .WIDTH(WIDTH),
        .DIGITS(DIGITS)
    ) converter (
        .clk(clk),
        .rst_n(rst_n),
        .start(fifo_read_enable),
        .binary(fifo_data_out),
```

```
        .bcd(bcd_out),
        .done(bcd_valid)
    );

    // Track converter state
    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            conversion_idle <= 1;
        end else begin
            if (fifo_read_enable) begin
                conversion_idle <= 0;
            end else if (bcd_valid) begin
                conversion_idle <= 1;
            end
        end
    end

endmodule
```

11.8.3 Troubleshooting Guide

11.8.3.1 Issue 1: Incorrect BCD Output

Symptoms: Output doesn't match expected decimal value **Causes**: - Insufficient DIGITS parameter - Width mismatch between binary input and internal registers - BCD digit corruption during processing

Debug Steps:

```
// Add internal signal monitoring
always_ff @(posedge clk) begin
    if (dut.r_fsm_main == ADD && dut.w_bcd_digit > 4) begin
        $display("ADD operation: digit=%d, new_value=%d",
                 dut.w_bcd_digit, dut.w_bcd_digit + 3);
    end
end
```

11.8.3.2 Issue 2: Conversion Never Completes

Symptoms: done signal never asserts **Causes**: - FSM stuck in loop - Incorrect loop termination condition - Clock or reset issues

Debug Steps:

```
// Monitor FSM state transitions
always_ff @(posedge clk) begin
    $display("State: %s, Loop: %d, Digit: %d",
             dut.r_fsm_main.name(), dut.r_loop_count,
dut.r_digit_index);
end
```

11.8.3.3 Issue 3: Timing Violations

Symptoms: Synthesis reports timing failures **Causes**: - Long combinational paths in ADD logic - High fan-out on BCD register - Insufficient pipeline stages

Solutions:

```
// Add pipeline registers
logic [DIGITS*4-1:0] bcd_pipe;
always_ff @(posedge clk) begin
    bcd_pipe <= r_bcd;
end

// Use parallel digit processing
// (Shown in optimization section)
```

11.9 Testing

11.9.1 Test Strategy

```
module tb_bin_to_bcd;

    // Test parameters
    parameter int WIDTH = 8;
    parameter int DIGITS = 3;
    parameter int MAX_VALUE = (1 << WIDTH) - 1;

    // DUT signals
    logic clk, rst_n, start, done;
    logic [WIDTH-1:0] binary;
    logic [DIGITS*4-1:0] bcd;

    // Test control
    logic [WIDTH-1:0] test_vector;
    logic [31:0] expected_decimal;
    logic [DIGITS*4-1:0] expected_bcd;

    // DUT instantiation
    bin_to_bcd #(
        .WIDTH(WIDTH),
        .DIGITS(DIGITS)
    ) dut (.*);

    // Clock generation
    initial clk = 0;
    always #5 clk = ~clk;
```

```
    // Test sequence
    initial begin
        rst_n = 0;
        start = 0;
        binary = 0;

        #100 rst_n = 1;

        // Test all possible values
        for (int i = 0; i <= MAX_VALUE; i++) begin
            test_conversion(i);
        end

        // Boundary tests
        test_conversion(0);
        test_conversion(1);
        test_conversion(MAX_VALUE);

        // Random tests
        repeat (1000) begin
            test_conversion($random % (MAX_VALUE + 1));
        end

        $display("All tests completed successfully!");
        $finish;
    end

    // Test task
    task test_conversion(input [WIDTH-1:0] value);
        begin
            binary = value;
            expected_decimal = value;
            expected_bcd = decimal_to_bcd(expected_decimal);

            @(posedge clk);
            start = 1;
            @(posedge clk);
            start = 0;

            wait(done);
            @(posedge clk);

            if (bcd !== expected_bcd) begin
                $error("Mismatch: binary=%d, expected_bcd=%h,
actual_bcd=%h",
                       value, expected_bcd, bcd);
```

```
            end else begin
                $display("PASS: binary=%d → bcd=%h", value, bcd);
            end
        end
    endtask

    // Reference function
    function [DIGITS*4-1:0] decimal_to_bcd(input [31:0] decimal);
        logic [31:0] temp;
        logic [DIGITS*4-1:0] result;
        begin
            temp = decimal;
            for (int i = 0; i < DIGITS; i++) begin
                result[i*4+:4] = temp % 10;
                temp = temp / 10;
            end
            decimal_to_bcd = result;
        end
    endfunction

endmodule
```

11.9.2 Coverage Model

```
covergroup bcd_conversion_cg;

    // Input value coverage
    cp_binary_value: coverpoint binary {
        bins zero = {0};
        bins small[] = {[1:9]};
        bins medium[] = {[10:99]};
        bins large[] = {[100:255]}; // For 8-bit
        bins max_val = {2**WIDTH - 1};
    }

    // State coverage
    cp_fsm_states: coverpoint dut.r_fsm_main {
        bins all_states[] = {IDLE, SHIFT, CK_S_IDX, ADD, CK_D_IDX,
BCD_DONE};
        bins state_transitions[] = (IDLE => SHIFT),
                                   (SHIFT => CK_S_IDX),
                                   (CK_S_IDX => ADD),
                                   (CK_S_IDX => BCD_DONE),
                                   (ADD => CK_D_IDX),
                                   (CK_D_IDX => ADD),
                                   (CK_D_IDX => SHIFT),
                                   (BCD_DONE => IDLE);
    }
```

```
    // BCD digit coverage
    cp_bcd_digits: coverpoint bcd {
        bins valid_bcd[] = {[12'h000:12'h999]}; // Valid BCD range
    }

    // Loop iteration coverage
    cp_loop_count: coverpoint dut.r_loop_count {
        bins all_counts[] = {[0:WIDTH-1]};
    }

    // Cross coverage
    cross_value_states: cross cp_binary_value, cp_fsm_states;

endcovergroup
```

11.9.3 Formal Verification Properties

```
// Properties for formal verification
module bin_to_bcd_properties #(
    parameter int WIDTH = 8,
    parameter int DIGITS = 3
);

    // Bind to DUT
    bind bin_to_bcd bin_to_bcd_properties #(WIDTH, DIGITS) props_inst
(.*);

    // Basic properties
    property conversion_eventually_completes;
        @(posedge clk) disable iff (!rst_n)
        start |-> ##[1:200] done; // Adjust bound based on max latency
    endproperty

    property bcd_output_valid;
        @(posedge clk) disable iff (!rst_n)
        done |-> valid_bcd(bcd);
    endproperty

    property conversion_correctness;
        logic [31:0] expected_decimal;
        @(posedge clk) disable iff (!rst_n)
        (start, expected_decimal = binary) |->
        ##[1:200] (done && (bcd_to_decimal(bcd) == expected_decimal));
    endproperty

    // Helper functions
```

```
    function bit valid_bcd(logic [DIGITS*4-1:0] bcd_val);
        for (int i = 0; i < DIGITS; i++) begin
            if (bcd_val[i*4+:4] > 9) return 0;
        end
        return 1;
    endfunction

    function [31:0] bcd_to_decimal(logic [DIGITS*4-1:0] bcd_val);
        logic [31:0] result = 0;
        for (int i = 0; i < DIGITS; i++) begin
            result += bcd_val[i*4+:4] * (10**i);
        end
        return result;
    endfunction

    // Assertions
    assert property (conversion_eventually_completes);
    assert property (bcd_output_valid);
    assert property (conversion_correctness);

endmodule
```

That covers the Binary to BCD converter end to end — algorithm, implementation, verification, and optimization. Double-Dabble is sequential by nature, but it gives you reliable, predictable conversion for a wide range of digital display applications.

11.10 References

- Couleur, J.F. “BIDEC - A Binary-to-Decimal or Decimal-to-Binary Converter.” *IRE Transactions on Electronic Computers*, EC-7(4), Dec. 1958. DOI: 10.1109/TEC.1958.5222665 — the publication conventionally cited as the origin of the shift-and-add-3 (“double dabble”) method this FSM implements.
- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 3 “Basic Combinational Circuits”, Sec. 3.12.1 Binary to BCD Conversion (textbook treatment; secondary source).
- Indexed in `rtl/common/references/README.md`.

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12 CAM Tag

Associative tag storage: you find entries by content, not by address.

12.1 Overview

The `cam_tag` module implements a Content Addressable Memory (CAM) for tag storage and lookup. Lookups are associative — you find data by content, not by address — which is exactly what cache implementations, translation lookaside buffers (TLBs), and routing tables need.

12.2 Parameters

Parameter	Type	Default	Description
ENABLE	int	1	Enable/disable CAM functionality
N	int	8	Width of each tag in bits
DEPTH	int	16	Number of tag entries in the CAM

12.3 Ports

12.3.1 Inputs

Port	Width	Description
clk	1	System clock
rst_n	1	Active-low asynchronous reset
tag_in_status	N	Tag to search for (lookup operation)
mark_valid	1	Signal to add a new valid tag
tag_in_valid	N	Tag value to mark as valid
mark_invalid	1	Signal to invalidate an existing tag
tag_in_invalid	N	Tag value to mark as

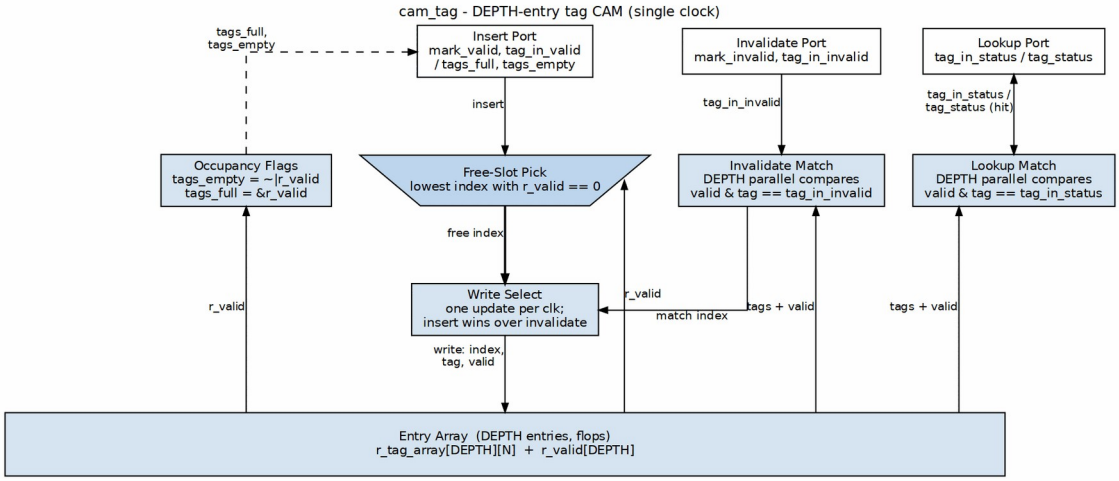
Port	Width	Description
		invalid

12.3.2 Outputs

Port	Width	Description
tags_empty	1	Indicates when CAM contains no valid entries
tags_full	1	Indicates when CAM is completely full
tag_status	1	Result of lookup operation (1 if tag found, 0 if not found)

12.4 Functional Description

12.4.1 Block diagram



cam_tag block diagram

Three requests enter at the top, one column each. An insert goes through the free-slot pick, the lowest-index entry whose valid bit is clear, and into the write select. An invalidate goes through DEPTH parallel compares and into the same write select. The write select applies one update per clock, and an insert wins over an invalidate. A lookup runs its own DEPTH parallel compares and returns `tag_status` straight back to its port. Every search reads the entry array straight up its own column. The occupancy flags reduce the valid vector and return `tags_full` and `tags_empty` to the insert side, where `tags_full` blocks an insert.

12.4.2 Tag Lookup

Continuously compares `tag_in_status` against all valid entries: - Returns 1 if tag found among valid entries - Returns 0 if tag not found or entry is invalid

12.4.3 Tag Insertion

When `mark_valid` is asserted and CAM is not full: 1. Store `tag_in_valid` at `w_next_valid_loc` 2. Set corresponding valid bit 3. Only occurs when `ENABLE != 0`

12.4.4 Tag Invalidation

When `mark_invalid` is asserted and matching tag found: 1. Clear the tag value at matching location 2. Clear corresponding valid bit

12.4.5 Enable Control

When `ENABLE = 0`: - Tag insertion is disabled - Tag lookup and invalidation still function - Useful for disabling CAM without losing existing data

12.4.6 Allocation Strategy

- **First Available**: Uses the **lowest-indexed** free location (see the search below). Functional behavior is correct either way; only the index order matters when matching allocations to waveforms.
- **Replacement**: No automatic replacement policy (must manually invalidate)
- **Overflow Protection**: Prevents insertion when full

12.4.7 Storage Arrays

```
logic [N-1:0]     r_tag_array [0:DEPTH-1];  // Tag storage array
logic [DEPTH-1:0] r_valid;                  // Valid bit for each
entry
```

12.4.8 Search Logic

Three parallel search operations run side by side:

12.4.8.1 1. Next Available Location Search

```
always_comb begin
    w_next_valid_loc = -1;
    for (int i=DEPTH-1; i >= 0; i--)
        if (r_valid[i] == 1'b0)
            w_next_valid_loc = i;
end
```

- Iterates from the highest index down to 0, overwriting `w_next_valid_loc` on every free slot — so the **last** assignment (index 0, if free) wins
- Returns the **lowest-indexed** free location (e.g., all slots free → slot 0)
- Returns -1 if no free locations available

12.4.8.2 2. Valid Tag Match Search

```
always_comb begin
    w_match_loc = -1;
    for (int i = 0; i < DEPTH; i++)
        if (r_valid[i] == 1'b1 && tag_in_status == r_tag_array[i])
            w_match_loc = i;
end
```

- Searches for exact tag match among valid entries
- Returns index of matching entry or -1 if no match

12.4.8.3 3. Invalid Tag Match Search

```
always_comb begin
    w_match_invalid_loc = -1;
    for (int i = 0; i < DEPTH; i++)
        if (r_valid[i] == 1'b1 && tag_in_invalid == r_tag_array[i])
            w_match_invalid_loc = i;
end
```

- Used for invalidation operations
- Finds valid entries that match the invalidation tag

12.4.9 State Management

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_valid <= 'b0;
        for (int i = 0; i < DEPTH; i++) begin
            r_tag_array[i] <= 'b0;
        end
    end else begin
        if (mark_valid && !tags_full && (ENABLE != 0)) begin
            r_tag_array[w_next_valid_loc] <= tag_in_valid;
            r_valid[w_next_valid_loc] <= 1'b1;
        end else if (mark_invalid && w_match_invalid_loc >= 0) begin
            r_tag_array[w_match_invalid_loc] <= 'b0;
            r_valid[w_match_invalid_loc] <= 1'b0;
        end
    end
end
```

12.4.10 Status Signals

- `tags_empty = ~|r_valid`: NOR of all valid bits
- `tags_full = &r_valid`: AND of all valid bits

- `tag_status`: Result of tag lookup operation

12.5 Related Modules

Nothing in the tree instantiates this module and it instantiates nothing: it is a leaf, used directly by whatever design needs it. Its nearest neighbours in `rtl/common/` are:

- (none in this category)

12.6 Timing Characteristics

This module is **sequential**: it contains clocked logic (via `always_ff` or the repository's `ALWAYS_FF_RST` macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

12.7 Usage Examples

Every parameter and port below is taken from the module declaration.

```
cam_tag #(
    .ENABLE                 (1),
    .N                      (8),
    .DEPTH                  (16)
) u_cam_tag (
    .clk                    (clk),
    .rst_n                  (rst_n),
    .tag_in_status          (tag_in_status),
    .mark_valid             (mark_valid),
    .tag_in_valid           (tag_in_valid),
    .mark_invalid           (mark_invalid),
    .tag_in_invalid         (tag_in_invalid),
    .tags_empty             (tags_empty),
    .tags_full              (tags_full),
    .tag_status             (tag_status)
);
```

12.8 Design Notes

12.8.1 Usage Considerations

- **Timing**: All lookups are combinational (single cycle)
- **Capacity**: Monitor `tags_full` before insertions
- **Conflicts**: No protection against duplicate tag insertion
- **Performance**: Search time is constant regardless of occupancy
- **Power**: All entries are searched simultaneously (high power for large depths)

12.9 Testing

`val/common/test_cam_tag.py` exercises this module. It collects 2 parameter cases at the default `REG_LEVEL`.

```
source env_python
pytest val/common/test_cam_tag.py -v
```

12.10 References

- Pagiamtzis, K., Sheikholeslami, A. “Content-Addressable Memory (CAM) Circuits and Architectures: A Tutorial and Survey.” *IEEE Journal of Solid-State Circuits*, 41(3), Mar. 2006. DOI: 10.1109/JSSC.2005.864128 — the standard reference for CAM organization (parallel compare of every stored word against the search key, match-line per entry); this module is a small register-based binary CAM.
- Sasao, T. *Memory-Based Logic Synthesis*, Springer, ch. 2 “Basic Elements”, §2.3 “Content Addressable Memory” — textbook treatment (secondary source).
- Indexed in `rtl/common/references/README.md` (local PDF: `ContentAddressableMemory__Pagiamtzis2006.pdf`)

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13 Clock Divider

Multiple divided clocks from one input clock, all tapped off a single shared counter so they come out phase-aligned.

WARNING: NOT recommended for functional clocks! Use PLL/MMCM/clock manager primitives instead. This module is intended for testbenches, debug outputs, and non-critical timing applications.

13.1 Overview

The `clock_divider` module generates multiple divided clock signals from a single input clock using configurable pick-off points from a shared master counter. Because every output taps the same counter, the divided clocks come out synchronized and phase-aligned — the behavior you need for multi-rate digital signal processing, communication systems, and power management applications.

```
module clock_divider #(
    parameter int N             = 4,  // Number of output clocks
    parameter int PO_WIDTH      = 8,  // Width of the Pick off
registers
    parameter int COUNTER_WIDTH = 64  // Width of the counter
) (
    input  logic                     clk,             // Input clock
signal
    input  logic                     rst_n,           // Active low
reset signal
    input  logic [(N*PO_WIDTH-1):0] pickoff_points,  // the pick off
point config registers
    output logic [           N-1:0] divided_clk      // Divided clock
signals
);
```

13.2 Parameters

Parameter	Type	Default	Description
N	int	4	Number of independently configurable output clocks. Range: 1 to 16 (practical limit). Determines number of output clock domains and resource usage (N registers + N muxes).

Parameter	Type	Default	Description
PO_WIDTH	int	8	Bit width of each pick-off point configuration register. Range: 4 to 8 bits. Must be > `$clog2(COUNTER_WIDTH)` to avoid truncation — equivalently `PO_WIDTH >= $clog2(COUNTER_WIDTH + 1)`, which ensures PO_WIDTH can hold the value COUNTER_WIDTH without truncation. Examples: CW=16 needs PO≥5, CW=32 needs PO≥6, CW=64 needs PO≥7. Larger width allows finer counter bit selection but uses more configuration storage. Typical: PO_WIDTH=8 for COUNTER_WIDTH up to 128.
COUNTER_WIDTH	int	64	Bit width of the master binary counter. Range: 2 to 64 bits (practical range). Determines maximum division ratio (2^COUNTER_WIDTH). Example: WIDTH=16 → max division ratio = 65536. Larger counters enable slower frequencies but consume more resources.

13.2.1 Parameter Relationships

- **Addressing Constraint**: `PO_WIDTH > $clog2(COUNTER_WIDTH)` (strictly greater — the RTL `$fatal`s at elaboration if `PO_WIDTH <= $clog2(COUNTER_WIDTH)`). E.g. `COUNTER_WIDTH=16` needs `PO_WIDTH ≥ 5`.
- **Division Range**: Each output can divide by 2^1 to 2^COUNTER_WIDTH
- **Configuration Width**: Total configuration bits = `N × PO_WIDTH`

13.3 Ports

13.3.1 Inputs

Port	Width	Description
clk	1	Master input clock (source to be divided)
rst_n	1	Active-low asynchronous reset
pickoff_points	N×PO_WIDTH	Packed pickoff point configuration array

pickoff_points Format:

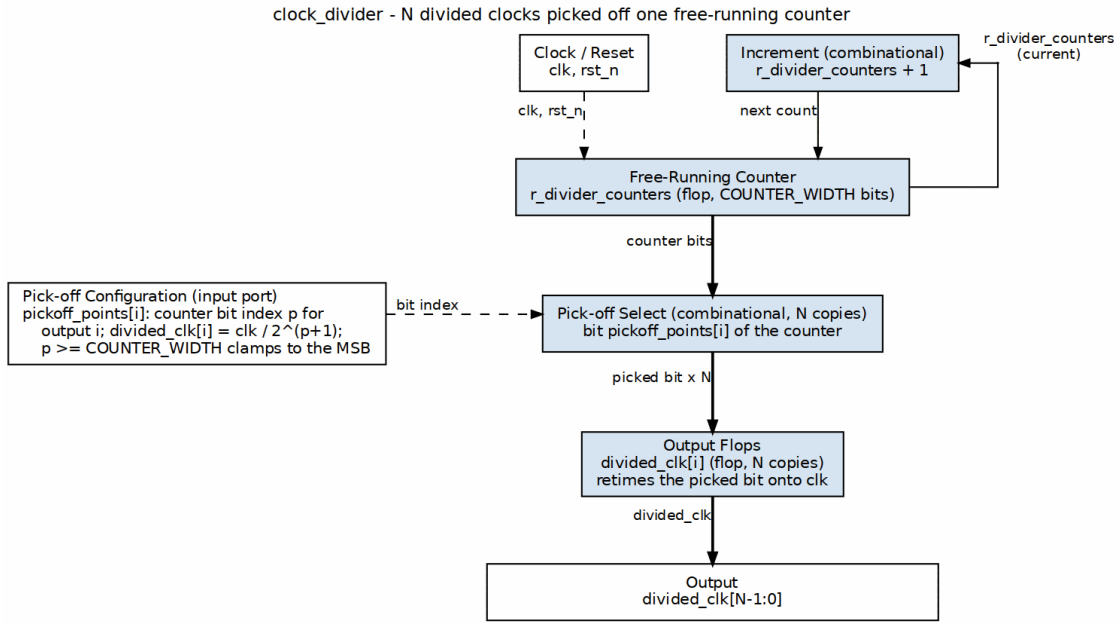
```
Bit Packing: {po[N-1], ..., po[1], po[0]}
Each po[i] (PO_WIDTH bits): Counter bit index to sample for output i
```

13.3.2 Outputs

Port	Width	Description
divided_clk	N	Divided clock outputs (registered)

13.4 Functional Description

13.4.1 Block diagram



clock_divider block diagram

The divided clocks are the main flow, drawn heavy. One free-running counter increments every clock and feeds back up the right edge. For each of the N outputs, a pick-off select takes counter bit `pickoff_points[i]`, which toggles at `clk / 2^(p+1)`, and an output flop retimes it onto `clk`. The pick-off configuration enters from the side as sideband.

13.4.2 Counter Operation

- Free-running binary counter increments every input clock cycle
- Wraps at 2^COUNTER_WIDTH (no overflow flag)
- Each bit toggles at half the frequency of the previous bit
    - Bit 0: clk/2 (toggles every cycle)
    - Bit 1: clk/4 (toggles every 2 cycles)
    - Bit N: clk/2^(N+1)

13.4.3 Pickoff Point Selection

- Each output `i` samples `counter[pickoff_points[i]]`

- Division ratio = 2^(pickoff_point+1)
- Example: pickoff=3 → samples counter[3] → divides by 16

13.4.4 Out-of-Range Handling

- If `pickoff_points[i] ≥ COUNTER_WIDTH`, clamps to `COUNTER_WIDTH-1` (MSB)
- Prevents illegal bit indexing
- MSB provides slowest possible division

13.4.5 Output Registration

- All `divided_clk` outputs are registered for glitch-free operation
- Adds 1 cycle latency but prevents combinational glitches
- Reset clears all outputs to 0 (low phase)

13.4.6 Division Ratios (100MHz Input Clock)

pickoff_point	Counter Bit	Division Ratio	Output Freq	Period
0	[0]	2	50MHz	20ns
1	[1]	4	25MHz	40ns
2	[2]	8	12.5MHz	80ns
3	[3]	16	6.25MHz	160ns
4	[4]	32	3.125MHz	320ns
5	[5]	64	1.5625MHz	640ns
10	[10]	2048	48.8kHz	20.5µs
15	[15]	65536	1.5kHz	655µs
19	[19]	1048576	95Hz	10.5ms
23	[23]	16777216	6Hz	168ms

13.4.7 Duty Cycle

- All divided clocks have 50% duty cycle
- Result of sampling counter bit (toggle behavior)
- High for 2^(pickoff_point) input cycles

- Low for 2^(pickoff_point) input cycles

13.4.8 Internal Signals

```
logic [COUNTER_WIDTH-1:0] r_divider_counters;  // Free-running binary
counter
localparam int ADDR_WIDTH = $clog2(COUNTER_WIDTH);  // Address width
```

13.5 Timing Characteristics

13.5.1 Latency

- **Output Latency**: 1 cycle (registered output)
- **Clock-to-Q**: Standard flip-flop delay
- **Propagation**: Counter increment → mux → output register

13.5.2 Phase Relationships

- All divided clocks share same counter → phase-locked to each other
- Divided clocks have phase offset relative to input clock
- Phase offset depends on counter reset value (0)

13.5.3 Jitter

- Input clock jitter propagates to outputs
- Additional quantization jitter: ±1 input clock cycle
- For low-jitter clocks: Use dedicated PLL/clock synthesis

13.5.4 Maximum Frequency

- **Limiting Factor**: Counter width and mux depth
- **Typical Performance**:
  - COUNTER_WIDTH=16: ~500 MHz (modern FPGAs)
  - COUNTER_WIDTH=32: ~400 MHz
  - COUNTER_WIDTH=64: ~300 MHz

13.5.5 Resource Scaling

COUNTER_WIDTH	N Outputs	FFs	LUTs (mux, est.)	fmax (Est.)
16	4	20	~12	500 MHz
32	4	36	~28	400 MHz
64	4	68	~52	300 MHz
16	8	24	~24	500 MHz

FF counts are exact (COUNTER_WIDTH counter bits + N output registers). The LUT column is `N * ceil((COUNTER_WIDTH-1)/5)` for the pickoff muxes on a 6-LUT architecture, excluding the clamp comparators; earlier revisions of this table listed 8/10/12, which did not scale with the mux width and understated the wide-counter cases by several times. fmax figures remain unsourced estimates.

13.6 Usage Examples

13.6.1 Four Debug Clocks

```
// Generate clk/4, clk/16, clk/256, clk/65536
logic [31:0] pickoff_cfg;
assign pickoff_cfg = {8'd15, 8'd7, 8'd3, 8'd1};  // {po3, po2, po1,
po0}


clock_divider #(
    .N(4),              // 4 outputs
    .PO_WIDTH(8),       // 8-bit pickoff point
    .COUNTER_WIDTH(16)  // 16-bit counter (max div = 65536)
) u_clk_div (
    .clk           (clk_100mhz),
    .rst_n         (rst_n),
    .pickoff_points(pickoff_cfg),
    .divided_clk   ({clk_1p5khz, clk_390khz, clk_6p25mhz, clk_25mhz})
);
```

13.6.2 Baud Rate Generator (Approximate)

```
// For 9600 baud from 100MHz:
// Ideal: 100MHz / 9600 ≈ 10417 (not power-of-2)
// Closest power-of-2: 2^13 = 8192 → 100MHz/8192 = 12.2kHz
// WARNING: 12.2kHz vs 9.6kHz is a ~27% error -- FAR beyond the ~2-3%
a UART
// can tolerate, and it is NOT a 16x-oversampling clock for 9600 baud
either.
// A power-of-2 divider CANNOT hit standard baud rates. For a real
UART use
// counter_load_clear (arbitrary modulus), not this divider.
```

```
logic [7:0] baud_pickoff = 8'd12;  // 2^13 = 8192 division
(illustrative only)

clock_divider #(
    .N(1),
    .PO_WIDTH(8),
    .COUNTER_WIDTH(16)
) u_baud_clk (
    .clk           (sys_clk),
    .rst_n         (rst_n),
    .pickoff_points(baud_pickoff),
    .divided_clk   (uart_baud_x16)  // 16× oversampling clock
);
```

Note: For precise baud rates (not power-of-2), use `counter_load_clear` instead.

13.6.3 Runtime-Programmable Divider

```
// APB-configurable clock divider
logic [7:0] cfg_pickoff_0, cfg_pickoff_1;

always_ff @(posedge pclk) begin
    if (pwrite && paddr == ADDR_PICKOFF_0) cfg_pickoff_0 <=
pwdata[7:0];
    if (pwrite && paddr == ADDR_PICKOFF_1) cfg_pickoff_1 <=
pwdata[7:0];
end

clock_divider #(
    .N(2),
    .PO_WIDTH(8),
    .COUNTER_WIDTH(24)
) u_cfg_clk_div (
    .clk           (core_clk),
    .rst_n         (core_rst_n),
    .pickoff_points({cfg_pickoff_1, cfg_pickoff_0}),
    .divided_clk   ({clk_out_1, clk_out_0})
);
```

13.6.4 Test Clock Generation

```
// Generate 8 test clocks for debug
clock_divider #(
    .N(8),              // 8 debug clocks
    .PO_WIDTH(8),
    .COUNTER_WIDTH(32)  // Support ultra-slow clocks
) u_test_clks (
    .clk           (ref_clk),
```

```
    .rst_n         (test_rst_n),
    .pickoff_points({8'd20, 8'd18, 8'd16, 8'd14, 8'd12, 8'd10, 8'd8,
8'd6}),
    .divided_clk   (test_clk_bus)
);
```

13.7 Design Notes

13.7.1 Changing Pickoff Points at Runtime

```
// To avoid glitches when changing pickoff configuration:
// 1. Assert rst_n (reset)
// 2. Change pickoff_points
// 3. Wait for configuration to settle
// 4. Deassert rst_n
// 5. Outputs start with new division ratios

// The sequence spans three clocks, so it is a small FSM in the logic
that owns
// the configuration registers. It cannot be written as one always_ff
with
// embedded `@(posedge cfg_clk)` statements: IEEE 1800 allows exactly
one event
// control on an always_ff and none inside its body, so that form does
not
// compile.

typedef enum logic [1:0] {CFG_IDLE, CFG_LOAD, CFG_RELEASE}
cfg_state_t;

cfg_state_t                 r_cfg_state;
logic [(N*PO_WIDTH)-1:0]    r_pickoff;
logic                       r_div_rst_n;

always_ff @(posedge cfg_clk or negedge cfg_rst_n) begin
    if (!cfg_rst_n) begin
        r_cfg_state <= CFG_IDLE;
        r_div_rst_n <= 1'b1;
        r_pickoff   <= DEFAULT_PICKOFF_POINTS;
    end else begin
        case (r_cfg_state)
            CFG_IDLE: if (cfg_write) begin
                r_div_rst_n <= 1'b0;         // 1. hold the divider in
reset
                r_cfg_state <= CFG_LOAD;
            end
            CFG_LOAD: begin
```

```
                r_pickoff   <= cfg_data;     // 2. safe now: outputs
are held low
                r_cfg_state <= CFG_RELEASE;
            end
            CFG_RELEASE: begin
                r_div_rst_n <= 1'b1;         // 3. restart on the new
ratios
                r_cfg_state <= CFG_IDLE;
            end
            default: r_cfg_state <= CFG_IDLE;
        endcase
    end
end

clock_divider #(
    .N            (N),
    .PO_WIDTH     (PO_WIDTH),
    .COUNTER_WIDTH(COUNTER_WIDTH)
) u_div (
    .clk           (cfg_clk),
    .rst_n         (r_div_rst_n),   // divider reset is driven by the
FSM above
    .pickoff_points(r_pickoff),
    .divided_clk   (divided_clk)
);
```

Note that `divided_clk` is held low for the two cycles the FSM spends in reset; downstream logic must tolerate the gap rather than assume a continuous clock. Treat these outputs as enables into a synchronous fabric, not as clocks driving a separate domain – see the warning under Synthesis Considerations.

13.7.2 Synthesis Considerations

Resource Utilization: - **Flip-Flops**: COUNTER_WIDTH (counter) + N (output registers) - **LUTs**: one COUNTER_WIDTH-to-1 mux per output (pickoff selection), plus a clamp comparator per output. On a 6-input LUT architecture a CW-to-1 bit mux costs about `ceil((CW-1)/5)` LUTs, so the mux cost scales with COUNTER_WIDTH – it is not a fixed overhead. clamp logic. These are analytic estimates, not synthesis results.

Timing Optimization: - **Critical Path**: Counter increment → mux → output register - **For high-frequency operation**: - Reduce COUNTER_WIDTH (if possible) - Pipeline counter output (add register stage) - Use dedicated clock resources (PLL/MMCM)

Power Optimization: - **Clock Gating**: Gate counter when no divided clocks needed - **Selective Outputs**: Only generate divided clocks actually used - **Example**:

```
logic counter_enable;
assign counter_enable = |divided_clk_enable;  // Only run when needed
```

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) r_divider_counters <= 0;
    else if (counter_enable) r_divider_counters <= r_divider_counters
+ 1;
end
```

13.7.3 Important Warnings

This is the part that bites people, so read it twice.

13.7.3.1 NOT for Functional Clocks

Do NOT use for: - Primary system clocks - Clocking synchronous logic - Production designs requiring precise timing

Use PLL/MMCM/Clock Manager instead: - Dedicated clock synthesis primitives - Clean clock tree - Proper static timing analysis (STA) - Phase-locked loops for low jitter - Precise frequency generation

13.7.3.2 Derived Clock Hazards

Using `divided_clk` as a clock creates a **derived clock**: - Complicates timing analysis - STA tools may not properly constrain - Potential setup/hold violations - Clock domain crossing issues

13.7.3.3 Only Power-of-2 Divisions

- This module ONLY supports power-of-2 division ratios (2, 4, 8, 16, ...)
- For arbitrary ratios (e.g., divide by 7, 100, 1000): Use `counter_load_clear`

13.7.3.4 Glitches During Configuration Changes

- Changing `pickoff_points` at runtime can cause glitches
- Always reset module when changing configuration
- Do NOT change configuration while outputs are actively used

13.8 Related Modules

- `counter_bin.sv` - Simple binary counter (related, not instantiated here — `clock_divider` keeps its own `r_divider_counters` array)
- `counter_freq_invariant.sv` - Time-based counter with 1MHz tick
- `counter_load_clear.sv` - Arbitrary division ratios (not power-of-2)
- `clock_pulse.sv` - Configurable pulse generator

- `clock_gate_ctrl.sv` - Clock gating for power management

13.9 Testing

13.9.1 Test Scenarios

1. **Single Output**: N=1 with various pickoff points
2. **Multiple Outputs**: N=4 with different divisions
3. **Out-of-Range Clamping**: pickoff ≥ COUNTER_WIDTH
4. **Runtime Changes**: Modify pickoff_points dynamically
5. **Reset Behavior**: All outputs low after reset
6. **Division Verification**: Measure output frequencies
7. **Edge Cases**:
   - pickoff=0 (divide by 2)
   - pickoff=COUNTER_WIDTH-1 (slowest division)
   - pickoff > COUNTER_WIDTH (clamping)

13.9.2 Coverage Points

```
covergroup clock_divider_cg @(posedge clk);
    cp_pickoff_range: coverpoint pickoff_points[7:0] {
        bins low = {[0:3]};
        bins mid = {[4:11]};
        bins high = {[12:15]};
        bins oob = {[16:255]};  // Out-of-bounds (will clamp)
    }

    cp_outputs: coverpoint divided_clk {
        bins all_outputs[] = {[1:$]};
    }
endcovergroup
```

13.10 References

- No peer paper — power-of-2 division by counter-bit pickoff is a standard technique with no defensible single origin (marked “standard technique” in the references index).

- Roy, S. *Advanced Digital System Design*, Springer, ch. 4 “Basic Sequential Circuits”, §4.11 “Clock Divider Circuits” and §4.11.4 “Programmable Clock Divider Circuit” — textbook treatment (secondary source).
- LaMeres, B.J. *Introduction to Logic Circuits & Logic Design with Verilog*, Springer, ch. 7 “Sequential Logic Design”, §7.3.1 “Toggle Flop Clock Divider” — textbook treatment (secondary source).
- Indexed in `rtl/common/references/README.md`

13.11 Navigation

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14 Clock Gate Controller

Stops the clock automatically when a block goes idle — one of the cheapest dynamic-power wins available to you.

14.1 Overview

The `clock_gate_ctrl` module is a clock gating controller that stops the clock automatically when a circuit block goes idle, significantly reducing dynamic power consumption. You get a configurable idle timeout, immediate wake-up capability, and safe clock gating through integrated clock gate (ICG) cells.

14.2 Module Interface

```
module clock_gate_ctrl #(
    parameter int IDLE_CNTR_WIDTH = 4   // the ONLY overridable
parameter
) (
    // Inputs
    input logic                         clk_in,
    input logic                         aresetn,
    input logic                         cfg_cg_enable,     // Global
clock gate enable
    input logic [IDLE_CNTR_WIDTH-1:0]   cfg_cg_idle_count, // Idle
countdown value
    input logic                         wakeup,            // Signal to
wake up the block

    // Outputs
    output logic          clk_out,
    output logic          gating            // clock gating indicator
);
    localparam int N = IDLE_CNTR_WIDTH;   // derived alias, NOT a
parameter
```

14.3 Parameters

Parameter	Type	Default	Description
IDLE_CNTR_WIDTH	int	4	Bit width of the idle timeout counter. Range: 2 to 16 bits (practical range). Determines maximum idle timeout (2^IDLE_CNTR_WIDTH - 1 cycles). Power trade-

Parameter	Type	Default	Description
			off: larger counters allow longer timeouts but consume more power. This is the ONLY overridable parameter.

14.3.1 Derived Parameters

Parameter	Default	Description
N	IDLE_CNTR_WIDTH	Internal alias for the counter width, used in port/signal widths. Declared as a `localparam int` in the module body, **not** a parameter — set only `IDLE_CNTR_WIDTH` when instantiating. `N` cannot be overridden: `#(.N(8))` is an elaboration error.

14.4 Ports

14.4.1 Inputs

Port	Width	Type	Description
`clk_in`	1	`logic`	Input clock to be gated
`aresetn`	1	`logic`	Active-low asynchronous reset
`cfg_cg_enable`	1	`logic`	Global clock gating enable control
`cfg_cg_idle_count`	N	`logic`	Idle timeout value (configuration)
`wakeup`	1	`logic`	Immediate wake-up signal

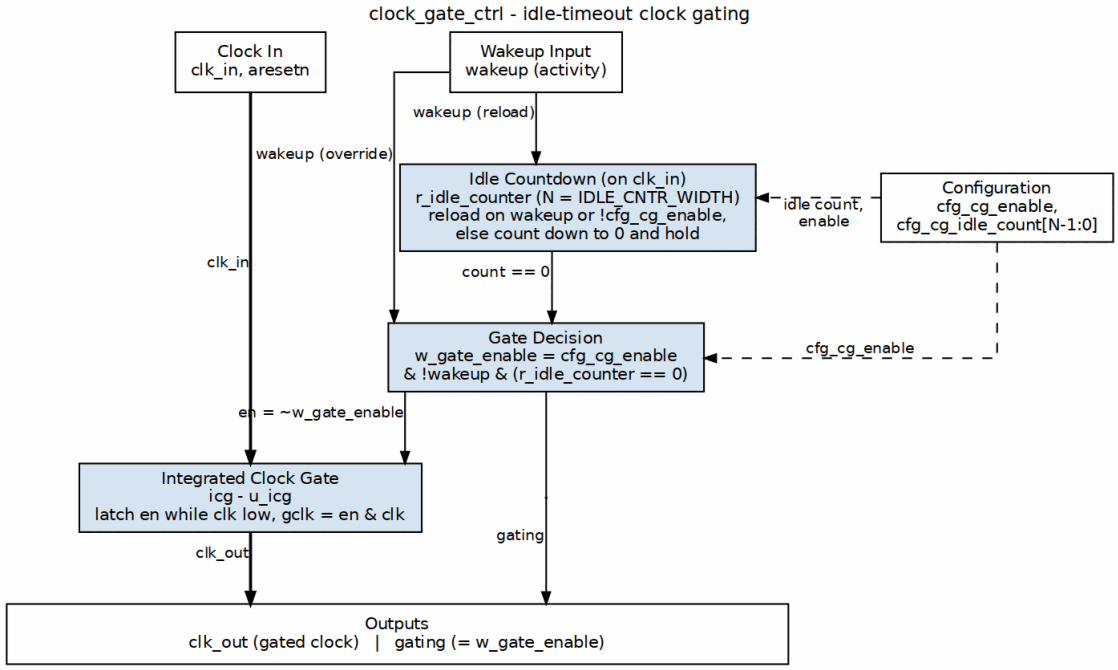
14.4.2 Outputs

Port	Width	Type	Description
`clk_out`	1	`logic`	Gated output clock

Port	Width	Type	Description
gating	1	logic	Clock gating status indicator

14.5 Functional Description

14.5.1 Block diagram



clock_gate_ctrl block diagram

The clock is the main flow, drawn heavy: `clk_in` runs straight down the left through the ICG and leaves as `clk_out`. Control runs down the center. A `wakeup` reloads the idle countdown, which otherwise counts down to zero and holds. The gate decision asserts `w_gate_enable` when gating is enabled, there is no wakeup, and the count has reached zero. The ICG then holds `clk_out` low, and `gating` reports the same decision. Configuration enters from the right as dashed sideband: the reload value and the enable.

14.5.2 Why Clock Gating?

Clock gating is one of the most effective power reduction techniques we have: - **Dynamic Power**: $P = C \times V^2 \times f \times \alpha$ (α = switching activity) - **Clock Networks**: Typically consume 20-40% of total chip power - **Gating Benefits**: Can reduce power by 50-90% in idle blocks - **Area Overhead**: Minimal (< 5% for typical designs)

14.5.3 Clock Gating Safety

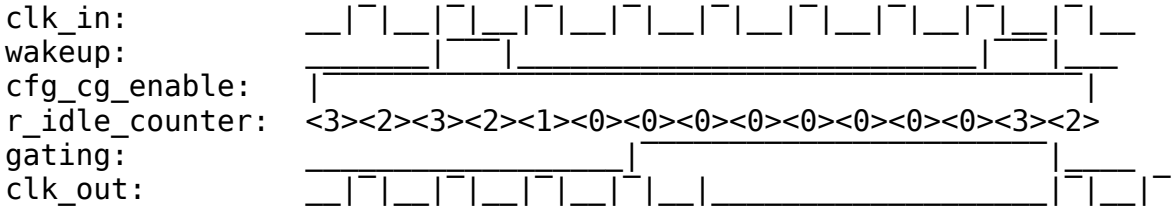
Proper clock gating requires: 1. **Glitch-Free**: No spurious clock edges during gating transitions 2. **Setup/Hold**: Adequate timing margins around gating decisions 3. **Reset Handling**: Proper behavior during reset conditions 4. **Tool Support**: Use of qualified ICG (Integrated Clock Gate) cells

14.5.4 State Machine Behavior

The controller operates as a countdown timer with three states:

1. **ACTIVE State**: Counter loaded with `cfg_cg_idle_count`
2. **COUNTDOWN State**: Counter decrements each cycle when no activity
3. **GATED State**: Counter at zero, clock is gated off

14.5.5 Timing Diagram



14.5.6 Idle Counter Logic

```
logic [N-1:0] r_idle_counter;

always_ff @(posedge clk_in or negedge aresetn) begin
    if (!aresetn) begin
        r_idle_counter <= cfg_cg_idle_count;
    end else begin
        if (wakeup || !cfg_cg_enable) begin
            // On wakeup or global disable, reset counter
            r_idle_counter <= cfg_cg_idle_count;
        end else if (r_idle_counter != 'h0) begin
            // Normal counting operation - decrement when not zero
            r_idle_counter <= r_idle_counter - 1'b1;
        end
        // When counter reaches zero, it stays at zero
    end
end
```

14.5.7 Gating Control Logic

```
wire w_gate_enable = cfg_cg_enable && !wakeup && (r_idle_counter ==
'h0);
```

```
// Instantiate the ICG cell
icg u_icg (
    .clk(clk_in),
    .en(~w_gate_enable),  // Enable is active high in ICG
    .gclk(clk_out)
);

assign gating = w_gate_enable;
```

14.5.8 The ICG Cell

```
module icg (
    input  logic clk,    // Input clock
    input  logic en,     // Enable (active high)
    output logic gclk    // Gated clock output
);

    // Implementation varies by technology library
    // Typically includes:
    // - Latch to avoid glitches
    // - AND gate for gating
    // - Proper setup/hold timing

endmodule
```

A simplified conceptual implementation (the real thing is technology-specific):

```
// Simplified ICG implementation (technology-specific)
module icg_generic (
    input  logic clk,
    input  logic en,
    output logic gclk
);

    logic en_latched;

    // Latch enable on falling edge to avoid glitches
    always_latch begin
        if (~clk)
            en_latched = en;
    end

    // Gate clock with latched enable
    assign gclk = clk & en_latched;

endmodule
```

14.5.9 Technology-Specific ICG Implementations

```
// Example for TSMC technology
module icg_tsmc (
    input  logic clk,
    input  logic en,
    output logic gclk
);

    // Technology-specific implementation
    CKGTPBUF_X1M_A9TH u_icg (
        .CK(clk),
        .E(en),
        .GCK(gclk)
    );

endmodule

// Example for Quartus technology
module icg_quartus (
    input  logic clk,
    input  logic en,
    output logic gclk
);

    // Quartus FPGA implementation
    logic en_reg;

    always_ff @(negedge clk) begin
        en_reg <= en;
    end

    assign gclk = clk & en_reg;

endmodule

// Generic behavioral model
module icg_behavioral (
    input  logic clk,
    input  logic en,
    output logic gclk
);

    logic en_latch;

    // Latch enable on falling edge to prevent glitches
```

```
    always_latch begin
        if (!clk)
            en_latch = en;
    end

    // Gate the clock
    assign gclk = clk & en_latch;

endmodule
```

14.5.10 Timing and Performance

14.5.11 Resource Utilization

Configuration	LUTs	FFs	ICG Cells	Area Overhead
4-bit counter	8	4	1	< 2%
8-bit counter	15	8	1	< 3%
16-bit counter	30	16	1	< 5%

14.6 Timing Characteristics

This module is **sequential**: it contains clocked logic (via `always_ff` or the repository's `ALWAYS_FF_RST` macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

14.7 Usage Examples

14.7.1 CPU Core Clock Gating

```
module cpu_clock_gating (
    input  logic        clk_cpu,
    input  logic        rst_n,
```

```
    // CPU control signals
    input  logic        cpu_active,
    input  logic        interrupt_pending,
    input  logic        debug_mode,

    // Power management
    input  logic [3:0]  idle_threshold,
    input  logic        power_save_enable,

    // Output clocks
    output logic        clk_cpu_gated,
    output logic        clk_debug,
    output logic        cpu_sleeping
);

    logic cpu_wakeup;
    logic debug_wakeup;

    // CPU wake-up conditions
    assign cpu_wakeup = cpu_active || interrupt_pending;

    // Debug interface always active in debug mode
    assign debug_wakeup = debug_mode || cpu_wakeup;

    // Main CPU clock gating
    clock_gate_ctrl #(
        .IDLE_CNTR_WIDTH(4)
    ) cpu_gate (
        .clk_in(clk_cpu),
        .aresetn(rst_n),
        .cfg_cg_enable(power_save_enable),
        .cfg_cg_idle_count(idle_threshold),
        .wakeup(cpu_wakeup),
        .clk_out(clk_cpu_gated),
        .gating(cpu_sleeping)
    );

    // Debug clock gating (shorter timeout)
    clock_gate_ctrl #(
        .IDLE_CNTR_WIDTH(3)
    ) debug_gate (
        .clk_in(clk_cpu),
        .aresetn(rst_n),
        .cfg_cg_enable(power_save_enable && !debug_mode),
        .cfg_cg_idle_count(3'd2), // Short timeout for debug
```

```
        .wakeup(debug_wakeup),
        .clk_out(clk_debug),
        .gating() // Unused
    );

endmodule
```

14.7.2 Peripheral Interface Clock Management

```
module peripheral_clock_manager (
    input  logic        clk_system,
    input  logic        rst_n,

    // Peripheral activity signals
    input  logic        uart_active,
    input  logic        spi_active,
    input  logic        i2c_active,
    input  logic        timer_active,

    // Configuration
    input  logic [7:0]  power_config,   // [7:4] = enable bits, [3:0]
= timeout

    // Gated clocks
    output logic        clk_uart,
    output logic        clk_spi,
    output logic        clk_i2c,
    output logic        clk_timer,

    // Status
    output logic [3:0]  peripheral_sleeping
);

    logic [3:0] idle_timeout;
    logic [3:0] gate_enables;

    assign idle_timeout = power_config[3:0];
    assign gate_enables = power_config[7:4];

    // UART clock gating
    clock_gate_ctrl #(4) uart_cg (
        .clk_in(clk_system),
        .aresetn(rst_n),
        .cfg_cg_enable(gate_enables[0]),
        .cfg_cg_idle_count(idle_timeout),
        .wakeup(uart_active),
        .clk_out(clk_uart),
```

```
        .gating(peripheral_sleeping[0])
    );

    // SPI clock gating
    clock_gate_ctrl #(4) spi_cg (
        .clk_in(clk_system),
        .aresetn(rst_n),
        .cfg_cg_enable(gate_enables[1]),
        .cfg_cg_idle_count(idle_timeout),
        .wakeup(spi_active),
        .clk_out(clk_spi),
        .gating(peripheral_sleeping[1])
    );

    // I2C clock gating
    clock_gate_ctrl #(4) i2c_cg (
        .clk_in(clk_system),
        .aresetn(rst_n),
        .cfg_cg_enable(gate_enables[2]),
        .cfg_cg_idle_count(idle_timeout),
        .wakeup(i2c_active),
        .clk_out(clk_i2c),
        .gating(peripheral_sleeping[2])
    );

    // Timer clock gating (different timeout)
    clock_gate_ctrl #(4) timer_cg (
        .clk_in(clk_system),
        .aresetn(rst_n),
        .cfg_cg_enable(gate_enables[3]),
        .cfg_cg_idle_count(4'd1), // Quick timeout for timers
        .wakeup(timer_active),
        .clk_out(clk_timer),
        .gating(peripheral_sleeping[3])
    );

endmodule
```

14.7.3 Memory Subsystem Clock Gating

```
module memory_clock_gating (
    input  logic        clk_memory,
    input  logic        rst_n,

    // Memory access signals
    input  logic        mem_read,
    input  logic        mem_write,
    input  logic        mem_refresh_req,
```

```
    input  logic        cache_miss,

    // Configuration
    input  logic [2:0]  mem_power_mode,  // 0=always on, 7=aggressive
gating

    // Gated clocks
    output logic        clk_mem_ctrl,    // Memory controller
    output logic        clk_mem_data,    // Data path
    output logic        clk_mem_refresh, // Refresh logic

    // Status
    output logic [2:0]  mem_gating_status
);

    logic [3:0] timeout_ctrl, timeout_data, timeout_refresh;
    logic mem_active, data_active, refresh_active;

    // Activity detection
    assign mem_active = mem_read || mem_write || cache_miss;
    assign data_active = mem_read || mem_write;
    assign refresh_active = mem_refresh_req;

    // Timeout configuration based on power mode
    always_comb begin
        case (mem_power_mode)
            3'd0: {timeout_ctrl, timeout_data, timeout_refresh} =
{4'd15, 4'd15, 4'd15}; // Always on
            3'd1: {timeout_ctrl, timeout_data, timeout_refresh} =
{4'd10, 4'd8, 4'd12};  // Conservative
            3'd2: {timeout_ctrl, timeout_data, timeout_refresh} =
{4'd8, 4'd6, 4'd10};   // Moderate
            3'd3: {timeout_ctrl, timeout_data, timeout_refresh} =
{4'd6, 4'd4, 4'd8};    // Balanced
            3'd4: {timeout_ctrl, timeout_data, timeout_refresh} =
{4'd4, 4'd2, 4'd6};    // Aggressive
            3'd5: {timeout_ctrl, timeout_data, timeout_refresh} =
{4'd2, 4'd1, 4'd4};    // Very aggressive
            3'd6: {timeout_ctrl, timeout_data, timeout_refresh} =
{4'd1, 4'd0, 4'd2};    // Maximum power save
            3'd7: {timeout_ctrl, timeout_data, timeout_refresh} =
{4'd0, 4'd0, 4'd1};    // Ultra power save
        endcase
    end

    // Memory controller clock gating
```

```
    clock_gate_ctrl #(4) ctrl_cg (
        .clk_in(clk_memory),
        .aresetn(rst_n),
        .cfg_cg_enable(mem_power_mode != 0),
        .cfg_cg_idle_count(timeout_ctrl),
        .wakeup(mem_active),
        .clk_out(clk_mem_ctrl),
        .gating(mem_gating_status[0])
    );

    // Data path clock gating
    clock_gate_ctrl #(4) data_cg (
        .clk_in(clk_memory),
        .aresetn(rst_n),
        .cfg_cg_enable(mem_power_mode != 0),
        .cfg_cg_idle_count(timeout_data),
        .wakeup(data_active),
        .clk_out(clk_mem_data),
        .gating(mem_gating_status[1])
    );

    // Refresh logic clock gating
    clock_gate_ctrl #(4) refresh_cg (
        .clk_in(clk_memory),
        .aresetn(rst_n),
        .cfg_cg_enable(mem_power_mode > 2), // Only gate in aggressive
modes
        .cfg_cg_idle_count(timeout_refresh),
        .wakeup(refresh_active),
        .clk_out(clk_mem_refresh),
        .gating(mem_gating_status[2])
    );

endmodule
```

14.7.4 Adaptive Clock Gating with Activity Monitoring

```
module adaptive_clock_gating #(
    parameter int ACTIVITY_WINDOW = 256  // Activity monitoring window
) (
    input  logic        clk_in,
    input  logic        rst_n,

    // Activity signals
    input  logic [7:0]  activity_vector,

    // Configuration
    input  logic        adaptive_enable,
```

```
    input  logic [3:0]  base_timeout,

    // Outputs
    output logic        clk_out,
    output logic        gating_active,
    output logic [7:0]  activity_level    // Activity percentage
);

    logic [15:0] activity_counter;
    logic [7:0] activity_window_counter;
    logic [3:0] adaptive_timeout;
    logic block_active;

    // Activity monitoring
    always_ff @(posedge clk_in or negedge rst_n) begin
        if (!rst_n) begin
            activity_counter <= 0;
            activity_window_counter <= 0;
            activity_level <= 0;
        end else begin
            // Count activity in current window
            if (|activity_vector)
                activity_counter <= activity_counter + 1;

            activity_window_counter <= activity_window_counter + 1;

            // Update activity level at end of window
            if (activity_window_counter == ACTIVITY_WINDOW - 1) begin
                activity_level <= activity_counter[15:8]; // Scale to
0-255
                activity_counter <= 0;
                activity_window_counter <= 0;
            end
        end
    end

    // Adaptive timeout based on activity level
    always_comb begin
        if (!adaptive_enable) begin
            adaptive_timeout = base_timeout;
        end else begin
            case (activity_level)
                8'h00: adaptive_timeout = base_timeout + 4'd8; // Very
low activity
                8'h01: adaptive_timeout = base_timeout + 4'd6; // Low
activity
                8'h02: adaptive_timeout = base_timeout + 4'd4; //
```

```
Medium-low activity
                8'h04: adaptive_timeout = base_timeout + 4'd2; //
Medium activity
                8'h08: adaptive_timeout = base_timeout;        //
Normal activity
                default: adaptive_timeout = (base_timeout > 2) ?
base_timeout - 2 : 0; // High activity
            endcase
        end
    end

    // Activity detection
    assign block_active = |activity_vector;

    // Clock gating controller
    clock_gate_ctrl #(4) adaptive_cg (
        .clk_in(clk_in),
        .aresetn(rst_n),
        .cfg_cg_enable(1'b1),
        .cfg_cg_idle_count(adaptive_timeout),
        .wakeup(block_active),
        .clk_out(clk_out),
        .gating(gating_active)
    );

endmodule
```

14.7.5 Hierarchical Clock Gating System

```
module hierarchical_clock_gating (
    input  logic        clk_root,
    input  logic        rst_n,

    // Global control
    input  logic        global_cg_enable,
    input  logic [3:0]  global_timeout,

    // Subsystem activity
    input  logic        cpu_active,
    input  logic        dsp_active,
    input  logic        peripheral_active,
    input  logic        memory_active,

    // Hierarchical clocks
    output logic        clk_cpu_domain,
    output logic        clk_dsp_domain,
    output logic        clk_peripheral_domain,
```

```
    output logic        clk_memory_domain,

    // Power monitoring
    output logic [3:0]  domain_gating_status,
    output logic [15:0] power_savings_estimate
);

    logic system_active;
    logic clk_system_gated;
    logic system_gated;

    // System-level activity detection
    assign system_active = cpu_active || dsp_active ||
peripheral_active || memory_active;

    // Top-level system clock gating
    clock_gate_ctrl #(4) system_cg (
        .clk_in(clk_root),
        .aresetn(rst_n),
        .cfg_cg_enable(global_cg_enable),
        .cfg_cg_idle_count(global_timeout + 4'd4), // Longer timeout
for system level
        .wakeup(system_active),
        .clk_out(clk_system_gated),
        .gating(system_gated)
    );

    // CPU domain gating
    clock_gate_ctrl #(4) cpu_cg (
        .clk_in(clk_system_gated),
        .aresetn(rst_n),
        .cfg_cg_enable(global_cg_enable && !system_gated),
        .cfg_cg_idle_count(global_timeout),
        .wakeup(cpu_active),
        .clk_out(clk_cpu_domain),
        .gating(domain_gating_status[0])
    );

    // DSP domain gating
    clock_gate_ctrl #(4) dsp_cg (
        .clk_in(clk_system_gated),
        .aresetn(rst_n),
        .cfg_cg_enable(global_cg_enable && !system_gated),
        .cfg_cg_idle_count(global_timeout - 1), // Aggressive for DSP
        .wakeup(dsp_active),
        .clk_out(clk_dsp_domain),
        .gating(domain_gating_status[1])
```

```
    );

    // Peripheral domain gating
    clock_gate_ctrl #(4) peripheral_cg (
        .clk_in(clk_system_gated),
        .aresetn(rst_n),
        .cfg_cg_enable(global_cg_enable && !system_gated),
        .cfg_cg_idle_count(global_timeout + 1), // Conservative for
peripherals
        .wakeup(peripheral_active),
        .clk_out(clk_peripheral_domain),
        .gating(domain_gating_status[2])
    );

    // Memory domain gating
    clock_gate_ctrl #(4) memory_cg (
        .clk_in(clk_system_gated),
        .aresetn(rst_n),
        .cfg_cg_enable(global_cg_enable && !system_gated),
        .cfg_cg_idle_count(global_timeout + 2), // Conservative for
memory
        .wakeup(memory_active),
        .clk_out(clk_memory_domain),
        .gating(domain_gating_status[3])
    );

    // Power savings estimation (simplified)
    logic [3:0] gated_domains;
    assign gated_domains = domain_gating_status[3:0];

    always_ff @(posedge clk_root or negedge rst_n) begin
        if (!rst_n) begin
            power_savings_estimate <= 0;
        end else begin
            // Rough power savings estimate (percentage)
            case ({system_gated, gated_domains})
                5'b1_0000: power_savings_estimate <= 16'd9000; // 90%
savings
                5'b0_1111: power_savings_estimate <= 16'd7000; // 70%
savings
                5'b0_1110: power_savings_estimate <= 16'd6000; // 60%
savings
                5'b0_1100: power_savings_estimate <= 16'd5000; // 50%
savings
                5'b0_1000: power_savings_estimate <= 16'd3000; // 30%
savings
                5'b0_0100: power_savings_estimate <= 16'd2000; // 20%
```

```
savings
                5'b0_0010: power_savings_estimate <= 16'd1500; // 15%
savings
                5'b0_0001: power_savings_estimate <= 16'd1000; // 10%
savings
                default:   power_savings_estimate <= 16'd0000; // 0%
savings
            endcase
        end
    end

endmodule
```

14.8 Design Notes

14.8.1 Glitch-Free Clock Switching

```
module glitch_free_clock_gate (
    input  logic clk_in,
    input  logic rst_n,
    input  logic gate_enable,
    input  logic wakeup,
    output logic clk_out
);

    logic gate_enable_sync;
    logic gate_request;

    // Synchronize gate enable to avoid glitches
    always_ff @(posedge clk_in or negedge rst_n) begin
        if (!rst_n) begin
            gate_enable_sync <= 0;
            gate_request <= 0;
        end else begin
            gate_enable_sync <= gate_enable;
            gate_request <= gate_enable_sync && !wakeup;
        end
    end

    // Use ICG with synchronized control
    icg u_icg (
        .clk(clk_in),
        .en(~gate_request),
        .gclk(clk_out)
    );

endmodule
```

14.8.2 Multi-Level Clock Gating

```
module multi_level_clock_gate #(
    parameter int LEVELS = 3
) (
    input  logic             clk_in,
    input  logic             rst_n,
    input  logic [LEVELS-1:0] level_enables,
    input  logic [LEVELS-1:0] level_wakeups,
    input  logic [4*LEVELS-1:0] level_timeouts,
    output logic             clk_out,
    output logic [LEVELS-1:0] level_gating_status
);

    logic [LEVELS:0] clk_levels; // Include input clock
    assign clk_levels[0] = clk_in;

    genvar i;
    generate
        for (i = 0; i < LEVELS; i++) begin : level_gates
            logic [3:0] timeout;
            assign timeout = level_timeouts[i*4+:4];

            clock_gate_ctrl #(4) level_cg (
                .clk_in(clk_levels[i]),
                .aresetn(rst_n),
                .cfg_cg_enable(level_enables[i]),
                .cfg_cg_idle_count(timeout),
                .wakeup(|level_wakeups[LEVELS-1:i]), // Wake if any
higher level active
                .clk_out(clk_levels[i+1]),
                .gating(level_gating_status[i])
            );
        end
    endgenerate

    assign clk_out = clk_levels[LEVELS];

endmodule
```

14.8.3 Measuring Power Savings

```
module power_analysis_wrapper (
    input  logic        clk_in,
    input  logic        rst_n,
    input  logic        cfg_cg_enable,
    input  logic [3:0]  cfg_cg_idle_count,
    input  logic        wakeup,
```

```
    output logic        clk_out,
    output logic        gating,
    output logic [15:0] power_savings_percent
);

    // Clock gate controller
    clock_gate_ctrl #(4) cg_ctrl (
        .clk_in(clk_in),
        .aresetn(rst_n),
        .cfg_cg_enable(cfg_cg_enable),
        .cfg_cg_idle_count(cfg_cg_idle_count),
        .wakeup(wakeup),
        .clk_out(clk_out),
        .gating(gating)
    );

    // Power monitoring
    logic [31:0] total_cycles;
    logic [31:0] gated_cycles;
    logic [31:0] ungated_cycles;

    always_ff @(posedge clk_in or negedge rst_n) begin
        if (!rst_n) begin
            total_cycles <= 0;
            gated_cycles <= 0;
            ungated_cycles <= 0;
        end else begin
            total_cycles <= total_cycles + 1;
            if (gating)
                gated_cycles <= gated_cycles + 1;
            else
                ungated_cycles <= ungated_cycles + 1;
        end
    end

    // Calculate power savings percentage
    always_ff @(posedge clk_in) begin
        if (total_cycles > 1000) begin // Avoid division by small
numbers
            power_savings_percent <= (gated_cycles * 100) /
total_cycles;
        end else begin
            power_savings_percent <= 0;
        end
    end

endmodule
```

14.8.4 Synthesis Constraints

```
# SDC constraints for clock gating
create_clock -name clk_in -period 10.0 [get_ports clk_in]

# Clock gating timing
set_case_analysis 0 [get_pins u_icg/en]
set_case_analysis 1 [get_pins u_icg/en]

# Setup/hold for gating control
set_input_delay -clock clk_in -max 2.0 [get_ports wakeup]
set_input_delay -clock clk_in -min 0.5 [get_ports wakeup]

# Clock gating cell timing
set_clock_gating_check -setup 0.2 -hold 0.1 [get_pins u_icg/en]

# UPF power intent for clock gating
create_power_domain PD_MAIN
create_power_domain PD_GATED

# Clock gating specification
set_domain_supply_net PD_MAIN -primary_power_net VDD -
primary_ground_net VSS
set_domain_supply_net PD_GATED -primary_power_net VDD -
primary_ground_net VSS

# Clock gating control
create_power_switch PS_GATED \
    -domain PD_GATED \
    -input_supply_port {vin VDD} \
    -output_supply_port {vout VDD_GATED} \
    -control_port {ctrl gating} \
    -on_state {on vin {!ctrl}}
```

14.8.5 Design Guidelines

Timeout selection:

```
// Guidelines for timeout values
parameter TIMEOUT_GUIDELINES = {
    "CPU cores: 4-16 cycles",
    "Memory controllers: 8-32 cycles",
    "Peripherals: 16-64 cycles",
    "Debug interfaces: 2-8 cycles",
    "Always-on domains: Disable gating"
};
```

Activity signal design:

```
// Good activity signal examples
logic cpu_active;
assign cpu_active = instruction_fetch ||
                    data_access ||
                    interrupt_processing ||
                    debug_access;

// Avoid overly broad activity signals
logic bad_activity;
assign bad_activity = |all_system_signals; // Too broad

// Avoid overly narrow activity signals
logic narrow_activity;
assign narrow_activity = specific_register_write; // Too narrow
```

Hierarchical gating strategy:

```
System Level (Longest timeout)
 CPU Domain (Medium timeout)
    ALU (Short timeout)
    Cache (Medium timeout)
    Debug (Very short timeout)
 DSP Domain (Short timeout)
 Peripheral Domain (Long timeout)
 Memory Domain (Medium timeout)
```

Verification checklist: - [ ] Verify gating occurs after correct timeout - [ ] Verify immediate wakeup functionality - [ ] Test global enable/disable - [ ] Check reset behavior - [ ] Validate power savings measurements - [ ] Test all timeout configurations - [ ] Verify clock integrity (no glitches) - [ ] Check setup/hold timing margins

14.8.6 Common Applications

1. **CPU Power Management**: Core, cache, and pipeline clock gating
2. **Peripheral Controllers**: UART, SPI, I2C, timer clock gating
3. **Memory Subsystems**: Controller, data path, refresh logic gating
4. **Graphics/Video**: Pixel processing, display controller gating
5. **Communication Blocks**: Baseband, RF, protocol stack gating
6. **Test and Debug**: Scan chains, JTAG, debug interface gating
7. **AI/ML Accelerators**: Compute engine, memory interface gating

14.8.7 Troubleshooting Guide

Common Issues: 1. **Clock Glitches**: Use qualified ICG cells, proper setup/hold margins 2. **Timing Violations**: Ensure adequate timing margins for gating signals 3. **Unexpected Gating**: Check activity signal connectivity and logic 4. **Power Not Saved**: Verify clock tree reaches gated logic 5. **Reset Issues**: Ensure proper reset sequencing and distribution

Debug Techniques: 1. **Simulation**: Monitor gating signals and clock activity 2. **Power Analysis**: Use tools to measure actual power savings 3. **Timing Analysis**: Check setup/hold violations on gating paths 4. **Logic Analyzer**: Capture real-time gating behavior in hardware 5. **Power Profiling**: Measure current consumption with/without gating

14.9 Related Modules

Read out of the RTL, not curated: these are the modules this one instantiates and the modules that instantiate it.

Instantiates: - `icg`

Instantiated by: - `amba_clock_gate_ctrl`

14.10 Testing

14.10.1 Test Bench

```
module tb_clock_gate_ctrl;

    parameter IDLE_CNTR_WIDTH = 4;

    logic clk_in, aresetn;
    logic cfg_cg_enable;
    logic [IDLE_CNTR_WIDTH-1:0] cfg_cg_idle_count;
    logic wakeup;
    logic clk_out, gating;

    // Clock generation
    initial clk_in = 0;
    always #5ns clk_in = ~clk_in; // 100MHz

    // DUT instantiation
    clock_gate_ctrl #(
        .IDLE_CNTR_WIDTH(IDLE_CNTR_WIDTH)
    ) dut (.*);

    // Test sequence
```

```
    initial begin
        // Initialize
        aresetn = 0;
        cfg_cg_enable = 0;
        cfg_cg_idle_count = 4'd5;
        wakeup = 0;

        #100ns aresetn = 1;

        // Test basic gating
        test_basic_gating();

        // Test immediate wakeup
        test_immediate_wakeup();

        // Test global disable
        test_global_disable();

        // Test different timeouts
        test_timeout_values();

        $display("All tests completed!");
        $finish;
    end

    // Test basic clock gating functionality
    task test_basic_gating();
        begin
            $display("Testing basic clock gating...");

            cfg_cg_enable = 1;
            cfg_cg_idle_count = 4'd3;

            // Wait for gating to occur
            wait_for_gating();

            // Verify clock is gated
            check_clock_gated();

            // Wake up and verify clock restoration
            wakeup = 1;
            #20ns wakeup = 0;
            check_clock_active();
        end
    endtask
```

```
    // Wait for clock gating to occur
    task wait_for_gating();
        int timeout_cycles;
        begin
            timeout_cycles = 0;
            while (!gating && timeout_cycles < 100) begin
                @(posedge clk_in);
                timeout_cycles++;
            end

            if (!gating) begin
                $error("Clock gating did not occur within timeout");
            end else begin
                $display("Clock gating occurred after %d cycles",
timeout_cycles);
            end
        end
    endtask

    // Check that clock is properly gated
    task check_clock_gated();
        logic prev_clk_out;
        int stable_cycles;
        begin
            prev_clk_out = clk_out;
            stable_cycles = 0;

            repeat (10) begin
                @(posedge clk_in);
                if (clk_out == prev_clk_out) begin
                    stable_cycles++;
                end else begin
                    stable_cycles = 0;
                    prev_clk_out = clk_out;
                end
            end

            if (stable_cycles >= 5) begin
                $display("PASS: Clock is properly gated");
            end else begin
                $error("FAIL: Clock is not properly gated");
            end
        end
    endtask

    // Check that clock is active
```

```
    task check_clock_active();
        int edge_count;
        begin
            edge_count = 0;

            repeat (20) begin
                @(posedge clk_in);
                if ($rose(clk_out)) edge_count++;
            end

            if (edge_count >= 8) begin
                $display("PASS: Clock is active after wakeup");
            end else begin
                $error("FAIL: Clock not properly restored after
wakeup");
            end
        end
    endtask

endmodule
```

14.10.2 Coverage Model

```
covergroup clock_gate_cg @(posedge clk_in);

    cp_idle_count: coverpoint cfg_cg_idle_count {
        bins short_timeout[] = {[0:3]};
        bins medium_timeout[] = {[4:10]};
        bins long_timeout[] = {[11:15]};
    }

    cp_counter_values: coverpoint dut.r_idle_counter {
        bins countdown[] = {[0:15]};
        bins zero = {0};
    }

    cp_gating_enable: coverpoint cfg_cg_enable {
        bins enabled = {1};
        bins disabled = {0};
    }

    cp_wakeup: coverpoint wakeup {
        bins active = {1};
        bins inactive = {0};
    }

    cp_gating_status: coverpoint gating {
```

```
        bins gated = {1};
        bins ungated = {0};
    }

    // Cross coverage
    cross_timeout_gating: cross cp_idle_count, cp_gating_status;
    cross_wakeup_gating: cross cp_wakeup, cp_gating_status;

endcovergroup
```

14.10.3 Formal Verification Properties

```
module clock_gate_properties;

    // Bind to DUT
    bind clock_gate_ctrl clock_gate_properties props_inst (.*);

    // Property: Immediate wakeup
    property immediate_wakeup;
        @(posedge clk_in) disable iff (!aresetn)
        wakeup |-> ##1 !gating;
    endproperty

    // Property: Gating after timeout
    property gating_after_timeout;
        @(posedge clk_in) disable iff (!aresetn)
        cfg_cg_enable && !wakeup && (dut.r_idle_counter == 0) |->
gating;
    endproperty

    // Property: Counter behavior
    property counter_countdown;
        @(posedge clk_in) disable iff (!aresetn)
        cfg_cg_enable && !wakeup && (dut.r_idle_counter > 0) |=>
        (dut.r_idle_counter == $past(dut.r_idle_counter) - 1);
    endproperty

    // Property: Global disable
    property global_disable;
        @(posedge clk_in) disable iff (!aresetn)
        !cfg_cg_enable |-> !gating;
    endproperty

    // Assertions
    assert property (immediate_wakeup);
    assert property (gating_after_timeout);
    assert property (counter_countdown);
```

```
    assert property (global_disable);

endmodule
```

Done right, the clock gate controller is essential power management: significant dynamic power reduction for minimal area overhead and design complexity. Getting it right means careful attention to timing, safety, and verification — that's what buys you reliable operation and the full power savings.

14.11 References

- Benini, L., Bogliolo, A., De Micheli, G. "A Survey of Design Techniques for System-Level Dynamic Power Management." *IEEE Transactions on VLSI Systems*, 8(3), June 2000. DOI: 10.1109/92.845896 — the fixed-timeout shutdown policy (power down after the component has been idle for a timeout) that this module's idle counter implements.
- Benini, L., De Micheli, G. "Automatic Synthesis of Low-Power Gated-Clock Finite-State Machines." *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 15(6), June 1996. DOI: 10.1109/43.503933 — the latch-plus-AND gated clock realized by the `icg` cell this module instantiates.
- Roy, S. *Advanced Digital System Design*, Springer, ch. 15 "Low-Power Digital System Design", §15.5.5 "Logic Level Power Down" — textbook treatment of disabling the clock to idle units (secondary source).
- Indexed in `rtl/common/references/README.md` (local PDFs: `DynamicPowerManagementTimeout__Benini2000.pdf`, `LatchBasedClockGating__Benini1996.pdf`)

14.12 Navigation

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15 clock_pulse

15.1 Overview

The `clock_pulse` module is a periodic pulse generator with a configurable period: it emits a single-cycle pulse every WIDTH clock cycles. It's the part you reach for whenever something in the system needs to happen on a schedule — timing generation, heartbeat signals, sampling triggers, periodic events of any kind.

15.2 Parameters

Parameter	Type	Default	Description
`WIDTH`	`int`	`10`	Period of the pulse generation in clock cycles

`WIDTH` can range from 2 to 2^31-1. The parameter is declared `int`, which is 32-bit SIGNED, so 2^32-1 is not representable. It sets the pulse frequency (f_clk / WIDTH). The pulse itself is always exactly 1 clock cycle wide, which makes the duty cycle 1/WIDTH — 10% for WIDTH=10.

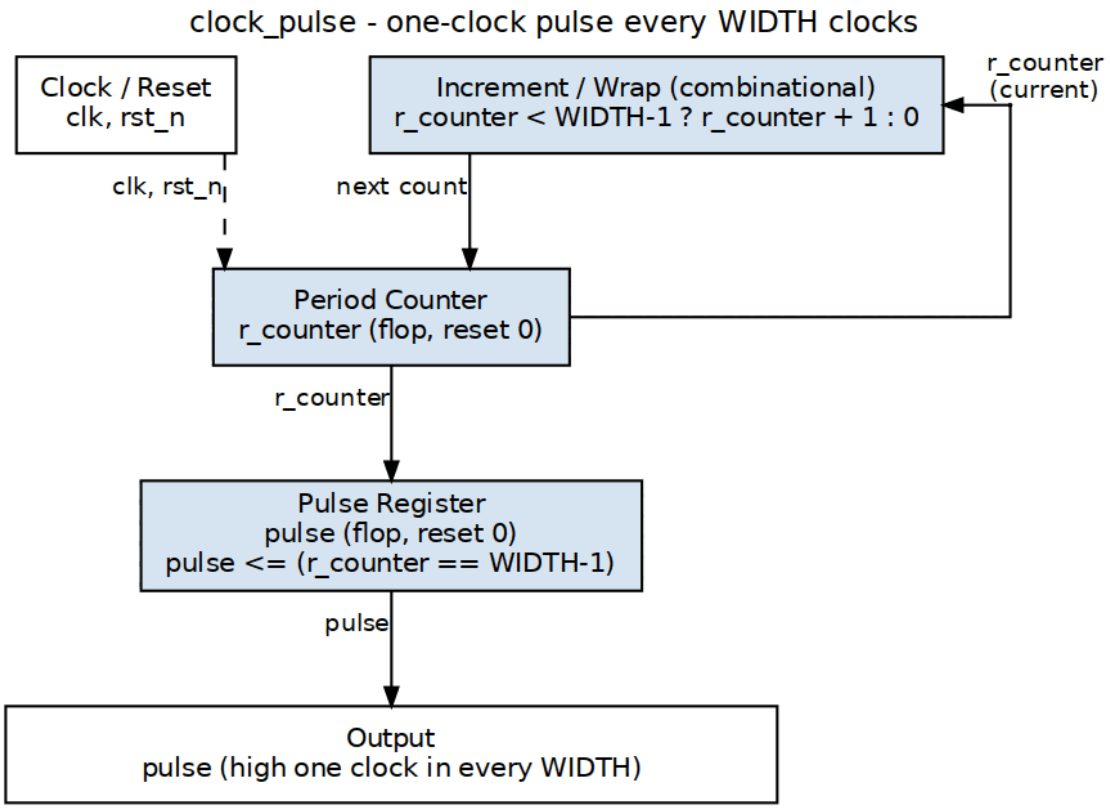
15.3 Ports

```
module clock_pulse #(
    parameter int WIDTH = 10  // PERIOD in clock cycles (the pulse
itself is
                          // always exactly 1 cycle wide)
) (
    input  logic clk,    // Input clock signal
    input  logic rst_n,  // Input reset signal
    output logic pulse   // Output pulse signal
);
```

Port	Direction	Width	Description
`clk`	Input	1	System clock input
`rst_n`	Input	1	Active-low asynchronous reset
`pulse`	Output	1	Periodic pulse output

15.4 Functional Description

15.4.1 Block diagram



clock_pulse block diagram

A period counter runs 0 to `WIDTH-1` and wraps, feeding back up the right edge into its increment logic. The pulse register samples the compare `r_counter == WIDTH-1`, so `pulse` is high for one clock in every `WIDTH`.

15.4.2 Internal Counter

```
// WIDTH is the PERIOD; the counter only needs to hold 0..WIDTH-1,
i.e.
// $clog2(WIDTH) bits (NOT WIDTH bits).
localparam int CW = (WIDTH < 2) ? 1 : $clog2(WIDTH);

logic [CW-1:0] r_counter;
logic [CW-1:0] w_width_minus_one;

// Properly sized period-1 constant
assign w_width_minus_one = CW'(WIDTH - 1);
```

15.4.3 Core Logic

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_counter <= 'b0;
        pulse     <= 'b0;
    end else begin
        if (r_counter < w_width_minus_one)
            r_counter <= r_counter + 1'b1;
        else
            r_counter <= 'b0;

        pulse <= (r_counter == w_width_minus_one);
    end
end
```

15.4.4 Operation Principles

1. **Counter**: Free-running counter from 0 to WIDTH-1
2. **Pulse Generation**: `pulse` is **registered** — `pulse <= (r_counter == WIDTH-1)` — so it asserts on the cycle **after** the counter reaches WIDTH-1, i.e. during the cycle when the counter has wrapped back to 0.
3. **Auto-Reset**: Counter wraps to 0 after reaching maximum
4. **Synchronous**: All operations synchronized to input clock
5. **Single Cycle**: Pulse duration is exactly one clock cycle

That second point is the one people get wrong, so let's be blunt about it: the registered comparison means `pulse` never coincides with the terminal count. It lands one cycle later, on the wrapped-to-zero count. Every timing diagram below follows from that.

15.5 Related Modules

Nothing in the tree instantiates this module and it instantiates nothing: it is a leaf, used directly by whatever design needs it. Its nearest neighbours in `rtl/common/` are:

- `clock_divider`
- `clock_gate_ctrl`

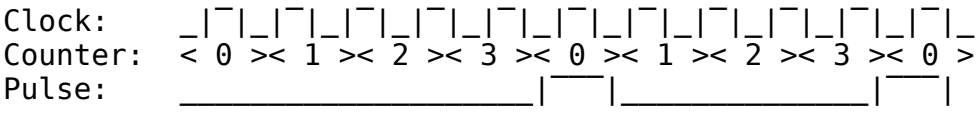
15.6 Timing Characteristics

15.6.1 Timing Characteristics

- **Period**: WIDTH clock cycles
- **Frequency**: f_clk / WIDTH
- **Pulse Width**: 1 clock cycle
- **Duty Cycle**: 1/WIDTH
- **Phase**: Because the comparison is registered, `pulse` is high one cycle **after** `r_counter == WIDTH-1` — that is, during the `r_counter == 0` cycle of each period, not on the WIDTH-1 count itself.

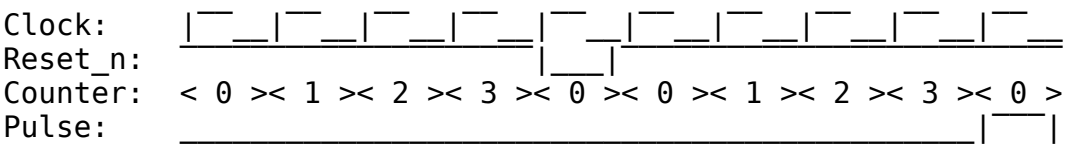
15.6.2 Basic Operation (WIDTH=4)

Watch where `pulse` actually lands — high during the `< 0 >` cell that immediately follows `< 3 >`, one cycle after the counter hit WIDTH-1. That's the registered comparison at work:



(The first `< 0 >` is the post-reset state, where `pulse` is still low.)

15.6.3 Reset Behavior



Two things worth seeing here, both consequences of the pulse being registered (`pulse <= (r_counter == w_width_minus_one)`):

- **The pulse that reset swallows.** The counter reaches 3 in the fourth cell, so a pulse would normally appear in the fifth — but reset is asserted exactly there, and reset forces `pulse <= 0`. That pulse never happens.
- **Recovery costs a full period.** After reset releases, the counter restarts from 0 and the first pulse appears in the cell *after* it next reaches 3 — the final cell above, not the one where the counter reads 3.

15.6.4 Different WIDTH Values

WIDTH	Period	Duty Cycle	Use Case
2	2 cycles	50%	Clock divider
4	4 cycles	25%	Sampling
10	10 cycles	10%	Timing reference
100	100 cycles	1%	Heartbeat
1000	1000 cycles	0.1%	Slow events

15.7 Usage Examples

Every parameter and port below is read from the module declaration.

```
clock_pulse #(
    .WIDTH                  (10)
) u_clock_pulse (
    .clk                    (clk),
    .rst_n                  (rst_n),
    .pulse                  (pulse)
);
```

15.8 Design Notes

15.8.1 Resource Utilization

WIDTH	Counter Bits	LUTs	FFs	Max Freq
8	3	8	4	500MHz
16	4	12	5	450MHz
32	5	18	6	400MHz
1024	10	28	11	350MHz

15.8.2 Synthesis Considerations

For high-frequency applications, pipeline the comparison:

```
// For high-frequency applications, pipeline the comparison (WIDTH >=
2).
// IMPORTANT: because the registered compare_result ALSO gates the
counter
// wrap, it must be produced from the WIDTH-2 threshold, not WIDTH-1.
Comparing
// against WIDTH-1 here makes the counter sequence 0..WIDTH,0 and
yields a
```

```
// period of WIDTH+1, not WIDTH. With WIDTH-2 the period is exactly
WIDTH and
// the pulse still lands on the r_counter==0 cycle, matching the base
module.
module clock_pulse_pipelined #(
    parameter int WIDTH = 1000
) (
    input  logic clk,
    input  logic rst_n,
    output logic pulse
);

    localparam int CW = (WIDTH < 2) ? 1 : $clog2(WIDTH);
    logic [CW-1:0] r_counter;
    logic          compare_result;

    // Register the comparison one count early so the pipelined wrap
keeps a
    // WIDTH-cycle period.
    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n) compare_result <= 1'b0;
        else        compare_result <= (r_counter == CW'(WIDTH - 2));
    end

    always_ff @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            r_counter <= 'b0;
            pulse     <= 1'b0;
        end else begin
            r_counter <= compare_result ? '0 : r_counter + 1'b1;
            pulse     <= compare_result;
        end
    end

endmodule
```

15.8.3 Design Considerations

1. **Choose Appropriate WIDTH**: Balance between resolution and resource usage
2. **Consider Reset Timing**: Ensure clean startup behavior
3. **Validate Timing**: Verify pulse frequency matches requirements
4. **Plan for Variation**: Consider process, voltage, temperature effects

5. **Monitor Resource Usage**: Large counters can impact timing and area
6. **Use Enables**: Add enable signals for conditional operation
7. **Document Timing**: Clearly specify pulse rates and relationships

15.8.4 Common Applications

1. **Timing References**: System heartbeats, watchdog timers
2. **Sampling Systems**: ADC triggers, data acquisition timing
3. **Communication**: Baud rate generation, protocol timing
4. **Memory Systems**: Refresh timing, access scheduling
5. **Test Equipment**: Pattern generation, stimulus timing
6. **Power Management**: Activity monitoring, timeout generation
7. **Display Systems**: Refresh rates, sync generation

That's the whole module. It's a small thing, but precise periodic timing shows up in nearly every design you'll ever ship — get this one right once, parameterize it well, and reuse it forever.

15.9 Testing

15.9.1 Comprehensive Test Bench

```
module tb_clock_pulse;

    parameter WIDTH = 10;

    logic clk, rst_n, pulse;

    // Clock generation
    initial clk = 0;
    always #5ns clk = ~clk; // 100MHz

    // DUT instantiation
    clock_pulse #(.WIDTH(WIDTH)) dut (.*);

    // Test sequence
    initial begin
        rst_n = 0;
        #100ns rst_n = 1;
```

```
        // Test basic pulse generation
        test_basic_pulse_generation();

        // Test reset behavior
        test_reset_behavior();

        // Test timing accuracy
        test_timing_accuracy();

        $display("All tests completed!");
        $finish;
    end

    // Test basic pulse generation
    task test_basic_pulse_generation();
        int pulse_count = 0;
        int cycle_count = 0;
        begin
            $display("Testing basic pulse generation (WIDTH=%0d)...",
WIDTH);

            // Monitor for several pulse periods. WIDTH*3 + 1 edges:
`@(posedge clk)`
            // resumes BEFORE the NBA update, so pulses are first
visible at edges
            // WIDTH+1, 2*WIDTH+1, 3*WIDTH+1 (see test_reset_behavior
for the trace).
            repeat (WIDTH * 3 + 1) begin
                @(posedge clk);
                cycle_count++;
                if (pulse) pulse_count++;
            end

            // Verify pulse count
            if (pulse_count == 3) begin
                $display("PASS: Correct number of pulses generated");
            end else begin
                $error("FAIL: Expected 3 pulses, got %0d",
pulse_count);
            end
        end
    endtask

    // Test reset behavior -- the two consequences of the registered
pulse
```

```
    task test_reset_behavior();
        int pulses_during_reset = 0;
        int cycles_to_first_pulse = 0;
        begin
            $display("Testing reset behavior...");

            // Line the counter up so a pulse is due, then assert
reset in
            // exactly that cycle. Reset forces pulse <= 0, so that
pulse is
            // swallowed and never appears.
            repeat (WIDTH - 1) @(posedge clk);
            rst_n = 0;
            repeat (WIDTH * 2) begin
                @(posedge clk);
                if (pulse) pulses_during_reset++;
            end
            if (pulses_during_reset == 0) begin
                $display("PASS: no pulse while reset is asserted");
            end else begin
                $error("FAIL: %0d pulse(s) during reset",
pulses_during_reset);
            end

            // Recovery costs a full period: the counter restarts from
0, so the
            // first pulse lands one cycle AFTER it next reaches
WIDTH-1.
            @(negedge clk);
            rst_n = 1;
            // Expect the first pulse on the (WIDTH+1)th sampled edge,
not the
            // WIDTHth. Three separate cycles are involved and it is
easy to
            // lose one: the counter reaches WIDTH-1 at edge WIDTH-1,
`pulse`
            // is NBA-assigned at edge WIDTH, and `@(posedge clk)`
resumes in
            // the Active region -- BEFORE that edge's NBA update --
so the
            // high value is first visible at edge WIDTH+1.
            forever begin
                @(posedge clk);
                cycles_to_first_pulse++;
                if (pulse) break;
                if (cycles_to_first_pulse > WIDTH * 2) begin
                    $error("FAIL: no pulse within two periods of reset
release");
```

```
                    break;
                end
            end
            if (cycles_to_first_pulse == WIDTH + 1) begin
                $display("PASS: first pulse %0d sampled edges after
release",
                         WIDTH + 1);
            end else begin
                $error("FAIL: expected first pulse at cycle %0d, got
%0d",
                       WIDTH + 1, cycles_to_first_pulse);
            end
        end
    endtask

    // Test timing accuracy
    task test_timing_accuracy();
        time pulse_times[10];
        time pulse_periods[9];
        int i;
        real avg_period, expected_period;
        begin
            $display("Testing timing accuracy...");

            expected_period = WIDTH * 10.0; // 10ns clock period

            // Capture pulse timing
            for (i = 0; i < 10; i++) begin
                @(posedge pulse);
                pulse_times[i] = $time;
            end

            // Calculate periods
            for (i = 0; i < 9; i++) begin
                pulse_periods[i] = pulse_times[i+1] - pulse_times[i];
            end

            // Calculate average period
            avg_period = 0;
            for (i = 0; i < 9; i++) begin
                avg_period += pulse_periods[i];
            end
            avg_period = avg_period / 9.0;

            // Check accuracy (within 1% tolerance)
            if (abs(avg_period - expected_period) < expected_period *
0.01) begin
```

```
                $display("PASS: Average period = %.1f ns (expected
%.1f ns)",
                         avg_period, expected_period);
            end else begin
                $error("FAIL: Average period = %.1f ns (expected %.1f
ns)",
                       avg_period, expected_period);
            end
        end
    endtask

    function real abs(real value);
        abs = (value >= 0) ? value : -value;
    endfunction

endmodule
```

15.9.2 Coverage Model

```
covergroup clock_pulse_cg @(posedge clk);

    cp_counter_values: coverpoint dut.r_counter {
        bins zero = {0};
        bins low[] = {[1:WIDTH/4]};
        bins mid[] = {[WIDTH/4+1:3*WIDTH/4]};
        bins high[] = {[3*WIDTH/4+1:WIDTH-2]};
        bins max = {WIDTH-1};
    }

    cp_pulse: coverpoint pulse {
        bins asserted = {1};
        bins deasserted = {0};
    }

    cp_reset: coverpoint rst_n {
        bins reset = {0};
        bins normal = {1};
    }

    // Transition coverage
    cp_pulse_edges: coverpoint pulse {
        bins rising = (0 => 1);
        bins falling = (1 => 0);
        bins stable_low = (0 => 0);
        bins stable_high = (1 => 1);
    }
```

```
    // Cross coverage
    cross_counter_pulse: cross cp_counter_values, cp_pulse;

endcovergroup
```

15.9.3 Formal Properties

```
module clock_pulse_properties;

    // Bind to DUT
    bind clock_pulse clock_pulse_properties props_inst (.*);

    // Property: reaching max count produces a pulse on the NEXT
cycle.
    // The comparison is registered (pulse <= r_counter == WIDTH-1),
so the
    // implication is |=> (next cycle), NOT |-> (same cycle).
    property pulse_after_max_count;
        @(posedge clk) disable iff (!rst_n)
        (dut.r_counter == WIDTH-1) |=> pulse;
    endproperty

    // Property: a pulse implies the PREVIOUS count was max
    property pulse_implies_prev_max;
        @(posedge clk) disable iff (!rst_n)
        pulse |-> $past(dut.r_counter == WIDTH-1);
    endproperty

    // Property: Counter wraps correctly
    property counter_wrap;
        @(posedge clk) disable iff (!rst_n)
        (dut.r_counter == WIDTH-1) |=> (dut.r_counter == 0);
    endproperty

    // Property: Counter increments
    property counter_increment;
        @(posedge clk) disable iff (!rst_n)
        (dut.r_counter < WIDTH-1) |=>
        (dut.r_counter == $past(dut.r_counter) + 1);
    endproperty

    // Property: Reset behavior
    property reset_behavior;
        @(posedge clk)
        !rst_n |=> (dut.r_counter == 0) && !pulse;
    endproperty
```

```
    // Assertions
    assert property (pulse_after_max_count);
    assert property (pulse_implies_prev_max);
    assert property (counter_wrap);
    assert property (counter_increment);
    assert property (reset_behavior);

endmodule
```

15.10 References

- No peer paper — a modulo-WIDTH counter emitting a one-cycle periodic pulse is a standard technique with no defensible single origin (marked "standard technique" in the references index).
- Roy, S. *Advanced Digital System Design*, Springer, ch. 4 "Basic Sequential Circuits", §4.6 "Synchronous Counter Design" — textbook treatment (secondary source).
- Indexed in `rtl/common/references/README.md`

15.11 Navigation

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16 Binary Counter with Load and Variable Increment

16.1 Overview

The `counter_bin_load` module takes basic binary counting and adds three distinct operating modes: standard +1 increment, variable-amount increment, and direct load. It's built specifically for FIFO pointer management, where drop/flush operations need to jump the read pointer by arbitrary amounts. The wraparound is FIFO-style — at 2×MAX, with an MSB toggle for full/empty detection. If you've ever had to discard a burst of FIFO entries mid-stream, this is the counter that makes it painless.

16.2 Module Interface

```
module counter_bin_load #(
    parameter int WIDTH = 5,     // Counter bit width
    parameter int MAX = 10       // Maximum count value (wraps at MAX-
1)
) (
    // Derived localparam (computed internally - not user-settable):
    // localparam int WRAP_BOUNDARY = 2 * MAX;  // Wraparound boundary

    input  logic             clk,               // Clock input
    input  logic             rst_n,             // Active-low
asynchronous reset
    input  logic             enable,            // Count +1 enable
    input  logic             add_enable,        // Variable increment
enable
    input  logic [WIDTH-1:0] add_value,         // Value to add
    input  logic             load,              // Load enable
(highest priority)
    input  logic [WIDTH-1:0] load_value,        // Value to load
    output logic [WIDTH-1:0] counter_bin_curr,  // Current count
(registered)
    output logic [WIDTH-1:0] counter_bin_next   // Next count
(combinational)
);
```

16.3 Parameters

16.3.1 User-Settable Parameters

Parameter	Type	Default	Description
WIDTH	int	5	Counter bit width (range: 2-64)
MAX	int	10	Maximum count value

Parameter	Type	Default	Description
			before wraparound (range: 2 to 2^(WIDTH-1))

WIDTH Guidelines: - **Minimum**: 2 bits (required for MSB inversion behavior) - **Typical**: FIFO address width + 1 (e.g., 4 for 8-entry FIFO: 3 addr + 1 MSB) - **Maximum**: 64 bits (practical limit for synthesis)

MAX Constraints: - Must fit within WIDTH-1 bits (excludes MSB) - Example: WIDTH=4 → MAX ≤ 8 (3 bits for addresses 0-7, 1 MSB for full/empty) - FIFO depth is typically MAX (e.g., MAX=8 → 8-entry FIFO)

16.3.2 Derived Localparam (Computed Internally)

Localparam	Computation	Description
WRAP_BOUNDARY	2 × MAX	Wraparound boundary for add operations

Note: WRAP_BOUNDARY is computed internally and cannot be overridden by users.

16.4 Ports

16.4.1 Inputs

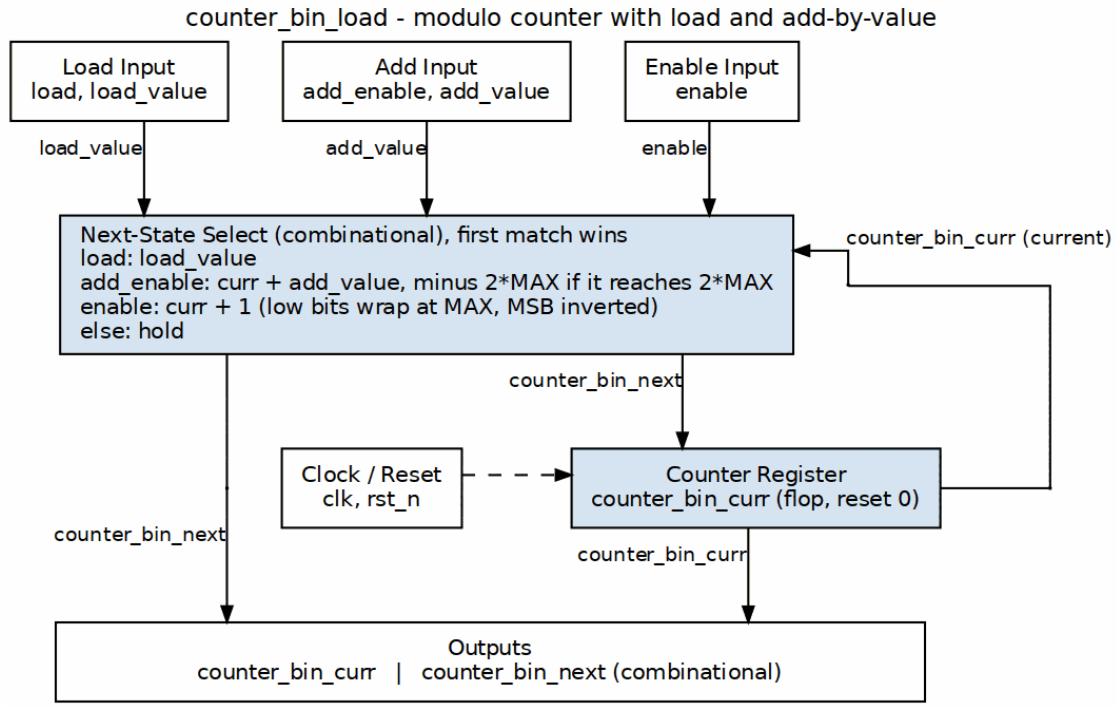
Port	Width	Description
clk	1	System clock (rising edge active)
rst_n	1	Active-low asynchronous reset
enable	1	Standard +1 increment enable
add_enable	1	Variable increment enable
add_value	WIDTH	Value to add when add_enable=1
load	1	Load enable (highest priority operation)
load_value	WIDTH	Value to load when load=1

16.4.2 Outputs

Port	Width	Description
counter_bin_curr	WIDTH	Current counter value (registered output)
counter_bin_next	WIDTH	Next counter value (combinational preview)

16.5 Functional Description

16.5.1 Block diagram



counter_bin_load block diagram

Three request ports feed one next-state select, and the first match wins: `load` takes `load_value`, `add_enable` adds `add_value` (wrapping by subtracting `2*MAX`), and `enable` increments with the same lap-bit wrap as `counter_bin`. With none asserted the register holds. The register value climbs the right edge back into the select, and the next value also leaves combinationally as `counter_bin_next`.

16.5.2 Operation Priority

The module implements a three-level priority hierarchy:

1. **Load Operation** (Highest Priority): `load = 1`

- Directly sets `counter_bin_next = load_value`
- Used for: Flush all, jump to specific position

2. **Variable Increment**: `add_enable = 1` (when `load = 0`)
   - Adds `add_value` to current count: `counter_bin_next = counter_bin_curr + add_value`
   - Wraps at 2×MAX: Subtracts 2×MAX if sum ≥ 2×MAX
   - Used for: Drop multiple entries, bulk operations
3. **Standard Increment**: `enable = 1` (when `load = 0` and `add_enable = 0`)
   - Adds 1 to current count: `counter_bin_next = counter_bin_curr + 1`
   - Special FIFO wraparound: MSB toggles at MAX-1, lower bits clear
   - Used for: Normal FIFO read/write pointer increment
4. **Hold**: All enables = 0
   - Maintains current value: `counter_bin_next = counter_bin_curr`

16.5.3 FIFO-Optimized Wraparound

Standard Increment Mode (enable=1):

```
For MAX=8, WIDTH=4:
Count sequence: 0, 1, 2, 3, 4, 5, 6, 7, 8 (toggle MSB, clear lower)
Binary:         0000 → 0001 → ... → 0111 → 1000 (MSB inverts, lower
bits = 0)
                0000 → 0001 → ... → 0111 → 1000 → 1001 → ... → 1111 →
0000
```

Key Feature: MSB toggles when lower bits reach MAX-1 - This creates a 2×MAX count range: 0 to (2×MAX - 1) - MSB different between write/read pointers → FIFO full or empty

Variable Increment Mode (add_enable=1):

```
Wraparound at 2×MAX:
Example: counter_bin_curr = 14, add_value = 5, MAX = 8, WRAP_BOUNDARY
= 16
Sum = 14 + 5 = 19 ≥ 16 → counter_bin_next = 19 - 16 = 3
```

Key Feature: Full 0 to (2×MAX - 1) range with modulo arithmetic

16.5.4 Full/Empty Detection Pattern

```
// In FIFO controller:
logic full, empty;
assign full  = (wr_ptr[WIDTH-2:0] == rd_ptr[WIDTH-2:0]) &&
               (wr_ptr[WIDTH-1] != rd_ptr[WIDTH-1]);  // MSB different

assign empty = (wr_ptr[WIDTH-2:0] == rd_ptr[WIDTH-2:0]) &&
               (wr_ptr[WIDTH-1] == rd_ptr[WIDTH-1]);  // MSB same
```

Why This Works: - Normal operation: wr_ptr advances, MSBs eventually differ → full - After wraparound: rd_ptr catches up, MSBs match again → empty - The MSB does lap-counter duty for the circular buffer — it's how you tell "wrapped, so full" apart from "caught up, so empty."

16.5.5 Priority Logic

```
always_comb begin
    counter_bin_next = counter_bin_curr;  // Default: hold

    if (load) begin
        // Highest priority: Direct load
        counter_bin_next = load_value;

    end else if (add_enable) begin
        // Variable increment with wraparound at 2*MAX
        w_sum_ext = {1'b0, counter_bin_curr} + {1'b0, add_value};

        if (w_sum_ext >= WRAP_BOUNDARY) begin
            counter_bin_next = w_sum_ext[WIDTH-1:0] -
WIDTH'(WRAP_BOUNDARY);
        end else begin
            counter_bin_next = w_sum_ext[WIDTH-1:0];
        end

    end else if (enable) begin
        // Standard +1 with MSB toggle at MAX-1
        if (counter_bin_curr[WIDTH-2:0] == (MAX - 1)) begin
            counter_bin_next = {~counter_bin_curr[WIDTH-1], {(WIDTH-1)
{1'b0}}};
        end else begin
            counter_bin_next = counter_bin_curr + 1;
        end
    end
    // else: hold (default assignment)
end
```

16.5.6 Registered Output

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n)
        counter_bin_curr <= 'b0;  // Reset to 0
    else
        counter_bin_curr <= counter_bin_next;
end
```

Reset Behavior: Counter initializes to 0 on `rst_n = 0`

16.5.7 Combinational Next-Value Preview

The `counter_bin_next` output gives you a one-cycle lookahead of the counter value. Handy for: - FIFO full/empty prediction - Pre-computing next address - Pipelined operations

16.6 Usage Examples

16.6.1 FIFO Read Pointer with Drop Support

```
// 8-entry FIFO with drop capability
localparam int FIFO_DEPTH = 8;
localparam int PTR_WIDTH = $clog2(FIFO_DEPTH) + 1;  // +1 for MSB

logic [PTR_WIDTH-1:0] rd_ptr, rd_ptr_next;
logic rd_en, drop_valid;
logic [PTR_WIDTH-1:0] drop_count;

counter_bin_load #(
    .WIDTH(PTR_WIDTH),      // 4 bits: 3 for addr (0-7) + 1 MSB
    .MAX(FIFO_DEPTH)        // Wrap at 8
) u_rd_ptr (
    .clk              (clk),
    .rst_n            (rst_n),
    .enable           (rd_en & !fifo_empty),        // Normal read
    .add_enable       (drop_valid),                 // Drop multiple
    .add_value        (drop_count),                 // How many to
drop
    .load             (flush_all),                  // Flush operation
    .load_value       (wr_ptr),                     // Jump to write
ptr
    .counter_bin_curr (rd_ptr),
    .counter_bin_next (rd_ptr_next)
);

// Use lower bits for memory address
assign rd_addr = rd_ptr[PTR_WIDTH-2:0];
```

16.6.2 FIFO Write Pointer

```
logic [PTR_WIDTH-1:0] wr_ptr, wr_ptr_next;
logic wr_en;

counter_bin_load #(
    .WIDTH(PTR_WIDTH),
    .MAX(FIFO_DEPTH)
) u_wr_ptr (
    .clk              (clk),
    .rst_n            (rst_n),
    .enable           (wr_en & !fifo_full),
    .add_enable       (1'b0),               // No variable increment
for write
    .add_value        ('0),
    .load             (flush_all),          // Reset to read pointer
    .load_value       (rd_ptr),
    .counter_bin_curr (wr_ptr),
    .counter_bin_next (wr_ptr_next)
);

assign wr_addr = wr_ptr[PTR_WIDTH-2:0];
```

16.6.3 Full/Empty Status Logic

```
// Full: Lower bits match, MSB different
assign fifo_full = (wr_ptr[PTR_WIDTH-2:0] == rd_ptr[PTR_WIDTH-2:0]) &&
                   (wr_ptr[PTR_WIDTH-1] != rd_ptr[PTR_WIDTH-1]);

// Empty: All bits match
assign fifo_empty = (wr_ptr == rd_ptr);

// Count calculation (handles wraparound)
logic [PTR_WIDTH-1:0] fifo_count;
assign fifo_count = (wr_ptr >= rd_ptr) ? (wr_ptr - rd_ptr) :
                                         (2*FIFO_DEPTH - rd_ptr +
wr_ptr);
```

16.6.4 Packet Drop Controller

```
// Drop N packets from FIFO
logic [7:0] packets_to_drop;
logic drop_start;

// State machine for multi-cycle drop
typedef enum logic [1:0] {
    IDLE,
    DROPPING
} drop_state_t;
```

```
drop_state_t drop_state;

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        drop_state <= IDLE;
        drop_valid <= 1'b0;
        drop_count <= '0;
    end else begin
        case (drop_state)
            IDLE: begin
                if (drop_start && packets_to_drop > 0) begin
                    drop_valid <= 1'b1;
                    drop_count <= packets_to_drop;
                    drop_state <= DROPPING;
                end
            end

            DROPPING: begin
                drop_valid <= 1'b0;  // Single-cycle pulse
                drop_state <= IDLE;
            end
        endcase
    end
end

// Connect to counter
counter_bin_load #(
    .WIDTH(PTR_WIDTH),
    .MAX(FIFO_DEPTH)
) u_rd_ptr (
    .clk        (clk),
    .rst_n      (rst_n),
    .enable     (rd_en & !fifo_empty),
    .add_enable (drop_valid),
    .add_value  (drop_count),
    .load       (1'b0),
    .load_value ('0),
    .counter_bin_curr(rd_ptr),
    .counter_bin_next(rd_ptr_next)
);
```

16.7 Timing Characteristics

16.7.1 Latency and Throughput

Property	Value
Increment latency	1 cycle (registered output)

Property	Value
Next-value availability	0 cycles (combinational `counter_bin_next`)
Throughput	1 operation per cycle
Reset recovery	1 cycle (asynchronous reset)

16.7.2 Operation Timing Diagrams

Standard Increment (enable=1):

```
Cycle:    0    1    2    3    4    5    6    7    8    9
clk:      _|‾|_|‾|_|‾|_|‾|_|‾|_|‾|_|‾|_|‾|_|‾|_|‾|_|‾|_
enable:   ______|‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾‾
curr:     0000  0000  0001  0010  0011  0100  0101  0110
next:     0000  0001  0010  0011  0100  0101  0110  0111
```

Variable Increment (add_enable=1, add_value=3):

```
Cycle:    0    1    2    3
clk:      _|‾|_|‾|_|‾|_|‾|_
add_en:   ______|‾‾|_______
add_val:  ------[3]-------
curr:     0010  0010  0101
next:     0010  0101  0101
                ↑ +3   ↑ add_en low -> hold (next = curr)
```

Load Operation (load=1, load_value=12):

```
Cycle:    0    1    2
clk:      _|‾|_|‾|_|‾|_
load:     ______|‾|___
load_val: ------[C]---
curr:     0011  0011  1100
next:     0011  1100  1100
                ↑ load  ↑ load low -> hold (next = curr)
```

16.7.3 Resource Utilization

Configuration	FFs	LUTs	Description
WIDTH=4, MAX=8	4	~12	Minimal (8-entry FIFO)
WIDTH=5, MAX=16	5	~16	Small (16-entry FIFO)
WIDTH=9, MAX=256	9	~30	Medium (256-entry FIFO)

Configuration	FFs	LUTs	Description
WIDTH=13, MAX=4096	13	~45	Large (4K-entry FIFO)

Area Scaling: Approximately linear with WIDTH due to adder and comparator

16.7.4 Maximum Frequency

- **Typical**: 300-500 MHz (modern FPGAs)
- **Limiting Factor**: Add operation with wraparound detection
- **Optimization**: Pipeline add path for >500 MHz

16.8 Testing

Test suite location: `val/common/test_counter_bin_load.py`

Key Test Scenarios: - Standard increment (enable=1) - Variable increment with various add_value - Load operation priority verification - Wraparound behavior at 2×MAX - MSB toggle at MAX-1 for standard increment - Full/empty condition verification - Priority hierarchy (load > add > enable) - Reset behavior

Test Command:

```
pytest val/common/test_counter_bin_load.py -v
```

16.9 Design Notes

16.9.1 When to Use counter_bin_load

Appropriate Use Cases: - FIFO read/write pointers with drop/flush - Circular buffer management with skip operations - Packet queue pointers with bulk discard - Ring buffer pointers with reset capability - Any counter needing jump-to-position capability

16.9.2 When to Use Alternatives

Use `counter_bin.sv` if: - Only need standard +1 increment - No load or variable increment required - Simpler interface preferred

Use `counter_load_clear.sv` if: - Need load/clear but not FIFO wraparound - Don't need MSB toggle behavior - Variable increment not required

16.9.3 Parameter Selection Guidelines

WIDTH Calculation:

```
// For FIFO with N entries:
localparam int FIFO_DEPTH = 16;
localparam int PTR_WIDTH = $clog2(FIFO_DEPTH) + 1;  // +1 for MSB
```

```
// Example: 16-entry FIFO
// FIFO_DEPTH = 16
// PTR_WIDTH = 4 + 1 = 5 bits
// Lower 4 bits = address (0-15)
// MSB = full/empty indicator
```

MAX Selection: - Set MAX equal to FIFO depth - MAX must fit in WIDTH-1 bits

16.9.4 Critical Paths

1. **Add Operation**: Extended addition + wraparound comparison
    - Longest path: `counter_bin_curr + add_value` → overflow detection → modulo
2. **Standard Increment**: Comparison + MSB toggle
    - Path: Lower bits comparison → MSB invert + clear
3. **Load Operation**: Direct assignment
    - Shortest path: Mux select

Optimization Tip: If the add operation is your critical path, pipeline it:

```
// Pipeline add operation
logic [WIDTH-1:0] r_add_result;
always_ff @(posedge clk) begin
    r_add_result <= counter_bin_curr + add_value;
end
```

16.9.5 Common Pitfalls

Anti-Pattern 1: Incorrect WIDTH for FIFO depth

```
// WRONG: WIDTH = $clog2(FIFO_DEPTH)
localparam int WIDTH = $clog2(16);  // = 4 bits
// Problem: No MSB for full/empty detection!

// RIGHT: WIDTH = $clog2(FIFO_DEPTH) + 1
localparam int WIDTH = $clog2(16) + 1;  // = 5 bits
// Lower 4 bits = address, MSB = full/empty indicator
```

Anti-Pattern 2: Ignoring operation priority

```
// WRONG: Trying to increment while load is active
load <= 1'b1;
enable <= 1'b1;  // This will be IGNORED!
```

```
// Result: Counter loads load_value, enable has no effect

// RIGHT: Ensure only one operation active
load <= drop_all;
enable <= rd_en & !drop_all;  // Mutually exclusive
```

Anti-Pattern 3: add_value exceeds 2×MAX

```
// WRONG: add_value larger than valid range
// MAX = 8, add_value = 20
// Result: Multiple wraparounds, unpredictable behavior

// RIGHT: Constrain add_value
assert property (@(posedge clk) add_enable |-> add_value < 2*MAX);
```

Anti-Pattern 4: Using wrong bits for memory address

```
// WRONG: Using full counter including MSB
assign mem_addr = wr_ptr;  // Includes MSB!
// Result: Addresses 0-15 appear as 0-15 and 16-31

// RIGHT: Use only lower bits
assign mem_addr = wr_ptr[WIDTH-2:0];  // Exclude MSB
```

Every one of these has bitten someone on a real project. Anti-Pattern 1 is the most common — the missing MSB doesn't show up until you try to distinguish full from empty, which is to say, at the worst possible moment.

16.10 Related Modules

- **counter_bin.sv** - Simple binary counter (no load or variable increment)
- **counter_load_clear.sv** - Counter with load/clear (no FIFO wraparound)
- **counter_freq_invariant.sv** - Time-based counter (millisecond/microsecond timing)
- **gaxi_drop_fifo_sync.sv** - Uses counter_bin_load for drop-capable FIFO
- **gaxi_fifo_sync.sv** - Standard synchronous FIFO with this counter

16.11 References

- Cummings, C.E. "Simulation and Synthesis Techniques for Asynchronous FIFO Design." *SNUG San Jose*, 2002 (canonical non-peer conference paper) — FIFO style #1: pointers one bit wider than the address, empty when the

pointers are equal, full when they differ only in the wrap (MSB) bit, the full/empty test this counter's wrap bit serves; load and add-value are repo extensions for drop/flush.

- Roy, S. *Advanced Digital System Design*, Springer, ch. 4 "Basic Sequential Circuits", §4.7.1 "Loadable Up Counter" — textbook treatment (secondary source).
- Indexed in `rtl/common/references/README.md` (PDF archived once, under `rtl/cdc/references/AsyncFifoGrayPointers__Cummings2002.pdf`)

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17 Binary Counter for FIFOs

17.1 Overview

The `counter_bin` module is a binary counter with one job: FIFO (First-In-First-Out) buffer addressing. Instead of a plain wrap, it inverts the MSB and clears the lower bits when it reaches the maximum value. That folded counting pattern is exactly what circular buffer management wants — the MSB becomes your lap marker.

17.2 Module Interface

```
module counter_bin #(
    parameter int WIDTH = 5,
    parameter int MAX   = 10
) (
    input  logic             clk,
    input  logic             rst_n,
    input  logic             enable,
    output logic [WIDTH-1:0] counter_bin_curr,
    output logic [WIDTH-1:0] counter_bin_next
);
```

17.3 Parameters

Parameter	Type	Default	Description
`WIDTH`	`int`	`5`	Total width of the counter in bits
`MAX`	`int`	`10`	Maximum count value before special wrap behavior

`WIDTH` must be ≥ 2 for proper MSB inversion operation, and it determines the counter's bit width and maximum representable value.

`MAX` ranges from 2 to `2^(WIDTH-1)` **inclusive**. The RTL computes `w_max_val = (MAX-1)` in `WIDTH-1` bits, and `MAX-1 = 2^(WIDTH-1)-1` fits exactly, so `MAX = 2^(WIDTH-1)` is legal (the doc's own validation asserts `MAX <= 2**(WIDTH-1)`, and both examples use it: `#(.WIDTH(4), .MAX(8))` and `#(.WIDTH(11), .MAX(1024))`). This matters because power-of-2 FIFO depths sit exactly at `MAX = 2^(WIDTH-1)`. The one hard constraint: `MAX-1` must fit within the lower `WIDTH-1` bits.

17.4 Ports

17.4.1 Inputs

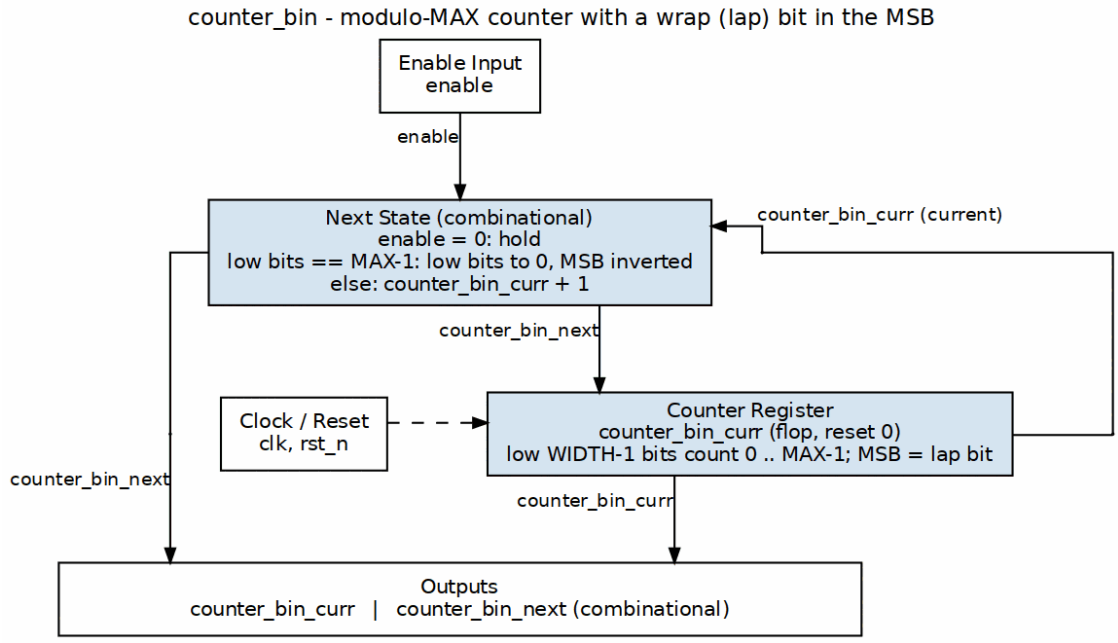
Port	Width	Description
`clk`	1	System clock input
`rst_n`	1	Active-low asynchronous reset
`enable`	1	Counter enable control

17.4.2 Outputs

Port	Width	Description
`counter_bin_curr`	WIDTH	Current counter value (registered)
`counter_bin_next`	WIDTH	Next counter value (combinational)

17.5 Functional Description

17.5.1 Block diagram



counter_bin block diagram

enable drives the next-state logic, which holds, increments, or wraps the low WIDTH-1 bits to 0 at MAX-1 while inverting the MSB. That MSB is the lap bit a FIFO compares to tell full from empty. The next value loads counter_bin_curr, which climbs the right edge back into the logic. The next value also leaves directly as the combinational counter_bin_next.

17.5.2 Internal Signals

```
logic [WIDTH-2:0] w_max_val;
assign w_max_val = (WIDTH-1)'(MAX - 1);
```

17.5.3 Special Wrap Logic

Here's the wrap behavior that makes this a FIFO counter rather than a generic one:

```
always_comb begin
    if (enable) begin
        if (counter_bin_curr[WIDTH-2:0] == w_max_val)
            counter_bin_next = {~counter_bin_curr[WIDTH-1], {(WIDTH -
1){1'b0}}};
        else
            counter_bin_next = counter_bin_curr + 1;
    end else begin
        counter_bin_next = counter_bin_curr;
    end
end
```

17.5.4 Normal Counting

When counter_bin_curr[WIDTH-2:0] < MAX-1, nothing exotic happens: - Counter increments normally by 1 - MSB remains unchanged - Standard binary counting behavior

17.5.5 Wrap Condition

When counter_bin_curr[WIDTH-2:0] == MAX-1: - MSB is inverted: ~counter_bin_curr[WIDTH-1] - Lower bits are cleared: {(WIDTH-1){1'b0}} - Creates a "folded" counting pattern

17.5.6 Enable Control

- When enable = 1: Counter operates normally
- When enable = 0: Counter holds current value

17.5.7 FIFO Application Context

17.5.8 Address Generation

This pattern earns its keep in FIFO addressing because: 1. **Circular Addressing**: Lower bits provide circular buffer addresses 2. **Overflow Detection**: MSB toggles indicate buffer wrap-around 3. **Full/Empty Logic**: Comparing MSBs between read/write pointers

17.5.9 FIFO Pointer Comparison

```
// FIFO empty when pointers exactly equal
assign fifo_empty = (rd_ptr == wr_ptr);

// FIFO full when addresses equal but MSBs differ
assign fifo_full = (rd_ptr[WIDTH-2:0] == wr_ptr[WIDTH-2:0]) &&
                   (rd_ptr[WIDTH-1] != wr_ptr[WIDTH-1]);
```

17.5.10 Advanced Variants

17.5.11 Address Extraction

```
// Extract memory address (lower bits only)
assign mem_addr = counter_bin_curr[WIDTH-2:0];

// Extract wrap flag (MSB only)
assign wrap_flag = counter_bin_curr[WIDTH-1];
```

17.5.12 Gray Code Conversion

For asynchronous FIFOs, convert to Gray code:

```
logic [WIDTH-1:0] gray_ptr;
bin2gray #(.WIDTH(WIDTH)) b2g (
    .binary(counter_bin_curr),
    .gray(gray_ptr)
);
```

17.6 Timing Characteristics

This module is **sequential**: it contains clocked logic (via `always_ff` or the repository's `ALWAYS_FF_RST` macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

17.6.1 Timing

- **Enable to Next**: Single logic level
- **Current to Next**: Increment/wrap logic depth

- **Critical Path**: Wrap detection and MSB inversion
- **Setup Time**: Standard flip-flop requirements
- **Clock-to-Q**: Standard flip-flop propagation
- **Reset Recovery**: One cycle after reset deassertion

17.6.2 Maximum Frequency

- **Limiting Factor**: Wrap detection and MSB inversion logic
- **Typical Performance**: 300-500 MHz in modern FPGAs
- **Optimization**: Consider registering intermediate signals for high speed

17.6.3 Power Consumption

- **Dynamic Power**: Proportional to toggle rate
- **Optimization**: Use enable signal to reduce unnecessary toggles
- **Clock Gating**: Gate clock when FIFO operations are idle

17.7 Usage Examples

17.7.1 1. Small FIFO Buffer (8 entries)

```
counter_bin #(
    .WIDTH(4),    // 4-bit counter
    .MAX(8)       // 8 entries (0-7)
) wr_ptr_counter (
    .clk(clk),
    .rst_n(rst_n),
    .enable(wr_enable),
    .counter_bin_curr(wr_ptr),
    .counter_bin_next(wr_ptr_next)
);
```

17.7.2 2. Large FIFO Buffer (1024 entries)

```
counter_bin #(
    .WIDTH(11),   // 11-bit counter (10 bits address + 1 MSB)
    .MAX(1024)    // 1024 entries (0-1023)
) rd_ptr_counter (
    .clk(clk),
    .rst_n(rst_n),
    .enable(rd_enable),
    .counter_bin_curr(rd_ptr),
```

```
    .counter_bin_next(rd_ptr_next)
);
```

17.7.3 3. Asymmetric FIFO

```
// Write pointer - 16 entries
counter_bin #(.WIDTH(5), .MAX(16)) wr_cnt (...);

// Read pointer - different size for width conversion
counter_bin #(.WIDTH(6), .MAX(32)) rd_cnt (...);
```

17.7.4 Counting Sequence Example

17.7.5 Configuration: WIDTH=4, MAX=6

```
Initial:  0000 (MSB=0, lower=000)
          0001
          0010
          0011
          0100
          0101  <- MAX-1 reached
Wrap:     1000  <- MSB inverted (0→1), lower bits cleared
          1001
          1010
          1011
          1100
          1101  <- MAX-1 reached again
Wrap:     0000  <- MSB inverted (1→0), lower bits cleared
```

17.7.6 Pattern Analysis

- **First Half**: MSB=0, counts 0 to MAX-1
- **Second Half**: MSB=1, counts 0 to MAX-1
- **Full Cycle**: 2×MAX states total ## Testing

17.7.7 Test Scenarios

1. **Basic Increment**: Verify normal counting behavior
2. **Wrap Boundary**: Test MSB inversion at MAX-1
3. **Enable Control**: Verify hold behavior when disabled
4. **Reset Operation**: Confirm proper reset to zero
5. **Full Sequence**: Complete cycle through all states

17.7.8 Coverage Points

```
covergroup counter_bin_cg @(posedge clk);
    cp_lower_bits: coverpoint counter_bin_curr[WIDTH-2:0] {
        bins all_values[] = {[0:MAX-1]};
    }
    cp_msb: coverpoint counter_bin_curr[WIDTH-1] {
        bins low = {0};
        bins high = {1};
    }
    cp_wrap: coverpoint (counter_bin_curr[WIDTH-2:0] == w_max_val);
    cp_enable: coverpoint enable;
endcovergroup
```

17.8 Design Notes

17.8.1 Resource Utilization

- **Flip-Flops**: WIDTH registers for current counter
- **LUTs**: Increment, wrap detection, and MSB inversion logic
- **Optimization**: Synthesis tools typically optimize the wrap logic efficiently

17.8.2 Timing Optimization

```
// Optional: Pipeline for high-frequency operation
logic [WIDTH-1:0] counter_bin_pipe;
always_ff @(posedge clk) begin
    counter_bin_pipe <= counter_bin_next;
end
```

17.8.3 Parameter Validation

```
initial begin
    assert (WIDTH >= 2)
        else $error("WIDTH must be at least 2 for MSB operation");
    assert (MAX <= (2**(WIDTH-1)))
        else $error("MAX too large for given WIDTH");
    assert (MAX > 0)
        else $error("MAX must be positive");
end
```

17.8.4 Common Issues and Solutions

Issue 1: Incorrect FIFO Size - **Problem**: MAX parameter doesn't match actual FIFO depth - **Solution**: Ensure MAX equals the number of FIFO entries

Issue 2: Width Mismatch - **Problem**: WIDTH too small for MAX value - **Solution**: Set WIDTH ≥ `$clog2(2*MAX)` for full range

Issue 3: Timing Violations - **Problem**: Wrap logic creates long combinational paths - **Solution**: Pipeline the next value calculation or reduce MAX

17.9 Related Modules

- **counter_bingray** - Provides both binary and Gray code outputs
- **counter** - Basic counter without wrap behavior
- **counter_load_clear** - Loadable counter with different wrap behavior
- Standard FIFO controllers that use this addressing scheme

17.10 References

- Cummings, C.E. “Simulation and Synthesis Techniques for Asynchronous FIFO Design.” *SNUG San Jose*, 2002 (canonical non-peer conference paper) — FIFO style #1: pointers one bit wider than the address, empty when the pointers are equal, full when they differ only in the wrap (MSB) bit, the full/empty test this counter’s MSB-inverting wrap exists to serve. The non-power-of-2 wrap (clear the low bits and invert the MSB at MAX-1) is repo practice.
- Roy, S. *Advanced Digital System Design*, Springer, ch. 4 “Basic Sequential Circuits”, §4.6 “Synchronous Counter Design” — textbook treatment (secondary source).
- Indexed in `rtl/common/references/README.md` (PDF archived once, under `rtl/cdc/references/AsyncFifoGrayPointers__Cummings2002.pdf`)

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18 Frequency Invariant Counter

18.1 Overview

The `counter_freq_invariant` module divides an arbitrary input clock down to a 1 MHz tick, so your design can count real time without knowing its clock frequency at compile time. You pick the division factor at run time through `freq_sel`, which indexes a lookup table built at elaboration time from the `MIN_FREQ_MHZ`, `MAX_FREQ_MHZ`, `NUM_FREQ_ENTRIES` and `FREQ_STRATEGY` parameters.

The whole module rests on one convenient identity: **the division factor and the clock frequency in MHz are the same number**. A 100 MHz clock has 100 cycles per microsecond, so dividing by 100 yields a 1 MHz tick. The LUT just stores frequencies in MHz and uses them directly as divisors. Elegant, once you see it.

One bit of history: an earlier revision of this module had a hardcoded 68-entry frequency table and a fixed 4-bit `freq_sel`. That table is gone; the LUT is now generated from parameters and `freq_sel` is sized to match `NUM_FREQ_ENTRIES`.

18.2 Module Interface

```
module counter_freq_invariant #(
    // User parameters
    parameter int COUNTER_WIDTH    = 16,     // Width of output
microsecond counter
    parameter int MIN_FREQ_MHZ     = 5,      // Lowest supported clock
(MHz)
    parameter int MAX_FREQ_MHZ     = 220,    // Highest supported
clock (MHz)
    parameter int NUM_FREQ_ENTRIES = 16,     // Number of LUT entries
    parameter int FREQ_STRATEGY    = 0,      // 0 = LINEAR, 1 = POW2
    parameter bit DEBUG_LUT        = 1'b0,   // Print LUT at time 0

    // Derived parameters (do not override)
    parameter int SEL_WIDTH     = (NUM_FREQ_ENTRIES > 1) ?
$clog2(NUM_FREQ_ENTRIES) : 1,
    parameter int DIV_WIDTH     = $clog2(MAX_FREQ_MHZ + 1),
    parameter int PRESCALER_MAX = 2 ** DIV_WIDTH
) (
    input  logic                       clk,
    input  logic                       rst_n,
    input  logic                       sync_reset_n,
    input  logic [SEL_WIDTH-1:0]       freq_sel,
    output logic [COUNTER_WIDTH-1:0]   o_counter,
    output logic                       tick
);
```

18.3 Parameters

counter_freq_invariant user parameters

Parameter	Type	Default	Description
COUNTER_WIDTH	int	16	Width of the microsecond counter. Wraps at 2^COUNTER_WIDTH.
MIN_FREQ_MHZ	int	5	Lowest clock frequency in the LUT, in MHz. Must be >= 1.
MAX_FREQ_MHZ	int	220	Highest clock frequency in the LUT, in MHz. Must be >= MIN_FREQ_MHZ.
NUM_FREQ_ENTRIES	int	16	Number of LUT entries. Must be >= 1. Sets `SEL_WIDTH`.
FREQ_STRATEGY	int	0	LUT distribution: 0 = LINEAR, 1 = POW2.
DEBUG_LUT	bit	1'b0	Print the generated LUT at simulation time 0. Simulation only.

Derived parameters – **do not override.** `SEL_WIDTH`, `DIV_WIDTH` and `PRESCALER_MAX` appear in the parameter list only so they can size ports and internal counters. They're computed from the user parameters:

Derived parameters at the default configuration

Derived	Expression	Value at defaults
SEL_WIDTH	`$clog2(NUM_FREQ_ENTRIES)`, min 1	4
DIV_WIDTH	`$clog2(MAX_FREQ_MHZ + 1)`	8
PRESCALER_MAX	`2 ** DIV_WIDTH`	256

Bad combinations get caught early: the three `$error` checks in the `param_check` initial block reject `MIN_FREQ_MHZ < 1`, `MAX_FREQ_MHZ < MIN_FREQ_MHZ`, and `NUM_FREQ_ENTRIES < 1` at elaboration.

18.4 Ports

18.4.1 Inputs

Port	Width	Description
clk	1	Input clock, any frequency covered by the LUT
rst_n	1	Active-low asynchronous reset
sync_reset_n	1	Synchronous run/reset. 0 holds the counter cleared; 1 runs.
freq_sel	SEL_WIDTH	LUT index selecting the division factor

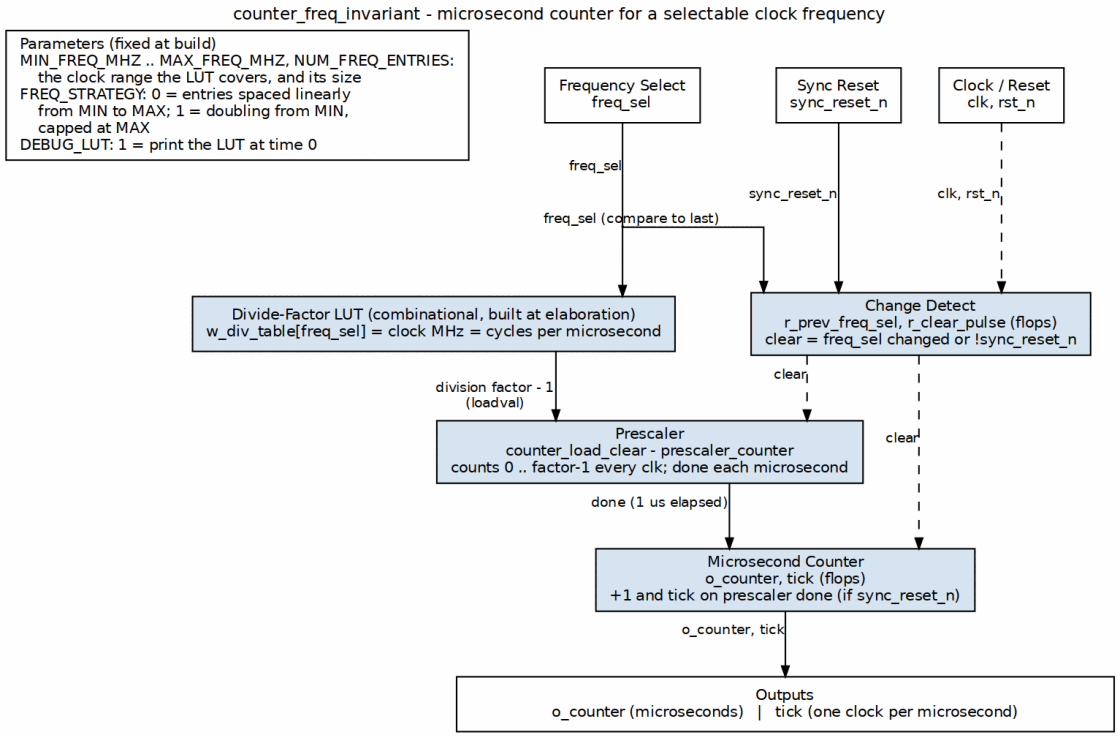
18.4.2 Outputs

Port	Width	Description
o_counter	COUNTER_WIDTH	Microsecond counter, wraps at 2^COUNTER_WIDTH
tick	1	Single-cycle pulse, once per microsecond

One naming gotcha: the output is `o_counter`, not `counter`.

18.5 Functional Description

18.5.1 Block diagram



counter_freq_invariant block diagram

`freq_sel` picks an entry from a divide-factor table built at elaboration, where each entry is the clock frequency in MHz, which equals clocks per microsecond. The prescaler (`counter_load_clear`) counts that many clocks and pulses `done` once per microsecond, and each `done` advances `o_counter` and fires `tick`. A change of `freq_sel`, or `sync_reset_n` low, raises the clear pulse, which restarts both the prescaler and the microsecond counter. The parameters legend says how the table entries are spaced.

```
        freq_sel ---> [ LUT mux ] ---> division_factor
                                |
clk, rst_n ---> [ change detect ] ---> clear_pulse
                                |
                                v
                  [ counter_load_clear ] ---> prescaler_done
                                |
                                v
                  [ counter + tick reg ] ---> o_counter, tick
```

18.5.2 LUT Generation Strategies

The LUT is built at elaboration time by a generate loop that evaluates a pure integer function per index, so synthesis infers a mux or ROM rather than any runtime arithmetic. That's what you want — no multiplier hiding in your timing report.

LINEAR (`FREQ_STRATEGY = 0`, default)

```
freq[i] = MIN_FREQ_MHZ + (MAX_FREQ_MHZ - MIN_FREQ_MHZ) * i /
(NUM_FREQ_ENTRIES - 1)
```

Uniform steps across the range. Integer division truncates. Good for FPGA bring-up where predictable coverage of the operating range matters.

POW2 (`FREQ_STRATEGY = 1`)

```
freq[i] = min(MIN_FREQ_MHZ * 2^i, MAX_FREQ_MHZ)
```

Doubling from the minimum, clamped at the maximum. Gives finer resolution at the low end and saturates once `MAX_FREQ_MHZ` is reached.

Default LUT (LINEAR, 5-220 MHz, 16 entries)

Default LINEAR LUT. Division factor equals the clock frequency in MHz.

freq_sel	Clock (MHz)	Cycles per us	freq_sel	Clock (MHz)	Cycles per us
0	5	5	8	119	119
1	19	19	9	134	134
2	33	33	10	148	148
3	48	48	11	162	162
4	62	62	12	177	177
5	76	76	13	191	191
6	91	91	14	205	205
7	105	105	15	220	220

Set `DEBUG_LUT` and the table your parameters actually produce prints at time 0 — much better than working it out by hand and being wrong.

Same table under POW2

Same range, same entry count, `FREQ_STRATEGY = 1`: you get 5, 10, 20, 40, 80, 160 MHz and then saturation at 220 MHz for indices 6 through 15. POW2 only earns its keep when `NUM_FREQ_ENTRIES` is close to `log2(MAX/MIN) + 1`; beyond that the remaining entries are duplicates.

18.5.3 Division Factor Lookup

```
logic [DIV_WIDTH-1:0] w_div_table [NUM_FREQ_ENTRIES];

generate
    for (genvar gi = 0; gi < NUM_FREQ_ENTRIES; gi++) begin :
gen_div_entry
        assign w_div_table[gi] = DIV_WIDTH'(freq_mhz_at_idx(gi));
    end
endgenerate

logic [DIV_WIDTH-1:0] w_division_factor;
assign w_division_factor = w_div_table[freq_sel];
```

18.5.4 Change Detection

A clear pulse fires whenever `freq_sel` changes or `sync_reset_n` is low. One subtlety worth noticing: `r_clear_pulse` resets to 1, not 0, so the counter starts life in the cleared state.

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_prev_freq_sel <= '0;
        r_clear_pulse   <= 1'b1;    // start in reset state
    end else begin
        r_prev_freq_sel <= freq_sel;
        r_clear_pulse   <= (freq_sel != r_prev_freq_sel) || !
sync_reset_n;
    end
end
```

18.5.5 Prescaler

```
counter_load_clear #(
    .MAX(PRESCALER_MAX)
) prescaler_counter (
    .clk       (clk),
    .rst_n     (rst_n),
    .clear     (r_clear_pulse),
    .increment (1'b1),
    .load      (1'b1),
    .loadval   (w_division_factor - DIV_WIDTH'(1)),
    .done      (w_prescaler_done),
    .count     ()
);
```

18.5.6 Counter and Tick

`tick` pulses on **every** prescaler completion, once per microsecond. It is not gated on the counter reaching its maximum — people assume otherwise, and it leads to off-by-2^N bugs in timeout logic. Don't.

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        o_counter <= '0;
        tick      <= 1'b0;
    end else begin
        if (r_clear_pulse) begin
            o_counter <= '0;
            tick      <= 1'b0;
        end else if (w_prescaler_done && sync_reset_n) begin
            o_counter <= o_counter + 1'b1;
            tick      <= 1'b1;
        end else begin
            tick <= 1'b0;
        end
    end
end
```

18.6 Timing Characteristics

18.6.1 Timing

counter_freq_invariant timing properties

Property	Value
Tick period	`division_factor` input clock cycles = 1 us
Counter wrap period	`2^COUNTER_WIDTH` microseconds
Latency	2 cycles (prescaler stage plus output register)
Reconfiguration	Immediate: a `freq_sel` change clears and reloads

At the default `COUNTER_WIDTH` of 16, the counter wraps every 65536 us, or about 65.5 ms. Size your timeout logic accordingly.

18.6.2 Resource Use

Resource use by element

Element	Size
Prescaler counter	`$clog2(PRESCALER_MAX)` bits
Microsecond counter	`COUNTER_WIDTH` bits
Change detection	`SEL_WIDTH + 1` flops
LUT	`NUM_FREQ_ENTRIES x DIV_WIDTH`

Element	Size
	bits of constant mux or ROM

18.6.3 Critical Path

The critical path runs through the prescaler increment and its terminal-count comparison. Raising `MAX_FREQ_MHZ` widens `DIV_WIDTH` and lengthens that path — something to watch if you're pushing frequency.

This module is **sequential**: it contains clocked logic (via `always_ff` or the repository's `ALWAYS_FF_RST` macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

18.7 Usage Examples

18.7.1 1. Microsecond timebase on an unknown FPGA clock

```
logic [15:0] usec_count;
logic        usec_tick;

counter_freq_invariant #(
    .COUNTER_WIDTH   (16),
    .MIN_FREQ_MHZ    (5),
    .MAX_FREQ_MHZ    (220),
    .NUM_FREQ_ENTRIES(16),
    .FREQ_STRATEGY   (0)
) u_timebase (
    .clk         (sys_clk),
    .rst_n       (sys_rst_n),
    .sync_reset_n(1'b1),
    .freq_sel    (cfg_freq_sel),   // 4 bits at these parameters
    .o_counter   (usec_count),
    .tick        (usec_tick)
);
```

18.7.2 2. ASIC range, 100 MHz to 1 GHz

```
counter_freq_invariant #(
    .COUNTER_WIDTH   (16),
    .MIN_FREQ_MHZ    (100),
    .MAX_FREQ_MHZ    (1000),
```

```
    .NUM_FREQ_ENTRIES(32),
    .FREQ_STRATEGY   (0)
) u_timer (
    .clk         (core_clk),
    .rst_n       (core_rst_n),
    .sync_reset_n(1'b1),
    .freq_sel    (cfg_freq_sel),   // 5 bits at NUM_FREQ_ENTRIES = 32
    .o_counter   (usec_count),
    .tick        (usec_tick)
);
```

`DIV_WIDTH` becomes `$clog2(1001)` = 10 bits here, and `PRESCALER_MAX` 1024.

18.7.33. Millisecond timeout built on the tick

```
logic [9:0] r_msec;

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n)                  r_msec <= '0;
    else if (usec_tick && usec_count[9:0] == 10'd0) r_msec <= r_msec +
1'b1;
end
```

A cheaper alternative is to watch a bit of `o_counter` directly: bit 9 toggles approximately every 512 us.

18.7.44. Gating the timer with sync_reset_n

```
// Hold the timer cleared while calibration is running
counter_freq_invariant #(
    .COUNTER_WIDTH(16)
) u_cal_timer (
    .clk         (clk),
    .rst_n       (rst_n),
    .sync_reset_n(!enter_calibration_mode),
    .freq_sel    (cfg_freq_sel),
    .o_counter   (cal_usec),
    .tick        (cal_tick)
);
```

18.8 Testing

Test: `val/common/test_counter_freq_invariant.py`

Scenarios worth covering:

1. Every `freq_sel` index produces the expected tick period for its LUT entry
2. Changing `freq_sel` mid-operation clears and restarts cleanly

3. Asynchronous reset and `sync_reset_n` both clear `o_counter` and `tick`
4. `o_counter` wraps correctly at 2^COUNTER_WIDTH
5. Both `FREQ_STRATEGY` values generate the expected LUT
6. `NUM_FREQ_ENTRIES = 1` degenerate case elaborates and runs

18.9 Design Notes

18.9.1 Choosing the LUT Range

`MIN_FREQ_MHZ` and `MAX_FREQ_MHZ` should bracket the clock frequencies the design will actually see. Widening the range without raising `NUM_FREQ_ENTRIES` coarsens the LINEAR steps, and a clock that falls between two LUT entries produces a tick that is off by the rounding error. At the defaults the step is about 14 MHz, so a 70 MHz clock selecting index 5 (76 MHz) ticks roughly 8% slow. That's not an error term you can wave away in a watchdog.

If you need an exact tick at one known frequency, there's a cleaner path: set `NUM_FREQ_ENTRIES` to 1 and `MIN_FREQ_MHZ` equal to `MAX_FREQ_MHZ` equal to that frequency. `SEL_WIDTH` clamps to 1 in that case and `freq_sel` should be tied to 0.

18.10 Related Modules

- **counter_bin** - basic binary counter
- **counter_load_clear** - programmable counter, used here as the prescaler
- **clock_divider** - integer clock division producing a divided clock, not a tick
- **clock_pulse** - configurable pulse generator

18.11 References

- No peer paper — a LUT-selected prescaler producing a fixed 1 us tick is a standard technique with no defensible single origin (marked "standard technique" in the references index).
- Roy, S. *Advanced Digital System Design*, Springer, ch. 4 "Basic Sequential Circuits", §4.11 "Clock Divider Circuits" and §4.11.4 "Programmable Clock Divider Circuit" — textbook treatment of programmable division (secondary source).
- Indexed in `rtl/common/references/README.md`

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19 Counter with Load and Clear

19.1 Overview

The `counter_load_clear` module is a counter with a loadable terminal count and a clear — the workhorse you reach for when a plain counter isn't quite enough. You get precise control over counting behavior through configurable match values, which makes it a natural fit for timing applications, protocol implementations, and configurable delay generation.

19.2 Module Interface

```
module counter_load_clear #(
    parameter int MAX = 32'd32
) (
    input logic                         clk,
    input logic                         rst_n,
    input logic                         clear,
    input logic                         increment,
    input logic                         load,
    input logic [$clog2(MAX)-1:0]       loadval,
    output logic [$clog2(MAX)-1:0]      count,
    output logic                        done
);
```

19.3 Parameters

Parameter	Type	Default	Description
MAX	int	32'd32	Maximum possible count value

MAX ranges from 2 to 2^32-1 (the RTL header states the same). MAX=1 gives `$clog2(1) = 0`, so `loadval` and `count` would be zero-width. The parameter determines the counter width and the maximum loadable value: counter width = `$clog2(MAX)`.

19.4 Ports

19.4.1 Inputs

Port	Width	Description
clk	1	System clock
rst_n	1	Active-low asynchronous reset
clear	1	Synchronous clear

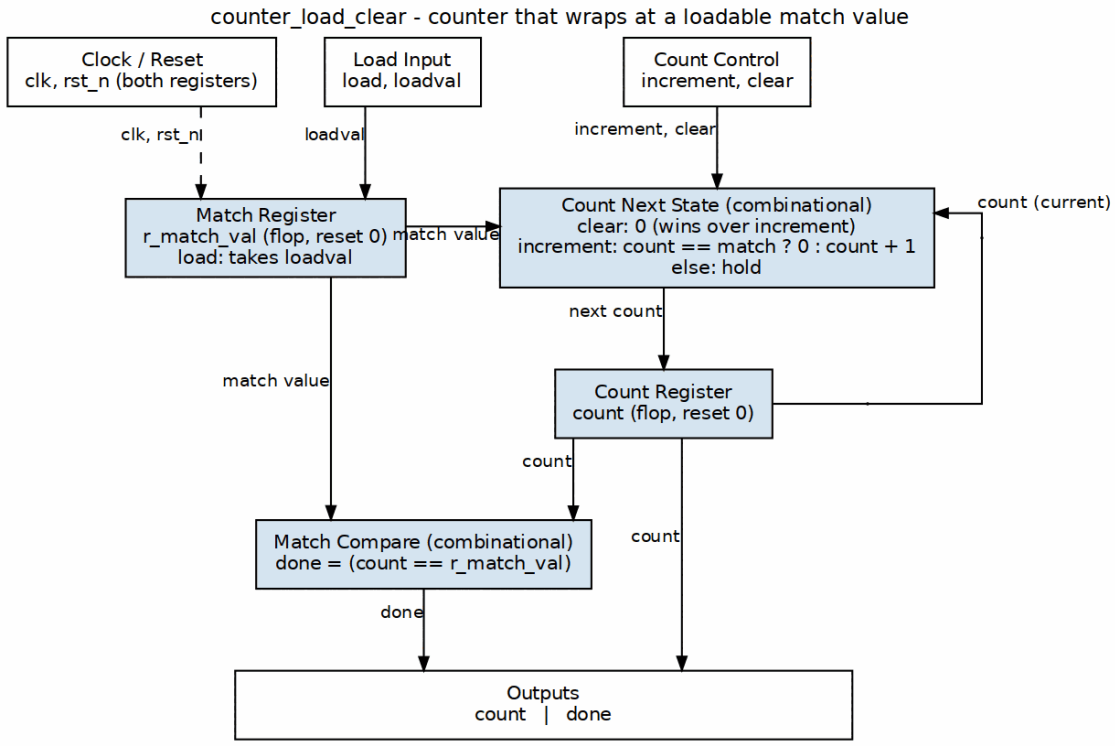
Port	Width	Description
		(highest priority)
`increment`	1	Enable counter increment
`load`	1	Load new match value
`loadval`	`$clog2(MAX)`	New terminal count value

19.4.2 Outputs

Port	Width	Description
`count`	`$clog2(MAX)`	Current counter value
`done`	1	Terminal count reached flag

19.5 Functional Description

19.5.1 Block diagram



counter_load_clear block diagram

`load` captures `loadval` into the match register. The count logic clears to 0 on `clear` (which wins), or on `increment` either adds one or wraps to 0 when the count equals the match value. The count register feeds back up the right edge into that logic. A compare raises `done` while the count equals the match value; `count` and `done` both leave in the output bar.

19.5.2 Internal Registers

```
logic [$clog2(MAX)-1:0] count;         // Current count value
logic [$clog2(MAX)-1:0] r_match_val;  // Terminal count register
```

19.5.3 Priority Logic

Control is priority-based, and the order matters:

1. **Asynchronous Reset** (Highest Priority)
    - Resets both `count` and `r_match_val` to 0
    - Independent of clock edge
2. **Load Operation** (High Priority)
    - Updates `r_match_val` with `loadval`
    - Can occur simultaneously with other operations
3. **Clear Operation** (Medium Priority)
    - Immediately sets `count` to 0
    - Overrides increment operation
4. **Increment Operation** (Lowest Priority)
    - Advances counter or wraps to 0 at terminal count
    - Only when `increment` is asserted

19.5.4 Main Control Logic

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        count <= 'b0;
        r_match_val <= 'b0;
    end else begin
        // Load has independent priority
        if (load) r_match_val <= loadval;
```

```
        // Clear has highest operational priority
        if (clear) begin
            count <= 'b0;
        end else if (increment) begin
            count <= (count == r_match_val) ? 'b0 : count + 'b1;
        end
    end
end
```

19.5.5 Done Signal Generation

```
assign done = (count == r_match_val);
```

19.5.6 Advanced Variants

19.5.7 1. Dual-Rate Timer

```
// Timer with fast/slow modes
logic fast_mode;
logic [15:0] timer_period;

always_comb begin
    timer_period = fast_mode ? 16'd99 : 16'd999;  // 100 vs 1000
cycles
end

counter_load_clear #(.MAX(65536)) dual_timer (
    .clk(clk),
    .rst_n(rst_n),
    .clear(timer_done),         // Auto-restart
    .increment(1'b1),
    .load(timer_done),          // Reload on completion
    .loadval(timer_period),
    .count(timer_value),
    .done(timer_done)
);
```

19.5.8 2. PWM Generator

```
// Simple PWM with configurable duty cycle
logic [7:0] duty_cycle;  // 0-255
logic pwm_out;

counter_load_clear #(.MAX(256)) pwm_counter (
    .clk(clk),
    .rst_n(rst_n),
    .clear(1'b0),
    .increment(1'b1),
```

```
    .load(1'b1),
    .loadval(8'd255),           // Always count 0-255
    .count(pwm_count),
    .done()
);

assign pwm_out = (pwm_count < duty_cycle);
```

19.5.9 3. Burst Generator

```
// Generate bursts of N pulses
logic [7:0] burst_length;
logic start_burst, burst_active, burst_complete;

counter_load_clear #(.MAX(256)) burst_counter (
    .clk(clk),
    .rst_n(rst_n),
    .clear(start_burst),
    .increment(burst_active),
    .load(start_burst),
    .loadval(burst_length - 1),
    .count(burst_count),
    .done(burst_complete)
);

// State machine for burst control
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        burst_active <= 1'b0;
    end else begin
        if (start_burst)
            burst_active <= 1'b1;
        else if (burst_complete)
            burst_active <= 1'b0;
    end
end
```

19.5.10 4. Communication Timeout

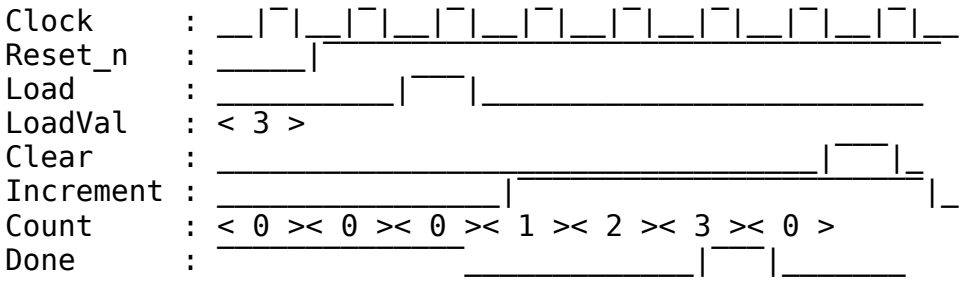
```
// UART receive timeout
logic rx_active, rx_timeout, char_received;
parameter RX_TIMEOUT_CYCLES = 1000;

counter_load_clear #(.MAX(2048)) rx_timer (
    .clk(clk),
    .rst_n(rst_n),
    .clear(char_received),      // Reset on new character
    .increment(rx_active),      // Only count when receiving
    .load(rx_active && !char_received),
```

```
    .loadval(RX_TIMEOUT_CYCLES-1),
    .count(),
    .done(rx_timeout)
);
```

19.6 Timing Characteristics

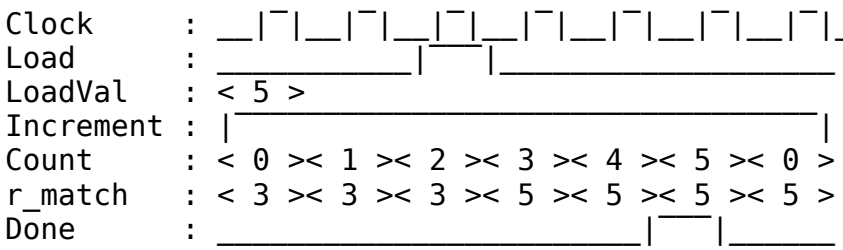
19.6.1 Basic Operation



done is combinational (`count == r_match_val`) and both reset to 0, so it is **high out of reset** and stays high until the load writes a non-zero match value. It is not a pulse that only appears at terminal count.

19.6.2 Load During Count

Load only writes `r_match_val`; it does **not** stall the count. With `increment` held high, `count` advances every cycle (including the load cycle). Here `load` asserts while `count == 2`, so `r_match_val` becomes 5 starting the next cycle — before `count` reaches the old match value 3 — so the counter never wraps at 3 and instead runs to the new match 5:



(If `load` had instead arrived after `count` already reached 3, the counter would have wrapped at 3 first — the outcome depends on load timing relative to the old match value.)

19.6.3 Maximum Frequency

- **Typical**: 300-500 MHz in modern FPGAs
- **Critical Path**: Increment + comparison logic
- **Optimization**: Consider pipelining for extreme speeds

19.6.4 Latency

- **Load Operation**: 1 cycle to update match value
- **Clear Operation**: 1 cycle to reset count
- **Done Signal**: Combinational (0 cycles)

19.7 Usage Examples

19.7.1 1. Basic Counting Mode

```
// Simple timer: count to 100
counter_load_clear #(.MAX(1000)) timer (
    .clk(clk),
    .rst_n(rst_n),
    .clear(1'b0),
    .increment(1'b1),
    .load(1'b1),
    .loadval(8'd99),    // Count 0-99 (100 cycles)
    .count(timer_count),
    .done(timer_done)
);
```

19.7.2 2. Configurable Delay Mode

```
// Variable delay generator
logic [7:0] delay_setting;
logic start_delay, delay_complete;

counter_load_clear #(.MAX(256)) delay_gen (
    .clk(clk),
    .rst_n(rst_n),
    .clear(start_delay),        // Start new delay
    .increment(1'b1),
    .load(start_delay),         // Load new value when starting
    .loadval(delay_setting),
    .count(),
    .done(delay_complete)
);
```

19.7.3 3. Retriggerable Timer Mode

```
// Watchdog timer that resets on activity
logic activity_detected, timeout;

counter_load_clear #(.MAX(10000)) watchdog (
    .clk(clk),
    .rst_n(rst_n),
```

```
    .clear(activity_detected),  // Reset on activity
    .increment(1'b1),
    .load(1'b1),
    .loadval(16'd9999),         // 10000 cycle timeout
    .count(),
    .done(timeout)
);
```

1. **Timer Modules**: Configurable timing generation
2. **Protocol Engines**: Timeout and retry mechanisms
3. **PWM Controllers**: Variable duty cycle generation
4. **Burst Generators**: Programmable pulse trains
5. **Watchdog Timers**: System monitoring and reset
6. **Clock Dividers**: Variable frequency division
7. **Test Pattern Generators**: Configurable test sequences

19.7.4 Design Patterns

19.7.5 Pattern 1: Auto-Reloading Timer

```
wire auto_reload = timer_done;
counter_load_clear timer (
    .clear(auto_reload),
    .load(auto_reload),
    .loadval(PERIOD-1),
    .done(timer_done),
    ...
);
```

19.7.6 Pattern 2: Conditional Counting

```
wire count_enable = condition_met && !paused;
counter_load_clear conditional_timer (
    .increment(count_enable),
    ...
);
```

19.7.7 Pattern 3: Dynamic Period Adjustment

```
logic period_changed;
wire reload_needed = timer_done || period_changed;

counter_load_clear adaptive_timer (
    .clear(reload_needed),
```

```
    .load(reload_needed),
    .loadval(current_period),
    ...
);
```

19.8 Testing

19.8.1 Test Scenarios

1. **Basic Counting**: Verify normal increment operation
2. **Terminal Count**: Test done signal generation
3. **Load Operation**: Verify match value updates
4. **Clear Priority**: Test clear overrides increment
5. **Simultaneous Operations**: Load + clear + increment
6. **Boundary Conditions**: Count = 0, count = match_val
7. **Reset Behavior**: Async reset during various operations

19.8.2 Coverage Model

```
covergroup load_clear_cg @(posedge clk);
    cp_count: coverpoint count {
        bins zero = {0};
        bins low = {[1:MAX/4]};
        bins mid = {[MAX/4+1:3*MAX/4]};
        bins high = {[3*MAX/4+1:MAX-1]};
    }

    cp_match_val: coverpoint r_match_val {
        bins zero = {0};
        bins low = {[1:MAX/4]};
        bins mid = {[MAX/4+1:3*MAX/4]};
        bins high = {[3*MAX/4+1:MAX-1]};
    }

    cp_operations: coverpoint {clear, increment, load} {
        bins clear_only = {3'b100};
        bins inc_only = {3'b010};
        bins load_only = {3'b001};
        bins clear_inc = {3'b110};
        bins clear_load = {3'b101};
        bins inc_load = {3'b011};
        bins all_ops = {3'b111};
```

```
        bins no_ops = {3'b000};
    }

    cp_done: coverpoint done;

    // Cross coverage
    cross_done_ops: cross cp_operations, done;
endcovergroup
```

19.8.3 Assertions

```
// Count stays within the match value ONLY for the load-then-count
pattern.
// This property does NOT hold in general: if a smaller match is
loaded while
// count is already above it (load 5, count up to 5, then load 2),
then
// count (5) > r_match_val (2) and the counter free-runs modulo
2^WIDTH until it
// happens to equal the new match again -- there is no terminal-count
clamp,
// only the equality wrap. Do not copy this assertion into a testbench
that
// dynamically REDUCES the terminal count; it will fail on correct
RTL.
property count_bounds;   // holds only when match is never reduced
below count
    @(posedge clk) disable iff (!rst_n)
    count <= r_match_val;
endproperty

// Done should be true when count equals match
property done_when_match;
    @(posedge clk) disable iff (!rst_n)
    (count == r_match_val) |-> done;
endproperty

// Clear should reset count to 0
property clear_resets_count;
    @(posedge clk) disable iff (!rst_n)
    clear |=> (count == 0);
endproperty

// Load should update match value
property load_updates_match;
    @(posedge clk) disable iff (!rst_n)
    load |=> (r_match_val == $past(loadval));
endproperty
```

```
assert property (count_bounds);
assert property (done_when_match);
assert property (clear_resets_count);
assert property (load_updates_match);
```

19.9 Design Notes

19.9.1 Resource Usage

- **Flip-Flops**: `$clog2(MAX)` for count + `$clog2(MAX)` for match = `2 × $clog2(MAX)`
- **LUTs**: Increment logic, comparator, and control logic

19.9.2 Timing Optimization

```
// For high-speed operation, pipeline the comparison
logic done_pipe;
always_ff @(posedge clk) begin
    done_pipe <= (count == r_match_val);
end
```

19.9.3 Parameter Validation

```
initial begin
    assert (MAX > 0) else $error("MAX must be positive");
    assert (MAX <= 2**30) else $warning("Large MAX may impact
synthesis");
end
```

19.10 Related Modules

- **counter** - Basic counter without load/clear
- **counter_freq_invariant** - Uses this module for prescaling
- Standard timer and PWM controller implementations

19.11 References

- No peer paper — a loadable match counter with synchronous clear is a standard technique with no defensible single origin (marked “standard technique” in the references index).

- Roy, S. *Advanced Digital System Design*, Springer, ch. 4 “Basic Sequential Circuits”, §4.7.1 “Loadable Up Counter” — textbook treatment (secondary source).
- Indexed in `rtl/common/references/README.md`

19.12 Navigation

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20 Basic Counter

20.1 Overview

The `counter` module is as simple as it gets: a configurable up-counter that raises a tick pulse when it reaches its maximum value. Use it for basic timing jobs where all you need is a periodic signal.

20.2 Module Interface

```
module counter #(
    parameter int MAX = 32767
) (
    input  logic clk,
    input  logic rst_n,
    output logic tick
);
```

20.3 Parameters

Parameter	Type	Default	Description
`MAX`	`int`	`32767`	Maximum count value before wrapping to zero

`MAX` can be any positive integer. The counter width is automatically calculated as `$clog2(MAX+1)`.

20.4 Ports

20.4.1 Inputs

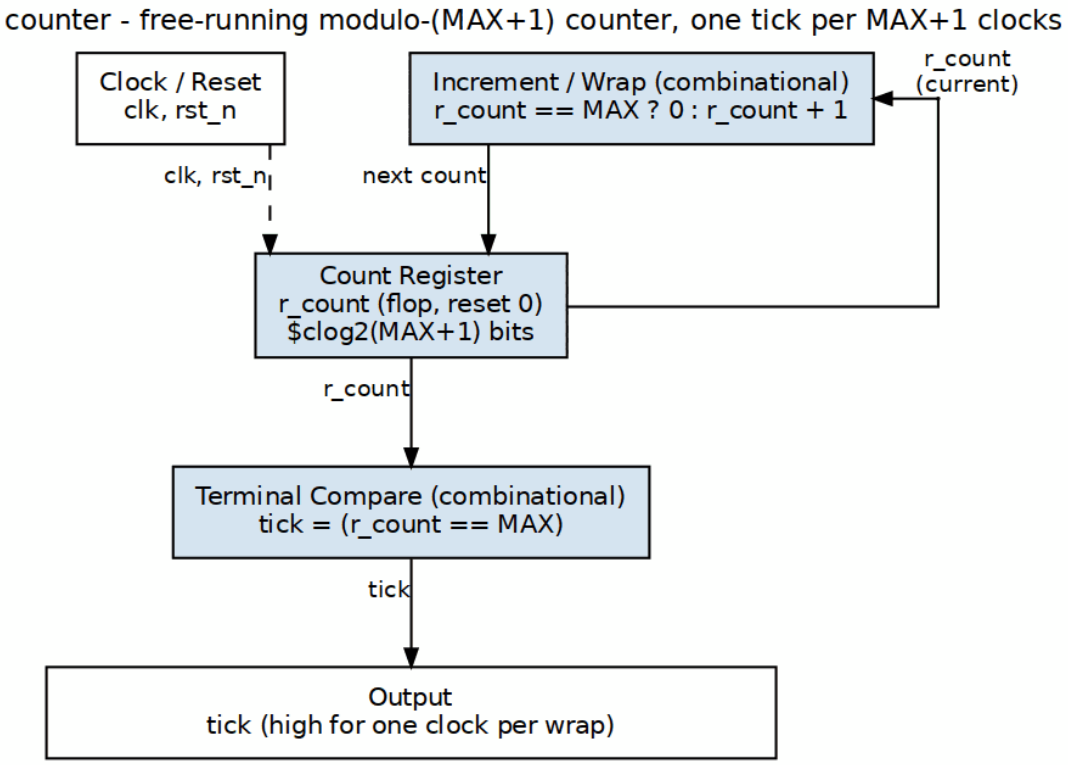
Port	Width	Description
`clk`	1	System clock input
`rst_n`	1	Active-low asynchronous reset

20.4.2 Outputs

Port	Width	Description
`tick`	1	Pulse output when counter reaches MAX

20.5 Functional Description

20.5.1 Block diagram



counter block diagram

The register `r_count` counts up by one every clock and wraps to 0 after it reaches `MAX`, so it repeats every `MAX+1` clocks. Its current value climbs the right edge back into the increment logic. A terminal compare raises `tick` for the one clock in each cycle where `r_count == MAX`.

20.5.2 Counter Register

```
logic [$clog2(MAX+1)-1:0] r_count;
```

- **Width**: Automatically sized to accommodate MAX value
- **Reset Value**: `'0` (zero)
- **Operation**: Increments every clock cycle until reaching MAX

20.5.3 State Machine Behavior

Behaviorally, it's a tiny state machine:

1. **Reset State**: `r_count = 0`, `tick = 0`

2. **Counting State**: `r_count` increments each cycle
3. **Maximum State**: `r_count = MAX`, `tick = 1` for one cycle
4. **Wrap State**: `r_count` returns to 0, cycle repeats

20.5.4 Main Counter Logic

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_count <= '0;
    end else begin
        if (r_count == MAX[$clog2(MAX+1)-1:0]) begin
            r_count <= '0;
        end else begin
            r_count <= r_count + 1'b1;
        end
    end
end
```

20.5.5 Tick Generation

```
assign tick = (r_count == MAX[$clog2(MAX+1)-1:0]);
```

20.5.6 Key Features

- **Automatic Width Sizing**: Counter width adapts to MAX parameter
- **Wrap-around Operation**: Automatically returns to zero after MAX
- **Single Cycle Tick**: Tick pulse lasts exactly one clock cycle
- **Asynchronous Reset**: Immediate reset capability

20.6 Timing Characteristics

20.6.1 Period Calculation

- **Total Period**: `MAX + 1` clock cycles
- **Tick Frequency**: `f_clk / (MAX + 1)`
- **Duty Cycle**: `1 / (MAX + 1)`

20.6.2 Examples

MAX Value	Counter Width	Period (cycles)	Tick Frequency
7	3 bits	8	f_clk/8

MAX Value	Counter Width	Period (cycles)	Tick Frequency
15	4 bits	16	f_clk/16
99	7 bits	100	f_clk/100
999	10 bits	1000	f_clk/1000

20.6.3 Resource Usage

- **LUTs**: Approximately `$clog2(MAX+1)` LUTs for the counter
- **FFs**: `$clog2(MAX+1)` flip-flops for the counter register
- **Logic Levels**: Single level for tick generation

20.6.4 Maximum Frequency

- **Maximum Frequency**: Limited by counter increment logic
- **Reset Recovery**: One clock cycle after reset deassertion
- **Propagation Delay**: Tick output has minimal combinational delay

20.6.5 Power Optimization

- **Clock Gating**: Consider gating when tick output not needed
- **Reset Strategy**: Asynchronous reset reduces reset tree loading
- **Parameter Selection**: Smaller MAX values reduce power consumption

20.7 Usage Examples

20.7.1 1. Basic Timer (1ms tick at 100MHz)

```
counter #(
    .MAX(99999)  // 100,000 cycles = 1ms at 100MHz
) timer_1ms (
    .clk(clk_100mhz),
    .rst_n(rst_n),
    .tick(tick_1ms)
);
```

20.7.2 2. Heartbeat Generator (1Hz at 50MHz)

```
counter #(
    .MAX(49999999)  // 50M cycles = 1 second at 50MHz
) heartbeat (
    .clk(clk_50mhz),
    .rst_n(rst_n),
```

```
    .tick(heartbeat_1hz)
);
```

20.7.3 3. Simple Divider (Divide by 8)

```
counter #(
    .MAX(7)  // 8 cycles total
) div8 (
    .clk(clk_in),
    .rst_n(rst_n),
    .tick(clk_div8)
);
```

1. **Periodic Timers**: Generating regular time intervals
2. **Clock Division**: Creating slower clock enables
3. **Timeout Generation**: Watchdog timers and timeouts
4. **Sampling Control**: Periodic data sampling triggers
5. **LED Blinking**: Visual indicators and status displays
6. **Protocol Timing**: Communication protocol timeouts

20.8 Testing

20.8.1 Test Scenarios

1. **Basic Counting**: Verify counting from 0 to MAX
2. **Wrap Behavior**: Confirm return to zero after MAX
3. **Tick Generation**: Verify single-cycle tick pulse
4. **Reset Functionality**: Test asynchronous reset operation
5. **Parameter Sweep**: Test various MAX values

20.8.2 Coverage Points

- Counter reaches MAX value
- Counter wraps from MAX to 0
- Tick pulse generation timing
- Reset during counting operation

20.9 Design Notes

20.9.1 Parameter Constraints

```
initial begin
    assert (MAX > 0) else $error("MAX must be positive");
    assert (MAX < 2**31) else $warning("Large MAX may impact
synthesis");
end
```

20.9.2 Optimization Hints

- **Binary MAX Values**: Powers of 2 minus 1 optimize better (7, 15, 31, etc.)
- **Tool Directives**: Consider synthesis attributes for timing-critical paths
- **Reset Inference**: Ensure reset is properly inferred as asynchronous

20.9.3 Limitations

1. **No Enable Control**: Always counting when not in reset
2. **Up-Counter Only**: Cannot count down or bidirectionally
3. **Fixed Increment**: Always increments by 1
4. **No Load Capability**: Cannot load arbitrary values
5. **Single Output**: Only provides tick signal, not count value

Know these going in. If you need any of them, you want `counter_load_clear` — don't bolt an enable onto this one and call it done.

20.10 Related Modules

- **counter_load_clear** - Adds load and clear functionality
- **counter_bin** - Specialized for FIFO applications
- **counter_freq_invariant** - Frequency-independent timing
- **clock_pulse** - Similar functionality with different interface

20.11 References

- No peer paper — a free-running modulo-(MAX+1) counter with terminal-count tick is a standard technique with no defensible single origin (marked "standard technique" in the references index).

- Roy, S. *Advanced Digital System Design*, Springer, ch. 4 “Basic Sequential Circuits”, §4.6 “Synchronous Counter Design” — textbook treatment (secondary source).
- Indexed in `rtl/common/references/README.md`

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21 Ring Counter

21.1 Overview

The `counter_ring` module is the classic ring counter: a single '1' bit circulating through a shift register. The result is a one-hot encoded sequence — exactly what you want for sequential enable signals, walking LED patterns, or multi-phase operations where precisely one stage may be active at a time.

21.2 Module Interface

```
module counter_ring #(
    parameter int WIDTH = 4
) (
    input  wire             clk,
    input  wire             rst_n,
    input  wire             enable,
    output reg  [WIDTH-1:0] ring_out
);
```

21.3 Parameters

Parameter	Type	Default	Description
`WIDTH`	`int`	`4`	Number of stages in the ring counter

`WIDTH` must be `>= 2`. At WIDTH=1 the rotate expression `{ring_out[0], ring_out[WIDTH-1:1]}` becomes the illegal reversed part-select `ring_out[0:1]` and the module does not elaborate. The counter generates WIDTH unique states, with a single '1' bit rotating through WIDTH positions.

21.4 Ports

21.4.1 Inputs

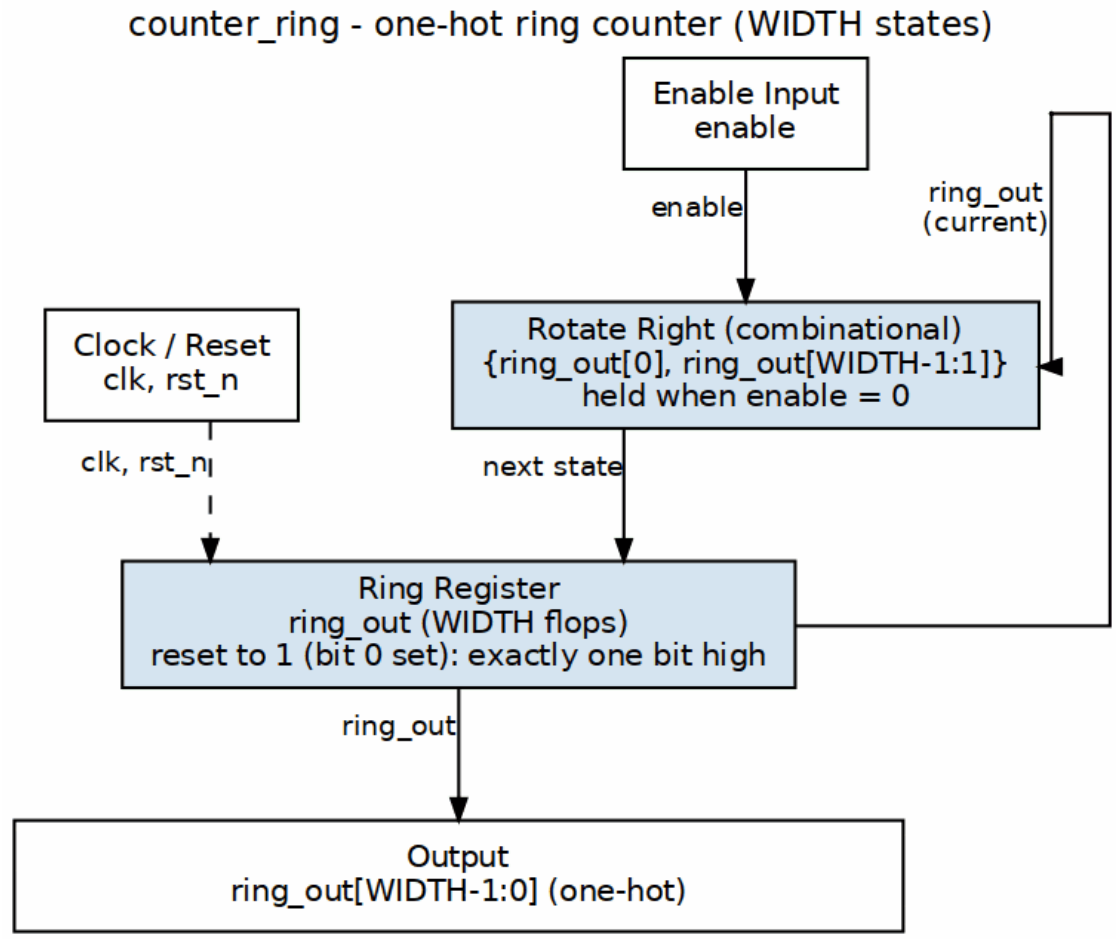
Port	Width	Description
`clk`	1	System clock input
`rst_n`	1	Active-low asynchronous reset
`enable`	1	Counter enable control

21.4.2 Outputs

Port	Width	Description
`ring_out`	WIDTH	Ring counter output (one-hot encoded)

21.5 Functional Description

21.5.1 Block diagram



counter_ring block diagram

A single 1 rotates one place right on each enabled clock. Reset sets bit 0, so exactly one bit is high from the first cycle. The register feeds back up the right edge into the rotate logic, and `ring_out` leaves as a one-hot vector cycling through `WIDTH` states.

21.5.2 Ring Counter Principle

A ring counter is just a circular shift register: - **Initialization**: Single '1' bit in LSB position - **Operation**: The '1' bit rotates right each clock cycle - **Feedback**: LSB connects back to MSB (no inversion) - **States**: WIDTH unique one-hot states

21.5.3 Mathematical Representation

For each clock cycle when enabled (`ring_out <= {ring_out[0], ring_out[WIDTH-1:1]}`):

```
ring_out[i] = ring_out[i+1]  for i = 0 to WIDTH-2
ring_out[WIDTH-1] = ring_out[0]
```

From reset `0001` this gives `0001 -> 1000 -> 0100 -> 0010 -> 0001`, matching the state tables below.

21.5.4 Core Logic

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        ring_out <= 'b1;  // Initialize with LSB set
    end else if (enable) begin
        // Right rotate operation
        ring_out <= {ring_out[0], ring_out[WIDTH-1:1]};
    end
end
```

21.5.5 Operation Breakdown

1. **Reset State**: `ring_out = 'b1` (only LSB set)
2. **Rotation**: `{ring_out[0], ring_out[WIDTH-1:1]}`
    - LSB becomes new MSB
    - All other bits shift right (toward the LSB)
3. **Enable Control**: Only advances when `enable` is high
4. **Circular**: After WIDTH cycles, returns to initial state

21.5.6 State Sequences

21.5.7 4-bit Ring Counter (WIDTH=4)

Step	ring_out	Binary	One-Hot Position
0	0001	Reset state	Position 0 active
1	1000	After 1 clock	Position 3

Step	ring_out	Binary	One-Hot Position
			active
2	0100	After 2 clocks	Position 2 active
3	0010	After 3 clocks	Position 1 active
4	0001	After 4 clocks	Back to start

21.5.8 3-bit Ring Counter (WIDTH=3)

Step	ring_out	Active Stage
0	001	Stage 0
1	100	Stage 2
2	010	Stage 1
3	001	Stage 0 (repeat)

21.5.9 8-bit Ring Counter Example

```
Step 0: 00000001 (Stage 0)
Step 1: 10000000 (Stage 7)
Step 2: 01000000 (Stage 6)
Step 3: 00100000 (Stage 5)
Step 4: 00010000 (Stage 4)
Step 5: 00001000 (Stage 3)
Step 6: 00000100 (Stage 2)
Step 7: 00000010 (Stage 1)
Step 8: 00000001 (Stage 0) - Cycle repeats
```

21.5.10 Properties and Characteristics

21.5.11 One-Hot Encoding

Ring counters hand you one-hot encoding for free: - **Exactly One Bit Set**: Always exactly one bit is ‘1’ - **Unique States**: Each state is distinctly different - **Easy Decode**: No additional decoding logic needed - **Glitch-Free**: Clean transitions between states

21.5.12 Self-Correcting Behavior

An honest word about self-correction — ring counters mostly don’t: - **All Zeros**: Will remain stuck (not self-correcting) - **Multiple Ones**: Will maintain multiple bits indefinitely - **Single One**: Will operate correctly

If an upset can land you in an illegal state, see the self-correcting variant under Advanced Variants.

21.5.13 Comparison with Other Counters

Counter Type	States	Decode Logic	One-Hot	Power
Binary	2^N	Complex	No	Low
Gray	2^N	Moderate	No	Low
Johnson	2×N	Moderate	No	Low
Ring	N	None	Yes	Medium

21.5.14 Advanced Variants

21.5.15 1. Self-Correcting Ring Counter

```
// Add error correction for stuck states
logic error_detected;
logic [WIDTH-1:0] corrected_ring;

assign error_detected = (ring_out == 'b0) || ($countones(ring_out) !=
1);

always_comb begin
    if (error_detected)
        corrected_ring = 'b1;  // Force to valid state
    else
        corrected_ring = ring_out;
end
```

21.5.16 2. Bidirectional Ring Counter

```
logic direction;  // 0 = right, 1 = left

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        ring_out <= 'b1;
    end else if (enable) begin
        if (direction)
            // Left rotate
            ring_out <= {ring_out[WIDTH-2:0], ring_out[WIDTH-1]};
        else
            // Right rotate
            ring_out <= {ring_out[0], ring_out[WIDTH-1:1]};
    end
end
```

21.5.17 3. Programmable Ring Counter

```
logic [WIDTH-1:0] init_pattern;
logic load_pattern;
```

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        ring_out <= 'b1;
    end else if (load_pattern) begin
        ring_out <= init_pattern;
    end else if (enable) begin
        ring_out <= {ring_out[0], ring_out[WIDTH-1:1]};
    end
end
```

21.6 Timing Characteristics

This module is **sequential**: it contains clocked logic (via `always_ff` or the repository's `ALWAYS_FF_RST` macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

21.6.1 Maximum Frequency

- **Typical**: 400-600 MHz in modern FPGAs
- **Limitation**: Shift register timing
- **Advantage**: Very simple logic path

21.6.2 Power Consumption

- **Dynamic**: Proportional to switching frequency
- **Optimization**: Higher switching than binary counters
- **Trade-off**: Simplicity vs. power efficiency

21.7 Usage Examples

21.7.1 1. Sequential Enable Generator

```
counter_ring #(
    .WIDTH(4)
```

```
) seq_enable (
    .clk(clk),
    .rst_n(rst_n),
    .enable(sequence_active),
    .ring_out(stage_enables)
);

// Use each bit as an enable for different stages
assign stage0_enable = stage_enables[0];
assign stage1_enable = stage_enables[1];
assign stage2_enable = stage_enables[2];
assign stage3_enable = stage_enables[3];

// Example: Pipeline stage control
always_ff @(posedge clk) begin
    if (stage0_enable) stage0_data <= input_data;
    if (stage1_enable) stage1_data <= process_stage0(stage0_data);
    if (stage2_enable) stage2_data <= process_stage1(stage1_data);
    if (stage3_enable) output_data <= process_stage2(stage2_data);
end
```

21.7.22. LED Chaser Display

```
// Create walking LED pattern
counter_ring #(
    .WIDTH(8)
) led_chaser (
    .clk(slow_clk),          // ~2Hz for visible effect
    .rst_n(rst_n),
    .enable(chase_enabled),
    .ring_out(led_pattern)
);

// Connect directly to LEDs
assign leds[7:0] = led_pattern;

// Optional: Bidirectional chaser
logic direction;
logic [7:0] led_display;

always_comb begin
    if (direction)
        led_display = led_pattern;
    else
        led_display = {led_pattern[0], led_pattern[7:1]}; // Reverse
end
```

21.7.33. Memory Bank Sequencer

```
counter_ring #(
    .WIDTH(4)
) bank_sequencer (
    .clk(clk),
    .rst_n(rst_n),
    .enable(memory_active),
    .ring_out(bank_select)
);

// Memory bank control
logic [3:0] bank_cs_n;
assign bank_cs_n = ~bank_select;  // Active low chip selects

// Example: Round-robin memory access
always_comb begin
    mem_addr = base_addr;
    mem_data_out = 32'h0;

    case (bank_select)
        4'b0001: begin
            mem_addr = bank0_addr;
            mem_data_out = bank0_data;
        end
        4'b0010: begin
            mem_addr = bank1_addr;
            mem_data_out = bank1_data;
        end
        4'b0100: begin
            mem_addr = bank2_addr;
            mem_data_out = bank2_data;
        end
        4'b1000: begin
            mem_addr = bank3_addr;
            mem_data_out = bank3_data;
        end
    endcase
end
```

21.7.44. State Machine Controller

```
// Simple state machine using ring counter
counter_ring #(
    .WIDTH(5)
) state_machine (
    .clk(clk),
    .rst_n(rst_n),
    .enable(state_advance),
```

```
    .ring_out(current_state)
);

// Decode states
assign state_idle    = current_state[0];
assign state_request = current_state[1];
assign state_wait    = current_state[2];
assign state_active  = current_state[3];
assign state_done    = current_state[4];

// State machine logic
always_ff @(posedge clk) begin
    case (1'b1)  // One-hot case statement
        state_idle:    state_advance <= start_request;
        state_request: state_advance <= request_granted;
        state_wait:    state_advance <= ready_signal;
        state_active:  state_advance <= operation_complete;
        state_done:    state_advance <= 1'b1;  // Always advance from
done
        default:       state_advance <= 1'b0;
    endcase
end
```

21.7.55. Clock Domain Multiplexer

```
// Sequentially select different clock domains
counter_ring #(
    .WIDTH(3)
) clk_mux_ctrl (
    .clk(control_clk),
    .rst_n(rst_n),
    .enable(mux_advance),
    .ring_out(clk_select)
);

// Clock multiplexer (use proper clock muxing in real design)
logic selected_clk;
always_comb begin
    case (clk_select)
        3'b001: selected_clk = clk_domain_0;
        3'b010: selected_clk = clk_domain_1;
        3'b100: selected_clk = clk_domain_2;
        default: selected_clk = 1'b0;
    endcase
end
```

21.8 Testing

21.8.1 Test Scenarios

1. **Reset Behavior**: Verify initialization to single '1' bit
2. **Rotation Sequence**: Check complete WIDTH-step cycle
3. **Enable Control**: Test hold behavior when disabled
4. **One-Hot Property**: Ensure exactly one bit set always
5. **Cycle Completion**: Verify return to initial state

21.8.2 Coverage Model

```
covergroup ring_cg @(posedge clk);
    cp_ring_state: coverpoint ring_out {
        bins valid_states[] = {1, 2, 4, 8, 16, 32, 64, 128}; // For
WIDTH=8
        bins invalid_states[] = default;
    }

    cp_enable: coverpoint enable {
        bins enabled = {1};
        bins disabled = {0};
    }

    cp_one_hot: coverpoint $countones(ring_out) {
        bins exactly_one = {1};
        bins zero = {0};
        bins multiple = {[2:WIDTH]};
    }

    // Verify complete cycle
    cp_transitions: coverpoint ring_out {
        bins cycle[] = (1 => 8), (8 => 4), (4 => 2), (2 => 1); // For
WIDTH=4
    }
endcovergroup
```

21.8.3 Assertions

```
// Exactly one bit should be set (except during reset)
property one_hot_property;
    @(posedge clk) disable iff (!rst_n)
    $countones(ring_out) == 1;
endproperty
```

```
// After reset, should start with LSB set
property reset_state;
    @(posedge clk)
    $rose(rst_n) |-> ring_out == 1;
endproperty

// When enabled, should rotate right
property rotation_property;
    @(posedge clk) disable iff (!rst_n)
    enable |=> ring_out == {$past(ring_out[0]), $past(ring_out[WIDTH-
1:1])};
endproperty

assert property (one_hot_property);
assert property (reset_state);
assert property (rotation_property);
```

21.9 Design Notes

21.9.1 Resource Utilization

- **Flip-Flops**: WIDTH registers
- **LUTs**: Minimal - just shift logic
- **Routing**: Simple interconnections

21.9.2 Optimization Tips

```
// Use SRL (Shift Register LUT) inference when appropriate
(* SRL_STYLE = "register" *) reg [WIDTH-1:0] ring_out;

// For very wide counters, consider using dedicated shift register
primitives
```

21.9.3 Timing Considerations

- **Clock Skew**: Ensure clean clock distribution
- **Setup/Hold**: Standard flip-flop requirements
- **Routing Delay**: Minimal due to simple structure

21.9.4 Common Pitfalls

1. Invalid Initial States

```
// Problem: Multiple bits set
ring_out <= 4'b1001;  // Invalid - two bits set

// Solution: Use proper initialization
ring_out <= 4'b0001;  // Valid - exactly one bit
```

2. Stuck at Zero

```
// Problem: All zeros state
if (ring_out == 'b0) begin
    ring_out <= 'b1;  // Force recovery
end
```

3. Width Mismatch

```
// Problem: Initialization doesn't match width
parameter WIDTH = 8;
ring_out <= 4'b0001;  // Wrong - should be 8'b00000001

// Solution: Use proper width
ring_out <= 'b1;      // Correct - sized automatically
```

21.10 Related Modules

- **counter_johnson** - Twisted-ring counter with 2×WIDTH states
- **counter_bingray** - Binary/Gray counter for different applications
- Standard shift register implementations
- One-hot state machines

21.11 References

- No peer paper — the one-hot ring (shift-register) counter is a standard technique with no defensible single origin (marked “standard technique” in the references index).
- Roy, S. *Advanced Digital System Design*, Springer, ch. 4 “Basic Sequential Circuits”, §4.9 “Shift Register Counters” (ring counter) — textbook treatment (secondary source).
- Indexed in `rtl/common/references/README.md`

21.12 Navigation

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22 Count Leading Zeros

22.1 Overview

The `count_leading_zeros` module counts how many consecutive zero bits sit at the beginning (MSB side) of a data word. This is fundamental computer arithmetic — you'll find it inside floating-point normalization, priority encoders, and half the bit-manipulation tricks in the book.

The scan starts at `data[WIDTH-1]` and proceeds downward, stopping at the first set bit:

```
data = 32'h8000_0000 -> clz =  0   (MSB is already set)
data = 32'h0000_0001 -> clz = 31   (31 zeros above the LSB)
data = 32'h0000_0000 -> clz = WIDTH
```

For the complementary count from the LSB upward, use **count_trailing_zeros**.

History - do not reintroduce input reversal. Earlier revisions of this module scanned from bit 0 upward, which computes *trailing* zeros, not leading zeros, despite the module name. Callers compensated by bit-reversing their input before connecting it. Both the module and every caller have been corrected: the module now performs a true MSB-down scan, and callers pass their operand straight in. Any reversal generate block or `w_*_reversed` wire feeding this module is stale and must not be re-added. If you actually want the LSB-up count, instantiate `count_trailing_zeros` instead of reversing the input.

22.2 Module Interface

```
module count_leading_zeros #(
    parameter int WIDTH = 32
) (
    input  logic [      WIDTH-1:0] data,
    output logic [$clog2(WIDTH):0] clz
);
```

22.3 Parameters

Parameter	Type	Default	Description
WIDTH	int	32	Bit width of the input data

`WIDTH` can be any positive integer ≥ 1; 8, 16, 32, and 64 are the common values for standard data widths. It determines the output width and algorithm complexity, and it's the only parameter.

22.4 Ports

22.4.1 Inputs

Port	Width	Description
`data`	WIDTH	Input data word to analyze

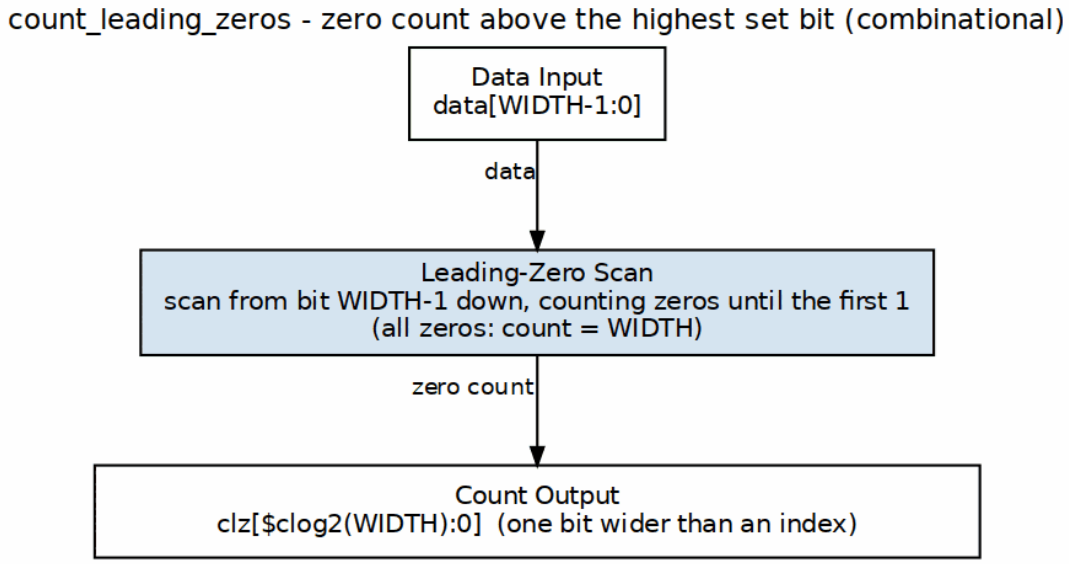
22.4.2 Outputs

Port	Width	Description
`clz`	`$clog2(WIDTH)+1`	Count of leading zeros

The output width `$clog2(WIDTH)+1` ensures the port can represent all possible counts: - **Range**: 0 to WIDTH (inclusive) - **Example**: For WIDTH=32, output is 6 bits (0-32 requires 6 bits) - **Special Case**: All zeros input produces CLZ = WIDTH

22.5 Functional Description

22.5.1 Block diagram



count_leading_zeros block diagram

`data` enters at the top and is scanned from bit `WIDTH-1` down, counting zeros until the first 1. This is a count, not a pick, so it is a plain block. The `clz` output is one bit wider than an index so that an all-zero input can report `WIDTH`.

22.5.2 Function-Based Approach

```
function automatic [$clog2(WIDTH):0] clz_func;
    input [WIDTH-1:0] input_data;
    logic found;
    begin
        clz_func = 0;
        found = 1'b0;
        // Scan from the MSB downward.
        for (int i = WIDTH - 1; i >= 0; i--) begin
            if (!input_data[i] && !found) begin
                clz_func += 1;
            end else begin
                found = 1'b1;  // Stop counting when first '1' found
            end
        end
    end
endfunction
```

22.5.3 Bit Scanning Process

The algorithm scans from MSB to LSB: 1. **Initialize**: `clz_func = 0`, `found = 0` 2. **Scan Loop**: For each bit position i from WIDTH-1 down to 0: - If `data[i] == 0` AND no '1' found yet: increment count - If `data[i] == 1`: set found flag, stop counting 3. **Result**: Final count is the number of zeros above the highest set bit

22.5.4 Why MSB-to-LSB Scanning?

"Leading" refers to the bits that lead the word when it is written out, which is the MSB side. The loop therefore starts at `data[WIDTH-1]` and walks down: - **Bit Order**: `data[0]` is LSB, `data[WIDTH-1]` is MSB - **Termination**: The `found` flag latches at the highest set bit, so all zeros below it are ignored - only the contiguous run above it is counted - **Count Logic**: `clz = (WIDTH-1) - position_of_highest_set_one`, and `clz = WIDTH` when no bit is set

Because the scan direction already matches the definition, the input must be fed in directly. Reversing `data` before connecting it turns this module back into a trailing zero counter - see the history note above.

22.5.5 Advanced Variants

22.5.6 1. Pipeline Version for High Speed

```
module count_leading_zeros_pipelined #(
    parameter int WIDTH = 32,
    parameter int STAGES = 2
) (
    input  logic                    clk,
    input  logic                    rst_n,
    input  logic [WIDTH-1:0]        data,
    output logic [$clog2(WIDTH):0] clz
```

```
);

// Stage 1: Partial counting
logic [WIDTH-1:0] data_s1;
logic [$clog2(WIDTH):0] partial_clz_s1;

always_ff @(posedge clk) begin
    data_s1 <= data;
    // Count leading zeros in upper half
    partial_clz_s1 <= clz_upper_half(data);
end

// Stage 2: Complete counting
always_ff @(posedge clk) begin
    if (partial_clz_s1 == WIDTH/2) begin
        // All upper bits are zero, count in lower half
        clz <= WIDTH/2 + clz_lower_half(data_s1);
    end else begin
        // Found '1' in upper half
        clz <= partial_clz_s1;
    end
end

endmodule
```

22.5.7 2. Hierarchical Implementation

```
// Divide and conquer approach for large widths
module count_leading_zeros_hierarchical #(
    parameter int WIDTH = 64
) (
    input  logic [WIDTH-1:0]       data,
    output logic [$clog2(WIDTH):0] clz
);

localparam int HALF_WIDTH = WIDTH/2;

logic [$clog2(HALF_WIDTH):0] clz_upper, clz_lower;
logic upper_all_zeros;

// Count in upper half
count_leading_zeros #(.WIDTH(HALF_WIDTH)) upper_clz (
    .data(data[WIDTH-1:HALF_WIDTH]),
    .clz(clz_upper)
);

// Count in lower half
```

```
count_leading_zeros #(.WIDTH(HALF_WIDTH)) lower_clz (
    .data(data[HALF_WIDTH-1:0]),
    .clz(clz_lower)
);

assign upper_all_zeros = (clz_upper == HALF_WIDTH);

// Combine results
assign clz = upper_all_zeros ? (HALF_WIDTH + clz_lower) : clz_upper;

endmodule
```

22.5.8 3. LUT-Based Implementation (Small Widths)

```
// Optimized for small widths using a wildcard case statement.
// casez is required so that '?' is treated as a don't-care.
module count_leading_zeros_lut #(
    parameter int WIDTH = 8
) (
    input  logic [WIDTH-1:0]       data,
    output logic [$clog2(WIDTH):0] clz
);

always_comb begin
    casez (data)
        8'b00000000: clz = 8;
        8'b00000001: clz = 7;
        8'b0000001?: clz = 6;
        8'b000001??: clz = 5;
        8'b00001???: clz = 4;
        8'b0001????: clz = 3;
        8'b001?????: clz = 2;
        8'b01??????: clz = 1;
        8'b1???????: clz = 0;
        default:     clz = 0;
    endcase
end

endmodule
```

22.6 Usage Examples

22.6.1 1. Floating-Point Normalization

```
// IEEE 754 floating-point normalization
logic [31:0] mantissa_raw;
logic [5:0] shift_amount;
logic [31:0] normalized_mantissa;
```

```
count_leading_zeros #(.WIDTH(32)) norm_clz (
    .data(mantissa_raw),
    .clz(shift_amount)
);

// Normalize mantissa by shifting left
assign normalized_mantissa = mantissa_raw << shift_amount;

// Adjust exponent accordingly
logic [7:0] adjusted_exponent;
assign adjusted_exponent = raw_exponent - shift_amount;
```

22.6.22. Priority Encoder

```
// Find highest priority request
logic [15:0] request_vector;
logic [4:0]  leading_zeros;
logic [3:0]  highest_priority;
logic any_request;

count_leading_zeros #(.WIDTH(16)) priority_enc (
    .data(request_vector),
    .clz(leading_zeros)
);

assign any_request = (leading_zeros != 16);
assign highest_priority = any_request ? (15 - leading_zeros) : 4'b0;

// Example: request_vector = 16'b0000_0100_1000_0000
// Highest set bit is bit 10
// CLZ = 5 (five leading zeros above bit 10)
// highest_priority = 15 - 5 = 10
```

22.6.33. Bit Width Calculation

```
// Determine minimum bits needed to represent a number
logic [31:0] number;
logic [5:0] min_bits_needed;

count_leading_zeros #(.WIDTH(32)) bit_width_calc (
    .data(number),
    .clz(leading_zeros)
);

assign min_bits_needed = (number == 0) ? 1 : (32 - leading_zeros);
```

```
// Examples:
// number = 0      → min_bits = 1
// number = 1      → min_bits = 1
// number = 255    → min_bits = 8
// number = 256    → min_bits = 9
// number = 65535  → min_bits = 16
```

22.6.44. Log2 Calculation

```
// Calculate floor(log2(x)) for x > 0
logic [31:0] input_value;
logic [4:0] log2_result;
logic valid;

count_leading_zeros #(.WIDTH(32)) log2_calc (
    .data(input_value),
    .clz(leading_zeros)
);

assign valid = (input_value != 0);
assign log2_result = valid ? (31 - leading_zeros) : 5'b0;

// Examples:
// input = 1    → log2 = 0
// input = 2    → log2 = 1
// input = 4    → log2 = 2
// input = 7    → log2 = 2 (floor)
// input = 8    → log2 = 3
```

22.6.55. Data Compression - Run Length Encoding

```
// Count leading zeros for compression
logic [63:0] data_block;
logic [6:0] zero_run_length;

count_leading_zeros #(.WIDTH(64)) rle_counter (
    .data(data_block),
    .clz(zero_run_length)
);

// Use in compression algorithm
if (zero_run_length > THRESHOLD) begin
    // Encode as zero run
    compressed_data = {RLE_CODE, zero_run_length};
end else begin
    // Encode as literal data
    compressed_data = {LITERAL_CODE, data_block};
end
```

1. **CPU/DSP Cores**: Instruction implementation (CLZ instruction)
2. **Floating-Point Units**: Mantissa normalization
3. **Network Processors**: Longest prefix matching
4. **Compression Engines**: Run-length encoding optimization
5. **Memory Controllers**: Priority arbitration
6. **Graphics Processors**: Texture coordinate processing
7. **Cryptographic Units**: Bit manipulation operations

22.6.6 Examples and Truth Tables

22.6.7 8-bit Examples (WIDTH=8)

The count is set by the **highest** set bit; bits below it never affect the result.

Input (data)	Binary	Highest set bit	Leading Zeros	CLZ Output
8'b00000000	00000000	None	8	8
8'b00000001	00000001	Bit 0	7	7
8'b00000010	00000010	Bit 1	6	6
8'b00000100	00000100	Bit 2	5	5
8'b00001000	00001000	Bit 3	4	4
8'b00010000	00010000	Bit 4	3	3
8'b00100000	00100000	Bit 5	2	2
8'b01000000	01000000	Bit 6	1	1
8'b10000000	10000000	Bit 7	0	0

Input (data)	Binary	Highest set bit	Leading Zeros	CLZ Output
8’b11111111	11111111	Bit 7	0	0
8’b10101010	10101010	Bit 7	0	0
8’b00010101	00010101	Bit 4	3	3
8’b00110000	00110000	Bit 5	2	2

22.6.8 32-bit Examples

Input	Hex	Leading Zeros	CLZ
32’h00000000	0x00000000	32	32
32’h00000001	0x00000001	31	31
32’h00000080	0x00000080	24	24
32’h00008000	0x00008000	16	16
32’h00800000	0x00800000	8	8
32’h80000000	0x80000000	0	0
32’hFFFFFFFF	0xFFFFFFFF	0	0

22.7 Timing Characteristics

22.7.1 Resource Utilization

WIDTH	LUTs (Typical)	Delay Levels	Max Frequency
8	15-20	3-4	500+ MHz
16	30-40	4-5	400+ MHz
32	60-80	5-6	300+ MHz
64	120-150	6-7	250+ MHz

22.7.2 Timing Characteristics

- **Combinational Delay**: O(log(WIDTH)) for tree implementations
- **Critical Path**: Through the priority encoding logic

- **Scalability**: Linear increase in logic for function implementation

22.8 Testing

22.8.1 Test Scenarios

1. **Boundary Cases**: All zeros, all ones, single bit patterns
2. **Random Patterns**: Comprehensive random testing
3. **Systematic Sweep**: Test all possible leading zero counts
4. **Corner Cases**: Maximum width values, alternating patterns

22.8.2 Coverage Model

```
covergroup clz_cg;
    cp_leading_zeros: coverpoint clz {
        bins zero_clz = {0};
        bins low_clz = {[1:WIDTH/4]};
        bins mid_clz = {[WIDTH/4+1:3*WIDTH/4]};
        bins high_clz = {[3*WIDTH/4+1:WIDTH-1]};
        bins all_zeros = {WIDTH};
    }

    cp_data_patterns: coverpoint data {
        bins all_zeros = {'0};
        bins all_ones = {'1};
        bins single_bit[] = {1, 2, 4, 8, 16, 32}; // For appropriate
WIDTH
        bins msb_set = {1 << (WIDTH-1)};
        bins mixed[] = {[1:2**(WIDTH-1)-1]};
    }
endcovergroup
```

22.8.3 Assertions

```
// CLZ should never exceed WIDTH
property clz_bounds;
    clz <= WIDTH;
endproperty

// All zeros should give CLZ = WIDTH
property all_zeros_case;
    (data == '0) |-> (clz == WIDTH);
endproperty

// MSB set should give CLZ = 0
```

```
property msb_set_case;
    data[WIDTH-1] |-> (clz == 0);
endproperty

// Relationship between CLZ and first '1' position
property clz_correctness;
    logic [WIDTH-1:0] shifted;
    assign shifted = data << clz;
    (data != '0) |-> shifted[WIDTH-1];
endproperty

assert property (clz_bounds);
assert property (all_zeros_case);
assert property (msb_set_case);
assert property (clz_correctness);
```

22.9 Design Notes

22.9.1 Tool-Specific Attributes

```
// Xilinx: Prevent SRL inference for better timing
(* SRL_STYLE = "register" *) logic [WIDTH-1:0] data_reg;

// Quartus: Use specific LUT implementation
// synthesis attribute LUT_SIZE of clz_func is 6;
```

22.9.2 Area vs. Speed Trade-offs

```
// For area optimization: Use iterative approach
// For speed optimization: Use tree-based or LUT approach
// For power optimization: Add enable signals and clock gating
```

22.10 Related Modules

- **count_trailing_zeros** - the LSB-up counterpart. Use CLZ when you care about the magnitude of a value (normalization, log2, bit width); use CTZ when you care about alignment or the lowest pending request.
- Population count (number of ‘1’ bits)
- Find first set (FFS) / Find last set (FLS)
- Priority encoders
- Barrel shifters

22.11 References

- Oklobdzija, V.G. “An Algorithmic and Novel Design of a Leading Zero Detector Circuit: Comparison with Logic Synthesis.” *IEEE Transactions on VLSI Systems*, 2(1), Mar. 1994. DOI: 10.1109/92.273153 — the hierarchical leading-zero-detector tree; this module is a behavioral priority scan, the synthesized baseline the paper compares against.
- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 12 “Floating Point Architectures”, Sec. 12.4 Leading Zero Counter (textbook treatment; secondary source).
- Indexed in `rtl/common/references/README.md` (local PDF: `LeadingZeroDetector__Oklobdzija1994.pdf`).

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23 count_trailing_zeros

23.1 Overview

The `count_trailing_zeros` module counts how many consecutive zero bits sit at the end (LSB side) of a data word. It's the mirror image of `count_leading_zeros`, and it's the natural primitive for alignment checks, lowest-set-bit extraction, and picking the least significant pending request out of a vector.

The scan starts at `data[0]` and walks upward, stopping at the first set bit:

```
data = 32'h0000_0001 -> ctz =  0   (bit 0 is already set)
data = 32'h8000_0000 -> ctz = 31   (31 zeros below the MSB)
data = 32'h0000_0000 -> ctz = WIDTH
```

For the complementary count from the MSB downward, use **count_leading_zeros**.

23.1.1 Choosing Between CLZ and CTZ

Question you are asking	Module
How big is this value? (normalization, log2, minimum bit width)	`count_leading_zeros`
How is this value aligned? (address alignment, burst size, power-of-two stride)	`count_trailing_zeros`
Which is the highest priority request set?	`count_leading_zeros`
Which is the lowest index request set?	`count_trailing_zeros`

Don't emulate one by bit-reversing the input to the other. Both modules exist; picking the right one costs no extra logic and keeps your intent visible in the netlist.

23.1.2 Module Declaration

```
module count_trailing_zeros #(
    parameter int WIDTH = 32
) (
    input  logic [      WIDTH-1:0] data,
    output logic [$clog2(WIDTH):0] ctz
);
```

23.2 Parameters

Parameter	Default	Description
`WIDTH`	32	Bit width of the input data

WIDTH can be any positive integer >= 1; 8, 16, 32, and 64 are the common values for standard data widths. It sets the output width and the algorithm complexity, and it's the only parameter.

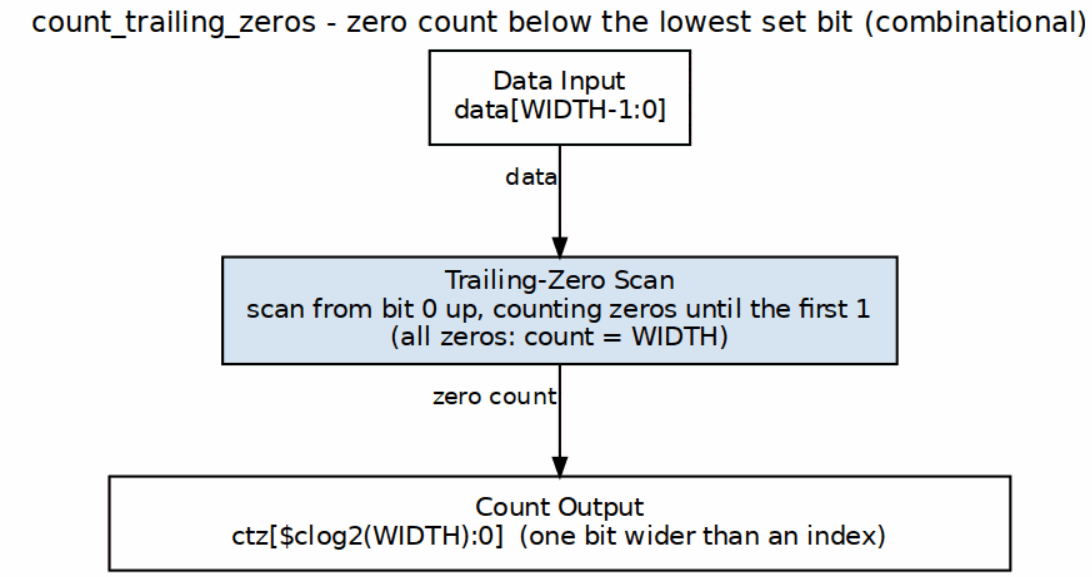
23.3 Ports

Port	Direction	Width	Description
data	Input	WIDTH	Input data word to analyze
ctz	Output	$clog2(WIDTH)+1	Count of trailing zeros

The output width $clog2(WIDTH)+1 lets the port represent every possible count: - **Range**: 0 to WIDTH (inclusive) - **Example**: For WIDTH=32, output is 6 bits (0-32 requires 6 bits) - **Special Case**: All zeros input produces CTZ = WIDTH

23.4 Functional Description

23.4.1 Block diagram



count_trailing_zeros block diagram

data enters at the top and is scanned from bit 0 up, counting zeros until the first 1. This is a count, not a pick, so it is a plain block. The ctz output is one bit wider than an index so that an all-zero input can report WIDTH.

23.4.2 Function-Based Approach

```
function automatic [$clog2(WIDTH):0] ctz_func;
    input [WIDTH-1:0] input_data;
    logic found;
    begin
        ctz_func = 0;
        found = 1'b0;
        // Scan from the LSB upward.
        for (int i = 0; i < WIDTH; i++) begin
            if (!input_data[i] && !found) begin
                ctz_func += 1;
            end else begin
                found = 1'b1;  // Stop counting when first '1' found
            end
        end
    end
endfunction
```

23.4.3 Bit Scanning Process

The algorithm scans from LSB to MSB: 1. **Initialize**: `ctz_func = 0`, `found = 0` 2. **Scan Loop**: For each bit position i from 0 to WIDTH-1: - If `data[i] == 0` AND no '1' found yet: increment count - If `data[i] == 1`: set found flag, stop counting 3. **Result**: Final count is the number of zeros below the lowest set bit

23.4.4 Why LSB-to-MSB Scanning?

"Trailing" means the bits that trail the word when it's written out — the LSB side. So the loop starts at `data[0]` and walks up: - **Bit Order**: `data[0]` is LSB, `data[WIDTH-1]` is MSB - **Termination**: The `found` flag latches at the lowest set bit, so every bit above it is ignored - only the contiguous run below it is counted - **Count Logic**: `ctz = position_of_lowest_set_one`, and `ctz = WIDTH` when no bit is set

23.4.5 8-bit Examples (WIDTH=8)

The count is set by the **lowest** set bit; bits above it never affect the result.

Input (data)	Binary	Lowest set bit	Trailing Zeros	CTZ Output
8'b00000000	00000000	None	8	8
8'b00000001	00000001	Bit 0	0	0
8'b00000010	00000010	Bit 1	1	1
8'b00000100	0000010	Bit 2	2	2

Input (data)	Binary	Lowest set bit	Trailing Zeros	CTZ Output
	0			
8’b00001000	00001000	Bit 3	3	3
8’b00010000	00010000	Bit 4	4	4
8’b00100000	00100000	Bit 5	5	5
8’b01000000	01000000	Bit 6	6	6
8’b10000000	10000000	Bit 7	7	7
8’b11111111	11111111	Bit 0	0	0
8’b10101010	10101010	Bit 1	1	1
8’b00110000	00110000	Bit 4	4	4

23.4.6 32-bit Examples

Input	Hex	Trailing Zeros	CTZ
32’h00000000	0x00000000	32	32
32’h00000001	0x00000001	0	0
32’h00000080	0x00000080	7	7
32’h00008000	0x00008000	15	15
32’h00800000	0x00800000	23	23
32’h80000000	0x80000000	31	31
32’hFFFFFFFF	0xFFFFFFFF	0	0

23.4.7 Relationship to CLZ

For a single-bit input `data == (1 << k)`, the two modules are complementary:

```
ctz = k
clz = WIDTH - 1 - k
ctz + clz = WIDTH - 1
```

That identity holds **only** for one-hot inputs. For general data the two counts are independent, because CLZ is decided by the highest set bit and CTZ by the lowest. For example `8'b00110000` gives `clz = 2` and `ctz = 4`, which sum to 6, not 7. Worth remembering before you write a check that assumes the identity holds for arbitrary data.

23.5 Timing Characteristics

23.5.1 Resource Utilization

WIDTH	LUTs (Typical)	Delay Levels	Max Frequency
8	15-20	3-4	500+ MHz
16	30-40	4-5	400+ MHz
32	60-80	5-6	300+ MHz
64	120-150	6-7	250+ MHz

23.5.2 Timing Characteristics

- **Combinational Delay**: O(log(WIDTH)) for tree implementations
- **Critical Path**: Through the priority encoding logic
- **Scalability**: Linear increase in logic for the function implementation

23.6 Usage Examples

23.6.1 Address Alignment Detection

```
// Determine the largest power-of-two boundary an address is aligned
to
logic [31:0] address;
logic [5:0]  alignment_log2;

count_trailing_zeros #(.WIDTH(32)) align_ctz (
    .data(address),
    .ctz(alignment_log2)
);

// alignment_log2 = 0  -> byte aligned only
// alignment_log2 = 2  -> 4-byte aligned
// alignment_log2 = 6  -> 64-byte aligned (cache line)
// alignment_log2 = 32 -> address is zero, aligned to everything
```

23.6.2 Maximum Burst Size for a DMA Transfer

```
// A burst may not cross its natural alignment boundary, so the
alignment of the
```

```
// start address caps the burst size.
logic [31:0] start_addr;
logic [5:0]  addr_align;
logic [5:0]  max_burst_log2;

count_trailing_zeros #(.WIDTH(32)) burst_ctz (
    .data(start_addr),
    .ctz(addr_align)
);

// Clamp against the protocol maximum (e.g. 4KB = 12)
assign max_burst_log2 = (addr_align > 6'd12) ? 6'd12 : addr_align;
```

23.6.3 Lowest-Index Request Arbitration

```
// Fixed priority arbiter favouring the lowest requester index
logic [15:0] request_vector;
logic [4:0]  trailing_zeros;
logic [3:0]  granted_index;
logic        any_request;

count_trailing_zeros #(.WIDTH(16)) req_ctz (
    .data(request_vector),
    .ctz(trailing_zeros)
);

assign any_request  = (trailing_zeros != 16);
assign granted_index = any_request ? trailing_zeros[3:0] : 4'b0;

// Example: request_vector = 16'b0000_0100_1000_0000
// Lowest set bit is bit 7
// CTZ = 7, so requester 7 is granted
```

23.6.4 Isolating and Clearing the Lowest Set Bit

```
// Iterate over set bits one at a time (software-style "x & -x" in
hardware)
logic [31:0] pending;
logic [5:0]  next_index;
logic [31:0] lowest_bit_mask;
logic [31:0] pending_next;

count_trailing_zeros #(.WIDTH(32)) pend_ctz (
    .data(pending),
    .ctz(next_index)
);

assign lowest_bit_mask = pending & (~pending + 32'd1);  // isolates
```

```
bit next_index
assign pending_next    = pending & ~lowest_bit_mask;    // clears it
```

23.6.5 Power-of-Two Detection

```
// A non-zero value is a power of two exactly when its only set bit is
the lowest one
logic [31:0] value;
logic [5:0]  value_ctz;
logic        is_power_of_two;

count_trailing_zeros #(.WIDTH(32)) pow2_ctz (
    .data(value),
    .ctz(value_ctz)
);

assign is_power_of_two = (value != 0) && (value == (32'd1 <<
value_ctz));

// value = 64   -> ctz = 6,  1<<6 = 64   -> true
// value = 96   -> ctz = 5,  1<<5 = 32   -> false
// value = 0    -> false by the explicit guard
```

23.7 Design Notes

23.7.1 Hierarchical Implementation

```
// Divide and conquer approach for large widths
module count_trailing_zeros_hierarchical #(
    parameter int WIDTH = 64
) (
    input  logic [WIDTH-1:0]       data,
    output logic [$clog2(WIDTH):0] ctz
);

localparam int HALF_WIDTH = WIDTH/2;

logic [$clog2(HALF_WIDTH):0] ctz_upper, ctz_lower;
logic lower_all_zeros;

// Count in lower half
count_trailing_zeros #(.WIDTH(HALF_WIDTH)) lower_ctz (
    .data(data[HALF_WIDTH-1:0]),
    .ctz(ctz_lower)
);

// Count in upper half
```

```
count_trailing_zeros #(.WIDTH(HALF_WIDTH)) upper_ctz (
    .data(data[WIDTH-1:HALF_WIDTH]),
    .ctz(ctz_upper)
);

assign lower_all_zeros = (ctz_lower == HALF_WIDTH);

// Combine results
assign ctz = lower_all_zeros ? (HALF_WIDTH + ctz_upper) : ctz_lower;

endmodule
```

23.7.2 LUT-Based Implementation (Small Widths)

```
// Optimized for small widths using a wildcard case statement.
// casez is required so that '?' is treated as a don't-care.
module count_trailing_zeros_lut #(
    parameter int WIDTH = 8
) (
    input  logic [WIDTH-1:0]       data,
    output logic [$clog2(WIDTH):0] ctz
);

always_comb begin
    casez (data)
        8'b???????1: ctz = 0;
        8'b??????10: ctz = 1;
        8'b?????100: ctz = 2;
        8'b????1000: ctz = 3;
        8'b???10000: ctz = 4;
        8'b??100000: ctz = 5;
        8'b?1000000: ctz = 6;
        8'b10000000: ctz = 7;
        default:     ctz = 8;   // all zeros
    endcase
end

endmodule
```

23.7.3 Mask-Based Implementation

```
// Isolate the lowest set bit, then one-hot encode its position.
// Often maps well to carry-chain logic on FPGAs.
logic [WIDTH-1:0] w_lowest_one;

assign w_lowest_one = data & (~data + 1'b1);
// w_lowest_one is one-hot (or all zero), so a plain one-hot encoder
yields ctz
```

23.7.4 Synthesis Trade-offs

```
// For area optimization: Use the iterative function approach
// For speed optimization: Use the hierarchical or LUT approach
// For power optimization: Add enable signals and clock gating
```

23.7.5 Common Applications

1. **CPU/DSP Cores**: Instruction implementation (CTZ / FFS instruction)
2. **Memory Controllers**: Address alignment and burst-size derivation
3. **DMA Engines**: Legal transfer size given a start address
4. **Schedulers and Arbiters**: Lowest-index pending request selection
5. **Allocators**: Free-list and bitmap scanning
6. **Interrupt Controllers**: Lowest pending interrupt vector

23.8 Related Modules

- **count_leading_zeros** - the MSB-down counterpart. See the selection table at the top of this page before choosing between them.
- Population count (number of '1' bits)
- Find first set (FFS) / Find last set (FLS)
- Priority encoders
- Barrel shifters

23.9 Testing

23.9.1 Test Scenarios

1. **Boundary Cases**: All zeros, all ones, single bit patterns
2. **Random Patterns**: Comprehensive random testing
3. **Systematic Sweep**: Test all possible trailing zero counts
4. **Corner Cases**: Maximum width values, alternating patterns
5. **Cross-Check**: For one-hot inputs, confirm `ctz + clz == WIDTH-1` against `count_leading_zeros`

23.9.2 Coverage Model

```
covergroup ctz_cg;
    cp_trailing_zeros: coverpoint ctz {
        bins zero_ctz  = {0};
        bins low_ctz   = {[1:WIDTH/4]};
        bins mid_ctz   = {[WIDTH/4+1:3*WIDTH/4]};
        bins high_ctz  = {[3*WIDTH/4+1:WIDTH-1]};
        bins all_zeros = {WIDTH};
    }

    cp_data_patterns: coverpoint data {
        bins all_zeros   = {'0};
        bins all_ones    = {'1};
        bins single_bit[] = {1, 2, 4, 8, 16, 32}; // For appropriate
WIDTH
        bins lsb_set     = {1};
        bins msb_set     = {1 << (WIDTH-1)};
        bins mixed[]     = {[1:2**(WIDTH-1)-1]};
    }
endcovergroup
```

23.9.3 Assertions

```
// CTZ should never exceed WIDTH
property ctz_bounds;
    ctz <= WIDTH;
endproperty

// All zeros should give CTZ = WIDTH
property all_zeros_case;
    (data == '0) |-> (ctz == WIDTH);
endproperty

// LSB set should give CTZ = 0
property lsb_set_case;
    data[0] |-> (ctz == 0);
endproperty

// Relationship between CTZ and the lowest '1' position
// Legal SVA: no continuous assigns inside a property body -- fold the
// expression in directly (or use a `let`)
property ctz_correctness;
    (data != '0) |-> ((data >> ctz) & 1'b1);
endproperty

// CTZ must select exactly the isolated lowest set bit
property ctz_isolates_lowest;
```

```
    (data != '0) |-> ((data & (~data + 1)) == (1 << ctz));
endproperty

// Concurrent asserts need a clocking event (or a default clocking
block);
// this module is combinational, so sample on the consumer's clock:
always @(posedge clk) begin
    assert property (ctz_bounds);
    assert property (all_zeros_case);
    assert property (lsb_set_case);
    assert property (ctz_correctness);
    assert property (ctz_isolates_lowest);
end
```

23.10 References

- Oklobdzija, V.G. “An Algorithmic and Novel Design of a Leading Zero Detector Circuit: Comparison with Logic Synthesis.” *IEEE Transactions on VLSI Systems*, 2(1), Mar. 1994. DOI: 10.1109/92.273153 — leading-zero-detector tree; the trailing-zero count is its mirror image (reverse the input).
- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 12 “Floating Point Architectures”, Sec. 12.4 Leading Zero Counter (textbook treatment; secondary source).
- Indexed in `rtl/common/references/README.md` (local PDF: `LeadingZeroDetector__Oklobdzija1994.pdf`).

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24 dataint_checksum

24.1 Overview

The `dataint_checksum` module is a simple accumulating checksum calculator. It keeps adding incoming data values into a running checksum you can use for basic data integrity verification. Nothing fancy — that's the point.

24.1.1 Module Declaration

```
module dataint_checksum #(
    parameter int WIDTH = 8
) (
    input  logic             clk,
    rst_n,
    input  logic             reset,
    input  logic             valid,
    input  logic [WIDTH-1:0] data,
    output logic [WIDTH-1:0] chksum
);
```

24.2 Parameters

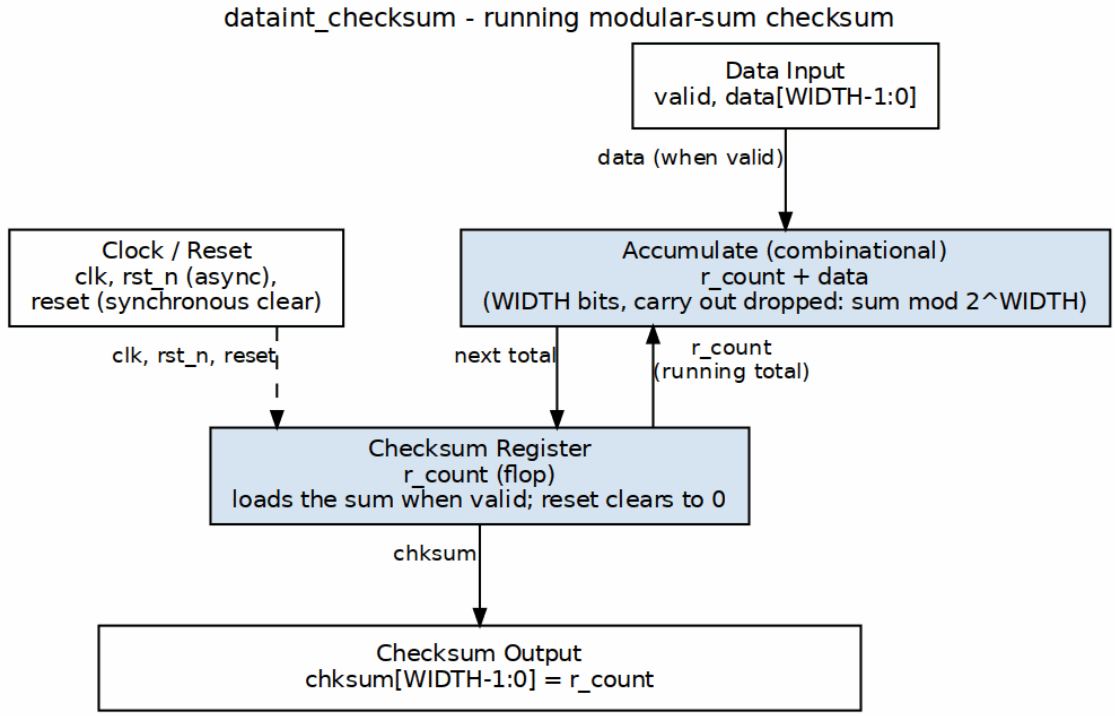
Parameter	Default	Description
`WIDTH`	8	Width of the data and checksum in bits

24.3 Ports

Port	Direction	Width	Description
`clk`	Input	1	Clock signal
`rst_n`	Input	1	Active-low asynchronous reset
`reset`	Input	1	Synchronous reset signal
`valid`	Input	1	Data valid signal - when high, data is accumulated
`data`	Input	WIDTH	Input data to be added to checksum
`chksum`	Output	WIDTH	Current checksum value

24.4 Functional Description

24.4.1 Block diagram



dataint_checksum block diagram

Data enters at the top and, on each cycle `valid` is high, the adder sums it with the running total in `r_count`. The register loads the new total, and the total climbs the right edge back into the adder. The sum is `WIDTH` bits wide with the carry out dropped, so the checksum is the sum modulo 2^WIDTH. The synchronous `reset` and the asynchronous `rst_n` both clear the total to 0, and `chksum` is the register itself.

24.4.2 Operation

The module keeps an internal counter (`r_count`) that accumulates incoming data: - On reset (either asynchronous `rst_n` or synchronous `reset`), the counter clears to zero - When `valid` is asserted, the input `data` is added to the current counter value - The checksum output is simply the current counter value

24.4.3 Reset Behavior

Two reset mechanisms, and you should know which one wins: 1. **Asynchronous Reset**: `rst_n` (active-low) immediately clears the counter 2. **Synchronous Reset**: `reset` (active-high) clears the counter on the next clock edge

24.4.4 Key Features

- **Simple Accumulation**: Uses basic addition for checksum calculation
- **Parameterizable Width**: Supports any data width through the `WIDTH` parameter
- **Dual Reset Support**: Both asynchronous and synchronous reset capabilities
- **Enable Control**: Data accumulation only occurs when `valid` is asserted

24.4.5 Special Considerations

- **Overflow Handling**: The checksum wraps around when the accumulated value exceeds the maximum representable in `WIDTH` bits — expected behavior for a checksum, not a bug
- **Continuous Output**: The checksum output is always valid and reflects the current accumulated value
- **Reset Priority**: Asynchronous reset (`rst_n`) takes precedence over synchronous reset (`reset`)

24.4.6 State Machine

No formal FSM here — it operates as a simple accumulator with two states: - **RESET**: Counter is zero - **ACCUMULATING**: Counter adds valid data inputs

24.5 Related Modules

Nothing in the tree instantiates this module and it instantiates nothing: it is a leaf, used directly by whatever design needs it. Its nearest neighbours in `rtl/common/` are:

- `dataint_crc`
- `dataint_crc_xor_shift`
- `dataint_crc_xor_shift_cascade`
- `dataint_ecc_hamming_decode_secded`
- `dataint_ecc_hamming_encode_secded`

24.6 Timing Characteristics

This module is **sequential**: it contains clocked logic (via `always_ff` or the repository's `ALWAYS_FF_RST` macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

24.7 Usage Examples

```
// 16-bit checksum calculator
dataint_checksum #(
    .WIDTH(16)
) checksum_calc (
    .clk(clk),
    .rst_n(rst_n),
    .reset(soft_reset),
    .valid(data_valid),
    .data(input_data),
    .chksum(calculated_checksum)
);
```

24.8 Design Notes

24.8.1 Common Applications

- Simple data integrity checking
- Packet checksum calculation
- Basic error detection in data streams
- Accumulation of data values for verification purposes

24.9 Testing

`val/common/test_dataint_checksum.py` exercises this module. It collects 4 parameter cases at the default `REG_LEVEL`.

```
source env_python
pytest val/common/test_dataint_checksum.py -v
```

24.10 References

- Maxino, T.C., Koopman, P.J. “The Effectiveness of Checksums for Embedded Control Networks.” *IEEE Transactions on Dependable and Secure Computing*, 6(1), Jan. 2009. DOI: 10.1109/TDSC.2007.70216 — measures this exact two’s-complement modular-sum checksum against one’s-complement, Fletcher, Adler and CRC; it is the weakest of the additive options at equal width.
- Fletcher, J.G. “An Arithmetic Checksum for Serial Transmissions.” *IEEE Transactions on Communications*, 30(1), Jan. 1982. DOI: 10.1109/TCOM.1982.1095369 — the stronger position-sensitive arithmetic checksum this module does not implement.
- Indexed in `rtl/common/references/README.md` (local PDF: `ChecksumEffectiveness__Maxino2009.pdf`).

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25 dataint_crc

25.1 Overview

The `dataint_crc` module is a generic Cyclic Redundancy Check (CRC) computation engine — configurable polynomials, data widths, and processing options, built for flexible, high-performance CRC calculation. It's a full CRC calculation engine for data integrity applications, communication protocols, and storage systems, with support for configurable polynomial widths, data widths, input/output reflection, and cascade processing for high-throughput applications.

25.1.1 Module Declaration

```
module dataint_crc #(
    parameter int DATA_WIDTH = 64,            // Input data width
    parameter int CRC_WIDTH = 64,             // CRC polynomial width
    parameter int REFIN = 1,                  // Reflect input data
    parameter int REFOUT = 1                  // Reflect output data
) (
    // Derived localparam (computed internally - not user-settable):
    parameter int CHUNKS = DATA_WIDTH / 8,      // Derived -- do NOT
override (port width dependency)
    // localparam int CW = CRC_WIDTH;           // CRC width alias
    // localparam int DW = DATA_WIDTH;          // Data width alias
    // localparam int CH = CHUNKS;              // Chunks alias
    input  logic [CRC_WIDTH-1:0]  POLY,             // CRC polynomial
    input  logic [CRC_WIDTH-1:0]  POLY_INIT,        // Initial CRC
value
    input  logic [CRC_WIDTH-1:0]  XOROUT,           // Output XOR mask
    input  logic                  clk,              // Clock
    input  logic                  rst_n,            // Active-low
reset
    input  logic                  load_crc_start,   // Load initial
CRC
    input  logic                  load_from_cascade, // Load from
cascade
    input  logic [    CHUNKS-1:0] cascade_sel,      // Cascade
selection (one-hot)
    input  logic [DATA_WIDTH-1:0] data,             // Input data
    output logic [ CRC_WIDTH-1:0] crc               // CRC output
);
```

25.2 Parameters

25.2.1 User-Settable Parameters

Parameter	Default	Description
DATA_WIDTH	64	Input data width in

Parameter	Default	Description
		bits (must be multiple of 8)
CRC_WIDTH	64	CRC polynomial width in bits (8, 16, 32, or 64)
REFIN	1	Reflect input data bytes (1=reflect, 0=direct)
REFOUT	1	Reflect output CRC (1=reflect, 0=direct)

25.2.2 Calculated Parameters (Do Not Override)

Parameter	Formula	Description
CHUNKS (CH)	DATA_WIDTH/8	Number of 8-bit processing chunks per cycle
CW	CRC_WIDTH	Convenience alias for CRC width
DW	DATA_WIDTH	Convenience alias for data width

Note: CW, DW, and CH are body localparams and cannot be overridden. CHUNKS is different: it is a derived `parameter` (the `cascade_sel` port needs it, and a body localparam elaborates too late for a port width), so it *can* be overridden at instantiation — and must not be. The RTL's own parameter comment says the same: derived, not configurable, leave the default alone.

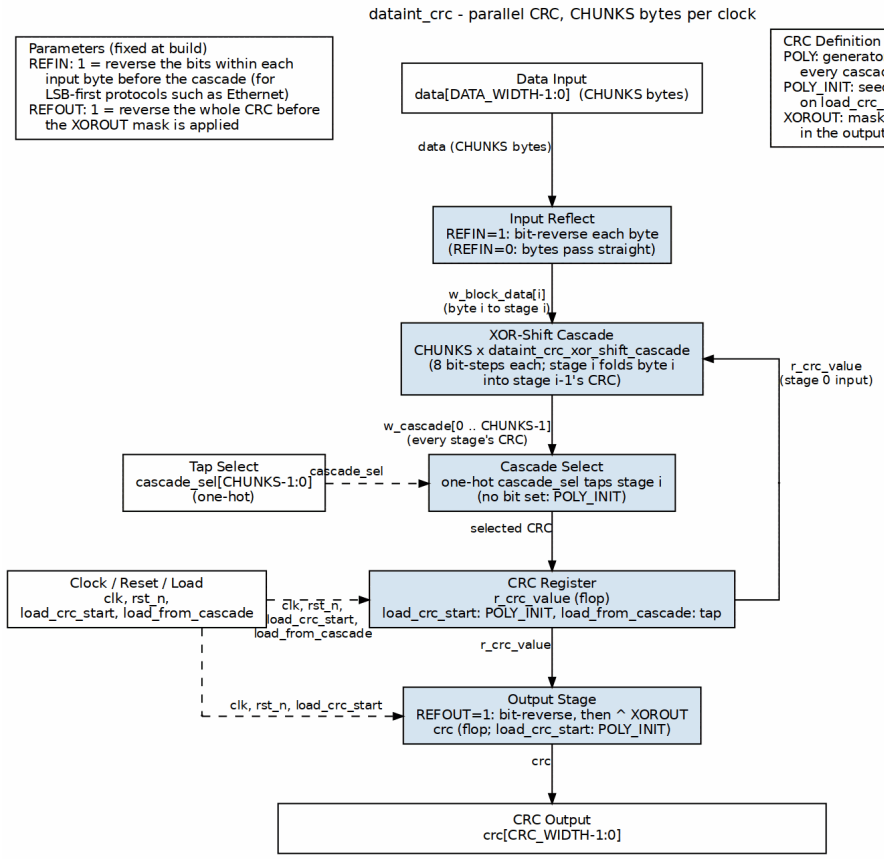
25.3 Ports

Port	Direction	Width	Description
POLY	Input	CRC_WIDTH	CRC polynomial coefficient
POLY_INIT	Input	CRC_WIDTH	Initial CRC value (seed)
XOROUT	Input	CRC_WIDTH	Final XOR mask for output
clk	Input	1	System clock

Port	Direction	Width	Description
rst_n	Input	1	Active-low asynchronous reset
load_crc_start	Input	1	Load initial CRC value
load_from_cascade	Input	1	Load CRC from cascade output
cascade_sel	Input	CHUNKS	One-hot cascade stage selection
data	Input	DATA_WIDTH	Input data for CRC calculation
crc	Output	CRC_WIDTH	Computed CRC value (registered)

25.4 Functional Description

25.4.1 Block diagram



dataint_crc block diagram

Data enters at the top and is split into CHUNKS bytes, each optionally bit-reversed when REFIN=1. The bytes feed a chain of CHUNKS dataint_crc_xor_shift_cascade stages. Stage 0 starts from the running CRC in r_crc_value, which climbs the right edge from the register, and each later stage folds one more byte into the previous stage's result. The one-hot cascade_sel taps the stage that matches the number of valid bytes, and load_from_cascade writes that tap into r_crc_value. The output stage reflects the register when REFOUT=1, XORs in XOROUT, and registers the result as crc. load_crc_start returns both registers to POLY_INIT. The two boxes at the top are legends. The build-time parameters REFIN and REFOUT select bit reflection; the runtime ports POLY, POLY_INIT and XOROUT define which CRC standard is computed, and each is described where it acts.

25.4.2 CRC Algorithm Implementation

The CRC calculation is parallel and runs through four stages:

1. **Input Data Conditioning**: Optional bit reflection per REFIN parameter

2. **Cascade Processing**: Parallel processing through multiple XOR-shift stages
3. **Output Conditioning**: Optional bit reflection per REFOUT parameter
4. **Final XOR**: Application of XOROUT mask to final result

25.4.3 Key Features

- **Parallel Processing**: Processes multiple bytes simultaneously for high throughput
- **Configurable Reflection**: Support for MSB-first and LSB-first bit ordering
- **Cascade Architecture**: Enables partial CRC computation and continuation
- **Standards Compliance**: Configurable for standard CRC algorithms (CRC-32, CRC-64, etc.)

25.4.4 Input Data Reflection

```
// Input reflection based on REFIN parameter
if (REFIN != 0) begin : gen_reflect_inputs
    for (genvar i = 0; i < CH; i++) begin : gen_ch_reflect
        for (genvar j = 0; j < 8; j++) begin : gen_bit_reflect
            assign w_block_data[i][j] = data[i*8+7-j];  // Bit
reversal
        end
    end
end else begin : gen_direct_assign_inputs
    for (genvar i = 0; i < CH; i++) begin : gen_ch_direct
        assign w_block_data[i] = data[i*8+:8];  // Direct assignment
    end
end
```

25.4.5 Cascade Processing Architecture

```
// Chain of XOR-shift cascade stages
for (i = 0; i < CH; i++) begin : gen_xor_shift_cascade
    if (i == 0) begin : gen_xor_cascade_0
        dataint_crc_xor_shift_cascade #(.CRC_WIDTH(CRC_WIDTH))
cascade_0 (
            .block_input(r_crc_value),     // Start with current CRC
            .poly(w_poly),
            .data_input(w_block_data[i]),
            .block_output(w_cascade[i])
        );
    end else begin : gen_xor_cascade_N
        dataint_crc_xor_shift_cascade #(.CRC_WIDTH(CRC_WIDTH))
cascade_N (
```

```
            .block_input(w_cascade[i-1]),  // Chain from previous
stage
            .poly(w_poly),
            .data_input(w_block_data[i]),
            .block_output(w_cascade[i])
        );
    end
end
```

25.4.6 Cascade Selection Logic

```
// Select appropriate cascade output based on data length
always_comb begin
    w_selected_cascade_output = POLY_INIT;  // Default
    for (int i = 0; i < CH; i++) begin
        if (cascade_sel[i]) begin
            w_selected_cascade_output = w_cascade[i];
        end
    end
end
```

25.5 Timing Characteristics

25.5.1 Throughput

Data Width	CRC Width	Max Frequency	Throughput
32-bit	32-bit	~400 MHz	12.8 Gbps
64-bit	32-bit	~350 MHz	22.4 Gbps
128-bit	64-bit	~300 MHz	38.4 Gbps

25.5.2 Latency

Property	Value
Pipeline stages	2 cycles (input conditioning + CRC computation)
Reset recovery	1 cycle
Cascade selection	0 cycles (combinational)

25.5.3 Critical Paths

1. **Cascade Chain**: Data flows through all cascade stages
2. **Output Reflection**: Bit reversal for REFOUT
3. **Register Setup**: Loading CRC state register

25.6 Usage Examples

25.6.1 Basic CRC Calculation

```
logic [63:0] input_data;
logic [31:0] crc_result;
logic data_valid, crc_ready;

dataint_crc #(
    .DATA_WIDTH(64),
    .CRC_WIDTH(32),
    .REFIN(1),
    .REFOUT(1)
) u_crc_engine (
    .POLY(32'h04C11DB7),
    .POLY_INIT(32'hFFFFFFFF),
    .XOROUT(32'hFFFFFFFF),
    .clk(clk),
    .rst_n(rst_n),
    .load_crc_start(data_valid),      // pulse: seed accumulator with
POLY_INIT
    .load_from_cascade(capture_crc),  // pulse: capture the cascade
result
    .cascade_sel(8'h80),              // ONE-HOT: select the 8-byte
(full) stage
    .data(input_data),
    .crc(crc_result)
);
```

The accumulator is only updated through the cascade path. Data reaches `crc` *only* via `load_from_cascade`; tying `load_from_cascade` to 0 (and using `cascade_sel = 8'hFF`) makes the engine reload `POLY_INIT` on every `load_crc_start` and emit a constant (`reflect(POLY_INIT) ^ XOROUT`), never a CRC of `input_data`. This is the mistake everyone makes with this module exactly once. A single-shot calculation must: (1) pulse `load_crc_start` to seed, (2) present `data`, then (3) pulse `load_from_cascade` with a **one-hot** `cascade_sel` selecting the stage for the number of valid bytes (`8'h80` for all 8). `cascade_sel` is one-hot, not a mask — `8'hFF` only works by highest-set-bit priority. See the streaming example below for the general pattern.

25.6.2 Streaming CRC with Cascade

```
// Process variable-length packets using cascade selection
logic [63:0] packet_data;
logic [7:0] byte_count;  // Number of valid bytes (1-8)
logic [7:0] cascade_select;
```

```
// Convert byte count to one-hot cascade selection
always_comb begin
    cascade_select = '0;
    if (byte_count > 0 && byte_count <= 8) begin
        cascade_select[byte_count-1] = 1'b1;
    end
end

dataint_crc #(
    .DATA_WIDTH(64),
    .CRC_WIDTH(32)
) u_stream_crc (
    .clk(clk),
    .rst_n(rst_n),
    .load_crc_start(packet_start),
    .load_from_cascade(packet_continue),
    .cascade_sel(cascade_select),
    .data(packet_data),
    .crc(streaming_crc),
    // ... other connections
);
```

25.6.3 Multi-Stage CRC Pipeline

```
// Pipeline multiple CRC engines for high throughput
logic [63:0] stage_data [0:3];
logic [31:0] stage_crc [0:3];

genvar stage;
generate
    for (stage = 0; stage < 4; stage++) begin : gen_crc_pipeline
        dataint_crc #(
            .DATA_WIDTH(64),
            .CRC_WIDTH(32)
        ) u_pipeline_crc (
            .clk(clk),
            .rst_n(rst_n),
            .data(stage_data[stage]),
            .crc(stage_crc[stage]),
            // ... configure for pipeline operation
        );
    end
endgenerate
```

25.6.4 CRC-32 (IEEE 802.3)

```
dataint_crc #(
    .DATA_WIDTH(32),
```

```
    .CRC_WIDTH(32),
    .REFIN(1),
    .REFOUT(1)
) u_crc32 (
    .POLY(32'h04C11DB7),
    .POLY_INIT(32'hFFFFFFFF),
    .XOROUT(32'hFFFFFFFF),
    // ... other connections
);
```

25.6.5 CRC-16 (CCITT)

```
dataint_crc #(
    .DATA_WIDTH(16),
    .CRC_WIDTH(16),
    .REFIN(0),
    .REFOUT(0)
) u_crc16 (
    .POLY(16'h1021),
    .POLY_INIT(16'hFFFF),
    .XOROUT(16'h0000),
    // ... other connections
);
```

25.6.6 CRC-64 (ECMA-182)

```
dataint_crc #(
    .DATA_WIDTH(64),
    .CRC_WIDTH(64),
    .REFIN(0),                          // ECMA-182 is NOT reflected
    .REFOUT(0)
) u_crc64 (
    .POLY(64'h42F0E1EBA9EA3693),
    .POLY_INIT(64'h0000000000000000), // init 0
    .XOROUT(64'h0000000000000000),    // xorout 0
    // ... other connections
);
```

25.7 Design Notes

25.7.1 Area Optimization

- **Polynomial Optimization**: Use polynomials with fewer terms
- **Width Reduction**: Match CRC_WIDTH to actual requirements
- **Cascade Stages**: Reduce CHUNKS for lower throughput applications

25.7.2 Power Optimization

- **Clock Gating**: Gate unused cascade stages
- **Data Gating**: Prevent unnecessary switching when not processing

Between the configurable polynomial, the reflection options, and the cascade architecture, `dataint_crc` covers most of the CRC standards you'll meet in practice — set it up once per protocol and let it run.

25.8 Related Modules

- **dataint_crc_xor_shift_cascade** - Building block for cascade stages
- **dataint_crc_xor_shift** - Simple XOR-shift CRC implementation
- **dataint_checksum** - Alternative integrity checking method
- **dataint_parity** - Simpler parity-based error detection

25.9 Testing

25.9.1 Setup Requirements

```
// Data must be stable before clock edge
timing_spec: assert property (
    @(posedge clk) $stable(data) throughout (load_crc_start)
);

// Cascade selection must be stable
cascade_timing: assert property (
    @(posedge clk) $stable(cascade_sel) throughout (load_from_cascade)
);
```

25.9.2 Verification Notes

- **Polynomial Validation**: Verify against known CRC test vectors
- **Cascade Testing**: Test all cascade selection combinations
- **Reflection Testing**: Verify bit ordering for REFIN/REFOUT modes
- **Standards Compliance**: Validate against published CRC standards

25.10 References

- Peterson, W.W., Brown, D.T. “Cyclic Codes for Error Detection.” *Proceedings of the IRE*, 49(1), Jan. 1961. DOI: 10.1109/JRPROC.1961.287814 — cyclic redundancy codes and their shift-register realization.
- Albertengo, G., Sisto, R. “Parallel CRC Generation.” *IEEE Micro*, 10(5), Oct. 1990. DOI: 10.1109/40.60527 — byte-parallel CRC lineage behind the per-cycle cascade of `DATA_WIDTH/8` stages.
- Campobello, G., Patanè, G., Russo, M. “Parallel CRC Realization.” *IEEE Transactions on Computers*, 52(10), Oct. 2003. DOI: 10.1109/TC.2003.1234528 — general parallel-CRC formulation; the RTL instead chains unrolled serial stages because `POLY` is a runtime input and no matrix can be precomputed.
- Williams, R.N. “A Painless Guide to CRC Error Detection Algorithms.” Version 3, Rocksoft Pty Ltd, Aug. 1993 (non-peer; canonical) — defines the WIDTH/POLY/INIT/REFIN/REFOUT/XOROUT parameter model this module exposes.
- Koopman, P., Chakravarty, T. “Cyclic Redundancy Code (CRC) Polynomial Selection for Embedded Networks.” *Proc. International Conference on Dependable Systems and Networks (DSN 2004)*, 2004. DOI: 10.1109/DSN.2004.1311885 — Hamming-distance-based guidance for choosing the runtime `POLY`.
- Moon, T.K. *Error Correction Coding*, 2nd ed., Wiley, 2021, ch. 4 “Cyclic Codes, Rings, and Polynomials” (Sec. 4.9 hardware background, Sec. 4.13 Binary CRC Codes) (textbook treatment; secondary source).
- Dally, W.J., Towles, B. *Principles and Practices of Interconnection Networks*, Morgan Kaufmann, 2004, ch. 21 “Error Control”, Sec. 21.3 Link Level Error Control (textbook treatment; secondary source).
- Indexed in `rtl/common/references/README.md` (local files: `CrcParameterModel__Williams1993.txt`, `CrcPolynomialSelection__Koopman2004.pdf`).

25.11 Navigation

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26 dataint_crc_xor_shift_cascade

26.1 Overview

Eight instances of `dataint_crc_xor_shift`, chained back to back so a full byte of data gets chewed through in a single combinational operation. Byte in, updated CRC out — no clock involved.

26.1.1 Module Declaration

```
module dataint_crc_xor_shift_cascade #(
    parameter int CRC_WIDTH = 32
) (
    input  [CRC_WIDTH-1:0] block_input,
    input  [CRC_WIDTH-1:0] poly,
    input  [          7:0] data_input,
    output [CRC_WIDTH-1:0] block_output
);
```

26.2 Parameters

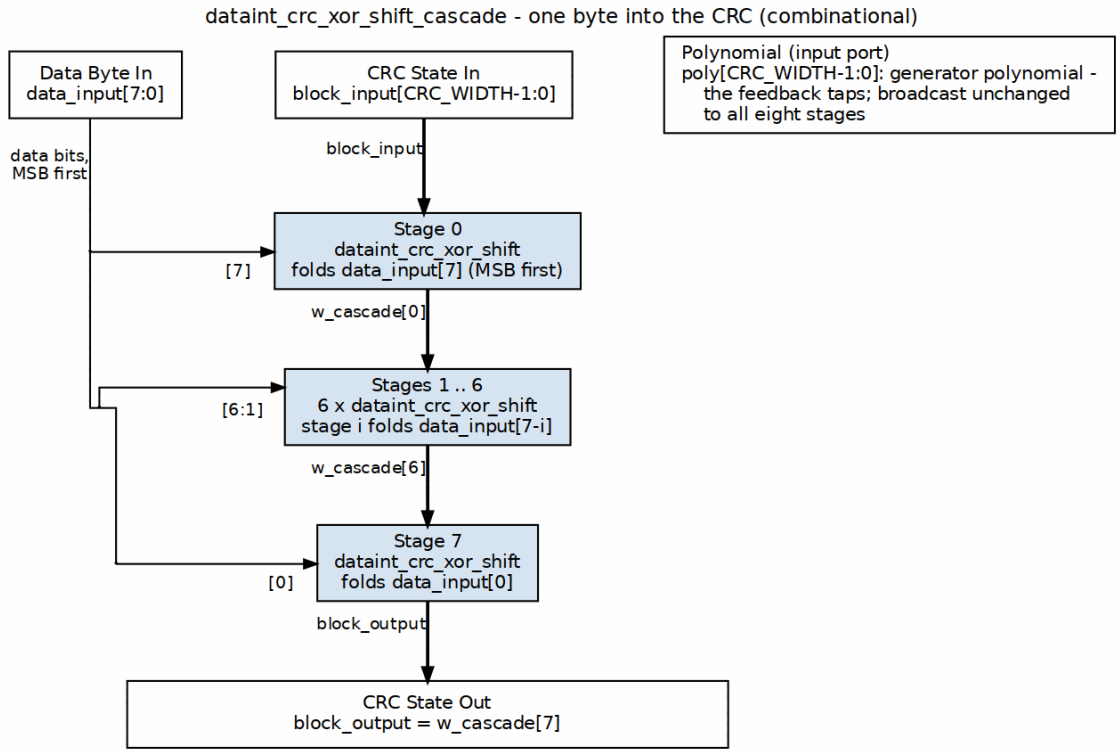
Parameter	Default	Description
CRC_WIDTH	32	Width of the CRC register and polynomial

26.3 Ports

Port	Direction	Width	Description
block_input	Input	CRC_WIDTH	Initial CRC state for the byte
poly	Input	CRC_WIDTH	CRC polynomial
data_input	Input	8	8-bit data byte to process
block_output	Output	CRC_WIDTH	Final CRC state after processing 8 bits

26.4 Functional Description

26.4.1 Block diagram



dataint_crc_xor_shift_cascade block diagram

The CRC state runs straight down the center through eight chained `dataint_crc_xor_shift` bit-steps, drawn as the first stage, the six middle stages and the last stage. The data byte enters from the left, most significant bit first: stage 0 folds `data_input[7]` and stage 7 folds `data_input[0]`. The polynomial goes unchanged to all eight stages, as its legend says. The last stage's state is `block_output`.

26.4.2 Cascade Operation

Eight CRC calculation stages, chained end to end: 1. **Stage 0**: Processes the MSB of the data byte against the initial CRC state 2. **Stages 1-6**: Each processes the next data bit against the previous stage's output 3. **Stage 7**: Processes the LSB and produces the final result

26.4.3 Data Bit Ordering

Data bits run MSB to LSB: - `data_input[7]` is processed first - `data_input[0]` is processed last - This matches typical serial transmission order

26.4.4 Architecture Overview

```
block_input → [Stage 0] → [Stage 1] → ... → [Stage 7] → block_output
              data[7]     data[6]           data[0]
```

26.4.5 Generate Block Structure

The cascade is built from a parameterized generate block:

```
generate
    for (i = 0; i < 8; i++) begin : gen_xor_shift_cascade
        if (i == 0) begin : gen_xor_shift_0
            // First stage uses block_input
            dataint_crc_xor_shift #(.CRC_WIDTH(CRC_WIDTH)) stage (
                .stage_input(block_input),
                .poly(poly),
                .new_bit(data_input[7-i]),
                .stage_output(w_cascade[i])
            );
        end else begin : gen_xor_shift_N
            // Subsequent stages chain previous output
            dataint_crc_xor_shift #(.CRC_WIDTH(CRC_WIDTH)) stage (
                .stage_input(w_cascade[i-1]),
                .poly(poly),
                .new_bit(data_input[7-i]),
                .stage_output(w_cascade[i])
            );
        end
    end
endgenerate
```

26.4.6 Intermediate Signals

- **`w_cascade[0:7]`**: Array holding intermediate CRC states
- **Fixed Width**: Each element is `CRC_WIDTH` bits wide
- **Sequential Chain**: Each stage feeds the next

26.4.7 Key Features

- **Parallel Byte Processing**: 8 bits in one combinational operation, no clock required — suitable for high-speed applications
- **Bit Order Handling**: MSB-first processing (network byte order), with the bit index calculated as `data_input[7-i]`, matching standard CRC bit processing order

- **Scalable Architecture**: Easy to modify for different byte sizes; generate blocks enable clean scaling, parameterized for different CRC widths

26.5 Timing Characteristics

26.5.1 Combinational Delay

Total delay is the sum of 8 CRC stage delays: - Each stage: ~2-3 gate delays - Total cascade: ~16-24 gate delays - May require pipelining for very high frequencies

Do that math against your clock period before you promise timing closure — 24 gate levels is not free at 400 MHz.

26.5.2 Critical Path

The critical path runs through all 8 stages:

```
block_input → Stage0 → Stage1 → ... → Stage7 → block_output
```

26.6 Usage Examples

26.6.1 Basic Usage

```
dataint_crc_xor_shift_cascade #(
    .CRC_WIDTH(16)
) crc_byte_processor (
    .block_input(current_crc_state),
    .poly(16'h1021),           // CRC-16-CCITT
    .data_input(data_byte),
    .block_output(updated_crc_state)
);
```

26.6.2 Multi-Byte Processing

```
// Process 4 bytes (32 bits) of data
wire [CRC_WIDTH-1:0] stage_output[0:3];

genvar i;
generate
    for (i = 0; i < 4; i++) begin : gen_multi_byte
        dataint_crc_xor_shift_cascade #(.CRC_WIDTH(CRC_WIDTH)) cascade
(
            .block_input(i == 0 ? initial_crc : stage_output[i-1]),
            .poly(polynomial),
            .data_input(data_bytes[i]),
            .block_output(stage_output[i])
        );
    end
endgenerate
```

```
assign final_crc = stage_output[3];
```

26.7 Design Notes

26.7.1 Performance Optimization

For high-frequency operation: - Consider adding pipeline registers between cascades - Balance latency vs. throughput requirements - You may need to pipeline within the 8-stage cascade

For area: - Each stage requires ~CRC_WIDTH XOR gates - Total area scales with CRC_WIDTH × 8 - Consider time-multiplexed alternatives for area-critical applications

26.7.2 Integration with the CRC System

This module is used by `dataint_crc` for: - Processing data in byte-sized chunks - Building wider data processing capability - Maintaining processing efficiency

On the interface with the parent module: - Multiple cascade instances handle wide data - Cascade selection allows intermediate results - Enables flexible data width support

26.7.3 Applications

- High-speed packet processing
- Real-time data integrity checking
- Communication protocol implementations
- Storage system error detection
- Building block for wider CRC processors

26.8 Related Modules

- **`dataint_crc_xor_shift`**: Single-bit processing building block
- **`dataint_crc`**: Parent CRC calculation module
- Used together to create flexible CRC calculation systems

26.9 Testing

No dedicated testbench for this module. It has no `val/**/test_dataint_crc_xor_shift_cascade.py`; it is exercised indirectly, through the tests of the modules that instantiate it:

- `dataint_crc` – `val/common/test_dataint_crc.py`

Indirect coverage exercises this module only in the configurations its parents elaborate. A parameter or mode no parent uses is untested.

26.10 References

- Peterson, W.W., Brown, D.T. “Cyclic Codes for Error Detection.” *Proceedings of the IRE*, 49(1), Jan. 1961. DOI: 10.1109/JRPROC.1961.287814 — cyclic-code shift-register division, unrolled here eight times per byte.
- Albertengo, G., Sisto, R. “Parallel CRC Generation.” *IEEE Micro*, 10(5), Oct. 1990. DOI: 10.1109/40.60527 — the k-bits-per-clock CRC derivation; this module unrolls the serial step directly instead of precomputing the transition matrix.
- Campobello, G., Patanè, G., Russo, M. “Parallel CRC Realization.” *IEEE Transactions on Computers*, 52(10), Oct. 2003. DOI: 10.1109/TC.2003.1234528 — later general parallel-CRC formulation for arbitrary k and polynomial.
- Moon, T.K. *Error Correction Coding*, 2nd ed., Wiley, 2021, ch. 4 “Cyclic Codes, Rings, and Polynomials” (Sec. 4.9 hardware background, Sec. 4.13 Binary CRC Codes) (textbook treatment; secondary source).
- Indexed in `rtl/common/references/README.md`.

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27 dataint_crc_xor_shift

27.1 Overview

The fundamental CRC step, applied to exactly one input bit: shift the current CRC state, and XOR in the polynomial when the feedback bit says to.

- **Single Bit Processing**: One bit per operation, no more
- **Combinational Logic**: Pure combinational implementation (no clock)
- **Parameterizable Width**: Any CRC width you need
- **Optimized Design**: Breaks the circular dependency with an intermediate signal

27.1.1 Module Declaration

```
module dataint_crc_xor_shift #(
    parameter int CRC_WIDTH = 32
) (
    input [CRC_WIDTH-1:0] stage_input,
    input [CRC_WIDTH-1:0] poly,
    input new_bit,
    output [CRC_WIDTH-1:0] stage_output
);
```

27.2 Parameters

Parameter	Default	Description
CRC_WIDTH	32	Width of the CRC register and polynomial

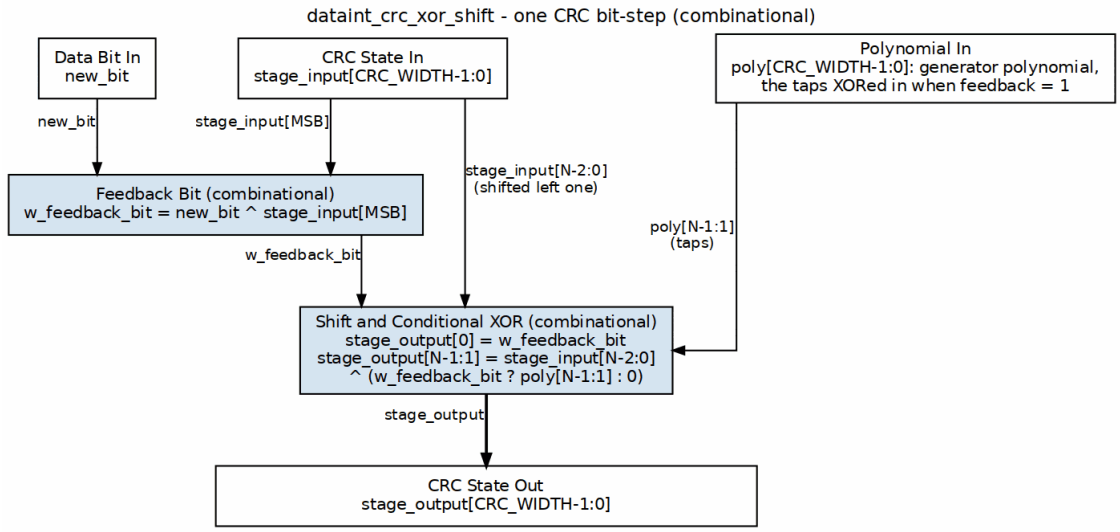
27.3 Ports

Port	Direction	Width	Description
stage_input	Input	CRC_WIDTH	Current CRC state
poly	Input	CRC_WIDTH	CRC polynomial
new_bit	Input	1	New data bit to process

Port	Direction	Width	Description
stage_output	Output	CRC_WIDTH	Updated CRC state

27.4 Functional Description

27.4.1 Block diagram



dataint_crc_xor_shift block diagram

One bit-step of a CRC, all combinational. The feedback bit is the incoming `new_bit` XORed with the MSB of the CRC state. The state shifts left one place, the feedback bit becomes the new bit 0, and when the feedback bit is 1 the polynomial taps `poly[N-1:1]` are XORed into the shifted state. `dataint_crc_xor_shift_cascade` chains eight of these to fold in a byte.

27.4.2 Core CRC Operation

This is the classic CRC shift-and-XOR, broken into three moves: 1. **Feedback Calculation**: XOR the new input bit with the MSB of the current CRC 2. **Shift Operation**: Shift the CRC register left by one position 3. **Polynomial XOR**: If the feedback bit is 1, XOR with the polynomial

27.4.3 Mathematical Foundation

The CRC operation follows this algorithm:

```
feedback = new_bit ⊕ CRC[MSB]
CRC_new[0] = feedback
CRC_new[MSB:1] = CRC[MSB-1:0] ⊕ (polynomial[MSB:1] & {WIDTH-
1{feedback}})
```

27.4.4 Circular Dependency Solution

A naive write-up of this has a circular dependency — you'd be computing the output from itself. The fix is to compute the feedback bit first, on its own, then reuse it:

```
logic feedback_bit;

// Calculate feedback first
assign feedback_bit = new_bit ^ stage_input[CRC_WIDTH-1];

// Then use it for output calculation
assign stage_output[0] = feedback_bit;
assign stage_output[CRC_WIDTH-1:1] = stage_input[CRC_WIDTH-2:0] ^
    ({CRC_WIDTH-1{feedback_bit}} & poly[CRC_WIDTH-1:1]);
```

27.4.5 Bit-wise Operations

- **LSB Assignment**: Always gets the feedback bit
- **MSBs Assignment**: Shifted previous bits XORed with the polynomial when feedback is active
- **Conditional XOR**: Bit replication builds the conditional XOR mask

27.4.6 Feedback Bit Logic

The feedback bit runs the whole show: - It decides whether the polynomial gets applied - It's the XOR of the input bit and the current MSB - It gates the polynomial application

27.4.7 Polynomial Application

The polynomial goes in conditionally: - Only bits [MSB:1] of the polynomial are used (bit 0 is implicit) - Applied through an AND mask built by replicating the feedback bit - XORed with the shifted CRC bits

27.4.8 No State Elements

This module is purely combinational: - No clock, no reset - Instantaneous response to input changes - Can be cascaded for multi-bit processing

27.4.9 Mathematical Verification

For a polynomial P(x) and data bit d: - Feedback = 0: it's a plain left shift - Feedback = 1: left shift, then XOR the polynomial

That's CRC polynomial division, done in gates.

27.5 Timing Characteristics

- **Latency**: Combinational (zero clock cycles)

- **Throughput**: One bit per clock when wrapped in a sequential system
- **Area**: Minimal — a few XOR gates and multiplexers
- **Power**: Low — just simple combinational logic

27.6 Usage Examples

You rarely instantiate this block alone — it's a building block.

27.6.1 Single Bit Processing

```
dataint_crc_xor_shift #(.CRC_WIDTH(16)) crc_stage (
    .stage_input(current_crc),
    .poly(crc_polynomial),
    .new_bit(data_bit),
    .stage_output(next_crc)
);
```

27.6.2 Cascaded for Multiple Bits

Chain as many instances as you have bits to process:

```
// Process 8 bits. Do NOT also `assign stage[0] = initial_crc;` --
stage[0] is
// driven by the i==0 instance below, and a second continuous driver
on the same
// wire is an elaboration error (X contention in simulation).
wire [CRC_WIDTH-1:0] stage[0:7];

genvar i;
generate
    for (i = 0; i < 8; i++) begin
        dataint_crc_xor_shift #(.CRC_WIDTH(CRC_WIDTH)) stage_inst (
            .stage_input(i == 0 ? initial_crc : stage[i-1]),
            .poly(polynomial),
            .new_bit(data_byte[7-i]),
            .stage_output(stage[i])
        );
    end
endgenerate

assign final_crc = stage[7];
```

27.7 Design Notes

27.7.1 Applications

- Building block for CRC calculation engines
- Serial CRC processors
- Component in parallel CRC architectures
- Educational CRC implementations
- Custom protocol CRC calculators

27.8 Related Modules

Read out of the RTL, not curated: these are the modules this one instantiates and the modules that instantiate it.

Instantiated by: - `dataint_crc_xor_shift_cascade`

27.9 Testing

No dedicated testbench for this module. It has no `val/**/test_dataint_crc_xor_shift.py`. It is exercised indirectly, through the tests of modules that instantiate it (directly or further up):

- `dataint_crc` – `val/**/test_dataint_crc.py`

Indirect coverage exercises this module only in the configurations those parents elaborate. A parameter or mode no parent uses is untested.

Treat any behaviour described on this page as unverified by simulation.

27.10 References

- Peterson, W.W., Brown, D.T. “Cyclic Codes for Error Detection.” *Proceedings of the IRE*, 49(1), Jan. 1961. DOI: 10.1109/JRPROC.1961.287814 — origin of cyclic codes for error detection and their shift-register encoder, the per-bit step this module implements.
- Albertengo, G., Sisto, R. “Parallel CRC Generation.” *IEEE Micro*, 10(5), Oct. 1990. DOI: 10.1109/40.60527 — byte-parallel CRC lineage; this one-bit stage is the unit the parallel forms unroll.

- Moon, T.K. *Error Correction Coding*, 2nd ed., Wiley, 2021, ch. 4 “Cyclic Codes, Rings, and Polynomials” (Sec. 4.9 hardware background, Sec. 4.13 Binary CRC Codes) (textbook treatment; secondary source).
- Indexed in `rtl/common/references/README.md`.

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28 dataint_ecc_hamming_decode_secded

28.1 Overview

A Hamming decoder with SECDED — Single Error Correction, Double Error Detection. It decodes Hamming-encoded data, detects and fixes single-bit errors, and raises a flag when two bits have gone bad (which it can see but not repair).

28.1.1 Module Declaration

```
module dataint_ecc_hamming_decode_secded #(
    parameter int WIDTH = 4,
    parameter int DEBUG = 0
) (
    input  logic                          clk,
    rst_n,
    input  logic                          enable,
    input  logic [WIDTH+ParityBits:0] hamming_data,
    output logic [         WIDTH-1:0] data,
    output logic                          error_detected,
    output logic                          double_error_detected
);
```

28.2 Parameters

Parameter	Default	Description
WIDTH	4	Width of original data in bits
DEBUG	0	Enable debug output (1=enabled, 0=disabled)

28.2.1 Calculated Parameters

Parameter	Formula	Description
ParityBits	$clog2(WIDTH + $clog2(WIDTH) + 1)	Number of Hamming parity bits
TotalWidth	WIDTH + ParityBits + 1	Total encoded data width

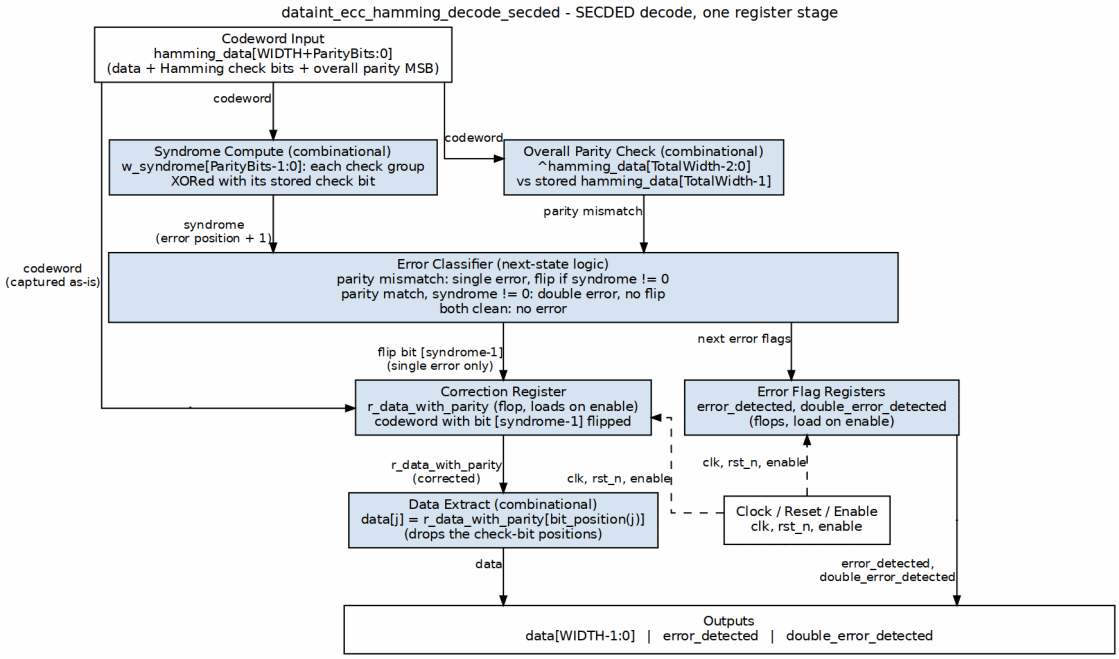
28.3 Ports

Port	Direction	Width	Description
clk	Input	1	Clock signal

Port	Direction	Width	Description
`rst_n`	Input	1	Active-low asynchronous reset
`enable`	Input	1	Enable decoding operation
`hamming_data`	Input	TotalWidth	Encoded data with ECC bits
`data`	Output	WIDTH	Decoded original data
`error_detected`	Output	1	Single or double error detected
`double_error_detected`	Output	1	Double error specifically detected

28.4 Functional Description

28.4.1 Block diagram



dataint_ecc_hamming_decode_secded block diagram

The codeword drops down the left edge into the correction register, the one register stage, which loads only when `enable` is high. Two combinational checks run on the same codeword: the syndrome (each Hamming check group XORed with its stored check bit) and the overall parity

against the stored SECDED bit. The classifier combines them. A parity mismatch is a single error: with a non-zero syndrome, bit `[syndrome-1]` is flipped as the codeword is captured; with a zero syndrome, the overall parity bit itself was hit and nothing is flipped. A parity match with a non-zero syndrome is a double error, flagged but not corrected. The flags are registered alongside the codeword, and a combinational extract drops the check-bit positions to produce `data`.

28.4.2 Decoding Process

1. **Syndrome Generation**: Calculate the parity check syndrome
2. **Overall Parity Check**: Verify the SECDED bit
3. **Error Classification**: Work out the error type and location
4. **Error Correction**: Fix single-bit errors
5. **Data Extraction**: Pull the corrected original data back out

28.4.3 Error Detection Logic

Overall Parity	Syndrome	Error Type	Action
Match	Zero	No Error	Pass data through
Mismatch	Non-zero	Single Error (Hamming)	Correct using syndrome
Mismatch	Zero	Single Error (SECDED)	No correction needed
Match	Non-zero	Double Error	Flag error, no correction

28.4.4 Key Functions

28.4.4.1 bit_position(k) Function

Tells you where data bit k ended up in the encoded stream:

```
function automatic integer bit_position(input integer k);
    integer j, pos;
    begin
        pos = k + 1;  // Account for parity bit at position 0
        for (j = 0; j < ParityBits; j++) begin
            if (pos >= (2 ** j)) pos = pos + 1;  // Skip parity
positions
        end
        bit_position = pos - 1;  // Convert to 0-based index
```

```
    end
endfunction
```

28.4.4.2 get_covered_bits(parity_bit) Function

Hands back a bitmask of every position a given parity bit covers:

```
function automatic [TotalWidth-1:0] get_covered_bits(input integer
parity_bit);
    integer j;
    begin
        get_covered_bits = 0;
        // FIXED: Exclude SECDED bit from Hamming parity calculations
        for (j = 0; j < TotalWidth - 1; j++) begin
            if (|(((j + 1) >> parity_bit) & 1))
                get_covered_bits[j] = 1'b1;
        end
    end
endfunction
```

28.4.5 Syndrome Generation

```
always_comb begin : create_syndrome_covered_bits
    if (enable) begin
        for (i = 0; i < ParityBits; i++) begin
            parity_pos = (2 ** i) - 1;  // Parity bit positions
            w_syndrome_in[i] = hamming_data[parity_pos];  // Stored
parity
            w_syndrome[i] = 1'b0;

            w_covered_bits = get_covered_bits(i);
            for (bit_index = 0; bit_index < TotalWidth; bit_index++)
begin
                // Calculate parity over covered bits (excluding
parity bit itself)
                if (w_covered_bits[bit_index] && (bit_index !=
parity_pos))
                    w_syndrome[i] = w_syndrome[i] ^
hamming_data[bit_index];
            end

            // Compare with stored parity to get syndrome
            w_syndrome[i] = w_syndrome[i] ^ w_syndrome_in[i];
        end
    end
end
```

28.4.6 Overall Parity Check

```
always_comb begin : check_overall_parity
    w_overall_parity_in = hamming_data[TotalWidth-1];  // SECDED bit
    if (enable) begin
        w_overall_parity = ^hamming_data[TotalWidth-2:0];  //
Calculated parity
    end else begin
        w_overall_parity = 'b0;
    end
end
```

28.4.7 Control States

There's no fancy FSM here — just `enable` gating two logical states:

1. **IDLE**: `enable = 0` - No processing
2. **DECODE**: `enable = 1` - Active decoding and error correction

28.4.8 Error Correction Logic

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_data_with_parity <= 'b0;
        error_detected <= 'b0;
        double_error_detected <= 'b0;
    end else if (enable) begin
        r_data_with_parity <= hamming_data;  // Default assignment
        error_detected <= 'b0;
        double_error_detected <= 'b0;

        // SECDED error detection logic
        if (w_overall_parity != w_overall_parity_in) begin
            error_detected <= 1'b1;

            if (w_syndrome != {ParityBits{1'b0}}) begin
                // Single-bit error in Hamming data - correct it
                r_data_with_parity[w_syndrome_0_based] <=
                    ~hamming_data[w_syndrome_0_based];
            end
            // Single-bit error in SECDED bit - no correction needed

        end else if (w_syndrome != {ParityBits{1'b0}}) begin
            // Double-bit error detected
            error_detected <= 1'b1;
            double_error_detected <= 1'b1;
        end
```

```
    end
end
```

28.4.9 Data Extraction

```
genvar j;
generate
    for (j = 0; j < WIDTH; j++) begin : gen_block
        assign data[j] = r_data_with_parity[bit_position(j)];
    end
endgenerate
```

That peels just the original data bits out of the corrected encoded word.

28.4.10 Error Handling Details

Syndrome interpretation: - **Syndrome** = **0**: No error in the Hamming bits - **Syndrome** ≠**0**: Points to the error bit position (1-based) - **Zero-based Syndrome**: `w_syndrome_0_based = w_syndrome - 1`

The SECDED bit is what lets you tell apart: - Single-bit errors (correctable) - Double-bit errors (detectable, but you're out of luck on correction) - No errors

The correction sequence: 1. Calculate the syndrome from the received parity bits 2. Check overall parity (the SECDED bit) 3. Single error detected? Flip the bit at the syndrome position 4. Double error detected? Flag it, but leave the data alone

28.4.11 Error Correction Capabilities

- **Single Error Correction**: Automatically fixes single-bit errors
- **Double Error Detection**: Flags double-bit errors but can't correct them
- **Error Location**: The syndrome points right at the bad bit
- **SECDED Protection**: The extra parity bit buys you double-error detection

28.4.12 Debug Support

Fair warning: the `DEBUG` parameter is a **no-op** today. The RTL's only nod to it is an empty `initial begin if (DEBUG != 0) begin ... end end` placeholder with no `$display` statements. Turn it on and you'll get exactly no output in the current implementation.

28.5 Timing Characteristics

- **Latency**: One clock cycle for error correction
- **Throughput**: One word per clock cycle
- **Detection Rate**: 100% for single and double errors

- **Critical Path**: Syndrome calculation and error correction
- **Pipeline Stages**: Consider pipelining if you're pushing frequency
- **Area**: Scales with data width and parity requirements
- **Power**: Moderate — the combinational parity math costs you

28.6 Usage Examples

28.6.1 Basic Decoder

```
dataint_ecc_hamming_decode_secded #(
    .WIDTH(8),
    .DEBUG(0)
) hamming_decoder (
    .clk(clk),
    .rst_n(rst_n),
    .enable(decode_enable),
    .hamming_data(encoded_input),
    .data(decoded_data),
    .error_detected(single_or_double_error),
    .double_error_detected(uncorrectable_error)
);
```

28.6.2 Error Handling System

```
always_ff @(posedge clk) begin
    if (error_detected) begin
        if (double_error_detected) begin
            // Log uncorrectable error
            error_log <= error_log + 1;
            // Request retransmission
            request_retransmit <= 1'b1;
        end else begin
            // Log corrected error
            corrected_errors <= corrected_errors + 1;
            // Continue with corrected data
        end
    end
end
```

28.7 Design Notes

28.7.1 Applications

- ECC memory systems

- Communication error correction
- Storage system integrity
- Real-time data processing
- Safety-critical systems
- Network protocol implementation

28.8 Related Modules

- **`dataint_ecc_hamming_encode_secded`**: Corresponding encoder
- Used together for complete ECC system implementation

28.9 Testing

`val/common/test_dataint_ecc_hamming_secded.py` covers this module. It builds an `ecc_secded_wrapper` instantiating the encoder and decoder together and drives injected single- and double-bit errors through the pair.

The test is named for the PAIR rather than the module, which is why an earlier audit looking only for `val/**/test_dataint_ecc_hamming_decode_secded.py` reported it as having no coverage. It does.

Treat any behaviour described on this page as unverified by simulation.

28.10 References

- Hamming, R.W. “Error Detecting and Error Correcting Codes.” *Bell System Technical Journal*, 29(2), Apr. 1950. DOI: 10.1002/j.1538-7305.1950.tb00463.x — syndrome-as-error-position decoding plus the overall-parity check that separates single from double errors.
- Hsiao, M.Y. “A Class of Optimal Minimum Odd-weight-column SEC-DED Codes.” *IBM Journal of Research and Development*, 14(4), July 1970. DOI: 10.1147/rd.144.0395 — the odd-weight-column SECDED alternative; this decoder is extended Hamming, not Hsiao.
- Moon, T.K. *Error Correction Coding*, 2nd ed., Wiley, 2021, Sec. 1.9 “Hamming Codes”; ch. 3 “Linear Block Codes” (Sec. 3.6 Binary Hamming Codes, Sec. 3.9 Modifications to Linear Codes, the extended code) (textbook treatment; secondary source).

- Indexed in `rtl/common/references/README.md` (local PDF: `HammingCode__Hamming1950.pdf`).

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29 dataint_ecc_hamming_encode_secded

29.1 Overview

The encoder half of a SECDED Hamming scheme (Single Error Correction, Double Error Detection). It generates the check bits that let the far end detect up to 2-bit errors and correct single-bit errors in the data.

29.1.1 Module Declaration

```
module dataint_ecc_hamming_encode_secded #(
    parameter int WIDTH = 4,
    parameter int DEBUG = 0
) (
    input  logic [     WIDTH-1:0] data,
    output logic [TotalWidth-1:0] encoded_data
);
```

29.2 Parameters

Parameter	Default	Description
WIDTH	4	Width of input data in bits
DEBUG	0	Enable debug output (1=enabled, 0=disabled)

29.2.1 Calculated Parameters

Parameter	Formula	Description
ParityBits	$clog2(WIDTH + $clog2(WIDTH) + 1)	Number of Hamming parity bits
TotalWidth	WIDTH + ParityBits + 1	Total encoded data width (includes SECDED bit)

29.2.2 Example Configurations

4-bit data (the common example): - Data bits: 4 - Parity bits: 3 (positions 0, 1, 3) - SECDED bit: 1 (position 7) - Total width: 8 bits

8-bit data: - Data bits: 8 - Parity bits: 4 (positions 0, 1, 3, 7) - SECDED bit: 1 (position 12) - Total width: 13 bits

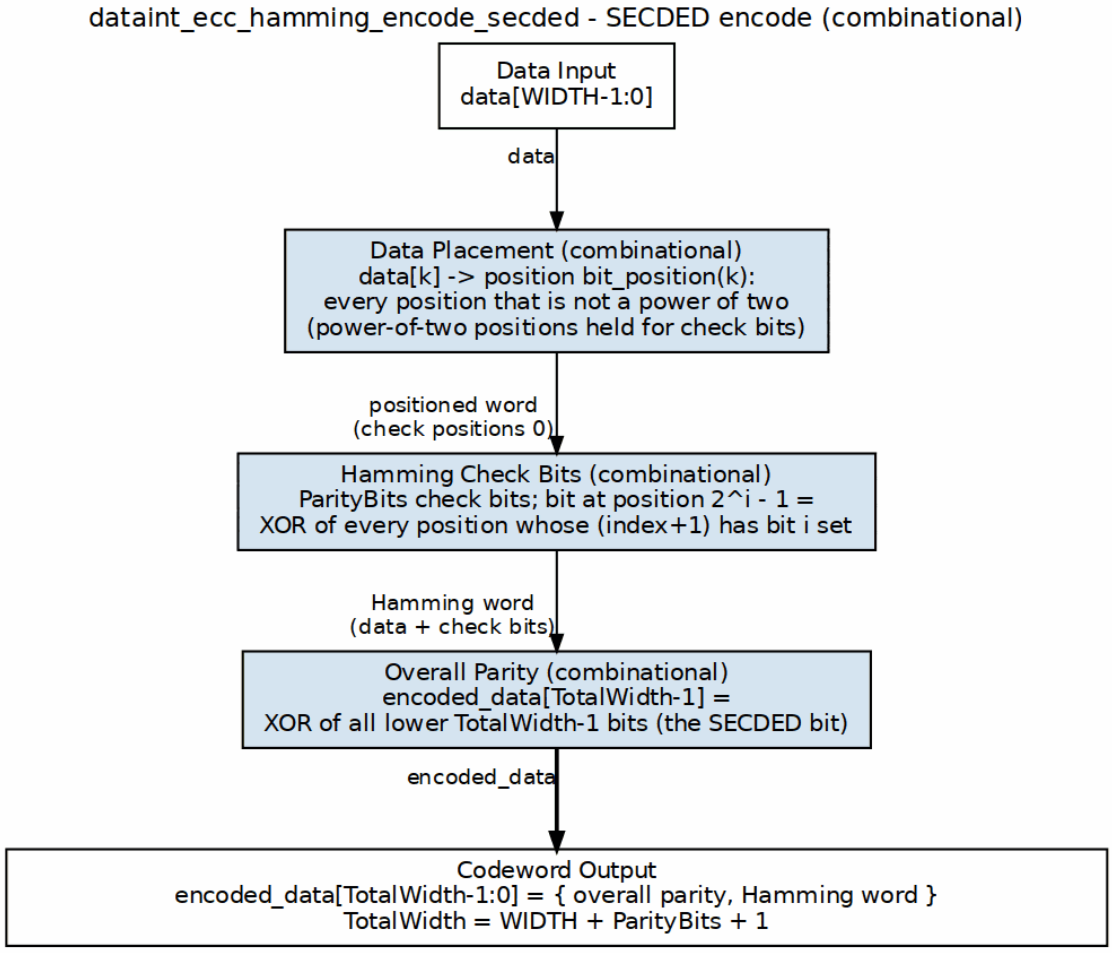
16-bit data: - Data bits: 16 - Parity bits: 5 (positions 0, 1, 3, 7, 15) - SECDED bit: 1 (position 21) - Total width: 22 bits

29.3 Ports

Port	Direction	Width	Description
`data`	Input	WIDTH	Original data to be encoded
`encoded_data`	Output	TotalWidth	Encoded data with parity and SECDED bits

29.4 Functional Description

29.4.1 Block diagram



dataint_ecc_hamming_encode_secded block diagram

The mirror of the SECDED decoder, all combinational. The data bits are first placed into every codeword position that is not a power of two, leaving the power-of-two positions empty. Each Hamming check bit at position 2^i - 1 is then the XOR of every position whose (index + 1) has bit i

set. Finally, the overall parity of all those bits is appended as the MSB, which is what lets the decoder tell a single error from a double error.

29.4.2 Hamming Code Structure

The encoded word carries three things: 1. **Data Bits**: Your original data, parked at specific positions 2. **Parity Bits**: Hamming parity bits at the power-of-2 positions (1, 2, 4, 8, ...) 3. **SECDED Bit**: One extra overall parity bit for double error detection

29.4.3 Bit Position Layout

```
Position: 1  2  3  4  5  6  7  8  9  10 11 12 ...
Content:  P1 P2 D1 P4 D2 D3 D4 P8 D5 D6 D7 D8 ...

Where: P = Parity bit, D = Data bit
```

29.4.4 Key Functions

29.4.4.1 bit_position(k) Function

Works out where data bit k belongs in the encoded stream:

```
function automatic integer bit_position(input integer k);
    integer j, pos;
    begin
        pos = k + 1;  // Start at k+1 to account for parity at
position 0
        for (j = 0; j < ParityBits; j++) begin
            if (pos >= (2 ** j)) pos = pos + 1;  // Skip parity
positions
        end
        bit_position = pos - 1;  // Convert to 0-based index
    end
endfunction
```

29.4.4.2 get_covered_bits(parity_bit) Function

Returns a bitmask of every position a specific parity bit covers:

```
function automatic [TotalWidth-1:0] get_covered_bits(input integer
parity_bit);
    integer j;
    begin
        get_covered_bits = 'b0;
        for (j = 0; j < TotalWidth; j++) begin
            // Check if position j+1 has parity_bit set in its binary
representation
            if (|(((j + 1) >> parity_bit) & 1))
                get_covered_bits[j] = 1'b1;
        end
```

```
    end
endfunction
```

29.4.5 Encoding Algorithm

29.4.5.1 Step 1: Data Placement

```
// Initialize with zeros
w_data_with_parity = {TotalWidth{1'b0}};

// Insert data bits into correct positions
for (i = 0; i < WIDTH; i++) begin
    w_data_with_parity[bit_position(i)] = data[i];
end
```

29.4.5.2 Step 2: Parity Calculation

```
// Calculate each Hamming parity bit
for (i = 0; i < ParityBits; i++) begin
    parity_pos = (2 ** i) - 1;  // Positions 0, 1, 3, 7, 15, ...
    w_data_with_parity[parity_pos] = 1'b0;  // Initialize

    // XOR all covered positions
    w_covered_bits = get_covered_bits(i);
    for (bit_index = 0; bit_index < TotalWidth; bit_index++) begin
        if (w_covered_bits[bit_index]) begin
            w_data_with_parity[parity_pos] =
                w_data_with_parity[parity_pos] ^
w_data_with_parity[bit_index];
        end
    end
end
```

29.4.5.3 Step 3: SECDED Bit Calculation

```
// Calculate overall parity for SECDED
w_data_with_parity[TotalWidth-1] = ^w_data_with_parity[TotalWidth-
2:0];
```

29.4.6 Error Correction Capabilities

Single error correction: - Can correct any single-bit error in the encoded data - Uses the Hamming parity bits to locate the error - The decoder flips the bad bit

Double error detection: - The SECDED bit is what spots double-bit errors - Can’t fix them, but it flags them - No silent data corruption

Error Type	Detection	Correction
No Error	Yes	N/A
Single Bit Error	Yes	Yes

Error Type	Detection	Correction
Double Bit Error	Yes	No
Triple+ Bit Error	Partial	No

29.4.7 Combinational Design

- **No Clock Required**: Purely combinational logic
- **Immediate Output**: Encoded data shows up as soon as the input settles
- **Stateless Operation**: No internal state, nothing to reset

29.4.8 Scalable Architecture

- **Parameterizable Width**: Any data width you like
- **Automatic Sizing**: The parity bit count is worked out for you
- **Efficient Layout**: Optimal bit positioning for the Hamming code

29.4.9 Debug Support

Note that `DEBUG` is a **no-op** in this module: the RTL never references it outside its parameter declaration, so there's no debug output to be had. The parameter is reserved for future use.

29.5 Timing Characteristics

This module is **purely combinational** – it contains no `always_ff` and no latch, so it holds no state and adds no clock cycles. Its outputs settle a propagation delay after its inputs, and it introduces no latency into a pipeline that instantiates it.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

29.6 Usage Examples

29.6.1 Basic Encoding

```
// 8-bit data encoder
dataint_ecc_hamming_encode_secded #(
    .WIDTH(8),
    .DEBUG(0)
) hamming_encoder (
    .data(input_data),
```

```
    .encoded_data(ecc_encoded_data)
);
```

29.6.2 With Debug

```
// Debug-enabled encoder for verification
dataint_ecc_hamming_encode_secded #(
    .WIDTH(4),
    .DEBUG(1)
) debug_encoder (
    .data(test_data),
    .encoded_data(encoded_output)
);
```

29.7 Design Notes

29.7.1 Applications

- Memory systems (ECC RAM)
- Communication protocols
- Storage systems
- Critical data processing
- Aerospace and automotive systems
- Any application requiring data integrity

29.8 Related Modules

- **`dataint_ecc_hamming_decode_secded`**: Corresponding decoder module
- Used together for complete SECDED error correction system

29.9 Testing

`val/common/test_dataint_ecc_hamming_secded.py` covers this module. It builds an `ecc_secded_wrapper` instantiating the encoder and decoder together and drives injected single- and double-bit errors through the pair.

The test is named for the PAIR rather than the module, which is why an earlier audit looking only for `val/**/test_dataint_ecc_hamming_encode_secded.py` reported it as having no coverage. It does.

Treat any behaviour described on this page as unverified by simulation.

29.10 References

- Hamming, R.W. “Error Detecting and Error Correcting Codes.” *Bell System Technical Journal*, 29(2), Apr. 1950. DOI: 10.1002/j.1538-7305.1950.tb00463.x – defines the power-of-two check-bit positions and the added overall parity bit (distance 4, SECDED) that this encoder produces.
- Hsiao, M.Y. “A Class of Optimal Minimum Odd-weight-column SEC-DED Codes.” *IBM Journal of Research and Development*, 14(4), July 1970. DOI: 10.1147/rd.144.0395 – the odd-weight-column SECDED alternative; this module is extended Hamming, not Hsiao.
- Moon, T.K. *Error Correction Coding*, 2nd ed., Wiley, 2021, Sec. 1.9 “Hamming Codes”; ch. 3 “Linear Block Codes” (Sec. 3.6 Binary Hamming Codes, Sec. 3.9 Modifications to Linear Codes, the extended code) (textbook treatment; secondary source).
- Indexed in `rtl/common/references/README.md` (local PDF: `HammingCode__Hamming1950.pdf`).

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30 dataint_parity

30.1 Overview

A generic parity generator and checker in one. It computes parity bits over data chunks and verifies incoming parity against them, supporting both even and odd schemes across as many data segments as you carve your bus into.

30.1.1 Module Declaration

```
module dataint_parity #(
    parameter int CHUNKS = 4,  // Number of chunks to check parity
    parameter int WIDTH  = 32  // Total width of the data
) (
    input  logic [ WIDTH-1:0] data_in,      // Data input
    input  logic [CHUNKS-1:0] parity_in,    // Parity input for
checking
    input  logic              parity_type,  // 1=even, 0=odd
    output logic [CHUNKS-1:0] parity,       // Generated parity bits
    output logic [CHUNKS-1:0] parity_err    // Error indicators
);
```

30.2 Parameters

Parameter	Default	Description
CHUNKS	4	Number of data chunks to process
WIDTH	32	Total width of input data in bits

30.2.1 Calculated Parameters

Parameter	Formula	Description
ChunkSize	WIDTH / CHUNKS	Size of each chunk in bits
ExtraBits	WIDTH % CHUNKS	Remaining bits if WIDTH not evenly divisible

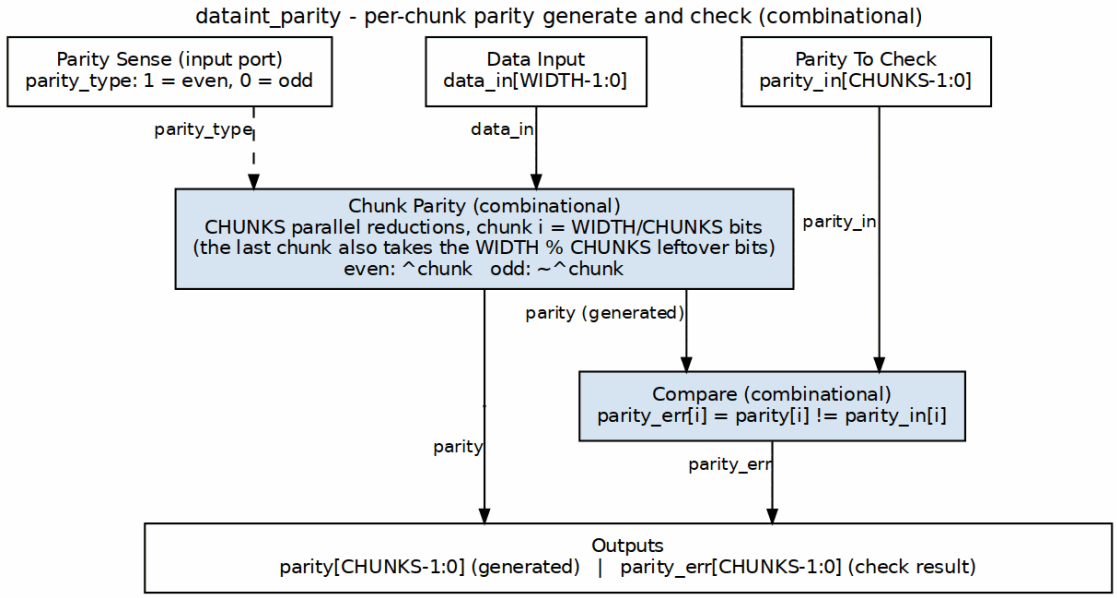
30.3 Ports

Port	Direction	Width	Description
data_in	Input	WIDTH	Input data to calculate/check parity

Port	Direction	Width	Description
`parity_in`	Input	CHUNKS	Expected parity bits for verification
`parity_type`	Input	1	Parity type (1=even, 0=odd)
`parity`	Output	CHUNKS	Calculated parity bits
`parity_err`	Output	CHUNKS	Error flags (1=parity mismatch)

30.4 Functional Description

30.4.1 Block diagram



dataint_parity block diagram

All combinational. `data_in` is split into `CHUNKS` chunks, and each chunk is reduced to one parity bit, XOR for even parity or XNOR for odd, as `parity_type` selects at run time. The generated bits leave as `parity`, which is the generate use. The same bits are compared with the supplied `parity_in` to produce `parity_err`, which is the check use.

30.4.2 Dual Operation Modes

The module does both jobs at once: 1. **Generates Parity**: Calculates parity for each data chunk 2. **Checks Parity**: Compares that calculated parity with the input parity

30.4.3 Parity Types

- **Even Parity** (`parity_type = 1`): The parity bit makes the total 1s even
- **Odd Parity** (`parity_type = 0`): The parity bit makes the total 1s odd

30.4.4 Data Chunking

Data gets divided into chunks, with a small wrinkle for widths that don't divide evenly: - Most chunks: `ChunkSize` bits each - The last chunk picks up any extra bits when `WIDTH % CHUNKS ≠ 0`

30.4.5 Generate Block Architecture

```
genvar i;
generate
    for (i = 0; i < CHUNKS; i++) begin : gen_parity
        // Calculate bounds for each chunk
        localparam int LowerBound = i * ChunkSize;
        localparam int UpperBound = (i < CHUNKS - 1) ?
            ((i + 1) * ChunkSize) - 1 : WIDTH - 1;

        // Calculate parity for this chunk
        wire calculated_parity = parity_type ?
            ^data_in[UpperBound:LowerBound] :      // Even parity
            ~^data_in[UpperBound:LowerBound];      // Odd parity

        assign parity[i] = calculated_parity;
        assign parity_err[i] = (calculated_parity != parity_in[i]);
    end
endgenerate
```

30.4.6 Chunk Boundary Calculation

Each chunk's bounds are worked out statically: - **Regular Chunks** (i < CHUNKS-1): Fixed-size chunks - **Last Chunk** (i = CHUNKS-1): Mops up any remainder bits

30.4.7 Parity Calculation Logic

For each chunk: 1. **XOR Reduction**: `^data_in[UpperBound:LowerBound]` 2. **Parity Type Application**: - Even parity: take the XOR result as-is - Odd parity: invert it

30.4.8 Error Detection

Error checking is just a compare — calculated against expected:

```
assign parity_err[i] = (calculated_parity != parity_in[i]);
```

30.4.9 Example Configurations

32-bit data, 4 chunks (8 bits each):

```
Chunk 0: data_in[7:0]    → parity[0], parity_err[0]
Chunk 1: data_in[15:8]   → parity[1], parity_err[1]
Chunk 2: data_in[23:16]  → parity[2], parity_err[2]
Chunk 3: data_in[31:24]  → parity[3], parity_err[3]
```

30-bit data, 4 chunks (uneven division):

```
Chunk 0: data_in[6:0]    (7 bits)  → parity[0], parity_err[0]
Chunk 1: data_in[13:7]   (7 bits)  → parity[1], parity_err[1]
Chunk 2: data_in[20:14]  (7 bits)  → parity[2], parity_err[2]
Chunk 3: data_in[29:21]  (9 bits)  → parity[3], parity_err[3]
```

30.4.10 Key Features

- **Scalable Architecture**: Parameterizable chunk count, flexible data width, uneven divisions handled for you
- **Simultaneous Operation**: Generation and checking run concurrently, each chunk minds its own business, and all chunks evaluate at the same time
- **Parity Flexibility**: You can flip the parity type during operation; all chunks share the same parity type; both common parity schemes covered

30.5 Related Modules

Nothing in the tree instantiates this module and it instantiates nothing: it is a leaf, used directly by whatever design needs it. Its nearest neighbours in `rtl/common/` are:

- `dataint_checksum`
- `dataint_crc`
- `dataint_crc_xor_shift`
- `dataint_crc_xor_shift_cascade`
- `dataint_ecc_hamming_decode_secded`

30.6 Timing Characteristics

- **Combinational Logic**: Zero clock delay

- **Critical Path**: XOR tree depth ≈ $\log_2$(ChunkSize)
- **Propagation Delay**: Minimal for typical chunk sizes
- **Area**: Grows linearly with CHUNKS and WIDTH; just XOR trees, so very little logic per chunk
- **Power**: Low — simple combinational logic, and power tracks the data switching

30.7 Usage Examples

30.7.1 Basic Parity Generation

```
dataint_parity #(
    .CHUNKS(8),
    .WIDTH(64)
) parity_gen (
    .data_in(input_data),
    .parity_in(8'b0),           // Not used for generation
    .parity_type(1'b1),         // Even parity
    .parity(generated_parity),
    .parity_err()               // Not used for generation
);
```

30.7.2 Parity Checking

```
dataint_parity #(
    .CHUNKS(4),
    .WIDTH(32)
) parity_check (
    .data_in(received_data),
    .parity_in(received_parity),
    .parity_type(1'b0),         // Odd parity
    .parity(recalc_parity),
    .parity_err(error_flags)
);

// Check for any errors
wire any_parity_error = |error_flags;
```

30.7.3 Error Correction System

```
logic [CHUNKS-1:0] error_location;
logic [WIDTH-1:0]  corrected_data;

dataint_parity #(.CHUNKS(CHUNKS), .WIDTH(WIDTH)) parity_checker (
    .data_in(received_data),
```

```
    .parity_in(received_parity),
    .parity_type(even_parity),
    .parity(recalc_parity),
    .parity_err(error_location)
);

// Simple error correction (single-bit errors only)
always_comb begin
    corrected_data = received_data;
    for (int i = 0; i < CHUNKS; i++) begin
        if (error_location[i]) begin
            // Flip bits in the erroneous chunk
            // (Simplified - real correction needs more sophisticated
logic)
        end
    end
end
```

30.8 Design Notes

30.8.1 Applications

- Memory parity checking
- Communication protocol verification
- Data bus integrity monitoring
- Storage system error detection
- Real-time parity generation and concurrent error checking
- Multi-segment data validation and protocol compliance verification
- Building block for ECC systems and larger error correction schemes
- Interface protection, debug and validation tools

Related application areas: - **ECC Systems**: Building block for Hamming codes - **Network Protocols**: Ethernet, UART parity - **Memory Systems**: DRAM/SRAM parity protection - **Communication**: RS-232, SPI, I2C error detection

30.8.2 Chunk Size Selection

- **Power of 2**: Usually the friendliest for alignment
- **Protocol Requirements**: Match whatever your system demands

- **Error Granularity**: Smaller chunks pin errors down tighter

30.8.3 Parity Type Selection

- **Even Parity**: The more common choice in digital systems
- **Odd Parity**: Gives you all-zeros detection
- **System Compatibility**: Match the existing standard

30.8.4 Width Considerations

- **Alignment**: Think about data bus alignment
- **Extra Bits**: Handle remainder bits sensibly
- **Performance**: Balance chunk count against granularity

30.9 Testing

`val/common/test_dataint_parity.py` exercises this module. It collects 16 parameter cases at the default `REG_LEVEL`.

```
source env_python
pytest val/common/test_dataint_parity.py -v
```

30.10 References

- Hamming, R.W. “Error Detecting and Error Correcting Codes.” *Bell System Technical Journal*, 29(2), Apr. 1950. DOI: 10.1002/j.1538-7305.1950.tb00463.x — the single parity check is the distance-2 starting point of Hamming’s construction; per-chunk even/odd parity itself is a standard technique with no single origin.
- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 3 “Basic Combinational Circuits”, Sec. 3.13 Parity Generators/Checkers (textbook treatment; secondary source).
- Indexed in `rtl/common/references/README.md` (local PDF: `HammingCode__Hamming1950.pdf`).

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31 debounce

31.1 Overview

Kills mechanical switch bounce by sampling button inputs on a regular tick and refusing to believe a button until the same state shows up for several consecutive samples.

31.2 Parameters

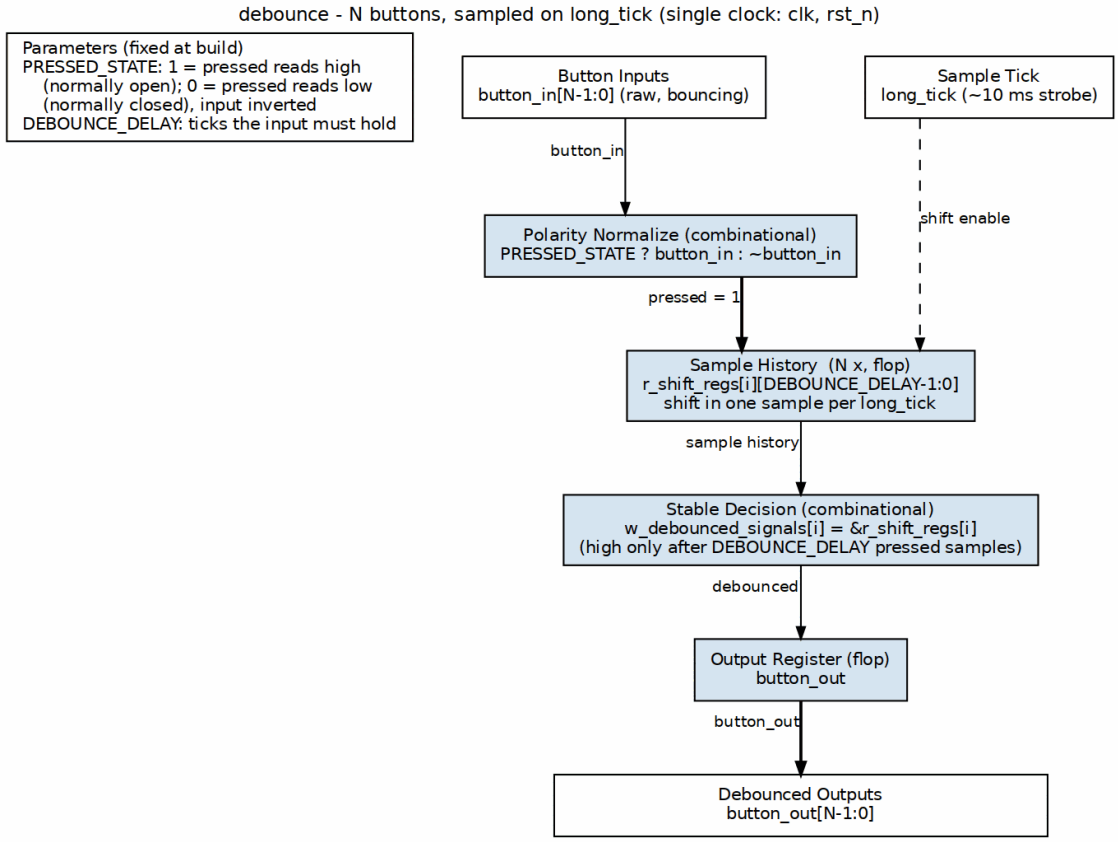
Parameter	Default	Description
N	4	Number of buttons/input signals
DEBOUNCE_DELAY	4	Number of consecutive stable samples required
PRESSED_STATE	1	Logic level when button is pressed (1 for normally open, 0 for normally closed)

31.3 Ports

Port	Direction	Width	Description
clk	Input	1	System clock signal
rst_n	Input	1	Active-low reset signal
long_tick	Input	1	~10ms sampling tick signal (controls sampling rate)
button_in	Input	N	Raw button input signals to be debounced
button_out	Output	N	Debounced button output signals

31.4 Functional Description

31.4.1 Block diagram



debounce block diagram

Each button is first normalized by `PRESSED_STATE`, so a press always reads as 1. On every `long_tick`, that sample shifts into the button's `DEBOUNCE_DELAY`-deep history register. The stable decision is an AND of the whole history, so the output goes high only after `DEBOUNCE_DELAY` consecutive pressed samples, and drops on the first released one. The result is registered as `button_out`. The `N` buttons are identical, independent channels.

31.4.2 Core Algorithm

The module tracks each button's recent history in a shift register:

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        for (int i = 0; i < N; i++) begin
            r_shift_regs[i] <= {DEBOUNCE_DELAY{1'b0}};
        end
    end else if (long_tick) begin
        for (int i = 0; i < N; i++) begin
```

```
            r_shift_regs[i] <= {
                r_shift_regs[i][DEBOUNCE_DELAY-2:0],
                PRESSED_STATE ? button_in[i] : ~button_in[i]
            };
        end
    end
end
```

31.4.3 Sampling Control

- Only samples inputs when `long_tick` fires
- Over-sampling invites false triggering — this avoids it
- Typical `long_tick` period: ~10ms

31.4.4 Button Type Support

- **Normally Open (NO)**: `PRESSED_STATE = 1`
    - Button reads ‘0’ when not pressed, ‘1’ when pressed
- **Normally Closed (NC)**: `PRESSED_STATE = 0`
    - Button reads ‘1’ when not pressed, ‘0’ when pressed
- The input inversion handles NC buttons transparently

31.4.5 Debounce Logic

- Keeps `DEBOUNCE_DELAY` samples per button
- The output only goes high when every sample in the shift register is ‘1’
- Which means the button has to sit stable in the pressed state for the whole delay

```
always_comb begin
    for (int i = 0; i < N; i++) begin
        // Output 1 when shift register shows stable pressed state
(all 1s)
        w_debounced_signals[i] = &r_shift_regs[i];
    end
end
```

31.4.6 Output Registration

- The final output is registered, so transitions come out clean and glitch-free

- Resets to all zeros on system reset

31.5 Timing Characteristics

31.5.1 Debounce Delay

- **Total delay** = `DEBOUNCE_DELAY` × `long_tick` period
- **Default**: 4 × 10ms = 40ms debounce time
- **Minimum stable time**: The button has to hold steady for the full delay period

31.5.2 Response Time

- **Press detection**: `DEBOUNCE_DELAY` ticks after the button stabilizes high (the output is `&r_shift_regs[i]`, so it needs `DEBOUNCE_DELAY` consecutive high samples to assert)
- **Release detection**: ~1 tick (+1 clk). The output is an AND-reduce, so the first `0` shifted in on release immediately clears it — release is **not** debounced beyond a single sample.
- **Asymmetric**: press is fully debounced; release propagates ~`DEBOUNCE_DELAY`× faster. Do NOT rely on symmetric debounce timing.

31.6 Usage Examples

Every parameter and port below is taken from the module declaration.

```
debounce #(
    .N                  (4),
    .DEBOUNCE_DELAY     (4),
    .PRESSED_STATE      (1)
) u_debounce (
    .clk                (clk),
    .rst_n              (rst_n),
    .long_tick          (long_tick),
    .button_in          (button_in),
    .button_out         (button_out)
);
```

31.7 Design Notes

31.7.1 Applications

- Mechanical pushbuttons and switches
- Rotary encoder inputs (with the delay set appropriately)
- Any digital input prone to contact bounce
- Both normally open and normally closed switch types

31.7.2 Design Considerations

- **Delay tuning**: Set `DEBOUNCE_DELAY` to match your switch's characteristics
- **Tick frequency**: `long_tick` should run much slower than the bounce duration
- **Multiple buttons**: All buttons share the same debounce parameters
- **Reset behavior**: All outputs go low on reset, no matter what the buttons are doing

31.8 Related Modules

Read out of the RTL, not curated: these are the modules this one instantiates and the modules that instantiate it.

Instantiated by: - `cdc_counter_display_top`

31.9 Testing

`val/common/test_debounce.py` exercises this module. It collects 4 parameter cases at the default `REG_LEVEL`.

```
source env_python
pytest val/common/test_debounce.py -v
```

31.10 References

- Ganssle, J. "A Guide to Debouncing." Ganssle Group, Aug. 2004 (rev. 4, Mar. 2014), ganssle.com/debouncing.htm — canonical non-peer

measurement study of contact bounce and of the sample-until-N-consecutive-equal-readings debounce this module implements.

- LaMeres, B.J. *Introduction to Logic Circuits & Logic Design with Verilog*, Springer, ch. 7 “Sequential Logic Design”, §7.3.3 “Switch Debouncing” — textbook treatment (secondary source).
- Merrick, R. *Getting Started with FPGAs*, No Starch Press, ch. 5 “Testing Your Code with Simulation”, “Project #4: Debouncing a Switch” — textbook treatment (secondary source).
- Indexed in `rtl/common/references/README.md`

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32 decoder

32.1 Overview

Converts an M-bit binary encoded input to a one-hot N-bit output where N = 2^M. Standard binary-to-one-hot decoding — nothing exotic.

32.2 Parameters

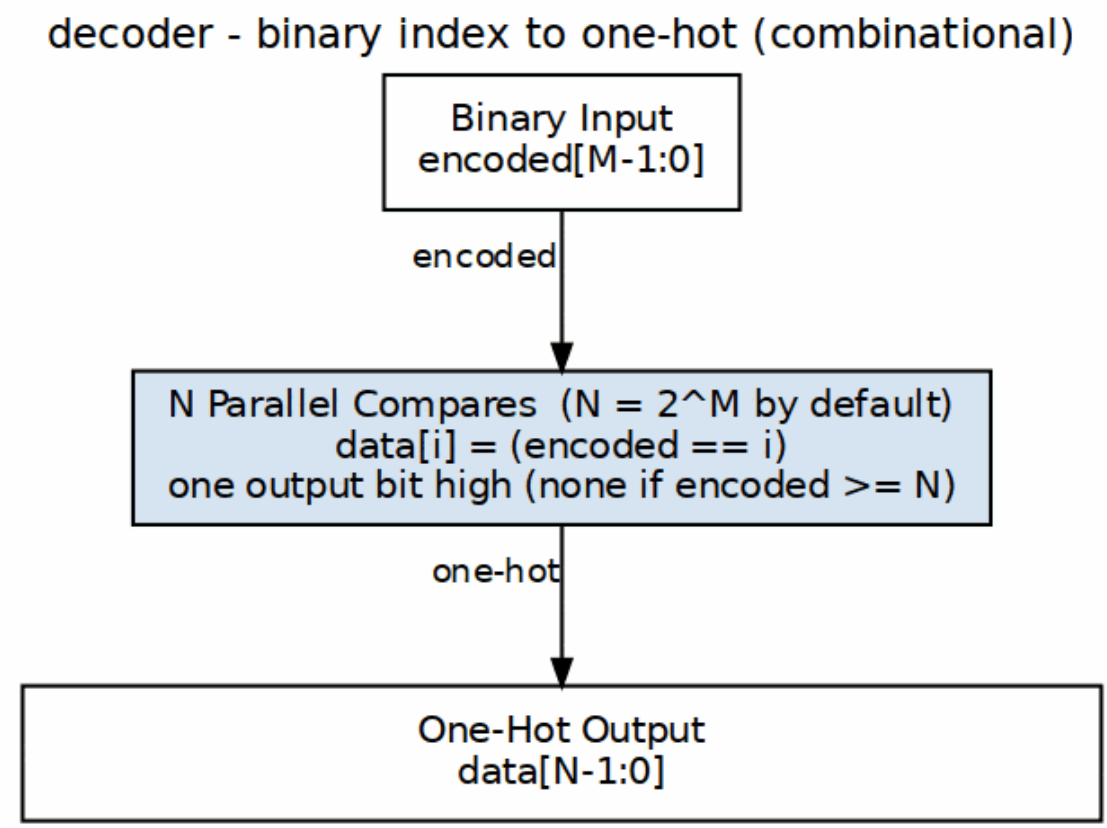
Parameter	Default	Description
`M`	4	Width of binary input
`N`	2^M = 16	Width of one-hot output

32.3 Ports

Port	Direction	Width	Description
`encoded`	Input	M	Binary encoded input value
`data`	Output	N	One-hot decoded output vector

32.4 Functional Description

32.4.1 Block diagram



decoder block diagram

The binary `encoded` input fans out to `N` parallel equality compares, one per output bit, and `data[i]` is high when `encoded == i`. With the default `N = 2^M` exactly one output bit is high; if `N` is set smaller, an `encoded` value of `N` or more leaves every output bit low.

32.4.2 Operation Principle

- Each output bit maps to one possible input value
- Only one output bit is high at any time (that's the one-hot part)
- Output bit `i` is high when the `encoded` input equals `i`

32.4.3 Truth Table Example (M=2, N=4)

encoded[1:0]	data[3]	data[2]	data[1]	data[0]
00	0	0	0	1

encoded[1:0]	data[3]	data[2]	data[1]	data[0]
01	0	0	1	0
10	0	1	0	0
11	1	0	0	0

32.4.4 Key Characteristics

- **Combinational logic**: No clock dependency, immediate response
- **One-hot output**: Exactly one bit high for valid inputs
- **Complete decoding**: All possible input combinations decoded
- **No default initialization**: because `N = 2^M` covers every input value, exactly one output is high at all times — the outputs are never all-zero, and the RTL contains no default assignment

32.4.5 Core Logic

A generate loop spins up one comparator per output bit:

```
genvar i;
generate
    for (i = 0; i < N; i++) begin : gen_DECODER_LOOP
        assign data[i] = (encoded == i) ? 1'b1 : 1'b0;
    end
endgenerate
```

32.4.6 Parameterization

- **Scalable width**: M sets the input width
- **Automatic sizing**: N falls out as 2^M
- **Generate loops**: Synthesizes cleanly for any size

32.5 Timing Characteristics

- **Setup/hold**: None (purely combinational)
- **Output changes**: Follow the inputs immediately

32.6 Usage Examples

32.6.1 Memory Address Decoding

```
decoder #(.M(2), .N(4)) addr_decoder (
    .encoded(address[1:0]),
    .data(chip_select)  // 4 chip selects
);
```

32.6.2 State Machine Output Decoding

```
decoder #(.M(3), .N(8)) state_decoder (
    .encoded(current_state),
    .data(state_outputs)  // 8 control signals
);
```

32.7 Design Notes

32.7.1 Applications

- **Address decoding**: Memory or register select signals
- **State machine outputs**: Decode binary state to control signals
- **Multiplexer control**: Generate select signals for data routing
- **Interrupt controllers**: Decode interrupt vectors
- **Bus decoding**: Generate chip select signals

32.7.2 Synthesis Considerations

- **Resource usage**: Typically lands in LUT-based logic
- **Propagation delay**: Single LUT delay in most FPGA architectures
- **Fan-out**: Each input bit feeds multiple output comparisons

32.7.3 Other Notes

- No error checking for invalid inputs (though all combinations are valid)
- Purely combinational: the generate loop drives every `data[i]` exactly once, so all outputs are defined at all times (no separate initialization needed — and none exists in the RTL)
- The generate loop structure scales efficiently to any required size

32.8 Related Modules

- **Encoder**: Performs inverse operation (one-hot to binary)
- **Priority Encoder**: Handles multiple simultaneous inputs
- **Multiplexer**: Often used together for data routing

32.9 Testing

`val/common/test_decoder.py` exercises this module. It collects 3 parameter cases at the default `REG_LEVEL`.

```
source env_python
pytest val/common/test_decoder.py -v
```

32.10 References

- Harris, S.L., Harris, D. *Digital Design and Computer Architecture, RISC-V Edition*, Morgan Kaufmann, 2022, Sec. 2.8.2 Decoders (textbook treatment; secondary source).
- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 3 “Basic Combinational Circuits”, Sec. 3.8 Decoders (textbook treatment; secondary source).
- LaMeres, B.J. *Introduction to Logic Circuits & Logic Design with Verilog*, 2nd ed., Springer, 2019, ch. 6 “MSI Logic”, Sec. 6.1.1 One-Hot Decoder (textbook treatment; secondary source).
- No peer paper: binary-to-one-hot decoding is a standard technique. Indexed in `rtl/common/references/README.md`.

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33 encoder

33.1 Overview

Converts a one-hot N-bit input back into its binary index — the inverse of a decoder. Feed it a one-hot vector, get out the position of the asserted bit in binary.

33.2 Parameters

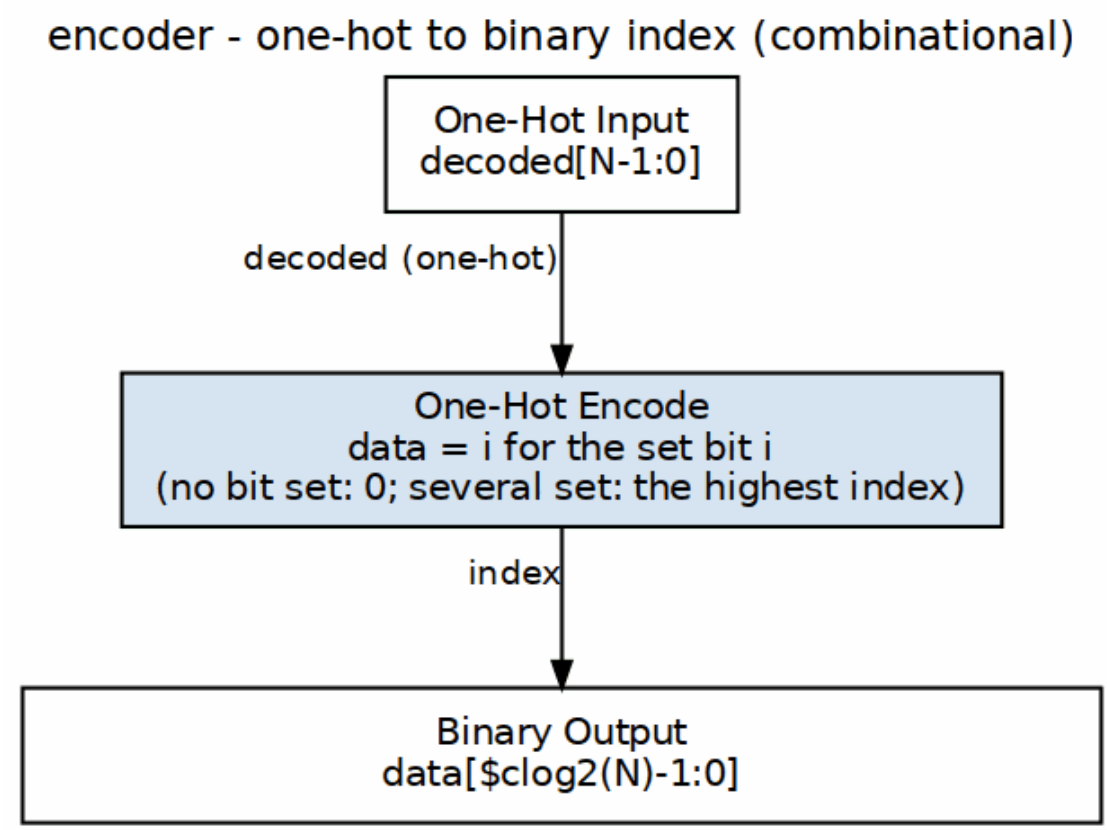
Parameter	Default	Description
N	8	Width of one-hot input vector

33.3 Ports

Port	Direction	Width	Description
decoded	Input	N	One-hot input vector
data	Output	$clog2(N)	Binary encoded output representing the position of the set bit

33.4 Functional Description

33.4.1 Block diagram



encoder block diagram

The one-hot `decoded` input enters at the top and the encoder returns the index of its set bit as `data`. It is drawn as a plain block, not a pick: the input is meant to be one-hot. With no bit set the output is 0, and with several bits set the loop leaves the highest index, so a multi-hot input needs `encoder_priority_enable` instead.

33.4.2 Operation Principle

- Scans the input vector from LSB to MSB
- Outputs the binary representation of the highest indexed set bit
- Which makes it a **priority encoder** when multiple bits are set (higher index wins)

33.4.3 Truth Table Example (N=8, output width=3)

decoded[7:0]	data[2:0]	Notes
00000001	000	Bit 0 set
00000010	001	Bit 1 set
00000100	010	Bit 2 set
00001000	011	Bit 3 set
00010000	100	Bit 4 set
00100000	101	Bit 5 set
01000000	110	Bit 6 set
10000000	111	Bit 7 set
00000000	000	No bits set
10000001	111	Multiple bits - highest wins

33.4.4 Key Characteristics

- **Priority encoding**: Higher-indexed bits take precedence
- **Zero default**: Outputs 0 when no input bits are set
- **Combinational**: Immediate response to input changes
- **Self-sizing**: Output width comes from `$clog2(N)`

33.4.5 Core Algorithm

A combinational always block runs the priority search:

```
always_comb begin
    data = 0;  // Default output when no bits set
    for (int i = 0; i < N; i++) begin
        if (decoded[i]) data = $clog2(N)'(i);
    end
end
```

33.4.6 Automatic Width Calculation

- **Output width**: `$clog2(N)` bits
- **Example**: N=8 → 3-bit output, N=16 → 4-bit output
- **Optimal sizing**: No wasted bits in the output

33.4.7 Priority Behavior

The for-loop gives you priority for free: - Later iterations overwrite earlier results - So the highest-indexed set bit decides the final output - For invalid one-hot inputs, it quietly behaves as a priority encoder

33.4.8 Behavioral Notes

- **For-loop order**: Creates LSB-to-MSB priority (higher index wins)
- **Default assignment**: Ensures output has known value for all inputs
- **Type casting**: `$clog2(N)'(i)` ensures proper bit width matching

33.5 Timing Characteristics

- **Propagation delay**: Depends on input width and synthesis
- **Critical path**: Runs through the for-loop comparisons
- **Setup/hold**: None (purely combinational)

33.6 Usage Examples

33.6.1 Interrupt Controller

```
encoder #(.N(8)) int_encoder (
    .decoded(interrupt_pending),
    .data(interrupt_id)
);
```

33.6.2 Arbiter Output Encoding

```
encoder #(.N(16)) grant_encoder (
    .decoded(grant_vector),
    .data(granted_id)
);
```

33.7 Design Notes

33.7.1 Applications

For valid one-hot inputs: - **Interrupt acknowledgment**: Convert interrupt vector to binary ID - **Resource arbitration**: Encode granted request to requester ID - **State machine encoding**: Convert one-hot state to binary - **Position encoding**: Find position of active element

For priority encoding: - **Multiple interrupt handling**: Encode highest priority interrupt - **Error reporting**: Report highest severity error condition - **Resource allocation**: Grant to highest priority requester

33.7.2 Input Validation

- **Assumes one-hot input** for standard encoder behavior
- **Handles multiple bits** gracefully (acts as priority encoder)
- **Zero input** produces zero output (whether that's what you want is up to you)

33.7.3 Synthesis Implications

- **Resource usage**: Typically synthesizes to multiplexer logic
- **Optimization**: Modern synthesizers chew through the for-loop just fine
- **Scalability**: Performance degrades gradually as N grows

33.8 Related Modules

- **Decoder**: Performs inverse operation (binary to one-hot)
- **Priority Encoder with Enable**: Enhanced version with enable control
- **Find First Set**: Similar functionality with different search order

33.9 Testing

`val/common/test_encoder.py` exercises this module. It collects 2 parameter cases at the default `REG_LEVEL`.

```
source env_python
pytest val/common/test_encoder.py -v
```

33.10 References

- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 3 "Basic Combinational Circuits", Sec. 3.9 Encoders (textbook treatment; secondary source).
- LaMeres, B.J. *Introduction to Logic Circuits & Logic Design with Verilog*, 2nd ed., Springer, 2019, ch. 6 "MSI Logic", Sec. 6.2.1 One-Hot Binary Encoder (textbook treatment; secondary source).
- No peer paper: the priority (highest-index-wins) encoder is a standard technique. Indexed in `rtl/common/references/README.md`.

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34 encoder_priority_enable

34.1 Overview

Encodes the highest priority (highest index) asserted bit in the input vector, with an enable control signal. Where the basic encoder gets its priority behavior as a side effect, this one searches MSB to LSB on purpose.

34.2 Parameters

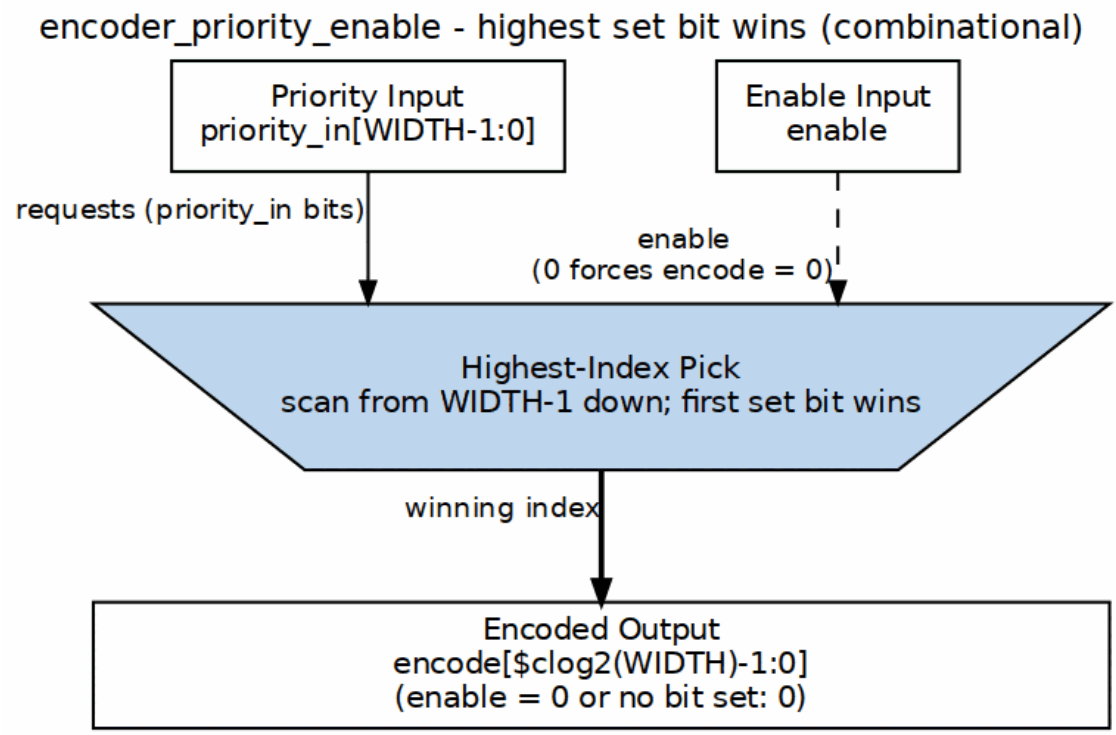
Parameter	Default	Description
WIDTH	8	Input vector width

34.3 Ports

Port	Direction	Width	Description
priority_in	Input	WIDTH	Priority input vector (higher index = higher priority)
enable	Input	1	Enable signal (active high)
encode	Output	$clog2(WIDTH)	Binary encoded output of highest priority bit position

34.4 Functional Description

34.4.1 Block diagram



encoder_priority_enable block diagram

`priority_in` enters at the top and the highest set bit wins. That is a genuine many-to-one pick, so it is drawn as the trapezium, with the winning index leaving its narrow edge as `encode`. `enable` is sideband: when it is low the output is forced to 0, which is also the result when no bit is set.

34.4.2 Operation Principle

- **True priority encoding**: Searches from MSB (highest priority) to LSB
- **Enable control**: Only runs when enable is asserted
- **Found flag**: Blocks multiple assignments, so the highest priority wins
- **Clean disable**: Outputs zero when disabled

34.4.3 Priority Resolution Example (WIDTH=8)

priority_in[7:0]	enable	encode[2:0]	Notes
00000001	1	000	Only bit 0 set
10000001	1	111	Bit 7 has

priority_in[7:0]	enable	encode[2:0]	Notes
			highest priority
01010101	1	110	Bit 6 is highest set
11111111	1	111	Bit 7 wins among all
10000001	0	000	Disabled - output zero
00000000	1	000	No bits set

34.4.4 Key Differences from the Basic Encoder

Feature	Basic Encoder	Priority Encoder with Enable
Search Direction	LSB to MSB	MSB to LSB
Priority	Higher index overwrites	True highest priority
Enable Control	None	Enable signal required
Found Flag	Implicit	Explicit prevention of overwrites

34.4.5 Core Algorithm

```
logic found;

always_comb begin
    // Default assignments to prevent latches
    encode = '0;
    found = 1'b0;

    if (enable == 1'b1) begin
        // Find the highest priority bit using found flag to prevent
overwrites
        for (int i = WIDTH-1; i >= 0; i--) begin
            if (priority_in[i] == 1'b1 && !found) begin
                encode = $clog2(WIDTH)'(i);
                found = 1'b1;
            end
        end
```

```
    end
end
```

34.4.6 True Priority Encoding

- **MSB-first search**: `for (int i = WIDTH-1; i >= 0; i--)`
- **Single assignment**: The found flag stops any overwrites
- **Highest wins**: The first match (highest index) sets the output

34.4.7 Enable Control

- **Conditional operation**: Only active when `enable = 1`
- **Clean disable**: Output forced to zero when disabled
- **Power savings**: Can gate operation when not needed

34.4.8 Latch Prevention

```
// Default assignments to prevent latches
encode = '0;
found = 1'b0;
```

- Gives every signal path a defined value
- Keeps the synthesizer from warning about incomplete assignments

34.5 Timing Characteristics

- **Propagation delay**: Depends on WIDTH and synthesis optimization
- **Critical path**: Through the priority resolution loop
- **Enable response**: Immediate when enable changes
- **Setup/hold**: None (purely combinational)

34.6 Usage Examples

34.6.1 Interrupt Controller

```
encoder_priority_enable #(.WIDTH(8)) irq_encoder (
    .priority_in(interrupt_requests),
    .enable(interrupt_enable),
    .encode(highest_priority_irq)
);
```

34.6.2 Resource Arbitration

```
encoder_priority_enable #(.WIDTH(16)) arbiter (
    .priority_in(request_vector),
    .enable(arbitration_enable),
    .encode(granted_requestor)
);
```

34.6.3 Error Reporting

```
encoder_priority_enable #(.WIDTH(4)) error_encoder (
    .priority_in({fatal_error, severe_error, warning, info}),
    .enable(error_reporting_on),
    .encode(error_level)
);
```

34.7 Design Notes

34.7.1 Applications

Multi-level interrupt controllers: - CPU interrupt prioritization - Nested interrupt handling - Real-time system scheduling

Bus arbitration: - Multi-master bus systems - DMA channel arbitration - Memory access prioritization

Error and exception handling: - Exception priority resolution - Multi-source error reporting - Diagnostic priority encoding

34.7.2 Priority Assignment

- **Bit assignment**: Higher index = higher priority
- **Planning**: Assign critical functions to higher-indexed bits
- **Documentation**: Write the priority hierarchy down — you'll thank yourself

34.7.3 Performance Scaling

- **Linear complexity**: Search time increases with WIDTH
- **Synthesis optimization**: Modern tools optimize priority encoding well
- **Resource usage**: Typically maps to priority encoder primitives

34.7.4 Enable Usage Patterns

- **Power management**: Disable when priority encoding not needed

- **Conditional operation**: Enable based on system state
- **Test/debug**: Disable during certain test modes

34.7.5 Synthesis Considerations

- **Resource efficient**: Maps well to FPGA priority encoder resources
- **Timing closure**: Generally meets timing with proper constraints
- **Area scaling**: Grows approximately linearly with WIDTH

34.8 Related Modules

- **Basic Encoder**: Simpler version without enable or true priority
- **Find Last Set**: Similar MSB-first search functionality
- **Arbiter**: Often uses priority encoders for grant generation

34.9 Testing

`val/common/test_encoder_priority_enable.py` exercises this module. It collects 2 parameter cases at the default `REG_LEVEL`.

```
source env_python
pytest val/common/test_encoder_priority_enable.py -v
```

34.10 References

- Chu, P.P. *FPGA Prototyping by SystemVerilog Examples*, Wiley, 2018, ch. 3 “RT-Level Combinational Circuit”, Sec. 3.12.3 Dual-priority encoder (textbook treatment; secondary source).
- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 3 “Basic Combinational Circuits”, Sec. 3.9 Encoders (textbook treatment; secondary source).
- No peer paper: the priority encoder with enable is a standard technique. Indexed in `rtl/common/references/README.md`.

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35 fifo_control

35.1 Overview

This is the shared brain that generates full/empty status flags for both FIFO variants (sync and async, the latter with either pointer encoding). All the tricky pointer arithmetic and mode-aware timing lives here.

35.2 Parameters

Parameter	Default	Description
ADDR_WIDTH	3	Address width
DEPTH	8	FIFO depth (default was 16 until common BUG-004 (was COMMON-014): 16 against ADDR_WIDTH=3 broke the module's own DEPTH <= 2^ADDR_WIDTH rule and wr_almost_full could never assert)
ALMOST_WR_MARGIN	1	Almost full threshold
ALMOST_RD_MARGIN	1	Almost empty threshold
REGISTERED	0	Output mode: 0=mux, 1=flop

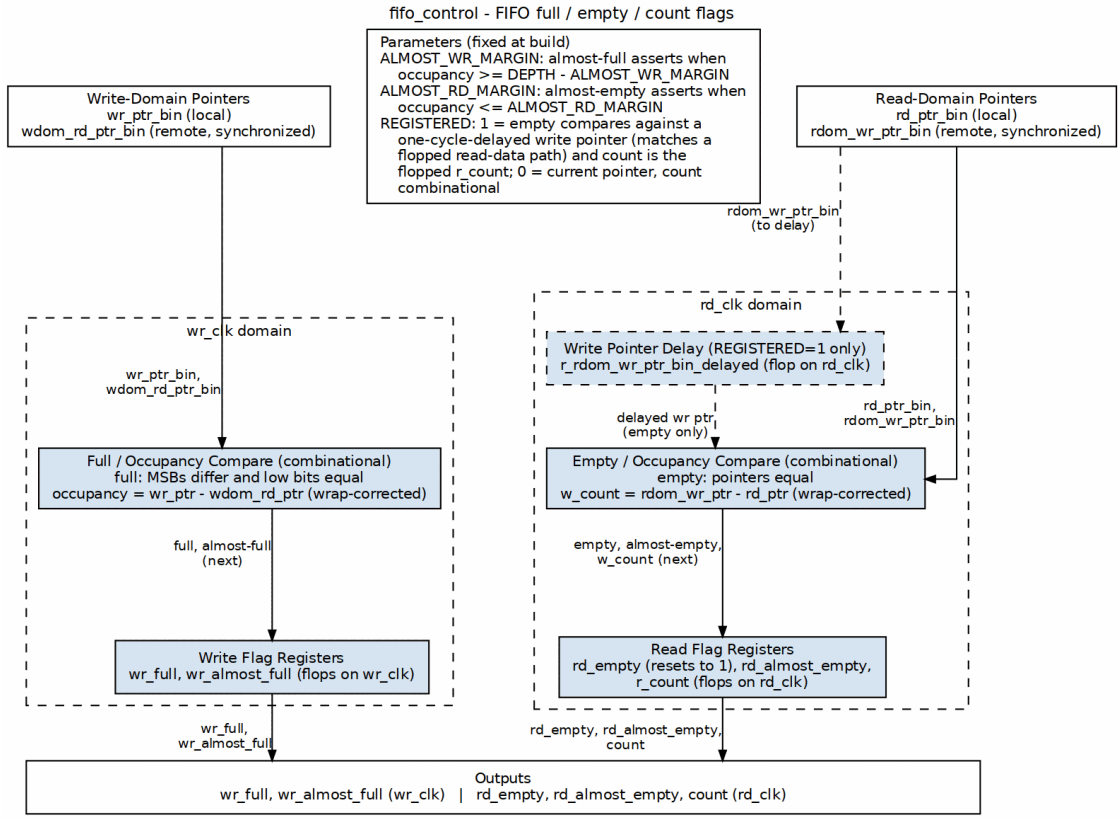
35.3 Ports

Port	Direction	Width	Description
wr_clk	Input	1	Write domain clock
wr_rst_n	Input	1	Write domain active-low reset
rd_clk	Input	1	Read domain clock
rd_rst_n	Input	1	Read domain active-low reset
wr_ptr_bin	Input	ADDR_WIDTH+1	Write pointer (binary, next value)
wdom_rd_ptr_bin	Input	ADDR_WIDTH+1	Read pointer synchronized to write

Port	Direction	Width	Description
			domain
rd_ptr_bin	Input	ADDR_WIDTH+1	Read pointer (binary, next value)
rdom_wr_ptr_bin	Input	ADDR_WIDTH+1	Write pointer synchronized to read domain
count	Output	ADDR_WIDTH+1	Current FIFO occupancy count
wr_full	Output	1	Write domain full flag
wr_almost_full	Output	1	Write domain almost full flag
rd_empty	Output	1	Read domain empty flag
rd_almost_empty	Output	1	Read domain almost empty flag

35.4 Functional Description

35.4.1 Block diagram



fifo_control block diagram

The flag logic shared by `fifo_sync` and `fifo_async`, drawn with the same two clock domains as `fifo_async`. Each side takes its own local pointer and its copy of the remote pointer. That copy is already synchronized by the caller, so nothing crosses inside this block. The write side compares them for full and almost-full and registers both on `wr_clk`. The read side compares for empty, almost-empty and the occupancy count, and registers them on `rd_clk`. `REGISTERED=1` adds the dashed write-pointer delay, so empty lines up with a flopped read-data path, and makes `count` the flopped `r_count`.

35.4.2 Dual-Domain Design

The module straddles two clock domains: - **Write domain**: Generates full and almost_full flags - **Read domain**: Generates empty and almost_empty flags

35.4.3 Pointer Arithmetic Foundation

All the status generation boils down to **pointer comparison with wraparound detection**:

```
// XOR the MSBs to detect wraparound condition
assign w_wdom_ptr_xor = wr_ptr_bin[AW] ^ wdom_rd_ptr_bin[AW];
assign w_rdom_ptr_xor = rd_ptr_bin[AW] ^ rdom_wr_ptr_bin[AW];
```

35.4.4 Full Detection Logic

```
assign w_wr_full_d = (w_wdom_ptr_xor &&
                      (wr_ptr_bin[AW-1:0] == wdom_rd_ptr_bin[AW-1:0]));
```

The algorithm: - **Condition 1**: The MSBs must differ (`w_wdom_ptr_xor = 1`) - **Condition 2**: The LSBs must be equal - **Meaning**: The write pointer has “lapped” the read pointer

A visual example (DEPTH=8, ADDR_WIDTH=3):

```
Pointers: [wrap_bit][address_bits]

Case 1: Not Full
wr_ptr  = 0_101  (address 5, no wrap)
rd_ptr  = 0_010  (address 2, no wrap)
→ MSBs same, not full

Case 2: Full
wr_ptr  = 1_010  (address 2, wrapped once)
rd_ptr  = 0_010  (address 2, not wrapped)
→ MSBs differ, addresses same = FULL
```

35.4.5 Almost Full Logic

Occupancy calculation:

```
assign w_almost_full_count = (w_wdom_ptr_xor) ?
    ((AW+1)'(D) - wdom_rd_ptr_bin[AW-1:0] + wr_ptr_bin[AW-1:0]) :
    (wr_ptr_bin[AW-1:0] - wdom_rd_ptr_bin[AW-1:0]);
```

Two cases handled: 1. **No wraparound** (`w_wdom_ptr_xor = 0`): - Count = `wr_ptr - rd_ptr` - Just a simple subtraction

2. **Wraparound** (`w_wdom_ptr_xor = 1`):
    - Count = `DEPTH - rd_ptr + wr_ptr`
    - Accounts for the circular buffer wrapping around

The threshold:

```
assign w_wr_almost_full_d = w_almost_full_count >= (AW+1)'(AFT);
// Where AFT = DEPTH - ALMOST_WR_MARGIN
```

35.4.6 Empty Detection Logic - Mode Aware

The critical innovation: mode-aware write pointer selection.

```
generate
    if (REGISTERED == 1) begin : gen_flop_mode
        // FLOP mode: Use previous cycle's write pointer
        logic [ADDR_WIDTH:0] r_rdom_wr_ptr_bin_delayed;

        always_ff @(posedge rd_clk or negedge rd_rst_n) begin
            if (!rd_rst_n) begin
                r_rdom_wr_ptr_bin_delayed <= '0;
            end else begin
                r_rdom_wr_ptr_bin_delayed <= rdom_wr_ptr_bin;
            end
        end

        assign w_wr_ptr_for_empty = r_rdom_wr_ptr_bin_delayed;
    end else begin : gen_mux_mode
        // MUX mode: Use current write pointer
        assign w_wr_ptr_for_empty = rdom_wr_ptr_bin;
    end
endgenerate
```

Why mode-aware detection?

In MUX mode (REGISTERED = 0): - **Data availability**: Immediate (combinational read) - **Write pointer**: Use the current value - **Reasoning**: Data is available the moment it's written

In FLOP mode (REGISTERED = 1): - **Data availability**: Delayed by 1 cycle (registered read) - **Write pointer**: Use the delayed value - **Reasoning**: Data isn't available until the next clock cycle

The empty detection algorithm:

```
assign w_rd_empty_d = (!w_rdom_ptr_xor_for_empty &&
                       (rd_ptr_bin[AW:0] == w_wr_ptr_for_empty[AW:0]));
```

- **Condition 1**: The MSBs must be the same (no wrap difference)
- **Condition 2**: All bits must be equal
- **Meaning**: The read pointer has caught up to the write pointer

35.4.7 Almost Empty Logic

Almost empty uses standard timing regardless of FIFO mode:

```
assign w_almost_empty_count = (w_rdom_ptr_xor) ?
    ((AW+1)'(D) - rd_ptr_bin[AW-1:0] + rdom_wr_ptr_bin[AW-1:0]) :
    (rdom_wr_ptr_bin[AW-1:0] - rd_ptr_bin[AW-1:0]);

assign w_rd_almost_empty_d = w_almost_empty_count <= (AW+1)'(AET);
// Where AET = ALMOST_RD_MARGIN
```

35.4.8 Count Generation

```
// Combinational occupancy...
logic [AW:0] w_count, r_count;
assign w_count = (w_rdom_ptr_xor) ?
    (rdom_wr_ptr_bin[AW-1:0] - rd_ptr_bin[AW-1:0] + (AW+1)'(D)) :
    (rdom_wr_ptr_bin[AW-1:0] - rd_ptr_bin[AW-1:0]);

// ...but the exported count is REGISTERED in flop mode (r_count is
w_count
// delayed one rd_clk); mux mode exports the combinational value
directly.
assign count = (REGISTERED == 1) ? r_count : w_count;
```

How to read the count: - **Range**: 0 to DEPTH - **Zero**: FIFO empty - **DEPTH**: FIFO full - **Uses**: Flow control, occupancy monitoring

35.4.9 Flag Registration

Write domain flags:

```
always_ff @(posedge wr_clk, negedge wr_rst_n) begin
    if (!wr_rst_n) begin
        wr_full <= 'b0;
        wr_almost_full <= 'b0;
    end else begin
        wr_full <= w_wr_full_d;
        wr_almost_full <= w_wr_almost_full_d;
    end
end
```

Read domain flags:

```
always_ff @(posedge rd_clk, negedge rd_rst_n) begin
    if (!rd_rst_n) begin
        rd_empty <= 'b1;          // Reset to empty
        rd_almost_empty <= 'b0;
    end else begin
        rd_empty <= w_rd_empty_d;
        rd_almost_empty <= w_rd_almost_empty_d;
    end
end
```

Reset behavior: - **Full flags**: Reset to 0 (not full) - **Empty flags**: Reset to 1 (empty) - **Almost flags**: Reset to 0

35.4.10 Width Casting Fix

```
// The occupancy count needs AW+1 bits, and D (= DEPTH) must be cast
to AW+1
```

```
// bits -- NOT AW bits. AW'(D) TRUNCATES: for depth=16, AW=4, AW'(16)
= 4'b0000
// (wrong!), whereas (AW+1)'(16) = 5'b10000 (correct). The count
register is
// widened to [AW:0] to hold the full range 0..DEPTH.
logic [AW:0] w_almost_full_count;

assign w_almost_full_count = (w_wdom_ptr_xor) ?
    ((AW+1)'(D) - wdom_rd_ptr_bin[AW-1:0] + wr_ptr_bin[AW-1:0]) :
    (wr_ptr_bin[AW-1:0] - wdom_rd_ptr_bin[AW-1:0]);
```

That `(AW+1)'(D)` cast (with the widened `[AW:0]` count) is the only thing standing between you and a truncated wraparound occupancy. Cast to `AW'(D)` instead and you reintroduce the truncation bug: e.g. DEPTH=16 with wr=2, rd=14 should give occupancy 16-14+2 = 4, but `AW'(16)` = 0 yields 0-14+2 (garbage).

35.5 Timing Characteristics

- **Minimal overhead**: Efficient pointer arithmetic
- **Low latency**: Single cycle flag updates
- **Resource efficient**: Shared across multiple FIFO types

35.6 Usage Examples

Every parameter and port below is taken from the module declaration.

```
fifo_control #(
    .ADDR_WIDTH           (3),
    .DEPTH                (8),
    .ALMOST_WR_MARGIN     (1),
    .ALMOST_RD_MARGIN     (1),
    .REGISTERED           (0)
) u_fifo_control (
    .wr_rst_n           (wr_rst_n),
    .rd_clk             (rd_clk),
    .rd_rst_n           (rd_rst_n),
    .wr_clk               (wr_clk),
    .wr_ptr_bin           (wr_ptr_bin),
    .wdom_rd_ptr_bin      (wdom_rd_ptr_bin),
    .rd_ptr_bin           (rd_ptr_bin),
    .rdom_wr_ptr_bin      (rdom_wr_ptr_bin),
    .count                (count),
    .wr_full              (wr_full),
    .wr_almost_full       (wr_almost_full),
    .rd_empty             (rd_empty),
```

```
    .rd_almost_empty       (rd_almost_empty)
);
```

35.7 Design Notes

35.7.1 Key Design Insights

Wraparound handling: - **MSB significance**: The extra bit is your wraparound detector - **Circular arithmetic**: Proper modulo DEPTH calculations - **Comparison logic**: XOR-based wraparound detection

Timing considerations: - **Synchronizer delay**: Accounted for in the conservative design - **Mode awareness**: Different timing for different output modes - **Flag updates**: Registered for clean transitions

Conservative design: - **Over-reporting**: Flags may assert slightly early - **Safety margin**: Prevents overflow/underflow - **Synchronizer latency**: Built into the safety margins

35.7.2 Applications

- **Status monitoring**: Provides comprehensive FIFO state
- **Flow control**: Enables back-pressure and rate matching
- **Debug/verification**: The count output earns its keep during debug

35.8 Related Modules

- **`fifo_sync`** and **`fifo_async`**: both FIFO variants share this status logic (binary or Johnson pointers). The former `fifo_async_div2` is retired — `fifo_async` with `USE_JOHNSON=1` replaces it.

35.9 Testing

No dedicated testbench for this module. It has no `val/**/test_fifo_control.py`; it is exercised indirectly, through the tests of the modules that instantiate it:

- `fifo_sync` – `val/common/test_fifo_sync.py`
- `gaxi_drop_fifo_sync` – `val/amba/test_gaxi_drop_fifo_sync.py`
- `gaxi_fifo_sync` – `val/amba/test_gaxi_fifo_sync.py`

Indirect coverage exercises this module only in the configurations its parents elaborate. A parameter or mode no parent uses is untested.

35.10 References

- Cummings, C.E. “Simulation and Synthesis Techniques for Asynchronous FIFO Design.” *SNUG San Jose*, 2002 (canonical non-peer conference paper) — FIFO style #1: pointers one bit wider than the address, empty when the pointers are equal, full when they differ only in the wrap (MSB) bit; this module computes the same flags from the binary pointers after they cross domains.
- Merrick, R. *Getting Started with FPGAs*, No Starch Press, ch. 6 “Common FPGA Modules”, “FIFO: First In, First Out” — textbook treatment of full/empty and almost-full/almost-empty flags (secondary source).
- Indexed in `rtl/common/references/README.md` (PDF archived once, under `rtl/cdc/references/AsyncFifoGrayPointers__Cummings2002.pdf`)

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36 fifo_sync

36.1 Overview

A synchronous First-In-First-Out buffer for when everything lives in a single clock domain. Depth, data width, and output modes are all configurable, and you get the full set of status signals.

36.2 Parameters

Parameter	Default	Description
MEM_STYLE	FIFO_AUTO	Memory implementation (FIFO_AUTO/SRL/BRAM) — see note below
REGISTERED	0	Output mode: 0=mux mode (combinational), 1=flop mode (registered)
DATA_WIDTH	4	Width of data bus
DEPTH	4	FIFO depth in words
ALMOST_WR_MARGIN	1	Almost full threshold
ALMOST_RD_MARGIN	1	Almost empty threshold

MEM_STYLE details: In this module every branch honours REGISTERED: with REGISTERED=0 the BRAM branch gives a *combinational* read (`assign rd_data = mem[r_rd_addr]`), not a registered one. Note that `fifo_async` differs here — its BRAM branch is unconditionally registered — so do not carry that assumption across. **Caveat:** MEM_STYLE=FIFO_BRAM with REGISTERED=0 asks for an asynchronous read from a block RAM, which real BRAM cannot do; synthesis will ignore the `ram_style`/`ramstyle` attribute and map to LUTRAM instead. Use REGISTERED=1 when you actually want block RAM.

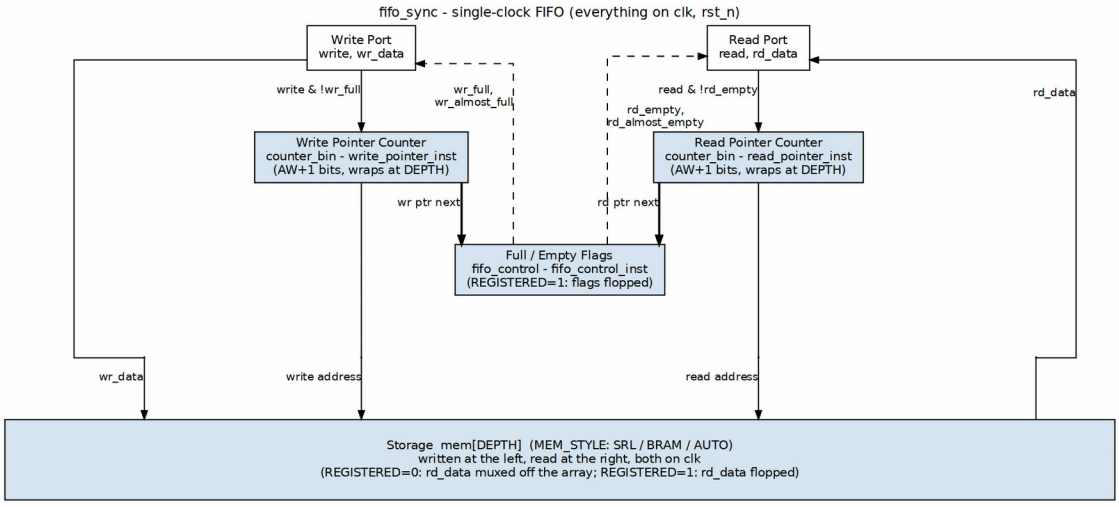
36.3 Ports

Port	Direction	Width	Description
clk	Input	1	System clock
rst_n	Input	1	Active-low reset
write	Input	1	Write enable signal
wr_data	Input	DATA_WIDTH	Data to write into FIFO

Port	Direction	Width	Description
`read`	Input	1	Read enable signal
`wr_full`	Output	1	Write domain full flag
`wr_almost_full`	Output	1	Write domain almost full flag
`rd_data`	Output	DATA_WIDTH	Data read from FIFO
`rd_empty`	Output	1	Read domain empty flag
`rd_almost_empty`	Output	1	Read domain almost empty flag

36.4 Functional Description

36.4.1 Block diagram



fifo_sync block diagram

Everything runs on one clock. Write data drops down the left into the storage array, at the address held by the write pointer counter; read data comes back up the right, from the read pointer's address. Both pointers go straight into the one `fifo_control` instance in the center, with no synchronizer and no crossing. It drives the full flags back to the write port and the empty flags back to the read port. `REGISTERED` selects whether `rd_data` is muxed straight off the array or flopped, and `MEM_STYLE` picks the storage resource.

36.4.2 Core Components

1. **Binary counters** for the read/write pointers
2. **Memory array** for data storage
3. **FIFO control logic** for status generation
4. **Output multiplexing** based on the REGISTERED parameter

36.4.3 Memory Organization

```
logic [DATA_WIDTH-1:0] mem [DEPTH];       // Memory array (declared
inside the
                                          // MEM_STYLE generate branch
so all
                                          // accesses share one scope)
assign r_wr_addr = r_wr_ptr_bin[AW-1:0];  // Write address
assign r_rd_addr = r_rd_ptr_bin[AW-1:0];  // Read address
```

36.4.4 Pointer Management

Binary counters drive both the read and write pointers:

```
counter_bin #(.WIDTH(AW + 1), .MAX(D)) write_pointer_inst (
    .clk(clk),
    .rst_n(rst_n),
    .enable(write && !wr_full),
    .counter_bin_curr(r_wr_ptr_bin),
    .counter_bin_next(w_wr_ptr_bin_next)
);
```

Pointer characteristics: - **Width**: `$clog2(DEPTH) + 1` bits (the extra bit catches wraps) - **Increment**: Only when the operation is valid (write && !full, read && !empty) - **Wraparound**: Automatic modulo DEPTH counting

36.4.5 Memory Operations

Write operation:

```
always_ff @(posedge clk) begin
    if (write && !wr_full) begin   // the !wr_full guard is REQUIRED:
it stops a
        mem[r_wr_addr] <= wr_data;  // write-while-full from
clobbering unread data
    end
end
```

Read operation — dual mode:

```
if (REGISTERED != 0) begin : g_flop
    logic [DATA_WIDTH-1:0] r_rd_data;
    `ALWAYS_FF_RST(clk, rst_n,
        if (!rst_n) r_rd_data <= '0;
        else        r_rd_data <= mem[r_rd_addr];
    )
    assign rd_data = r_rd_data;

end else begin : g_mux
    // Mux mode - non-registered output
    assign rd_data = mem[r_rd_addr];
end
```

36.4.6 Status Flag Generation

Status comes out of the shared `fifo_control` module: - **Full detection**: Based on pointer comparison with wraparound handling - **Almost full**: When remaining space ≤ `ALMOST_WR_MARGIN` - **Empty detection**: When read pointer equals write pointer - **Almost empty**: When available data ≤ `ALMOST_RD_MARGIN`

36.4.7 Operating Modes

Mux mode (REGISTERED = 0): - **Read latency**: 0 cycles (combinational output) - **Data availability**: Immediate after write - **Use case**: Low-latency applications - **Timing**: Read data changes combinationally with the address

Flop mode (REGISTERED = 1): - **Read latency**: 1 cycle (registered output) - **Data availability**: 1 cycle after read enable - **Use case**: High-speed designs, timing closure - **Timing**: Read data stable for the full clock cycle

36.4.8 Functional Behavior

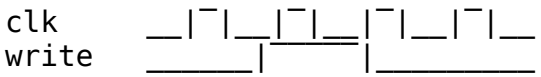
Write operations: - **Condition**: `write && !wr_full` - **Action**: Store data at the write pointer, bump the pointer - **Blocking**: Writes ignored when the FIFO is full - **Status**: Full flags track the new occupancy

Read operations: - **Condition**: `read && !rd_empty` - **Action**: Advance the read pointer (data handling depends on mode) - **Blocking**: Reads ignored when the FIFO is empty - **Status**: Empty flags track the new occupancy

Reset behavior: - **Pointers**: Reset to 0 - **Flags**: Full flags → 0, Empty flags → 1 - **Data**: Read data cleared in flop mode - **Memory**: Contents undefined, but that doesn't matter

36.5 Timing Characteristics

36.5.1 Write Sequence



```
wr_data ======[ D0 ]=========
wr_full ______________________|¯  (when FIFO becomes full)
```

36.5.2 Read Sequence - Mux Mode

```
clk     __|¯|__|¯|__|¯|__|¯|__
read    _______|     |__________
rd_data ======[ D0 ]=========  (immediate)
```

36.5.3 Read Sequence - Flop Mode

```
clk     __|¯|__|¯|__|¯|__|¯|__
read    _______|     |__________
rd_data ===========[ D0 ]=====  (1 cycle delay)
```

36.5.4 Performance Characteristics

- **Throughput**: 1 operation per clock cycle (when not full/empty)
- **Latency**: 0 cycles (mux) or 1 cycle (flop) for read
- **Occupancy**: 0 to DEPTH words
- **Efficiency**: 100% bandwidth utilization possible

36.6 Usage Examples

Every parameter and port below is taken from the module declaration.

```
fifo_sync #(
    .MEM_STYLE             (FIFO_AUTO),
    .REGISTERED            (0),
    .DATA_WIDTH            (4),
    .DEPTH                 (4),
    .ALMOST_WR_MARGIN      (1),
    .ALMOST_RD_MARGIN      (1)
) u_fifo_sync (
    .rst_n               (rst_n),
    .clk                   (clk),
    .write                 (write),
    .wr_data               (wr_data),
    .wr_full               (wr_full),
    .wr_almost_full        (wr_almost_full),
    .read                  (read),
    .rd_data               (rd_data),
    .rd_empty              (rd_empty),
    .rd_almost_empty       (rd_almost_empty)
);
```

36.7 Design Notes

36.7.1 Mode Selection Guidelines

- **Choose Mux Mode when**: Latency is critical, moderate clock speeds
- **Choose Flop Mode when**: High clock speeds, timing closure issues
- **Performance impact**: Flop mode costs you 1 cycle of latency but buys fmax

Typical use cases: - **Low latency applications (mux mode)**: Data streaming with minimal delay, real-time processing pipelines, clock domain buffers - **High speed applications (flop mode)**: High-frequency designs, timing-critical paths, pipelined architectures

36.7.2 Sizing Considerations

- **Depth**: Must accommodate worst-case burst sizes
- **Almost flags**: Set margins based on producer/consumer response times
- **Data width**: Should match datapath requirements

36.7.3 Error Detection

Note: the current RTL does **not** contain any runtime `$display` overflow/underflow checks. The `!wr_full` write guard is the only overflow protection. If you need write-while- full / read-while-empty telemetry, add assertions in your own testbench.

36.8 Related Modules

- **fifo_async**: For clock domain crossing applications
- **fifo_control**: Shared status flag generation logic
- **counter_bin**: Binary counter implementation

36.9 Testing

36.9.1 WaveDrom Visualization

The WaveDrom test generates detailed timing diagrams for synchronous FIFO operation:

```
# Generate synchronous FIFO waveforms (single clock domain)
pytest val/common/test_fifo_sync_wavedrom.py -v
```

Waveform scenarios generated:

1. **Write-Fill-Read-Empty Cycle**
    - Basic synchronous FIFO operation
    - Simple binary pointer management
    - No CDC complexity (single clock domain)
2. **Back-to-Back Operations**
    - Maximum throughput demonstration
    - Minimal inter-transaction delays
    - Sequential write and read bursts
3. **Simultaneous Write-Read (Ping-Pong)**
    - Read and write in same cycle
    - Steady-state FIFO operation
    - Demonstrating simultaneous access capability
4. **Flag Transitions**
    - Full/empty flag behavior at boundaries
    - Almost-full/almost-empty thresholds
    - Flow control signaling

Key characteristics vs. async FIFOs:

- **Single Clock Domain**: No CDC complexity, simpler design
- **Binary Pointers**: Direct addressing, no Gray code conversion
- **Zero CDC Latency**: Flags update immediately (no synchronization delay)
- **Simultaneous Access**: Can read and write in same cycle

Comparison tests:

- `test_fifo_async_wavedrom.py` - Gray code CDC (power-of-2 depths)

36.9.2 Functional Test Suite

- `val/common/test_fifo_buffer.py` - Full functional verification
- `val/common/test_fifo_sync_wavedrom.py` - WaveDrom timing diagrams

Run the tests:

```
# Full functional test (basic/medium/full levels)
pytest val/common/test_fifo_buffer.py -v

# WaveDrom waveform generation
pytest val/common/test_fifo_sync_wavedrom.py -v
```

36.10 References

- Cummings, C.E. “Simulation and Synthesis Techniques for Asynchronous FIFO Design.” *SNUG San Jose*, 2002 (canonical non-peer conference paper) — FIFO style #1: pointers one bit wider than the address, empty when the pointers are equal, full when they differ only in the wrap (MSB) bit; this module applies the same wrap-bit full/empty test to single-clock pointers held in `counter_bin`.
- Merrick, R. *Getting Started with FPGAs*, No Starch Press, ch. 6 “Common FPGA Modules”, “FIFO: First In, First Out” — textbook treatment (secondary source).
- Indexed in `rtl/common/references/README.md` (PDF archived once, under `rtl/cdc/references/AsyncFifoGrayPointers__Cummings2002.pdf`)

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37 find_first_set

37.1 Overview

Finds the index of the least significant bit (LSB) that's set to '1' in the input vector — a priority encoder that plays favorites with the low bits.

37.2 Parameters

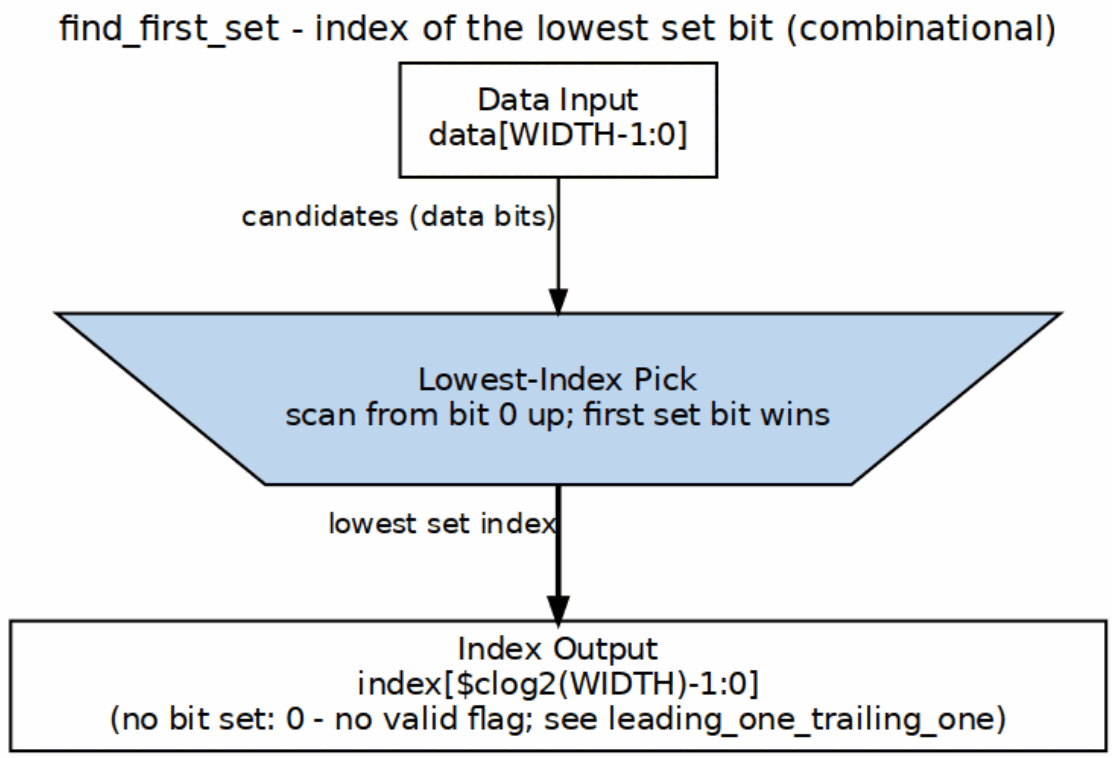
Parameter	Default	Description
`WIDTH`	32	Width of input data vector — the module's only parameter

37.3 Ports

Port	Direction	Width	Description
`data`	Input	WIDTH	Input data vector to search
`index`	Output	$clog2(WIDTH)	Index of the first (lowest) set bit

37.4 Functional Description

37.4.1 Block diagram



find_first_set block diagram

`data` enters at the top and its lowest set bit wins. That is a genuine many-to-one pick, so it is drawn as the trapezium, with the winning index leaving its narrow edge as `index`. There is no valid output: no bit set and bit 0 set both give 0. `leading_one_trailing_one` adds the `valid` flag.

37.4.2 Search Strategy

- **Direction**: LSB to MSB (left to right, ascending index)
- **Priority**: Lower indices have higher priority
- **First match**: Stops at the first '1' bit encountered
- **Found flag**: Prevents multiple assignments

37.4.3 Truth Table Examples (WIDTH=8)

data[7:0]	index[2:0]	Description
00000001	000	Bit 0 is first set

data[7:0]	index[2:0]	Description
00000010	001	Bit 1 is first set
00000100	010	Bit 2 is first set
00001000	011	Bit 3 is first set
00010000	100	Bit 4 is first set
00100000	101	Bit 5 is first set
01000000	110	Bit 6 is first set
10000000	111	Bit 7 is first set
00000000	000	No bits set - default
10000001	000	Multiple bits - LSB wins
11111111	000	All bits set - bit 0 wins

37.4.4 Key Characteristics

- **LSB priority**: Multiple bits set? You get the lowest index
- **Zero default**: Returns 0 when no bits are set
- **Deterministic**: Same input always produces same output
- **Combinational**: Immediate response to input changes

37.4.5 Core Algorithm

```
localparam int N = $clog2(WIDTH);
logic w_found;

always_comb begin
    index = {N{1'b0}}; // Default value if no bit is set
    w_found = 1'b0;

    for (int i = 0; i < WIDTH; i++) begin
        if (data[i] && !w_found) begin
            index = i[N-1:0]; // Ensure correct bit width
            w_found = 1'b1;
        end
    end
end
```

37.4.6 Width Parameterization

- **Automatic sizing**: Output width calculated as `$clog2(WIDTH)`
- **Examples**:
  - WIDTH=8 → index[2:0] (3 bits)
  - WIDTH=16 → index[3:0] (4 bits)
  - WIDTH=32 → index[4:0] (5 bits)

37.4.7 Found Flag Logic

```
if (data[i] && !w_found) begin
    index = i[N-1:0];
    w_found = 1'b1;
end
```

- **Purpose**: Keeps the first found result from being overwritten
- **Mechanism**: Once `w_found` is set, no further assignments occur
- **Result**: Only the lowest index is captured

37.4.8 Bit Width Handling

```
index = i[N-1:0]; // Ensure correct bit width
```

- **Type safety**: Explicit width matching keeps synthesis warnings away
- **Truncation**: Safely handles loop variable width vs. output width

37.5 Timing Characteristics

37.5.1 Propagation Delay

- **Worst case**: Multiple LUT delays (higher bits set)
- **Average case**: Depends on typical input patterns

37.5.2 Critical Path

- **Path length**: Increases with WIDTH
- **Optimization**: Modern synthesizers create efficient priority encoders
- **Scaling**: Generally logarithmic complexity in hardware

37.6 Usage Examples

37.6.1 Interrupt Processing

```
find_first_set #(.WIDTH(8)) irq_ffs (
    .data(interrupt_pending),
    .index(lowest_priority_irq)
);
```

Process the lowest-numbered interrupt first — fair servicing of interrupt sources.

37.6.2 Resource Allocation

```
find_first_set #(.WIDTH(16)) resource_ffs (
    .data(available_resources),
    .index(allocated_resource_id)
);
```

Allocate the lowest-numbered available resource; round-robin allocation when combined with masking.

37.6.3 Error Detection

```
find_first_set #(.WIDTH(4)) error_ffs (
    .data(error_status),
    .index(first_error_type)
);
```

Report the first error condition detected, prioritizing error handling by position.

37.6.4 Bit Manipulation

```
find_first_set #(.WIDTH(32)) bit_scan (
    .data(search_pattern),
    .index(first_set_position)
);
```

Bit scanning operations, pattern analysis and parsing.

37.6.5 Round-Robin Arbitration

```
// Mask off already-serviced requests
assign masked_requests = pending_requests & ~service_mask;

find_first_set #(.WIDTH(N)) arbiter_ffs (
    .data(masked_requests),
    .index(next_grant)
);

// Update mask to exclude granted request
assign next_mask = service_mask | (1 << next_grant);
```

37.6.6 Free List Management

```
find_first_set #(.WIDTH(POOL_SIZE)) free_finder (
    .data(free_pool_bitmap),
    .index(allocated_entry)
);

// Mark entry as allocated
assign updated_bitmap = free_pool_bitmap & ~(1 << allocated_entry);
```

37.6.7 Packet Classification

```
find_first_set #(.WIDTH(NUM_RULES)) rule_matcher (
    .data(rule_match_vector),
    .index(matched_rule_id)
);
```

37.7 Design Notes

37.7.1 Algorithm Comparison

Versus find_last_set:

Aspect	Find First Set	Find Last Set
Search direction	LSB → MSB	MSB → LSB
Priority	Lower index wins	Higher index wins
Use case	Fair allocation	Priority allocation

Versus a priority encoder:

Aspect	Find First Set	Priority Encoder
Priority direction	LSB has priority	MSB has priority
Application	Round-robin systems	Hierarchical systems

37.7.2 Design Considerations

Input validation: - **All zeros**: Returns 0 (may need validation in application) - **Valid range**: All input patterns handled gracefully - **No error conditions**: Always produces a result

Performance optimization: - **Synthesis**: Let tools optimize the priority encoding - **Pipelining**: Consider registering for high-speed applications - **Parallel**: Can process multiple vectors simultaneously

Width selection: - **Power of 2**: Often most efficient for synthesis - **Arbitrary width**: Supported but may have overhead - **Large widths**: Consider hierarchical implementation

37.7.3 Synthesis Considerations

- **LUT usage**: Typically maps to priority encoder primitives
- **Efficiency**: Good utilization for most FPGA architectures
- **Scaling**: Linear increase in resources with WIDTH
- **Critical paths**: May need pipeline registers for large WIDTH
- **Clock frequency**: Generally achieves high fmax
- **Constraints**: Standard combinational timing constraints apply

37.8 Related Modules

- **find_last_set**: MSB-first search variant
- **encoder**: Basic priority encoder
- **encoder_priority_enable**: Enhanced priority encoder with enable

37.9 Testing

37.9.1 Test Scenarios

- **Single bit**: Each bit position individually
- **Multiple bits**: Various combinations
- **All zeros**: Edge case handling
- **All ones**: Maximum case behavior

37.9.2 Coverage Metrics

- **Bit coverage**: Each input bit position
- **Pattern coverage**: Representative bit combinations
- **Boundary cases**: Min/max index values

37.10 References

- Oklobdzija, V.G. “An Algorithmic and Novel Design of a Leading Zero Detector Circuit: Comparison with Logic Synthesis.” *IEEE Transactions on VLSI Systems*, 2(1), Mar. 1994. DOI: 10.1109/92.273153 — leading-one/zero

detector structure; find-first-set is the LSB-side mirror and is otherwise a standard priority scan.

- Chu, P.P. *FPGA Prototyping by SystemVerilog Examples*, Wiley, 2018, ch. 3 “RT-Level Combinational Circuit”, Sec. 3.12.3 Dual-priority encoder (textbook treatment; secondary source).
- Indexed in `rtl/common/references/README.md` (local PDF: `LeadingZeroDetector__Oklobdzija1994.pdf`).

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38 find_last_set

38.1 Overview

Finds the index of the most significant bit (MSB) that's set to '1' in the input vector — the mirror image of find_first_set, favoring the higher-indexed bits.

38.2 Parameters

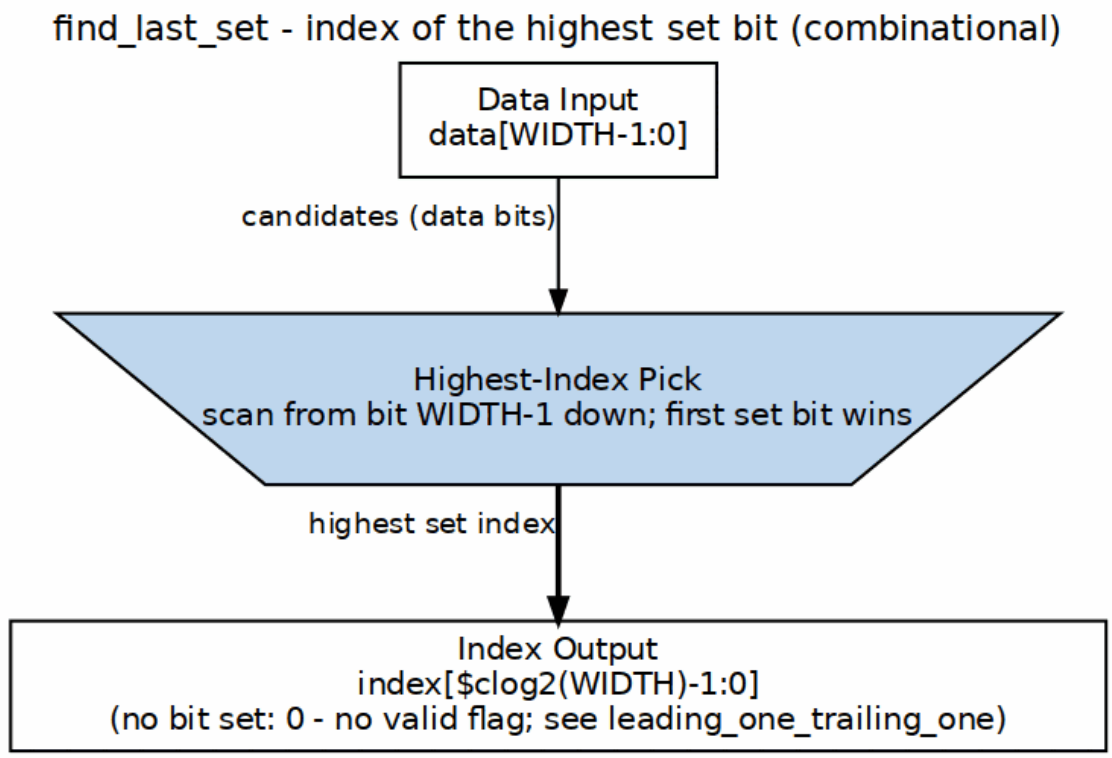
Parameter	Default	Description
WIDTH	32	Width of input data vector — the module's only parameter

38.3 Ports

Port	Direction	Width	Description
data	Input	WIDTH	Input data vector to search
index	Output	$clog2(WIDTH)	Index of the last (highest) set bit

38.4 Functional Description

38.4.1 Block diagram



find_last_set block diagram

`data` enters at the top and its highest set bit wins. That is a genuine many-to-one pick, so it is drawn as the trapezium, with the winning index leaving its narrow edge as `index`. There is no valid output: no bit set and bit 0 set both give 0. `leading_one_trailing_one` adds the `valid` flag.

38.4.2 Search Strategy

- **Direction**: MSB to LSB (right to left, descending index)
- **Priority**: Higher indices have higher priority
- **First match**: Stops at the first '1' bit encountered from the MSB
- **Found flag**: Prevents multiple assignments

38.4.3 Truth Table Examples (WIDTH=8)

data[7:0]	index[2:0]	Description
10000000	111	Bit 7 is last set

data[7:0]	index[2:0]	Description
01000000	110	Bit 6 is last set
00100000	101	Bit 5 is last set
00010000	100	Bit 4 is last set
00001000	011	Bit 3 is last set
00000100	010	Bit 2 is last set
00000010	001	Bit 1 is last set
00000001	000	Bit 0 is last set
00000000	000	No bits set - default
10000001	111	Multiple bits - MSB wins
11111111	111	All bits set - bit 7 wins

38.4.4 Key Characteristics

- **MSB priority**: Multiple bits set? You get the highest index
- **Zero default**: Returns 0 when no bits are set (same as find_first_set)
- **Deterministic**: Same input always produces same output
- **Combinational**: Immediate response to input changes

38.4.5 Core Algorithm

```
localparam int N = $clog2(WIDTH);
logic w_found;

always_comb begin
    index = {N{1'b0}}; // Default value if no bit is set
    w_found = 1'b0;

    for (int i = WIDTH - 1; i >= 0; i--) begin
        if (data[i] && !w_found) begin
            index = i[N-1:0]; // Ensure correct bit width
            w_found = 1'b1;
        end
    end
end
```

38.4.6 Reverse Loop Implementation

```
for (int i = WIDTH - 1; i >= 0; i--) begin
```

- **MSB-first**: Starts from the highest bit index
- **Decrementing**: Walks toward the LSB
- **Early termination**: Stops at the first match found

38.4.7 Width Parameterization

Same auto-sizing as `find_first_set`: - **Output width**: `$clog2(WIDTH)` bits - **Examples**: - WIDTH=8 → index[2:0] (3 bits) - WIDTH=16 → index[3:0] (4 bits) - WIDTH=32 → index[4:0] (5 bits)

38.4.8 Found Flag Logic

```
if (data[i] && !w_found) begin
    index = i[N-1:0];
    w_found = 1'b1;
end
```

- **Single assignment**: Prevents overwriting the highest found bit
- **Priority enforcement**: Only the first match (highest index) is captured

38.5 Timing Characteristics

38.5.1 Propagation Delay

- **Worst case**: Multiple LUT delays (LSB only set)
- **Average case**: Depends on typical bit patterns

38.5.2 Critical Path Analysis

```
MSB check → MSB-1 check → ... → LSB check → Output
```

- **Path length**: Linear with WIDTH in the worst case
- **Typical case**: Shorter path when higher bits are set
- **Optimization**: Synthesis tools create efficient implementations

38.6 Usage Examples

38.6.1 Interrupt Priority Handling

```
find_last_set #(.WIDTH(8)) irq_priority (
    .data(interrupt_pending),
```

```
    .index(highest_priority_irq)
);
```

Service the highest-priority interrupt first — preemptive interrupt handling.

38.6.2 Leading Zero Count

```
// Find position of leading 1, then calculate leading zeros
find_last_set #(.WIDTH(32)) leading_one (
    .data(input_value),
    .index(msb_position)
);

assign leading_zero_count = (input_value == 0) ? 32 : (31 -
msb_position);
```

38.6.3 Bit Width Calculation

```
find_last_set #(.WIDTH(64)) width_calc (
    .data(value),
    .index(msb_pos)
);

assign required_bits = msb_pos + 1; // Minimum bits needed to
represent value
```

38.6.4 Cache Line Selection

```
find_last_set #(.WIDTH(16)) cache_priority (
    .data(cache_line_priorities),
    .index(selected_cache_line)
);
```

Select the highest priority cache line for replacement — LRU policy implementation.

38.6.5 Hierarchical Priority Encoding

```
// For very wide inputs, use hierarchical approach
find_last_set #(.WIDTH(8)) level0_fls [7:0] (
    .data(data_chunks),  // 8 chunks of 8 bits each
    .index(chunk_indices)
);

find_last_set #(.WIDTH(8)) level1_fls (
    .data(chunk_valid),  // Which chunks have set bits
    .index(active_chunk)
);

// Combine results
assign final_index = {active_chunk, chunk_indices[active_chunk]};
```

38.6.6 Conditional Priority Selection

```
// Select between different priority schemes
assign priority_vector = use_high_priority ? high_pri_requests :
                         use_fair_mode ? fair_requests :
                         normal_requests;

find_last_set #(.WIDTH(N)) flexible_priority (
    .data(priority_vector),
    .index(selected_request)
);
```

38.6.7 Dynamic Width Processing

```
// Process variable-width data
logic [MAX_WIDTH-1:0] padded_data;
assign padded_data = {{(MAX_WIDTH-actual_width){1'b0}}, input_data};

find_last_set #(.WIDTH(MAX_WIDTH)) variable_fls (
    .data(padded_data),
    .index(raw_index)
);

// Adjust result for actual width
assign adjusted_index = (raw_index >= actual_width) ? 0 : raw_index;
```

38.7 Design Notes

38.7.1 Algorithm Comparison

Find last set vs. find first set:

Aspect	Find Last Set	Find First Set
Search Direction	MSB → LSB	LSB → MSB
Priority	Higher index wins	Lower index wins
Loop Direction	`i--` (decrementing)	`i++` (incrementing)
Use Case	Priority systems	Fair allocation
Default Output	0 (same)	0 (same)

Typical use case patterns:

Application Type	Preferred Algorithm
Interrupt Controllers	Find Last Set (priority)
Round-Robin Arbiters	Find First Set (fairness)
Error Reporting	Find Last Set (severity)

Application Type	Preferred Algorithm
Resource Allocation	Find First Set (efficiency)
Bit Manipulation	Either (depends on need)

38.7.2 Zero Input Handling

Both modules return 0 for zero input:

```
// data = 8'b00000000
// Both find_first_set and find_last_set return 3'b000
```

Your application may need explicit zero detection:

```
assign valid_result = |data;  // OR-reduce to detect any bits set
assign result_index = valid_result ? index : INVALID_INDEX;
```

38.7.3 Performance Scaling

- **Small WIDTH (<16)**: Excellent performance
- **Medium WIDTH (16-64)**: Good performance, may need pipelining
- **Large WIDTH (>64)**: Consider hierarchical implementation

38.7.4 Synthesis Optimization

Modern synthesis tools recognize these patterns: - **Priority encoder primitives**: Often map to dedicated hardware - **LUT optimization**: Efficient resource utilization - **Timing optimization**: Automatic critical path optimization

38.8 Related Modules

- **find_first_set**: LSB-priority search
- **encoder_priority_enable**: Enhanced priority encoder
- **leading_one_trailing_one**: Position detection for Johnson counters

38.9 Testing

38.9.1 Comprehensive Test Coverage

```
// Test single bit at each position
for (int i = 0; i < WIDTH; i++) begin
    test_vector = 1 << i;
    expected_result = i;
    // Verify find_last_set returns i
end
```

```
// Test multiple bits with known MSB
test_vector = 8'b10101010;
expected_result = 7;  // MSB position

// Test edge cases
test_vector = 8'b00000000;  // All zeros
expected_result = 0;        // Default
```

38.9.2 Random Testing

```
// Random bit patterns
repeat (1000) begin
    random_vector = $random();
    expected = calculate_msb_position(random_vector);
    // Verify against reference function
end
```

38.10 References

- Oklobdzija, V.G. “An Algorithmic and Novel Design of a Leading Zero Detector Circuit: Comparison with Logic Synthesis.” *IEEE Transactions on VLSI Systems*, 2(1), Mar. 1994. DOI: 10.1109/92.273153 — leading-one detection structure; this module’s MSB-first priority scan computes the same index behaviorally.
- Chu, P.P. *FPGA Prototyping by SystemVerilog Examples*, Wiley, 2018, ch. 3 “RT-Level Combinational Circuit”, Sec. 3.12.3 Dual-priority encoder (textbook treatment; secondary source).
- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 12 “Floating Point Architectures”, Sec. 12.4 Leading Zero Counter (textbook treatment; secondary source).
- Indexed in `rtl/common/references/README.md` (local PDF: `LeadingZeroDetector__Oklobdzija1994.pdf`).

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39 hex_to_7seg

39.1 Overview

Converts 4-bit hexadecimal values (0-F) to 7-segment display patterns for common anode displays — the thing you reach for when you want human-readable output for debugging, user interfaces, and embedded systems.

39.2 Parameters

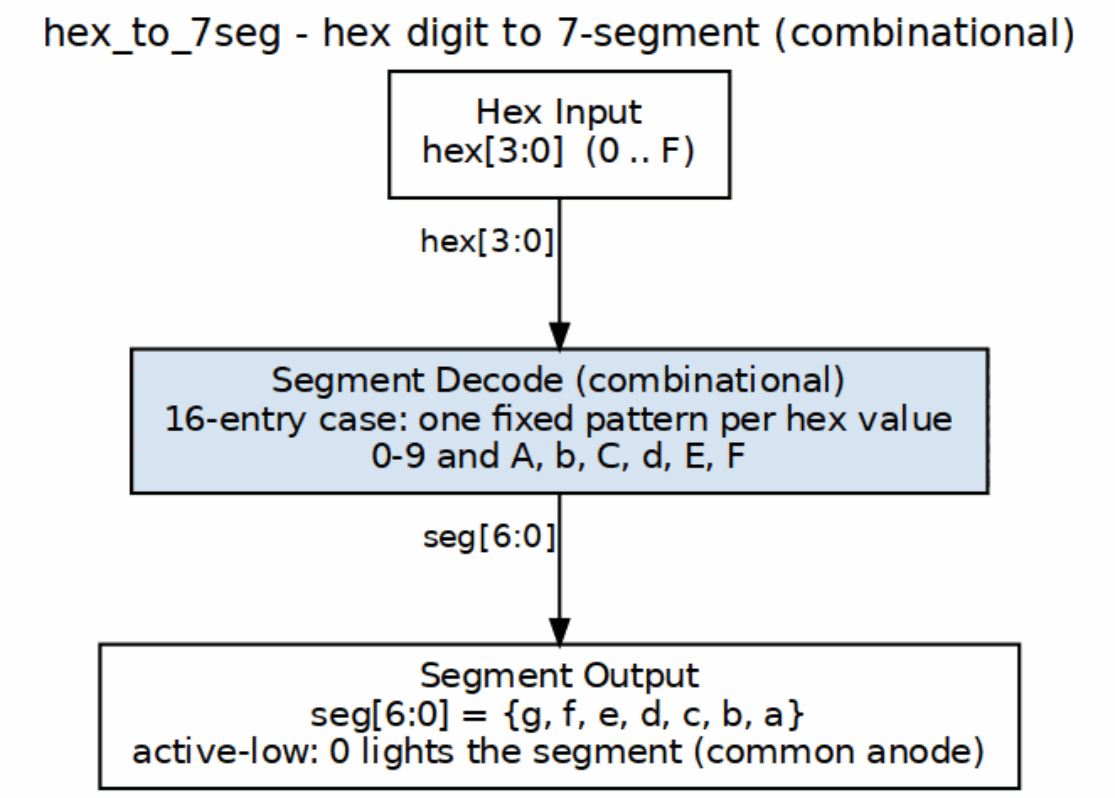
This module has no parameters — fixed functionality for standard 7-segment displays.

39.3 Ports

Port	Direction	Width	Description
hex	Input	4	4-bit hexadecimal input (0x0 to 0xF)
seg	Output	7	7-segment display pattern

39.4 Functional Description

39.4.1 Block diagram



hex_to_7seg block diagram

A single combinational case statement maps the 4-bit `hex` value to a fixed 7-bit pattern. The output is ordered `{g, f, e, d, c, b, a}` and is active-low, so a 0 lights the segment on a common-anode display.

39.4.2 Segment Naming Convention

```
 aaa
f   b
f   b
 ggg
e   c
e   c
 ddd
```

39.4.3 Bit Assignment

- **seg[6:0] = {g, f, e, d, c, b, a}**

- **Common Anode**: 0 = segment ON, 1 = segment OFF
- **Active Low**: LED segments illuminate when driven low

39.4.4 Character Patterns

Complete truth table:

Hex	Display	seg[6:0]	Segments ON	Pattern
0x0	**0**	1000000	abcdef	Classic 0
0x1	**1**	1111001	bc	Right side
0x2	**2**	0100100	abged	Snake pattern
0x3	**3**	0110000	abgcd	Right-heavy
0x4	**4**	0011001	fgbc	Left + right
0x5	**5**	0010010	afgcd	S-curve
0x6	**6**	0000010	afgdce	Almost all
0x7	**7**	1111000	abc	Top + right
0x8	**8**	0000000	all	All segments
0x9	**9**	0011000	abgcf	Top + middle + right, no tail
0xA	**A**	0001000	abcefg	Letter A
0xB	**b**	0000011	fgedc	Lowercase b
0xC	**C**	1000110	afed	Letter C
0xD	**d**	0100001	bgedc	Lowercase d
0xE	**E**	0000110	afged	Letter E
0xF	**F**	0001110	afge	Letter F

Visual representation:

```
 ###   #   ###  ###  # #  ###  ###  ###  ###  ###
# # # #     #    # # #  #    #      # #   # #   #
# #   #   ###  ###  ###  ###  ###    #  ###  ###
# #   #  #      #    #    # #   #    # #   #    #
```

```
 ###    #  ###  ###    #  ###  ###    #  ###    #
  0     1    2    3    4    5    6    7    8    9

 ###  ###   ##  ###  ###  ###
# # #     #   #   # #    #

### ###   #   #   # ###  ###
# # #   # #   #   # #    #
# # ###   ##  ###  ###  #
  A     b    C    d    E    F
```

39.4.5 Case Statement Structure

```
always_comb begin
    casez (hex)
        4'h0: seg = 7'b1000000;  // 0: abcdef (not g)
        4'h1: seg = 7'b1111001;  // 1: bc
        4'h2: seg = 7'b0100100;  // 2: abged
        4'h3: seg = 7'b0110000;  // 3: abgcd
        4'h4: seg = 7'b0011001;  // 4: fgbc
        4'h5: seg = 7'b0010010;  // 5: afgcd
        4'h6: seg = 7'b0000010;  // 6: afgdce
        4'h7: seg = 7'b1111000;  // 7: abc
        4'h8: seg = 7'b0000000;  // 8: all segments
        4'h9: seg = 7'b0011000;  // 9: abgcf
        4'ha: seg = 7'b0001000;  // A: abcefg (not d)
        4'hb: seg = 7'b0000011;  // b: fgedc (not ab)
        4'hc: seg = 7'b1000110;  // C: afed (not bcg)
        4'hd: seg = 7'b0100001;  // d: bgedc (not af)
        4'he: seg = 7'b0000110;  // E: afged (not bc)
        4'hf: seg = 7'b0001110;  // F: afge (not bcd)
        default: seg = 7'bxxxxxxx;  // Invalid - undefined output
    endcase
end
```

39.5 Timing Characteristics

- **Combinational logic**: The case statement maps to a LUT
- **No clock dependency**: Immediate response to input changes

On display update rate: - **Electronic response**: Immediate (propagation delay) - **Visual response**: Limited by LED rise/fall times (~microseconds) - **Human perception**: >30Hz refresh rate for flicker-free display

39.6 Usage Examples

39.6.1 Debug Displays

```
// Display processor status
hex_to_7seg status_display (
    .hex(processor_status[3:0]),
    .seg(status_segments)
);
```

39.6.2 Multi-Digit Displays

```
// 4-digit hex display
hex_to_7seg digit0 (.hex(value[3:0]),   .seg(seg0));
hex_to_7seg digit1 (.hex(value[7:4]),   .seg(seg1));
hex_to_7seg digit2 (.hex(value[11:8]),  .seg(seg2));
hex_to_7seg digit3 (.hex(value[15:12]), .seg(seg3));
```

39.6.3 Memory/Register Viewers

```
// Display memory contents
hex_to_7seg mem_display (
    .hex(memory_data[address]),
    .seg(memory_segments)
);
```

39.6.4 Error Code Display

```
// Show error codes
hex_to_7seg error_display (
    .hex(error_code),
    .seg(error_segments)
);
```

39.7 Design Notes

39.7.1 Design Choices

Lowercase vs. uppercase: - **A**: Uppercase (more readable) - **b, d**: Lowercase (distinguishable from 8, 0) - **C**: Uppercase (standard representation) - **E, F**: Uppercase (clear representation)

Common anode configuration: - **Active low outputs**: 0 = LED on, 1 = LED off - **Current sourcing**: Display provides +V, controller sinks current - **Logic inversion**: Many controllers expect active-high, requiring external inversion

39.7.2 Display Technology Integration

Common anode connection:

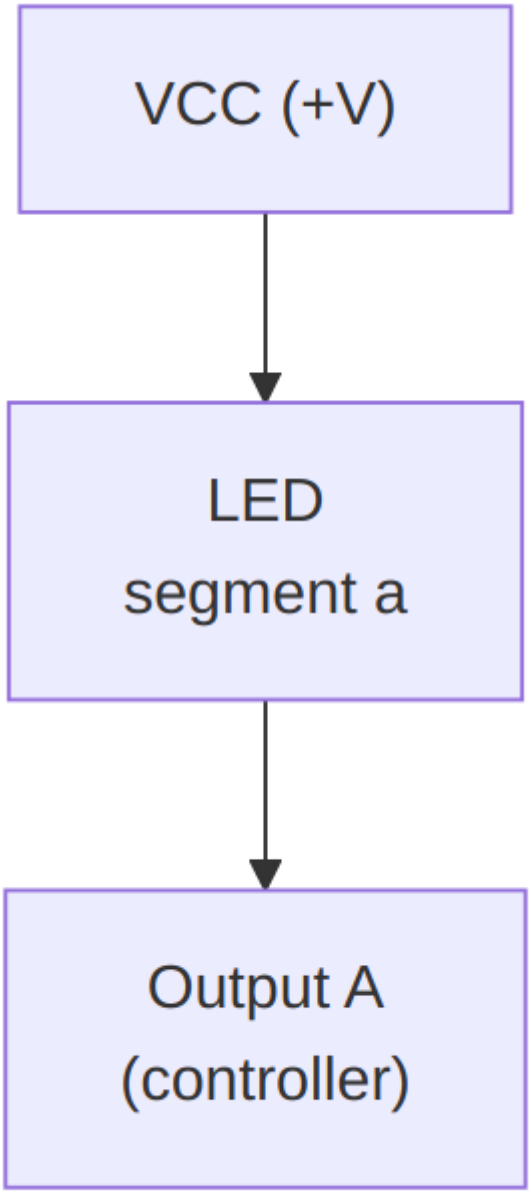


Diagram 2

Running common cathode instead? Invert all outputs:

```
// For common cathode displays:
assign seg_common_cathode = ~seg;
```

Current limiting:

```
Controller → [Resistor] → LED Segment → Common Anode (+V)
```

Typical resistor values: 220Ω - 1kΩ depending on LED specifications.

39.7.3 Design Variations

Extended character set — some implementations add extra characters:

```
// Extended characters (implementation dependent)
4'h10: seg = 7'b1111111;  // Blank/space
4'h11: seg = 7'b0111111;  // Dash/minus
4'h12: seg = 7'b1110111;  // Underscore
```

Alternative fonts for better readability:

```
// Alternative 'b' (uppercase-like)
4'hb: seg = 7'b0000111;  // More symmetric b
```

Brightness control:

```
// PWM brightness control
logic [6:0] seg_raw;
hex_to_7seg decoder (.hex(hex_in), .seg(seg_raw));

// Apply PWM for brightness
assign seg_out = brightness_enable ? seg_raw : 7'b1111111;
```

39.7.4 Multiplexed Display Support

Time-division multiplexing:

```
// 4-digit multiplexed display
logic [1:0] digit_select;
logic [3:0] current_hex;

// Select current digit data
always_comb begin
    case (digit_select)
        2'b00: current_hex = value[3:0];    // Digit 0
        2'b01: current_hex = value[7:4];    // Digit 1
        2'b10: current_hex = value[11:8];   // Digit 2
        2'b11: current_hex = value[15:12];  // Digit 3
    endcase
end

hex_to_7seg display_decoder (
    .hex(current_hex),
    .seg(segments)
);

// Digit enable signals (one active at a time)
assign digit_enable = ~(4'b0001 << digit_select);
```

Refresh rate calculation:

```
// For 4-digit display at 1kHz total refresh:
// Each digit displayed for 250µs
// Switching frequency: 4kHz

localparam REFRESH_DIVIDER = CLOCK_FREQ / 4000;
```

39.7.5 Common Issues and Debugging

Segment mapping errors: - **Bit order**: Ensure {g,f,e,d,c,b,a} ordering matches hardware - **Active level**: Verify common anode vs. common cathode - **Pin assignment**: Check physical connections match bit assignments

Display quality issues: - **Ghost segments**: Check current limiting and multiplexing timing - **Dim display**: Verify current levels and refresh rates - **Flicker**: Ensure adequate refresh rate (>30Hz visible, >60Hz preferred)

Character readability: - **Ambiguous characters**: 6 vs b, 8 vs B, 0 vs O - **Context**: Use surrounding digits/letters for disambiguation - **Alternative fonts**: Consider modified patterns for better distinction

39.8 Related Modules

- **BCD to 7-segment**: For decimal-only displays
- **ASCII to 7-segment**: Extended character set support
- **Display multiplexer**: Time-division multiplexing controller
- **PWM brightness controller**: Dynamic brightness adjustment

39.9 Testing

39.9.1 Exhaustive Testing

```
// Test all 16 hex values
for (int i = 0; i < 16; i++) begin
    hex_input = i[3:0];
    #1;
    expected_segments = reference_table[i];
    assert(seg_output == expected_segments);
end
```

39.9.2 Visual Verification

- **Physical testing**: Connect to actual 7-segment display

- **Simulation**: Use testbench with visual output
- **Reference comparison**: Compare with known-good implementations

39.9.3 Pattern Validation

```
// Verify no segments are permanently on/off
logic [6:0] all_segments_used = '0;
for (int i = 0; i < 16; i++) begin
    all_segments_used |= ~reference_table[i];  // Accumulate active
segments
end
assert(all_segments_used == 7'b1111111);  // All segments used
somewhere
```

39.10 References

- Chu, P.P. *FPGA Prototyping by SystemVerilog Examples*, Wiley, 2018, ch. 3 “RT-Level Combinational Circuit”, Sec. 3.10.1 Hexadecimal digit to seven-segment LED decoder (textbook treatment; secondary source).
- LaMeres, B.J. *Introduction to Logic Circuits & Logic Design with Verilog*, 2nd ed., Springer, 2019, ch. 6 “MSI Logic”, Sec. 6.1.2 7-Segment Display Decoder (textbook treatment; secondary source).
- No peer paper: hex-to-seven-segment decoding is a standard technique. Indexed in `rtl/common/references/README.md`.

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40 icg

40.1 Overview

An Integrated Clock Gating (ICG) cell: a latch-based clock gate that lets you selectively kill clock signals to cut dynamic power consumption. It's one of the fundamental power management moves in modern ASIC and FPGA designs, and for good reason.

40.1.1 Module Declaration

```
module icg(
    input  logic en,
    input  logic clk,
    output logic gclk
);
```

40.2 Parameters

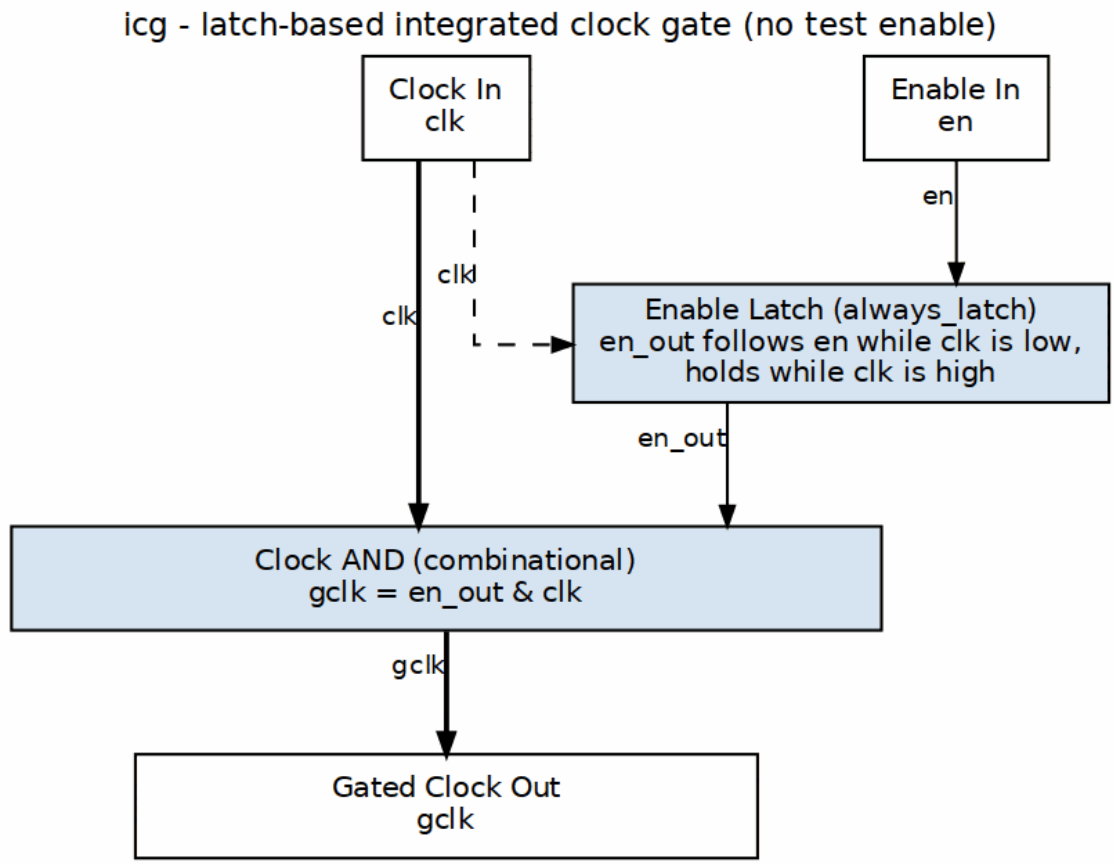
This module has no parameters.

40.3 Ports

Port	Direction	Width	Description
en	Input	1	Enable signal - when high, allows clock to pass through
clk	Input	1	Input clock signal to be gated
gclk	Output	1	Gated clock output - clk when en=1, low when en=0

40.4 Functional Description

40.4.1 Block diagram



icg block diagram

The clock is the main flow: `clk` goes straight through the AND gate to `gclk`. The enable latch is transparent while `clk` is low and holds while it is high, so `en_out` can only change while `clk` is low, and the AND never cuts a high phase short or adds a glitch. There is no test-enable input.

40.4.2 Clock Gating Operation

The ICG does clock gating the safe way:

1. **Latches Enable Signal**: Uses a level-sensitive latch, transparent while `clk` is low, to capture the enable signal
2. **Prevents Glitches**: Ensures the enable signal is stable during clock high periods
3. **Generates Gated Clock**: ANDs the latched enable with the input clock

40.4.3 Implementation Details

```
logic en_out;

// Latch-based approach - enable is latched on clock low
always_latch begin
    if (!clk) begin
        en_out = en;
    end
end

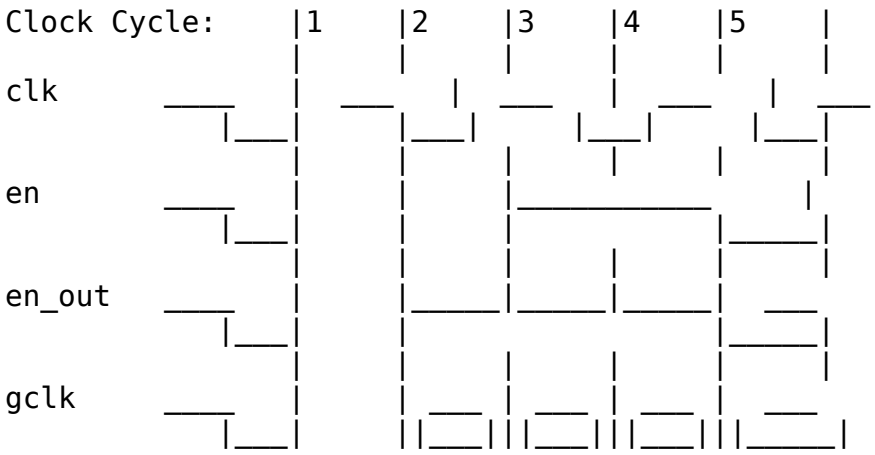
// Generate gated clock
assign gclk = en_out && clk;
```

40.5 Timing Characteristics

40.5.1 Critical Timing Requirements

Characteristic	Description
Setup Time	en must be stable before clk rising edge
Hold Time	en must remain stable after clk rising edge
Propagation Delay	Delay from clk to gclk through AND gate
Enable Latching	en is latched when clk is low

40.5.2 Timing Diagram



40.6 Usage Examples

40.6.1 Basic Register Bank Gating

```
// Clock gating for a register bank
logic bank_enable;
logic gated_clk;

icg u_bank_icg (
    .en   (bank_enable),
    .clk  (sys_clk),
    .gclk (gated_clk)
);

// Register bank using gated clock
always_ff @(posedge gated_clk or negedge rst_n) begin
    if (!rst_n) begin
        register_bank <= '0;
    end else begin
        register_bank <= next_register_values;
    end
end
```

40.6.2 CPU Pipeline Stage Gating

```
// Pipeline stage clock gating
logic pipe_stage_valid;
logic stage_clk;

icg u_pipe_icg (
    .en   (pipe_stage_valid),
    .clk  (cpu_clk),
    .gclk (stage_clk)
);

// Pipeline stage registers
always_ff @(posedge stage_clk or negedge rst_n) begin
    if (!rst_n) begin
        pipe_stage_regs <= '0;
    end else begin
        pipe_stage_regs <= pipe_stage_inputs;
    end
end
```

40.6.3 Memory Interface Gating

```
// Memory controller clock gating
logic mem_access_active;
logic mem_clk;
```

```
icg u_mem_icg (
    .en   (mem_access_active),
    .clk  (system_clk),
    .gclk (mem_clk)
);

assign mem_access_active = mem_read_req || mem_write_req || mem_busy;
```

40.7 Design Notes

40.7.1 Power Optimization Benefits

The ICG cuts dynamic power on three fronts:

- **Clock Tree Power**: Eliminates switching activity in downstream clock networks
- **Register Power**: Prevents unnecessary flip-flop transitions when data is not changing
- **Combinational Power**: Reduces spurious switching in logic fed by gated registers

Clock gating removes switching on the gated clock branch while a block is idle. The actual saving is **design- and technology-specific** — it depends on the activity factor, gating ratio, process node, and clock-tree structure — so no fixed percentage applies here. Quantify it from synthesis/power analysis of your own design rather than a generic figure. (Earlier revisions of this page quoted 20-40% / 30-60% / 10-30% numbers that had no source or stated conditions.)

40.7.2 Safety Requirements

- **Glitch-Free Operation**: Latch-based design prevents glitches on gated clock
- **Setup/Hold Timing**: Enable signal must meet timing requirements relative to clock
- **Clock Domain**: All logic using `gclk` must be in the same clock domain

40.7.3 Enable Signal Guidelines

```
// Good: Enable changes during clock low period
always_ff @(posedge clk) begin
    if (rst_n) begin
        enable_reg <= new_enable_value;  // Changes on clock edge
    end
```

```
end

// Caution: Ensure enable is stable during clock high
assign icg_enable = enable_reg && additional_condition;
```

40.7.4 Implementation Alternatives

Many ASIC libraries provide optimized ICG cells:

```
// Using library ICG cell (example)
CLOCK_GATE_ICG u_lib_icg (
    .CLK    (clk),
    .EN     (en),
    .GCLK   (gclk)
);
```

For FPGA targets, use dedicated clock control primitives:

```
// Xilinx BUFGCE example
BUFGCE u_bufgce (
    .I  (clk),
    .CE (en),
    .O  (gclk)
);
```

40.7.5 Synthesis Guidelines

Technology mapping: - **ASIC**: Use library-specific ICG cells for optimal area/power/timing - **FPGA**: Map to dedicated clock control primitives (BUFGCE, etc.) - **Timing Closure**: Account for additional setup/hold requirements

Clock tree synthesis:

```
# Example synthesis constraints for ICG
set_case_analysis 0 [get_pins u_icg/en]  ;# For power analysis
set_clock_gating_check -setup 0.1 [get_pins u_icg/gclk]
set_clock_gating_check -hold 0.1 [get_pins u_icg/gclk]
```

40.7.6 Design Integration

Where you put the ICG in the clock tree matters:

```
Main Clock → ICG → Local Clock Tree → Registers
           ↑
     Enable Logic
```

Power domain considerations: - **Retention**: Ensure enable logic remains powered during clock gating - **Isolation**: Properly isolate gated domains in multi-voltage designs - **Power Sequencing**: Coordinate with power management units

The `icg` module is a fundamental building block of power-efficient digital design: significant power savings through controlled clock gating, without giving up design safety or functionality.

40.8 Related Modules

- `clock_gate_ctrl`: Enhanced clock gating controller with additional features
- `clock_divider`: Clock frequency division functionality
- `reset_sync`: Reset synchronization for gated clock domains

40.9 Testing

40.9.1 Functional Tests

- **Enable/Disable**: Verify correct gating behavior for enable transitions
- **Clock Integrity**: Ensure gated clock maintains proper duty cycle when enabled
- **Reset Behavior**: Verify proper operation during reset sequences

40.9.2 Timing Tests

```
// Timing verification example
initial begin
    // Test enable setup/hold timing
    @(negedge clk);
    en = 1'b1;  // Enable during clock low
    @(posedge clk);
    // Verify gclk follows clk

    @(negedge clk);
    en = 1'b0;  // Disable during clock low
    @(posedge clk);
    // Verify gclk remains low
end
```

40.9.3 Power Analysis

- **Activity Monitoring**: Track switching activity reduction in gated domains
- **Power Estimation**: Measure power savings using EDA power analysis tools
- **Efficiency Metrics**: Calculate gating efficiency (% time clock is disabled)

40.10 References

- Benini, L., De Micheli, G. “Automatic Synthesis of Low-Power Gated-Clock Finite-State Machines.” *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 15(6), June 1996. DOI: 10.1109/43.503933 — Fig. 1(b) and Sec. II give exactly this structure: a latch transparent while the clock is low captures the enable, and an AND gate combines it with the clock so enable glitches cannot reach the gated clock.
- Benini, L., Siegel, P., De Micheli, G. “Saving Power by Synthesizing Gated Clocks for Sequential Circuits.” *IEEE Design & Test of Computers*, 11(4), 1994. DOI: 10.1109/54.329451 — the earlier paper introducing gated-clock synthesis for sequential circuits.
- Roy, S. *Advanced Digital System Design*, Springer, ch. 15 “Low-Power Digital System Design”, §15.5.5 “Logic Level Power Down” — textbook treatment of disabling the clock to idle units (secondary source).
- Indexed in `rtl/common/references/README.md` (local PDF: `LatchBasedClockGating__Benini1996.pdf`)

40.11 Navigation

- **← Back to rtl-common Index**
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41 leading_one_trailing_one

41.1 Overview

Finds both ends of your set bits at once: the position of the most significant bit (leading one) and the least significant bit (trailing one) in an input data vector. You get index outputs, one-hot vector representations of both positions, and status flags for the edge cases.

- Finds the highest set bit position (leading one)
- Finds the lowest set bit position (trailing one)
- Provides one-hot vector outputs for both positions
- Includes validity and edge case detection
- Parameterizable width with automatic index sizing

41.2 Parameters

Parameter	Default	Description
`WIDTH`	8	Width of input data vector — the module's only parameter

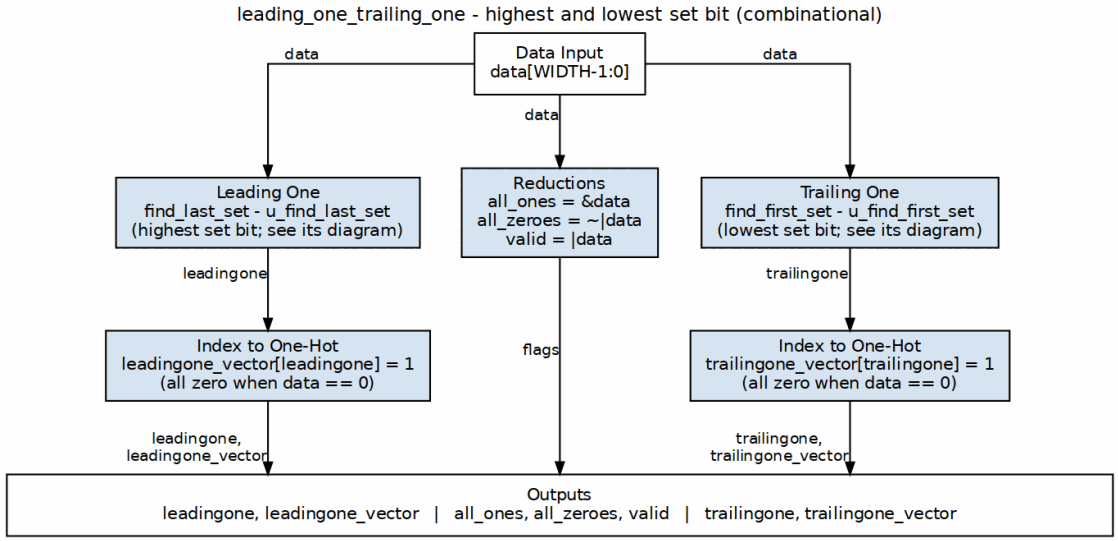
41.3 Ports

Port	Direction	Width	Description
`data`	Input	WIDTH	Input data vector to analyze
`leadingone`	Output	$clog2(WIDTH)	Index of the most significant set bit
`leadingone_vector`	Output	WIDTH	One-hot vector with leading one position marked
`trailingone`	Output	$clog2(WIDTH)	Index of the least significant set bit
`trailingone_vector`	Output	WIDTH	One-hot vector with trailing one position marked
`all_zeroes`	Output	1	High when input data is

Port	Direction	Width	Description
			all zeros
`all_ones`	Output	1	High when input data is all ones
`valid`	Output	1	High when input data contains at least one set bit

41.4 Functional Description

41.4.1 Block diagram



leading_one_trailing_one block diagram

The same `data` feeds three columns. On the left, a `find_last_set` instance finds the leading (highest) one; on the right, a `find_first_set` instance finds the trailing (lowest) one. Each index is expanded back into a one-hot vector, which is all zero when `data` is zero. In the centre, the reductions produce `all_ones`, `all_zeroes` and `valid`. The two pickers are drawn as instance boxes; their own diagrams show the pick.

41.4.2 Submodules

Internally it's two specialized finder modules working opposite ends:

1. **find_last_set**: Locates the position of the most significant set bit
   - Connected to `leadingone` output
   - Implements priority encoding from MSB to LSB

2. **find_first_set**: Locates the position of the least significant set bit
    - Connected to `trailingone` output
    - Implements priority encoding from LSB to MSB

41.4.3 One-Hot Vector Generation

```
always_comb begin
    leadingone_vector = '0;
    trailingone_vector = '0;

    // Only set vector bits if there is valid data
    if (|data) begin
        if (int'(leadingone) < WIDTH) begin
            leadingone_vector[leadingone] = 1'b1;
        end

        if (int'(trailingone) < WIDTH) begin
            trailingone_vector[trailingone] = 1'b1;
        end
    end
end
```

41.4.4 Status Flag Logic

- **all_ones**: Uses reduction AND (`&data`) to detect when all bits are set
- **all_zeroes**: Uses reduction OR negation (`~(|data)`) to detect when no bits are set
- **valid**: Uses reduction OR (`|data`) to indicate presence of at least one set bit

41.4.5 Special Implementation Notes

1. **Index Width Calculation**: Uses `$clog2(WIDTH)` to automatically size index outputs for any input width
2. **Bounds Checking**: Includes explicit bounds checking (`int'(leadingone) < WIDTH`) to prevent out-of-range indexing
3. **Zero Input Handling**: When input is all zeros:
    - One-hot vectors remain all zeros
    - Index values are deterministically **0** — `find_first_set`/`find_last_set` initialize `index = '0` and never

reassign it when no bit is set (the RTL header documents “0 if all zeros”). They are not X/undefined.

- `valid` flag indicates invalid state (use it to qualify the indices)

4. **Type Casting**: Uses `int'()` casting for index comparisons to ensure proper integer arithmetic

41.5 Timing Characteristics

This module is **purely combinational** – it contains no `always_ff` and no latch, so it holds no state and adds no clock cycles. Its outputs settle a propagation delay after its inputs, and it introduces no latency into a pipeline that instantiates it.

Timing closure is therefore a question of the surrounding logic’s slack, not of this module’s cycle count. No synthesis figures are quoted; none have been measured.

41.6 Usage Examples

41.6.1 8-bit Example

```
Input:  data = 8'b00101000
Output: leadingone = 3'd5
        leadingone_vector = 8'b00100000
        trailingone = 3'd3
        trailingone_vector = 8'b00001000
        all_zeroes = 1'b0
        all_ones = 1'b0
        valid = 1'b1
```

41.6.2 Edge Cases

```
Input:  data = 8'b00000000
Output: leadingone = 3'd0   (deterministic, not X; qualify with valid)
        leadingone_vector = 8'b00000000
        trailingone = 3'd0   (deterministic, not X; qualify with
valid)
        trailingone_vector = 8'b00000000
        all_zeroes = 1'b1
        all_ones = 1'b0
        valid = 1'b0

Input:  data = 8'b11111111
Output: leadingone = 3'd7
        leadingone_vector = 8'b10000000
        trailingone = 3'd0
```

```
trailingone_vector = 8'b00000001
all_zeroes = 1'b0
all_ones = 1'b1
valid = 1'b1
```

41.7 Design Notes

41.7.1 Applications

- Priority arbitration systems
- Bit manipulation operations
- Data compression algorithms
- Network packet processing
- Cache management systems

41.8 Related Modules

- **find_last_set**: Locates the most significant set bit; drives the `leadingone` output
- **find_first_set**: Locates the least significant set bit; drives the `trailingone` output

41.9 Testing

`val/common/test_leading_one_trailing_one.py` exercises this module. It collects 2 parameter cases at the default `REG_LEVEL`.

```
source env_python
pytest val/common/test_leading_one_trailing_one.py -v
```

41.10 References

- Oklobdzija, V.G. “An Algorithmic and Novel Design of a Leading Zero Detector Circuit: Comparison with Logic Synthesis.” *IEEE Transactions on VLSI Systems*, 2(1), Mar. 1994. DOI: 10.1109/92.273153 — leading-one detection, the function of the leading side of this module (built on `find_last_set`).

- Chu, P.P. *FPGA Prototyping by SystemVerilog Examples*, Wiley, 2018, ch. 3 “RT-Level Combinational Circuit”, Sec. 3.12.3 Dual-priority encoder (textbook treatment; secondary source).
- Indexed in `rtl/common/references/README.md` (local PDF: `LeadingZeroDetector__Oklobdzija1994.pdf`).

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42 pwm

42.1 Overview

Generates multiple independent pulse width modulation signals with configurable duty cycles, periods, and repeat counts. Each channel runs as its own state machine with precise timing control and completion detection.

- Multiple independent PWM channels (parameterizable)
- Individual duty cycle, period, and repeat count control per channel
- Edge-triggered start control with automatic edge detection
- Completion status reporting per channel
- Support for continuous operation (infinite repeat)
- Precise timing with configurable counter width

42.2 Parameters

Parameter	Default	Description
`WIDTH`	8	Bit width of counters and timing parameters
`CHANNELS`	4	Number of independent PWM channels

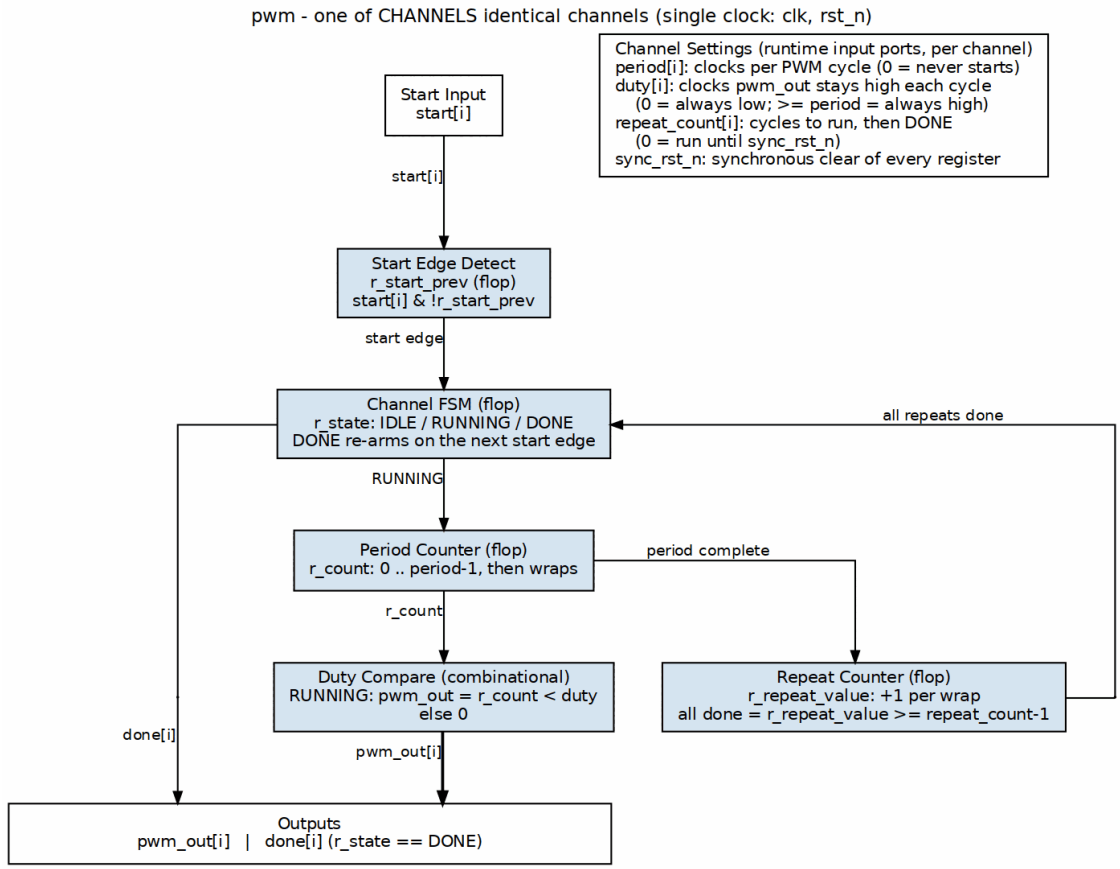
42.3 Ports

Port	Direction	Width	Description
`clk`	Input	1	System clock
`rst_n`	Input	1	Active-low asynchronous reset
`sync_rst_n`	Input	1	Active-low **synchronous** reset (functional — all state/counter/edge-detect blocks reset on `!sync_rst_n`). Must be

Port	Direction	Width	Description
			driven; leaving it unconnected gives X in sim / spurious resets in hardware.
`start`	Input	CHANNELS	Start trigger for each channel (edge-triggered)
`duty`	Input	CHANNELS* WIDTH	Concatenated duty cycle values for all channels
`period`	Input	CHANNELS* WIDTH	Concatenated period values for all channels
`repeat_count`	Input	CHANNELS* WIDTH	Concatenated repeat counts (0 = infinite)
`done`	Output	CHANNELS	Completion status for each channel
`pwm_out`	Output	CHANNELS	PWM output signals for each channel

42.4 Functional Description

42.4.1 Block diagram



pwm block diagram

Each of the `CHANNELS` channels is independent; the figure shows one. A rising edge on `start[i]` moves the channel FSM to RUNNING, and the period counter counts 0 to `period-1` and wraps. While running, `pwm_out` is high when the count is below `duty`. Each wrap increments the repeat counter, and when `repeat_count` periods have run (never, if `repeat_count` is 0) the FSM moves to DONE, which drives `done[i]` and waits for the next start edge.

42.4.2 State Machine

Each PWM channel runs a 3-state finite state machine:

```
typedef enum logic [1:0] {
    IDLE = 2'b00,     // Waiting for start trigger
    RUNNING = 2'b01,  // Actively generating PWM
    DONE = 2'b10      // Completed all repeats
} state_t;
```

42.4.3 State Transitions

IDLE → RUNNING: - **Condition**: `w_start_edge && local_period > 0` - **Action**: Reset counters and begin PWM generation

RUNNING → DONE: - **Condition**: `w_period_complete && w_all_repeats_done` - **Action**: Stop PWM generation and assert done flag

DONE → RUNNING: - **Condition**: `w_start_edge` - **Action**: Restart PWM generation with reset counters

Any State → IDLE: - **Condition**: System reset - **Action**: Initialize all registers

42.4.4 Per-Channel Parameter Extraction

```
localparam int EndIdx = (i + 1) * WIDTH - 1;
assign local_duty = duty[EndIdx-:WIDTH];
assign local_period = period[EndIdx-:WIDTH];
assign local_repeat = repeat_count[EndIdx-:WIDTH];
```

42.4.5 Edge Detection for Start Signal

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) r_start_prev <= 1'b0;
    else r_start_prev <= start[i];
end
assign w_start_edge = start[i] && !r_start_prev;
```

42.4.6 PWM Output Generation Logic

```
always_comb begin
    case (r_state)
        RUNNING: begin
            if (local_duty == 0) begin
                // 0% duty cycle
                pwm_out[i] = 1'b0;
            end else if (local_duty >= local_period) begin
                // 100% duty cycle (duty >= period)
                pwm_out[i] = 1'b1;
            end else begin
                // Normal case: PWM high for first 'duty' cycles
                pwm_out[i] = (r_count < local_duty);
            end
        end
        default: begin
            pwm_out[i] = 1'b0;
        end
    endcase
end
```

42.4.7 Counter Management

Main counter (period timing): - Increments each clock cycle during RUNNING state - Resets to 0 when reaching `local_period - 1` - Controls PWM frequency

Repeat counter: - Increments each time a period completes - Compared against `repeat_count` to determine completion - Special case: `repeat_count == 0` means infinite operation

42.4.8 Critical Control Signals

Period completion detection:

```
assign w_period_complete = (r_count == local_period - 1) && (r_state
== RUNNING);
```

All repeats done detection:

```
assign w_all_repeats_done = (local_repeat == 0) ? 1'b0 :  // 0 means
infinite
                            (r_repeat_value >= local_repeat - 1'b1);
```

The `- 1'b1` is load-bearing. The check fires in the same cycle that `r_repeat_value` increments at a period boundary, so a bare `>= local_repeat` emits `local_repeat + 1` periods — one too many. With `local_repeat = 1`: the corrected form is done at the first boundary (`0 >= 0`), while the bare form needs a second (`0 >= 1` false, then `1 >= 1`). The `local_repeat == 0` branch also keeps the `- 1` from underflowing, since 0 means infinite.

42.4.9 Special Implementation Notes

1. **Edge-Triggered Start Control**

- Prevents continuous restart while start signal is held high
- Enables precise control of PWM initiation
- Each channel has independent edge detection

2. **Duty Cycle Edge Cases**

- **0% Duty**: Output permanently low during RUNNING state
- **100% Duty**: Output permanently high when `duty >= period`
- **Normal Operation**: High for first `duty` clock cycles of each period

3. **Infinite Repeat Mode**

- `repeat_count = 0` enables continuous operation

- PWM runs indefinitely. A `start` pulse while RUNNING is **ignored** — the FSM leaves RUNNING only on `w_period_complete && w_all_repeats_done`, which never fires in continuous mode — so a running channel cannot be restarted or stopped by `start`. Use reset, or a non-zero `repeat_count`, to end it.
- Useful for continuous motor control or LED dimming

4. **Simultaneous Multi-Channel Operation**

- All channels operate independently
- No inter-channel dependencies or conflicts
- Scalable to any number of channels via parameters

5. **Reset Behavior**

- Two resets: `rst_n` (asynchronous) and `sync_rst_n` (synchronous). Either one returns the channel FSM to IDLE and clears counters.
- Asynchronous reset immediately stops all PWM outputs
- All internal counters and state machines reset to IDLE
- Clean startup guaranteed after reset deassertion

42.5 Timing Characteristics

42.5.1 Single Period Example (WIDTH=8, duty=3, period=8)

```
Clock:  0 1 2 3 4 5 6 7 0 1 2 3 4 5 6 7 ...
Count:  0 1 2 3 4 5 6 7 0 1 2 3 4 5 6 7 ...
PWM:    1 1 1 0 0 0 0 0 1 1 1 0 0 0 0 0 ...
```

42.5.2 Multi-Repeat Example (repeat_count=2)

```
Period: 1     2     DONE
PWM:    ____
```

42.6 Usage Examples

42.6.1 Motor Speed Control

```
// 50% duty cycle, 1kHz frequency (assuming 1MHz clock)
// WIDTH >= 10 needed for these values (default 8 truncates silently)
duty = 500;
```

```
period = 1000;
repeat_count = 0;  // Continuous operation
```

42.6.2 LED Dimming Sequence

```
// Fade effect with 10 brightness levels
duty = brightness_level * 10;
period = 100;
repeat_count = fade_duration;
```

42.6.3 Servo Motor Control

```
// 20ms period with 1-2ms pulse width
duty = servo_position;     // 1000-2000 for 1-2ms
period = 20000;  // needs WIDTH >= 15 -- at the default 8, 20000
truncates to 32           // 20ms at 1MHz clock
repeat_count = 1;         // Single pulse
```

42.7 Design Notes

42.7.1 Applications

- Motor speed control
- LED brightness control
- Servo motor positioning
- Audio signal generation
- Power supply switching
- Heating element control

42.8 Related Modules

Read out of the RTL, not curated: these are the modules this one instantiates and the modules that instantiate it.

Instantiated by: - `arbiter_rr_pwm_monbus` - `arbiter_wrr_pwm_monbus`

42.9 Testing

`val/common/test_pwm.py` exercises this module. It collects 2 parameter cases at the default `REG_LEVEL`.

```
source env_python
pytest val/common/test_pwm.py -v
```

42.10 References

- No peer paper — counter-compare pulse-width modulation is a standard technique with no defensible single origin (marked “standard technique” in the references index).
- Harris, D.M., Harris, S.L. *Digital Design and Computer Architecture, RISC-V Edition*, Morgan Kaufmann, ch. e9 “Embedded I/O Systems”, §9.3 (PWM peripheral) — textbook treatment (secondary source).
- Indexed in `rtl/common/references/README.md`

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43 rtl-common Quick Start Guide

Version: 1.0 **Last Updated:** 2025-09-30 **Status:** Stable, Production-Ready

---

43.1 Overview

The Common RTL Library is **48 reusable building blocks** for FPGA and ASIC designs. Every module is technology-agnostic, fully parameterizable, and verified with comprehensive CocoTB test suites. This guide is the fast path in.

Quick Stats: - **48 modules** across 10 categories (arithmetic split out to `rtl/math/`, clock-crossing to `rtl/cdc/`) - **Technology agnostic** (FPGA/ASIC portable) - **Well documented** (inline + external docs) - **Every module has a CocoTB test** under `val/common/`

43.2 Module Categories

Counts live in one place — the count table in index.md. Here's the browse-oriented view of what's in each category:

Category	Examples
Clock/Reset	clock_divider, clock_gate_ctrl, clock_pulse
Counters	counter, counter_bin, counter_bin_load, counter_freq_invariant, counter_load_clear, counter_ring
Data Integrity	dataint_crc, dataint_ecc_hamming, dataint_parity, dataint_checksum
Conversion & Encoding	bin_to_bcd, decoder, encoder, encoder_priority_enable, hex_to_7seg
Bit Ops & Search	count_leading_zeros, find_first_set, find_last_set, leading_one_trailing_one
Shifters & LFSRs	shifter_barrel, shifter_lfsr, shifter_lfsr_fibonacci, shifter_lfsr_galois
Arbiters	arbiter_round_robin, arbiter_round_robin_weighted,

Category	Examples
	arbiter_priority_encoder
FIFOs	fifo_sync, fifo_control (`fifo_async` moved to `rtl/cdc/`)
CAM	cam_tag
Miscellaneous	pwm, sort

Arithmetic (`math_*`) split out into `rtl/math/`; its docs are the rtl-math book.

Full per-module docs: docs/markdown/rtl-common/

43.3 Quick Start

43.3.1 1. Browse Available Modules

```
# List all modules
ls rtl/common/*.sv

# List by category
ls rtl/common/counter*.sv
ls rtl/common/arbiter*.sv
ls rtl/common/dataint*.sv

# Search for functionality
find rtl/common/ -name "*.sv" | xargs grep "fifo\|counter\|arbiter"
```

43.3.2 2. Check Module Interface

```
# View module header and parameters
head -50 rtl/common/counter_bin.sv

# Extract module definition
grep "^module\|parameter\|input\|output" rtl/common/counter_bin.sv
```

43.3.3 3. Review Test Examples

```
# Check if test exists
ls val/common/test_counter_bin.py

# View test for usage examples
cat val/common/test_counter_bin.py
```

43.3.4.4. Integrate into Your Design

```
// Example: Instantiate binary counter
counter_bin #(
    .WIDTH(11),          // 10 count bits + the wrap flag in the MSB
    .MAX  (1000)
) u_my_counter (
    .clk              (my_clk),
    .rst_n            (my_rst_n),
    .enable           (count_enable),
    .counter_bin_curr (count_value),
    .counter_bin_next (count_value_next)
);
```

43.4 Common Use Cases

43.4.1 Use Case 1: Simple Event Counter

Need: Count events, wrapping at a maximum

Module: `counter_bin.sv` — a FIFO-pointer counter. The low `WIDTH-1` bits count `0..MAX-1`; on wrap they clear and the MSB toggles, so a read/write pointer pair can tell full from empty. There is no overflow output.

```
counter_bin #(
    .WIDTH(9),           // 8 count bits + the wrap flag
    .MAX  (255)
) u_event_counter (
    .clk              (clk),
    .rst_n            (rst_n),
    .enable           (event_detected),
    .counter_bin_curr (event_count),
    .counter_bin_next (event_count_next)
);
```

43.4.2 Use Case 2: Timeout Timer

Need: A time base that stays correct when the clock frequency changes

Module: `counter_freq_invariant.sv` — a microsecond tick generator, not a timeout timer. It has no timeout parameter and no timeout output; build the timeout yourself by counting `tick`.

```
counter_freq_invariant #(
    .COUNTER_WIDTH   (16),
    .MIN_FREQ_MHZ    (5),
    .MAX_FREQ_MHZ    (220),
```

```
    .NUM_FREQ_ENTRIES(16)
) u_us_tick (
    .clk          (clk),
    .rst_n        (rst_n),
    .sync_reset_n (sync_clear_n),
    .freq_sel     (freq_sel),      // Picks the prescaler for the
actual clock
    .o_counter    (microseconds),
    .tick         (us_tick)        // One pulse per microsecond
);

// Timeout = count us_tick pulses and compare against your own
threshold.
```

43.4.3 Use Case 3: Multi-Master Arbitration

Need: Fair arbitration between 4 bus masters

Module: `arbiter_round_robin.sv`

```
arbiter_round_robin #(
    .CLIENTS     (4),   // 4 requesters -- N is the DERIVED
$clog2(CLIENTS)
    .WAIT_GNT_ACK(0)    // 1 = hold the grant until grant_ack
) u_bus_arbiter (
    .clk        (clk),
    .rst_n      (rst_n),
    .block_arb  (1'b0),
    .request    (master_requests[3:0]),  // One bit per master
    .grant_ack  ('0),
    .grant_valid(grant_valid),
    .grant      (master_grants[3:0]),    // One-hot grant
    .grant_id   (master_grant_idx),
    .last_grant (last_grant)
);
```

43.4.4 Use Case 4: CRC-32 (Ethernet)

Need: Calculate Ethernet CRC-32 for packet data

Module: `dataint_crc.sv`

```
dataint_crc #(
    .DATA_WIDTH(32),
    .CRC_WIDTH (32),
    .REFIN     (1),
```

```
    .REFOUT    (1)
) u_crc32 (
    .POLY             (32'h04C11DB7),  // CRC-32 Ethernet -- a PORT,
not a param
    .POLY_INIT        (32'hFFFFFFFF),
    .XOROUT           (32'hFFFFFFFF),
    .clk              (clk),
    .rst_n            (rst_n),
    .load_crc_start   (crc_restart),
    .load_from_cascade(data_valid),
    .cascade_sel      (4'b1000),       // One-hot over DATA_WIDTH/8
bytes
    .data             (packet_data),
    .crc              (crc_result)
);

// The polynomial, init and xorout are WIRED, so the CRC can be
retuned at run
// time. 250 configurations are exercised by the test suite.
```

43.4.5 Use Case 5: Memory ECC (Single Error Correction)

Need: Protect memory with error correction

Modules: `dataint_ecc_hamming_encode_secded.sv` + `dataint_ecc_hamming_decode_secded.sv`

```
// Encoder (write path) -- purely combinational
dataint_ecc_hamming_encode_secded #(
    .WIDTH(64)            // parameter is WIDTH, not DATA_WIDTH
) u_ecc_encoder (
    .data         (mem_write_data),
    .encoded_data (mem_data_with_ecc)   // one codeword: data + parity
+ SECDED bit
);

// Decoder (read path) -- CLOCKED, unlike the encoder
dataint_ecc_hamming_decode_secded #(
    .WIDTH(64)
) u_ecc_decoder (
    .clk                  (clk),
    .rst_n                (rst_n),
    .enable               (1'b1),
    .hamming_data         (mem_read_data_with_ecc),
    .data                 (corrected_data),
    .error_detected       (single_bit_error),   // Correctable
```

```
    .double_error_detected (double_bit_error)    // Detectable only
);
```

43.4.6 Use Case 6: Clock Domain Crossing (CDC)

Need: Safely cross a signal from one clock domain to another

Module: `cdc_synchronizer.sv` (for data) or `sync_pulse.sv` – both in `rtl/cdc/` (for pulses). There is no `sync_2ff` module — earlier revisions of this guide named one, but it has never existed in the tree.

```
// Option 1: Synchronize multi-bit data (rtl/cdc/cdc_synchronizer.sv)
// NOTE: a plain multi-flop synchronizer is only safe for signals
whose bits
// can be sampled independently -- a single-bit flag, or a quasi-
static value
// held stable across the crossing. For a multi-bit value that changes
as a
// unit, use Gray coding (counter_bingray) or a handshake/async FIFO
instead.
cdc_synchronizer #(
    .WIDTH(8),
    .FLOP_COUNT(3)
) u_data_sync (
    .clk        (dst_clk),
    .rst_n      (dst_rst_n),
    .async_in   (src_data),     // From source clock domain
    .sync_out   (dst_data)      // In destination clock domain
);

// Option 2: Synchronize single pulse
sync_pulse u_pulse_sync (
    .i_src_clk  (src_clk),
    .i_src_rst_n(src_rst_n),
    .i_pulse    (src_pulse),    // Single-cycle pulse
    .i_dst_clk  (dst_clk),
    .i_dst_rst_n(dst_rst_n),
    .o_pulse    (dst_pulse)     // Single-cycle pulse in dst domain
);
```

43.5 Module Selection Guide

43.5.1 "I need a counter..."

Requirement	Module	Notes
Simple up counter	`counter_load_clear.sv`	Plain counting to a value (counter_bin is a FIFO-POINTER counter – its MSB is a wrap flag, not a count bit)
With load/clear	`counter_load_clear.sv`	Explicit control
Time-based timeout	`counter_freq_invariant.sv`	Frequency-independent
Gray code	`counter_bingray.sv` (in `rtl/cdc/`)	For CDC
Ring counter	`counter_ring.sv`	Circular/sequential
Johnson counter	`counter_johnson.sv` (in `rtl/cdc/`)	2N states with N FFs

43.5.2 "I need an arbiter..."

Requirement	Module	Notes
Fair round-robin	`arbiter_round_robin.sv`	Most versatile
Weighted QoS	`arbiter_round_robin_weighted.sv`	Assign priorities
Fixed priority	`arbiter_priority_encoder.sv`	Lowest index wins
Minimal logic	`arbiter_round_robin_simple.sv`	Smallest area

43.5.3 "I need CRC..."

Standard	Module	Configuration
Any CRC standard	`dataint_crc.sv`	Drive the POLY input port (a PORT, not a parameter)
CRC-32 (Ethernet)	`dataint_crc.sv`	POLY port = 32'h04C11DB7
CRC-16-CCITT	`dataint_crc.sv`	POLY port = 16'h1021

Standard	Module	Configuration
CRC-8	`dataint_crc.sv`	POLY port = 8'h07
Custom	`dataint_crc_xor_shift.sv`	Build custom

The validated CRC configurations (250 standards) are the `crc_parameters` table in `bin/TBClasses/common/crc_testing.py`, which drives `val/common/test_dataint_crc.py`.

43.5.4 "I need a FIFO..."

For production designs: → Use `rtl/amba/gaxi/gaxi_fifo_sync.sv` (solid, well-tested)

For learning/simple cases: → Check `rtl/common/` for basic FIFO examples

---

43.6 Integration Checklist

When you integrate one of these modules into your design:

- ☐ **Search first** - Verify no better alternative exists
- ☐ **Read module header** - Understand parameters and constraints
- ☐ **Check parameter ranges** - Ensure your values are valid
- ☐ **Review test** - See `val/common/test_{module}.py` for examples
- ☐ **Match reset convention** - the port is `rst_n`, active-low async (`clock_gate_ctrl` uses `aresetn`); no rtl/common module exposes `i_rst_n`
- ☐ **Verify port widths** - Match parameter-dependent widths
- ☐ **Lint your design** - Run `verilator --lint-only` on top level
- ☐ **Test integration** - Create simple testbench
- ☐ **Check timing** - Verify no timing violations

43.7 Testing Your Integration

43.7.1 Option 1: Quick Functional Test

```
# Run existing module test to verify baseline
pytest val/common/test_{module}.py -v
```

```
# Example:
pytest val/common/test_counter_bin.py -v
```

43.7.2 Option 2: Create Your Own Test

```
# val/your_subsystem/test_your_design.py
import cocotb
from cocotb.clock import Clock
from cocotb.triggers import RisingEdge

@cocotb.test()
async def test_your_design(dut):
    """Test your design that uses common modules"""

    # Start clock
    clock = Clock(dut.clk, 10, units="ns")
    cocotb.start_soon(clock.start())

    # Reset
    dut.rst_n.value = 0
    await RisingEdge(dut.clk)
    await RisingEdge(dut.clk)
    dut.rst_n.value = 1

    # Your test logic here
    # ...
```

43.7.3 Option 3: Waveform Debug

```
# Run test with VCD waveform dump
pytest val/common/test_counter_bin.py -v --vcd=waves.vcd

# View waveforms
gtkwave waves.vcd

# Or use saved waveform config
gtkwave val/common/GTKW/counter_bin.gtkw
```

43.8 Common Pitfalls and How to Avoid Them

These four show up in design reviews again and again. All of them are cheap to avoid once you know what to look for.

43.8.1 Pitfall 1: Wrong Reset Polarity

Wrong:

```
counter_bin u_counter (
    .rst_n(my_positive_reset),  // ERROR: Inverted!
    // ...
);
```

Right:

```
counter_bin u_counter (
    .rst_n(~my_positive_reset),  // Invert if you have positive reset
    // OR better: use active-low reset throughout design
    .rst_n(my_rst_n),
    // ...
);
```

43.8.2 Pitfall 2: Parameter Mismatch

Wrong:

```
counter_bin #(
    .WIDTH(16)
    // Forgot MAX - uses the default of 10
) u_counter (
    .counter_bin_curr(count[7:0])  // ERROR: Width mismatch!
);
```

Right:

```
counter_bin #(
    .WIDTH(8),          // Match output width
    .MAX  (200)
) u_counter (
    .counter_bin_curr(count[7:0])  // Correct width
);
```

43.8.3 Pitfall 3: Clock Domain Crossing Without Sync

Wrong:

```
// Signal crosses from clk_a to clk_b domain
always_ff @(posedge clk_b)
    r_data <= signal_from_clk_a;  // ERROR: Metastability!
```

Right:

```
// Use proper synchronizer (rtl/cdc/cdc_synchronizer.sv)
cdc_synchronizer #(.WIDTH(8), .FLOP_COUNT(3)) u_sync (
    .clk      (clk_b),
    .rst_n    (rst_n_b),
```

```
    .async_in (signal_from_clk_a),
    .sync_out (signal_in_clk_b)
);
```

43.8.4 Pitfall 4: Creating New Module Without Searching

Wrong:

```
// "I need a counter, let me write one..."
module my_new_counter (...);
  // Reinventing the wheel
endmodule
```

Right:

```
# "I need a counter, does one exist?"
ls rtl/common/counter*.sv
# → Found counter_bin.sv, counter_load_clear.sv, ...
# → Use existing module!
```

43.9 Performance Considerations

43.9.1 Area Optimization

Small area: - Use `arbiter_round_robin_simple.sv` instead of full `arbiter_round_robin.sv` - Minimize WIDTH parameters - Use FFs instead of RAMs for small storage (<16 entries)

Large area (more features): - Use full-featured modules - Use RAM-based FIFOs for large buffers

43.9.2 Timing Optimization

Critical path reduction: - Enable output pipelining where available - Break long combinational chains - Use registered versions of modules

Example:

```
arbiter_round_robin #(
    .CLIENTS(16)        // N is derived ($clog2(CLIENTS)); do not
override it
) u_arbiter (
    // ...
);
```

43.9.3 Power Optimization

Clock gating: - Use `clock_gate_ctrl.sv` to gate inactive blocks - Gate counter enables when not counting

Example:

```
clock_gate_ctrl #(
    .IDLE_CNTR_WIDTH(4)
) u_gate (
    .clk_in           (clk),
    .aresetn          (aresetn),        // NOTE: this module uses
aresetn
    .cfg_cg_enable    (cg_enable),
    .cfg_cg_idle_count(4'd8),           // Cycles idle before gating
    .wakeup           (block_active),
    .clk_out          (clk_gated),
    .gating           (clk_is_gated)
);

// Use clk_gated for power-sensitive logic
```

43.10 Getting Help

43.10.1 Documentation

- **This file** - Quick start and common use cases
- `docs/markdown/rtl-common/index.md` - Detailed specifications and module catalog
- `rtl/common/CLAUDE.md` - AI assistance guide for this subsystem
- `/CLAUDE.md` - Repository-wide guidance
- `/PRD.md` - Master project requirements

43.10.2 Examples

- **Tests:** `val/common/test_*.py` - Working examples for every module
- **Usage:** Search rtl/amba/ and projects/components/ for integration examples

43.10.3 Commands

```
# Find usage examples of a module
grep -r "counter_bin" rtl/amba/ projects/components/

# See how tests use the module
cat val/common/test_counter_bin.py
```

```
# Check module parameters
grep "parameter" rtl/common/counter_bin.sv
```

43.10.4 For Claude Code Users

See `rtl/common/CLAUDE.md` for AI-specific guidance including: - Module search strategies - Common integration patterns - Debugging tips - Anti-patterns to avoid

---

43.11 Quick Command Reference

```
# List all common modules
ls rtl/common/*.sv

# Search for specific functionality
find rtl/common/ -name "*.sv" | xargs grep -i "keyword"

# Run test for a module
pytest val/common/test_{module}.py -v

# Run all common tests
pytest val/common/ -v

# Lint a module
verilator --lint-only rtl/common/{module}.sv

# View waveforms
gtkwave waves.vcd

# Count lines of code
wc -l rtl/common/*.sv

# Find instantiations of a module
grep -r "module_name" rtl/
```

43.12 What's Next?

43.12.1 Explore Other Subsystems

- **`rtl/amba/`** - AMBA protocol infrastructure (AXI, APB, AXIS monitors)
- **`projects/components/dma-ip/rapids/`** - Rapid AXI Programmable In-band Descriptor System (example accelerator)

- **`bin/TBClasses/`** - Verification infrastructure

43.12.2 Create Your Own Modules

1. Follow the patterns in the existing modules
2. Use consistent naming (`category_function.sv`)
3. Add comprehensive header comments
4. Create a test in `val/common/`
5. Update this quickstart and the count table in index.md if you add a category

43.12.3 Contribute

- Report issues
- Suggest improvements
- Share integration examples
- Document lessons learned

Questions? Check `docs/markdown/rtl-common/index.md` for detailed specifications or `rtl/common/CLAUDE.md` for AI assistance guidance.

Version: 1.0 **Last Updated:** 2025-09-30 **Maintained By:** RTL Design Sherpa Project

44 reverse_vector

44.1 Overview

The `reverse_vector` module flips the bit order of an input vector end for end: MSB swaps with LSB, second MSB with second LSB, and so on down the line. It's a purely combinational operation — no clock, no state, just wires with an opinion about ordering.

- Combinational bit reversal operation
- Parameterizable vector width
- Zero propagation delay (combinational logic only)
- No clock or reset required

44.2 Parameters

Parameter	Type	Default	Description
`WIDTH`	—	32	Width of the input/output vectors

44.3 Ports

44.3.1 Inputs

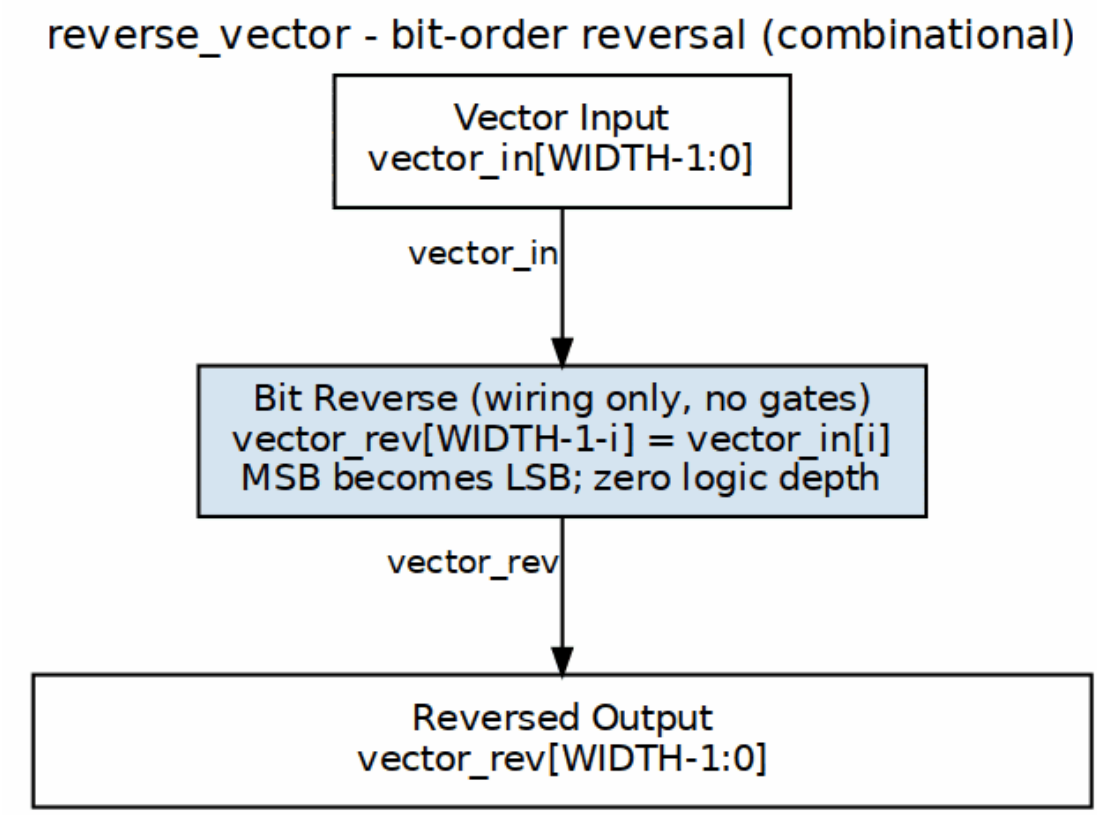
Port	Width	Description
`vector_in`	WIDTH	Input vector to be reversed

44.3.2 Outputs

Port	Width	Description
`vector_rev`	WIDTH	Bit-reversed output vector

44.4 Functional Description

44.4.1 Block diagram



reverse_vector block diagram

The input vector drops straight through one wiring-only block into the output: bit `i` lands on bit `WIDTH-1-i`. There are no gates, so the block adds no logic depth; it costs only routing.

44.4.2 Bit Reversal Logic

```
always_comb begin
    for (integer i = 0; i < WIDTH; i++) begin
        vector_rev[(WIDTH-1)-i] = vector_in[i];
    end
end
```

A simple for-loop maps each input bit to its mirrored output position: - Input bit 0 → Output bit (WIDTH-1) - Input bit 1 → Output bit (WIDTH-2) - Input bit i → Output bit (WIDTH-1-i) - Input bit (WIDTH-1) → Output bit 0

44.4.3 8-bit Vector Reversal

```
Input:  vector_in  = 8'b10110001
Output: vector_rev = 8'b10001101

Bit mapping:
vector_in[7] = 1 → vector_rev[0] = 1
vector_in[6] = 0 → vector_rev[1] = 0
vector_in[5] = 1 → vector_rev[2] = 1
vector_in[4] = 1 → vector_rev[3] = 1
vector_in[3] = 0 → vector_rev[4] = 0
vector_in[2] = 0 → vector_rev[5] = 0
vector_in[1] = 0 → vector_rev[6] = 0
vector_in[0] = 1 → vector_rev[7] = 1
```

44.4.4 4-bit Vector Reversal

```
Input:  vector_in  = 4'b1011
Output: vector_rev = 4'b1101

Bit mapping:
vector_in[3] = 1 → vector_rev[0] = 1
vector_in[2] = 0 → vector_rev[1] = 0
vector_in[1] = 1 → vector_rev[2] = 1
vector_in[0] = 1 → vector_rev[3] = 1
```

44.5 Timing Characteristics

There is no clock and no state here, so timing in the usual sense doesn't apply. The output follows the input after gate and wire delay, and that's the whole story:

- **Propagation Delay**: Minimal (wire delay only), < 1ns typical
- **Critical Path**: Direct wire connections
- **Setup/Hold**: Not applicable (combinational)
- **Maximum Frequency**: Unlimited by this module — limited by surrounding logic
- **Throughput**: One operation per clock cycle (if clocked)

44.6 Usage Examples

44.6.1 Endianness Conversion

```
// Convert between big-endian and little-endian bit ordering
little_endian_data = reverse_vector(big_endian_data);
```

44.6.2 Serial Communication

```
// Reverse bit order for serial transmission protocols
// that send LSB first instead of MSB first
serial_data = reverse_vector(parallel_data);
```

44.6.3 Cryptographic Operations

```
// Bit permutation operations in encryption algorithms
permuted_key = reverse_vector(encryption_key);
```

44.6.4 Signal Processing

```
// Bit-reversed addressing for FFT algorithms
reversed_address = reverse_vector(sequential_address);
```

44.6.5 Data Format Conversion

```
// Convert between different data representation formats
converted_data = reverse_vector(input_format);
```

44.6.6 Mirror Pattern Generation

```
// Create mirror patterns for display or test applications
mirror_pattern = reverse_vector(original_pattern);
```

44.6.7 Instantiation with Different Widths

```
// 16-bit reversal
reverse_vector #(.WIDTH(16)) rev16 (
    .vector_in(data_16bit),
    .vector_rev(reversed_16bit)
);

// 64-bit reversal
reverse_vector #(.WIDTH(64)) rev64 (
    .vector_in(data_64bit),
    .vector_rev(reversed_64bit)
);
```

44.6.8 Pipeline Integration

```
// Use in data processing pipeline
always_ff @(posedge clk) begin
    stage1_data <= input_data;
    stage2_data <= reverse_vector_inst.vector_rev;
    stage3_data <= process(stage2_data);
end
```

44.6.9 Conditional Reversal

```
// Conditional bit reversal based on control signal
assign output_data = reverse_enable ?
```

```
                    reverse_vector_inst.vector_rev :
                    vector_in;
```

44.7 Design Notes

44.7.1 1. Combinational Operation

- No clock signal required
- No sequential logic elements
- Output changes immediately when input changes
- Zero propagation delay (limited only by gate delays)

44.7.2 2. Parameterizable Width

The module works with any vector width specified by the WIDTH parameter: - Automatically scales to any desired width - Synthesis tools optimize for the specific width - No hardcoded bit indices

44.7.3 3. For-Loop Synthesis

Synthesis unrolls the for-loop completely, leaving one wire assignment per bit position. What you get: - Parallel bit assignments (no sequential operation) - Optimal timing characteristics - Efficient hardware implementation

44.7.4 4. Bidirectional Property

Reverse twice and you're back where you started:

```
original_vector == reverse(reverse(original_vector))
```

44.7.5 Power Consumption

- **Static Power**: Minimal (no storage elements)
- **Dynamic Power**: Only during input transitions
- **Switching Activity**: Proportional to input bit toggles

44.7.6 Area Utilization

- **Logic Elements**: Zero (pure routing)
- **Routing Resources**: WIDTH wire connections
- **Memory**: None required
- **Scalability**: Linear with WIDTH

44.7.7 Hardware Implementation

What synthesis actually builds: - Direct wire connections from input bits to output bits - No logic gates required (just routing) - Minimal area overhead - Excellent timing characteristics

44.7.8 Optimization

```
// Synthesis tools will optimize this to simple wire assignments:
assign vector_rev[WIDTH-1] = vector_in[0];
assign vector_rev[WIDTH-2] = vector_in[1];
// ...
assign vector_rev[1] = vector_in[WIDTH-2];
assign vector_rev[0] = vector_in[WIDTH-1];
```

44.7.9 Performance Metrics

Area: - **Gate Count**: 0 (wire-only implementation) - **Equivalent Gates**: WIDTH (for routing complexity) - **Memory Bits**: 0

Power: - **Leakage**: Negligible - **Dynamic**: Proportional to toggle rate - **Peak Power**: During simultaneous bit transitions

44.7.10 Common Pitfalls

Simulation warnings:

```
// Use 'integer' instead of 'int' for compatibility
for (integer i = 0; i < WIDTH; i++) begin
```

Synthesis issues:

```
// Ensure WIDTH is properly parameterized
localparam int W = WIDTH;  // Local parameter for clarity
```

Timing closure:

```
// Add pipeline stage if needed for timing
always_ff @(posedge clk) begin
    vector_rev_reg <= vector_rev;
end
```

44.8 Related Modules

44.8.1 Byte Reversal

If what you actually need is byte-level endianness conversion, a separate module is the better tool:

```
// Reverse byte order instead of bit order
assign byte_reversed = {data[7:0], data[15:8], data[23:16],
data[31:24]};
```

44.8.2 Configurable Reversal

An enhanced version could add an enable:

```
assign vector_out = enable ? vector_rev : vector_in;
```

44.8.3 Multi-Width Reversal

Support multiple data widths in a single module:

```
generate
    case (WIDTH)
        8: assign vector_rev = {vector_in[0], vector_in[1], ...};
        16: assign vector_rev = {vector_in[0], vector_in[1], ...};
        32: assign vector_rev = {vector_in[0], vector_in[1], ...};
    endcase
endgenerate
```

44.9 Testing

44.9.1 Test Cases

```
// Test all zeros
vector_in = '0;
assert(vector_rev == '0);

// Test all ones
vector_in = '1;
assert(vector_rev == '1);

// Test alternating pattern
vector_in = 8'b10101010;
assert(vector_rev == 8'b01010101);

// Test double reversal
temp = reverse_vector_inst1.vector_rev;
assert(reverse_vector_inst2.vector_rev == vector_in);
```

44.9.2 Coverage Points

- All bit positions exercised
- Boundary conditions (all 0s, all 1s)
- Random patterns
- Double reversal verification

44.10 References

- No peer paper: bit reversal is a standard technique (it is the REFIN/REFOUT operation of the CRC parameter model; see `dataint_crc.md`). Indexed in `rtl/common/references/README.md`.

44.11 Navigation

- **← Back to rtl-common Index**
- **← Back to Main Documentation Index**

45 shifter_barrel

45.1 Overview

The `shifter_barrel` module is a combinational barrel shifter: any shift amount, any supported mode, done in a single clock cycle. It covers logical left/right shifts with and without wrap-around, arithmetic right shifts, and a no-shift pass-through, all selected by a 3-bit control signal.

- Single-cycle shift operations for any shift amount
- Multiple shift modes: logical, arithmetic, and rotational
- Configurable shift amount up to data width
- Wrap-around (rotation) capabilities
- Purely combinational logic for maximum speed

45.2 Parameters

Parameter	Type	Default	Description
`WIDTH`	—	8	Width of the data bus

45.3 Ports

45.3.1 Inputs

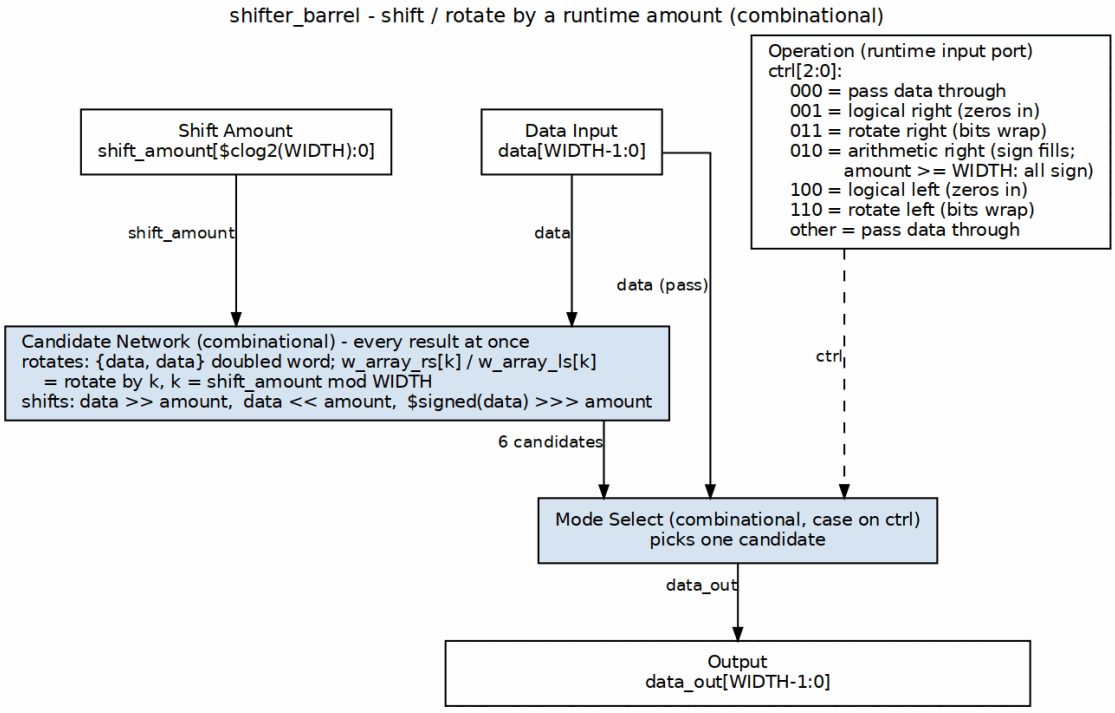
Port	Width	Description
`data`	WIDTH	Input data to be shifted
`ctrl`	3	Control signal for shift operation mode
`shift_amount`	$clog2(WIDTH)+1	Number of positions to shift

45.3.2 Outputs

Port	Width	Description
`data_out`	WIDTH	Shifted output data

45.4 Functional Description

45.4.1 Block diagram



shifter_barrel block diagram

Data and shift amount enter at the top and feed one candidate network that computes every result at once: rotates come from taps on the doubled word `{data, data}` (rotate amount reduced mod `WIDTH`), and plain shifts use >>, << and >>>. The mode select picks one candidate, or the unshifted data, according to `ctrl`; the legend at the top right gives each `ctrl` code. Everything is combinational.

45.4.2 Control Signal Encoding

The 3-bit `ctrl` signal determines the shift operation:

ctrl	Operation	Description
3'b000	No shift	Output equals input
3'b001	Logical Right Shift (no wrap)	Fill with zeros from left
3'b010	Arithmetic Right Shift	Preserve sign bit (MSB)
3'b011	Logical Right Shift	Rotate right

ctrl	Operation	Description
	(wrap)	
3'b100	Logical Left Shift (no wrap)	Fill with zeros from right
3'b110	Logical Left Shift (wrap)	Rotate left
Others	No shift	Default case

45.4.3 Shift Amount Modulation

```
logic [$clog2(WIDTH)-1:0] w_shift_amount_trunc, w_shift_amount_mod;
assign w_shift_amount_trunc = shift_amount[$clog2(WIDTH)-1:0];
assign w_shift_amount_mod = (32'(w_shift_amount_trunc) >= WIDTH)
                          ? $clog2(WIDTH)'(32'(w_shift_amount_trunc) -
WIDTH)
                          : w_shift_amount_trunc;
```

`shift_amount_mod` folds the amount into [0, WIDTH) for the wrap (rotate) paths — **only they use it**. The fold is truncate-to-$clog2(WIDTH)-bits then one conditional subtract of WIDTH: plain truncation is modulo a power of two, which for non-power-of-2 widths could reach past the rotate arrays (an out-of-bounds X before 2026-08-21; exact for all widths now). The three no-wrap paths take the RAW amount, with IEEE saturation for amounts at or beyond the width: logical right/left by `>= WIDTH` gives 0, arithmetic right gives all-sign-bits. (Before 2026-08-20 the no-wrap paths carried a `shift_amount_mod == 0` passthrough guard, so shifting by exactly WIDTH returned the *input* — and the arithmetic path shifted by the modded amount, wrong for every amount `>= WIDTH`. Found by a doc review round, sim-confirmed, fixed with directed WIDTH-boundary tests and an un-tautologized formal model that fails on the old RTL.)

45.4.4 Wrap-Around Implementation

```
logic [WIDTH*2-1:0] w_data_double;
assign w_data_double = {data, data};

// Generate lookup values for rotating shifts
genvar i;
generate
    for (i = 0; i < WIDTH; i++) begin : gen_unrolled_shifts
        assign w_array_rs[i] = w_data_double[WIDTH-1+i:i];      //
Right shift
        assign w_array_ls[i] = w_data_double[WIDTH*2-1-i:WIDTH-i]; //
Left shift
    end
endgenerate
```

45.4.5 Pre-computed Rotation Arrays

Two lookup arrays do the heavy lifting for rotation: - **w_array_rs[WIDTH]**: Pre-computed right rotation results for all possible shift amounts - **w_array_ls[WIDTH]**: Pre-computed left rotation results for all possible shift amounts

Pre-computing means there's no barrel-shifting network to evaluate at runtime — any rotation amount is a constant-time array lookup.

45.4.6 Main Shift Logic

```
always_comb begin
    case (ctrl)
        3'b000: // No shift
            data_out = data;

        3'b001: // Logical Right Shift (no wrap)
            data_out = data >> shift_amount;

        3'b010: // Arithmetic Right Shift (preserve sign)
            // if/else, NOT a ternary: a mixed-signedness ternary
silently
            // degrades >>> to a logical shift
            if (shift_amount >= ($clog2(WIDTH)+1)'(WIDTH))
                data_out = {WIDTH{data[WIDTH-1]}};
            else
                data_out = $signed(data) >>> shift_amount;

        3'b011: // Logical Right Shift with wrap
            data_out = w_array_rs[w_shift_amount_mod];

        3'b100: // Logical Left Shift (no wrap)
            data_out = data << shift_amount;

        3'b110: // Logical Left Shift with wrap
            data_out = w_array_ls[w_shift_amount_mod];

        default:
            data_out = data;
    endcase
end
```

45.5 Timing Characteristics

45.5.1 8-bit Examples (WIDTH=8)

45.5.1.1 Logical Right Shift (no wrap)

```
Input:  data = 8'b11010110, shift_amount = 3, ctrl = 3'b001
Output: 8'b00011010
```

45.5.1.2 Arithmetic Right Shift

```
Input:  data = 8'b11010110, shift_amount = 3, ctrl = 3'b010
Output: 8'b11111010  (sign extended)
```

45.5.1.3 Right Rotation

```
Input:  data = 8'b11010110, shift_amount = 3, ctrl = 3'b011
Output: 8'b11011010  (bits wrap around)
```

45.5.1.4 Logical Left Shift (no wrap)

```
Input:  data = 8'b11010110, shift_amount = 3, ctrl = 3'b100
Output: 8'b10110000
```

45.5.1.5 Left Rotation

```
Input:  data = 8'b11010110, shift_amount = 3, ctrl = 3'b110
Output: 8'b10110110  (bits wrap around)
```

45.6 Usage Examples

Barrel shifters show up anywhere bits need to move fast. Typical homes for this one:

- ALU implementations
- Cryptographic operations
- Signal processing
- Bit manipulation engines
- Network packet processing
- Data alignment circuits

45.7 Design Notes

45.7.1 1. Combinational Design

- All operations complete in zero clock cycles

- No state machines or sequential logic
- Suitable for high-frequency operations

45.7.2 2. Arithmetic Right Shift

The arithmetic shift uses SystemVerilog's >>> operator with a `$signed()` cast, so sign extension behaves correctly for two's complement numbers.

45.7.3 4. Double-Width Concatenation

The `{data, data}` concatenation is the oldest trick in the book: paste the data next to itself and a rotation becomes a plain slice out of a 2×-wide window. Any rotation amount, same cost.

45.7.4 5. Generate Block Optimization

The generate block pre-computes every possible rotation at compile time, so runtime is just a lookup. You spend a little area, you get back maximum speed.

45.7.5 Area Considerations

- **Lookup Arrays**: Require WIDTH × WIDTH bits of combinational logic
- **Multiplexers**: Large mux structures for final output selection
- **Total**: Approximately O(WIDTH^2) area complexity

45.7.6 Speed Characteristics

- **Propagation Delay**: Constant regardless of shift amount
- **Critical Path**: Through lookup array and final output mux
- **Frequency**: Limited by combinational delay, not shift complexity

45.8 Related Modules

Read out of the RTL, not curated: these are the modules this one instantiates and the modules that instantiate it.

Instantiated by: - `math_bf16_adder` - `math_ieee754_2008_fp32_adder`

45.9 Testing

`val/common/test_shifter_barrel.py` exercises this module. It collects 3 parameter cases at the default `REG_LEVEL`.

```
source env_python
pytest val/common/test_shifter_barrel.py -v
```

45.10 References

- Pillmeier, M.R., Schulte, M.J., Walters, E.G., III. “Design Alternatives for Barrel Shifters.” *Proc. SPIE* 4791, Advanced Signal Processing Algorithms, Architectures, and Implementations XII, Dec. 2002. DOI: 10.1117/12.452034 — compares mux-based, data-reversal and mask-based shift/rotate structures; this module selects rotates as fixed slices of `{data,data}` and leaves the plain shifts to synthesis.
- Chu, P.P. *FPGA Prototyping by SystemVerilog Examples*, Wiley, 2018, ch. 3 “RT-Level Combinational Circuit”, Sec. 3.10.3 Barrel shifter and Sec. 3.12.1-3.12.2 multi-function / parameterized barrel shifters (textbook treatment; secondary source).
- Harris, S.L., Harris, D. *Digital Design and Computer Architecture, RISC-V Edition*, Morgan Kaufmann, 2022, Sec. 5.2.5 Shifters and Rotators (textbook treatment; secondary source).
- Indexed in `rtl/common/references/README.md`.

45.11 Navigation

- **← Back to rtl-common Index**
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46 shifter_beat_pack

Module: `shifter_beat_pack.sv` **Location:** `rtl/common/` **Status:** Production Ready

46.1 Overview

`shifter_beat_pack` is a bit-granular packing/aligning shifter. It accepts fixed-width `CHUNK_BITS` entries on a push handshake, accumulates them in a multi-chunk staging register, and drains **configurable-width "beats"** out the low end on a pop handshake. The beat width is a runtime input (bytes minus one), so the same instance can emit different beat sizes across bursts.

All of the shift/load/mux logic lives inside this one module so that callers — an aligner that needs to repack DFI cycles into DRAM beats, say — don't have to invent their own shift-and-compensate scheme. Push and pop can occur in the same cycle; the next-state logic applies the pop first and the push second, and a single non-blocking assignment commits the result, so no internal last-assignment race is possible.

This module exists for a problem that shows up constantly: repacking a stream of one fixed width into beats of another (often runtime-selected) width. Data arrives one whole chunk at a time and leaves as beats whose width is chosen at run time; the module holds the partial-beat residue between cycles and shifts it down as beats drain, so the caller sees clean valid/ready handshakes on both sides.

Use Cases:

- Packing DFI cycles into DRAM beats in a memory-controller aligner
- Width adaptation where ingress chunk width and egress beat width differ and the beat width is runtime-configurable
- Any packer that must hold partial-beat residue while new fixed-width data lands

Key Benefit: Centralizes the shift, load, and residue-compensation logic behind two simple handshakes, so callers get runtime-variable beat repacking without hand-rolling their own barrel shifter.

Feature summary:

- **Bit-granular staging:** A `DEPTH_CHUNKS × CHUNK_BITS` register packs chunks and drains beats at bit resolution
- **Runtime beat width:** `cfg_beat_bytes_m1` selects the beat size (bytes − 1) per burst without re-elaboration

- **Independent ingress / egress sizing:** Chunk width and beat cap are separate parameters
- **Same-cycle push + pop:** Pop-then-push ordering with one NBA eliminates internal races
- **Elaboration-time contract checks:** `$error` guards enforce `DEPTH_CHUNKS >= 2` and `MAX_BEAT_BITS < STORAGE_BITS`
- **Status outputs:** `empty` and `count_bits_o` expose occupancy to the wrapper

46.2 Parameters

Parameter	Type	Default	Description
`CHUNK_BITS`	int	128	Width of one push entry (ingress granularity)
`MAX_BEAT_BYTES`	int	16	Maximum beat width in bytes; caps what reaches `pop_data`
`CFG_BITS`	int	8	Width of the runtime `cfg_beat_bytes_m1` field. 8 bits gives beat_bytes ∈ 1..256; 4 bits gives 1..16, etc.
`DEPTH_CHUNKS`	int	2	Staging-register depth in chunks. Default 2 = "current + prefetched-next". Must be `>= 2`. Larger depths absorb more before draining but widen the pop-side barrel shifter (timing cost).
`MAX_BEAT_BITS`	int	`MAX_BEAT_BYTES * 8`	Derived (do not override): maximum beat width in bits
`STORAGE_BITS`	int	`DEPTH_CHUNKS * CHUNK_BITS`	Derived (do not override): total staging-

Parameter	Type	Default	Description
			register width
COUNT_BITS	int	$clog2(STORAGE_BITS + 1)	Derived (do not override): width of the bit-occupancy counter
IDX_BITS	int	$clog2(STORAGE_BITS)	Derived (do not override): index width for r_data part-selects

46.3 Ports

Functionally the ports form a clock/reset pair, a runtime configuration input, a push (ingress) handshake, a pop (egress) handshake, and two status outputs. Grouped by direction:

46.3.1 Inputs

Port	Width	Description
clk	1	Clock
rst_n	1	Active-low reset (via reset_defs.svh macros)
cfg_beat_bytes_m1	CFG_BITS	Beat width in bytes − 1 (0 → 1 byte). Held stable per burst.
push_valid	1	New chunk valid (push/ingress handshake)
push_data	CHUNK_BITS	The chunk to pack
pop_ready	1	Consumer accepts a beat (pop/egress handshake)

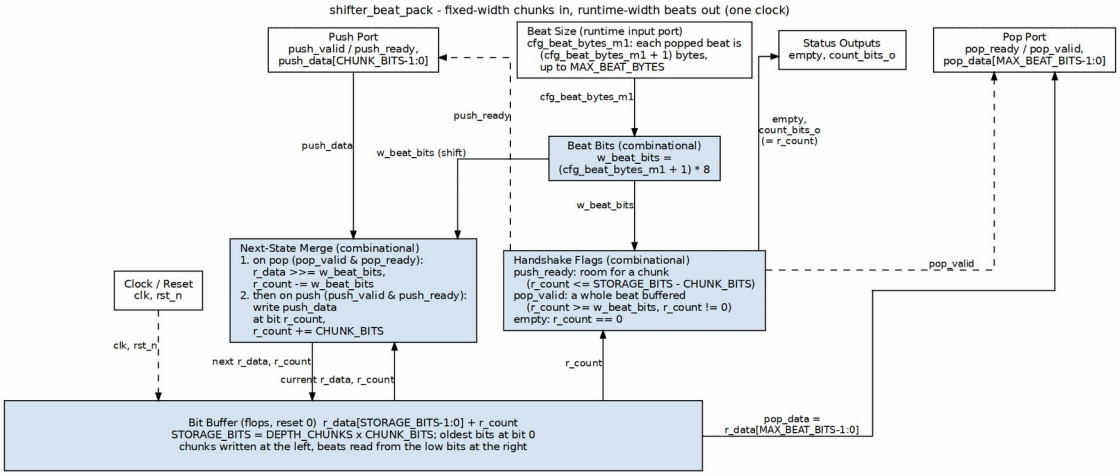
46.3.2 Outputs

Port	Width	Description
push_ready	1	Room for another whole chunk (post-push count <= STORAGE_BITS)
pop_valid	1	At least one whole beat is available
pop_data	MAX_BEAT_BITS	Low MAX_BEAT_BITS of the staging register; consumer reads only the low (cfg_beat_bytes_m1+1)*8 bits
empty	1	Staging register holds no bits (r_count

Port	Width	Description
		== 0)
count_bits_o	COUNT_BITS	Current occupancy in bits

46.4 Functional Description

46.4.1 Block diagram



shifter_beat_pack block diagram

Fixed `CHUNK_BITS` chunks enter down the left, and runtime-sized beats leave from the low bits of the bit buffer up the right edge. The beat size comes from `cfg_beat_bytes_m1` (legend at the top). The next-state merge reads the buffer and count straight up its own column, applies a pop (shift right by the beat width) and then a push (write the chunk at bit `r_count`), all in one clock. The handshake flags read `r_count` and return `push_ready` and `pop_valid` dashed to their ports.

46.4.2 Runtime Beat Width

The runtime beat width in bits comes from the byte-minus-one config field, widened by 4 bits so the counter math can represent the largest beat any `CFG_BITS`-wide value can encode:

```
localparam int BEAT_BITS_W = CFG_BITS + 4;
assign w_beat_bits = ({..., cfg_beat_bytes_m1} + 1) << 3;   //
(bytes_m1 + 1) * 8
```

`MAX_BEAT_BYTES` caps what actually appears on `pop_data`; `w_beat_bits` is used only for the occupancy accounting.

46.4.3 Handshake and Status

The combinational handshakes are driven directly from the bit-occupancy counter `r_count`:

- `empty` = `(r_count == 0)`

- `push_ready` = there is room for another whole chunk, i.e. `r_count <= STORAGE_BITS - CHUNK_BITS`
- `pop_valid` = the register holds at least one whole beat, i.e. `r_count >= w_beat_bits` and non-zero
- `pop_data` = the low `MAX_BEAT_BITS` of `r_data` (a plain slice, well-defined because a beat is always strictly narrower than the staging window)

When the configured beat is narrower than `MAX_BEAT_BITS`, the upper bits of `pop_data` are stale bytes belonging to the next beat. That's fine — the consumer already knows the width from `cfg` and reads only the low valid bits.

46.4.4 Next-State: Pop First, Push Second

Each cycle the combinational block computes next-state `w_v_data` / `w_v_count` in two ordered steps:

1. **Pop** (when `pop_valid && pop_ready`): shift the register down by `w_beat_bits` and subtract that many bits from the count, draining the low beat.
2. **Push** (when `push_valid && push_ready`): land the new chunk at `[w_v_count +: CHUNK_BITS]` using the *post-pop* occupancy, so the chunk sits right above the still-valid bits whether or not a pop fired, then add `CHUNK_BITS` to the count.

```
if (pop_valid && pop_ready) begin
    w_v_data  = w_v_data >> w_beat_bits;
    w_v_count = w_v_count - COUNT_BITS'(w_beat_bits);
end
if (push_valid && push_ready) begin
    w_v_data[w_v_count[IDX_BITS-1:0] +: CHUNK_BITS] = push_data;
    w_v_count = w_v_count + COUNT_BITS'(CHUNK_BITS);
end
```

One non-blocking assignment then writes `r_data` / `r_count` from `w_v_data` / `w_v_count`, so same-cycle push + pop has no internal last-assignment race.

46.4.5 Elaboration-Time Contract

Two `$error` guards enforce the sizing rules that keep forward progress guaranteed:

- `DEPTH_CHUNKS >= 2`: a single-chunk register cannot hold partial-beat residue while a new chunk lands.

- `MAX_BEAT_BITS < STORAGE_BITS`: a beat must be strictly narrower than the staging window.

46.4.6 Sizing Guidance

Pick `CHUNK_BITS` and `MAX_BEAT_BYTES` so any runtime beat width fits in the `2N` (or deeper) storage, i.e. `cfg_beat_bytes × 8 <= DEPTH_CHUNKS × CHUNK_BITS`. Ingress chunk width and egress beat cap are independent, so callers can right-size both.

46.5 Timing Characteristics

This module is **sequential**: it contains clocked logic (via `always_ff` or the repository's `ALWAYS_FF_RST` macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

46.6 Usage Examples

```
// Repack 128-bit DFI cycles into runtime-sized DRAM beats (up to 16
bytes).
shifter_beat_pack #(
    .CHUNK_BITS     (128),
    .MAX_BEAT_BYTES (16),
    .CFG_BITS       (8),
    .DEPTH_CHUNKS   (2)
) u_beat_pack (
    .clk               (clk),
    .rst_n             (rst_n),
    .cfg_beat_bytes_m1 (beat_bytes - 8'd1),   // e.g. 8 bytes -> 7

    // ingress: one 128-bit chunk per handshake
    .push_valid        (dfi_valid),
    .push_ready        (dfi_ready),
    .push_data         (dfi_cycle),

    // egress: one beat per handshake
    .pop_valid         (beat_valid),
    .pop_ready         (beat_ready),
    .pop_data          (beat_data),           // read low
(beat_bytes*8) bits
```

```
    // status
    .empty              (pack_empty),
    .count_bits_o       (pack_bits)
);
```

46.7 Design Notes

- **Bit occupancy, not entry count.** `r_count` tracks bits, not entries, because chunks land whole but beats drain at a runtime-variable bit width.
- **Pop-then-push ordering is deliberate.** Draining first frees the low bits so the incoming chunk lands at the correct post-pop position, which is what makes same-cycle push + pop safe.
- **`pop_data` upper bits may be stale.** Only the low `(cfg_beat_bytes_m1+1)*8` bits are meaningful; the consumer must mask to the configured width.
- **Depth vs. timing.** Increasing `DEPTH_CHUNKS` lets the packer absorb more data before draining but widens the pop-side barrel shifter, degrading timing.
- **Reset macros.** The module uses the project `reset_defs.svh` `ALWAYS_FF_RST` / `RST_ASSERTED` macros for the registered state.

46.8 Related Modules

- **Used by:** memory-controller aligners that repack DFI cycles into DRAM beats
- **Uses:** nothing beyond the `reset_defs.svh` reset macros — the module is self-contained
- **See also:** fifo_sync - Whole-entry synchronous FIFO (contrast: this module repacks at bit granularity)

46.9 Testing

`val/common/test_shifter_beat_pack.py` exercises this module. It collects 12 parameter cases at the default `REG_LEVEL`.

```
source env_python
pytest val/common/test_shifter_beat_pack.py -v
```

46.10 References

- No peer paper: chunk-to-beat packing (a width gearbox) is a standard technique. Indexed in `rtl/common/references/README.md`.

Last Updated: 2026-07-15

46.11 Navigation

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47 shifter_lfsr_fibonacci

47.1 Overview

The `shifter_lfsr_fibonacci` module is a Fibonacci Linear Feedback Shift Register — the textbook form, where all the feedback funnels into a single point at the most significant bit. It's the complement to the Galois LFSR, and the two come with genuinely different implementation trade-offs.

- Fibonacci (external XOR) LFSR architecture
- Configurable tap positions and polynomial
- Right-shift operation with feedback to MSB
- Seed loading and cycle detection
- Non-zero state enforcement
- Parameterizable width and tap configuration

47.2 Parameters

Parameter	Type	Default	Description
`WIDTH`	—	8	Width of the LFSR register
`TAP_INDEX_WIDTH`	—	12	Width of each tap index
`TAP_COUNT`	—	4	Number of feedback taps
`TIW`	—	—	Shorthand for TAP_INDEX_WIDTH

47.3 Ports

47.3.1 Inputs

Port	Width	Description
`clk`	1	System clock
`rst_n`	1	Active-low

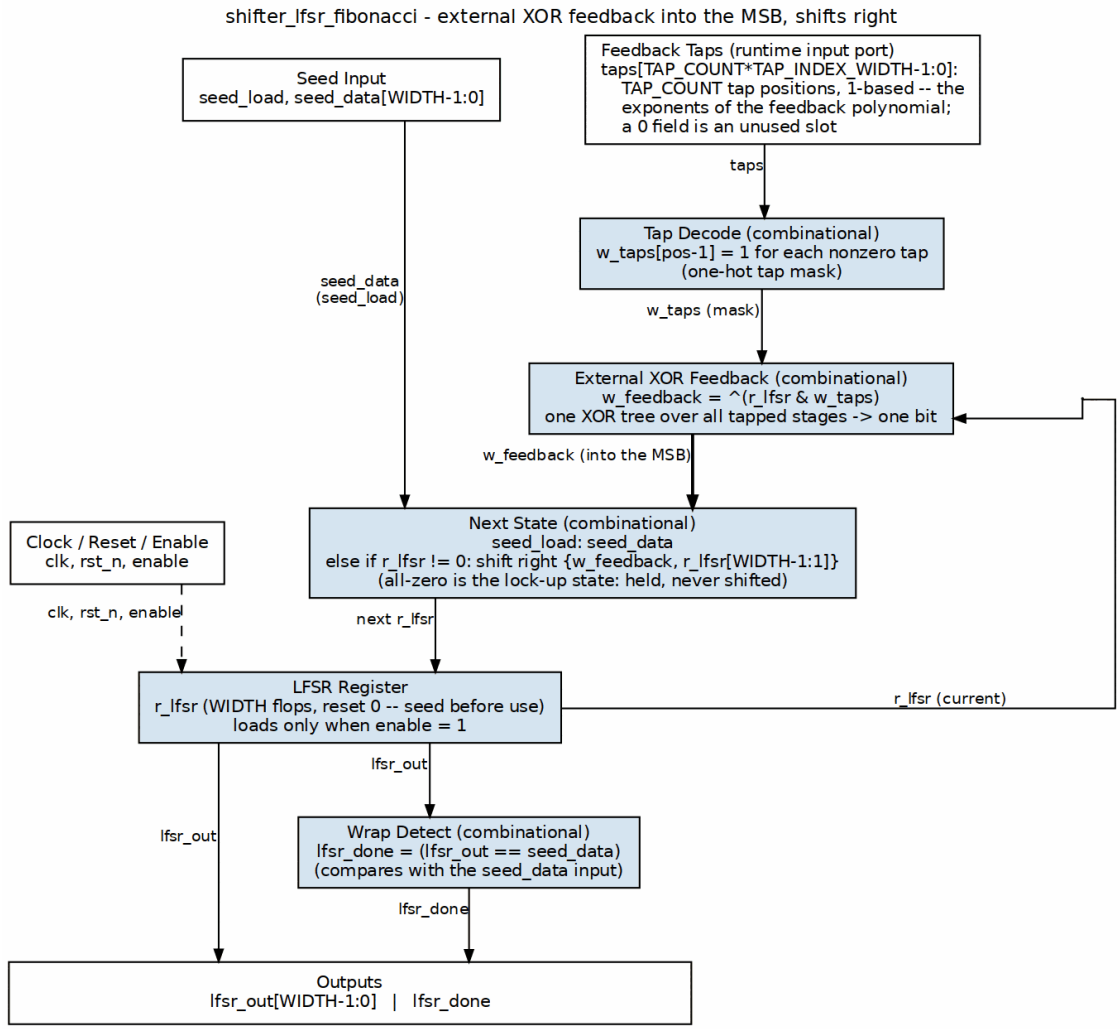
Port	Width	Description
		asynchronous reset
enable	1	Enable LFSR operation
seed_load	1	Load seed value into LFSR
seed_data	WIDTH	Seed value for LFSR initialization
taps	TAP_COUNT*TIW	Concatenated tap positions

47.3.2 Outputs

Port	Width	Description
lfsr_out	WIDTH	Current LFSR value
lfsr_done	1	High when LFSR returns to seed value

47.4 Functional Description

47.4.1 Block diagram



shifter_lfsr_fibonacci block diagram

This is the Fibonacci form: one XOR tree OUTSIDE the shift chain folds the tapped bits of the current register into a single feedback bit, which enters the MSB as the register shifts right. An all-zero register is the lock-up state and is held rather than shifted, so seed a non-zero value. Compare the Galois diagram, where the XORs sit inside the chain instead.

47.4.2 Fibonacci LFSR Characteristics

- **Feedback Location**: Single XOR gate feeding MSB
- **Data Path**: Simple right-shift with external feedback

- **XOR Gate Count**: One multi-input XOR gate
- **Critical Path**: Through feedback XOR to MSB

47.4.3 Fibonacci vs Galois at a Glance

Fibonacci vs Galois LFSR structure

	Fibonacci (this module)	Galois (`shifter_lfsr_galois`)
Where the XOR sits	one XOR tree outside the chain	one XOR per tap, inside the chain
Feedback bit	XOR of all tapped bits, into the MSB	bit 0 shifted out, toggles each tap
Critical path	through the XOR tree	one XOR between stages

47.4.4 Tap Processing (Same as Standard LFSR)

```
// Split concatenated tap positions
always_comb begin
    for (int i = 0; i < TAP_COUNT; i++)
        w_tap_positions[i] = taps[i*TIW+:TIW];
end

// Convert to bit mask
always_comb begin
    w_taps = 'b0;
    for (int i = 0; i < TAP_COUNT; i++)
        if (w_tap_positions[i] > 0)
            w_taps[w_tap_positions[i]-1'b1] = 1'b1;
end
```

47.4.5 Fibonacci Feedback Calculation

```
assign w_feedback = ^(r_lfsr & w_taps);
```

Key Difference: XOR (^) here, not the XNOR (~^) the standard LFSR module uses.

47.4.6 Right-Shift with MSB Feedback

```
always_ff @(posedge clk or negedge rst_n) begin
    if (~rst_n) begin
        r_lfsr <= 'b0;  // initialization to all 0's
    end else begin
        if (enable) begin
            if (seed_load) begin
                r_lfsr <= seed_data;  // Load seed
            end else if (|r_lfsr) begin // Only shift if we have non-
```

```
zero value
                    // Fibonacci LFSR: Shift right, feedback to MSB
                    r_lfsr <= {w_feedback, r_lfsr[WIDTH-1:1]};
                end
            end
        end
    end
```

47.5 Timing Characteristics

47.5.1 4-bit Fibonacci LFSR Example

Configuration: - WIDTH = 4 - Polynomial: $x^4 + x^3 + 1$, which **this module encodes as taps `[4,1]`** — see the tap-direction note under “Polynomial Examples” below - Seed: 4’b1001

47.5.2 Sequence Generation

Tap position `p` selects bit `p-1`, feedback is the XOR of the tapped bits, and the register shifts right with the feedback entering the MSB (`{w_feedback, r_lfsr[WIDTH-1:1]}`):

```
Cycle | LFSR | Tap bits (4,1) | Feedback | Next LFSR
------|------|----------------|----------|----------
0     | 1001 | 1,1            | 1^1=0    | 0100
1     | 0100 | 0,0            | 0^0=0    | 0010
2     | 0010 | 0,0            | 0^0=0    | 0001
3     | 0001 | 0,1            | 0^1=1    | 1000
4     | 1000 | 1,0            | 1^0=1    | 1100
5     | 1100 | 1,0            | 1^0=1    | 1110
...
```

The sequence runs the full period of 15 and returns to the seed, at which point `lfsr_done` asserts. Reset loads all zeros, and the `|r_lfsr` guard means the register cannot leave zero on its own — a `seed_load` is required to start it. Getting the taps wrong costs the full period, and how it fails depends on the seed: sweeping all 15 non-zero seeds at WIDTH=4 with `[4,3]`, three walk to zero and freeze under the `|r_lfsr` guard (seeds 1..3), three revisit their seed so `lfsr_done` DOES assert (seeds 6, 11, 13), and the remaining nine settle into a cycle that never revisits the seed.

47.6 Usage Examples

- Pseudo-random sequence generation (same maximal-length guarantees as the Galois form; the STATE SEQUENCES differ – see the Usage Note and the Galois page)
- CRC calculation (with appropriate polynomial)
- Data scrambling and encryption
- Test pattern generation

- Spread spectrum communications
- Error detection and correction

47.7 Design Notes

47.7.1 1. XOR vs XNOR Feedback

- **Fibonacci LFSR**: Uses XOR (^) for feedback calculation
- **Standard LFSR**: Uses XNOR (~^) for feedback calculation
- Both produce maximal-length sequences with appropriate polynomials

47.7.2 2. Non-Zero State Protection

```
end else if (|r_lfsr) begin // Only shift if we have non-zero value
```

This guard stops the LFSR from shifting when it's sitting at all zeros — a state it could never leave on its own. The sequence would lock at zero permanently.

47.7.3 3. Right-Shift Architecture

```
r_lfsr <= {w_feedback, r_lfsr[WIDTH-1:1]};
```

- Feedback enters at MSB position
- Data shifts right (towards LSB)
- LSB data is discarded each cycle

47.7.4 4. Polynomial Compatibility

Your tap positions need to correspond to a primitive polynomial appropriate for Fibonacci LFSR implementation. The polynomial form is:

```
P(x) = x^n + x^(tap1) + x^(tap2) + ... + x^(tapk) + 1
```

47.7.5 5. Reset to All Zeros

Unlike some LFSR implementations that reset to all ones, this module resets to all zeros and relies on seed loading for proper initialization.

47.7.6 Advantages of Fibonacci LFSR

- **Simpler Data Path**: Only one shift register with external feedback
- **Easy to Understand**: Classical LFSR textbook implementation
- **Single XOR Gate**: All feedback logic concentrated in one location

47.7.7 Disadvantages of Fibonacci LFSR

- **Fan-out**: Feedback XOR gate may have high fan-out for many taps
- **Critical Path**: All tapped bits must route to single XOR gate
- **Timing**: May be slower than distributed Galois implementation

47.7.8 When to Use Fibonacci LFSR

- **Educational Purposes**: Easier to understand and debug
- **Few Taps**: When polynomial has only 2-3 taps
- **Area Constraints**: When minimizing XOR gate count is critical
- **Legacy Compatibility**: When interfacing with existing Fibonacci LFSR systems

47.7.9 Polynomial Examples for Fibonacci LFSR

The tap numbers below are specific to this module's shift direction. The tap column published for Galois LFSRs – and the table in the `shifter_lfsr.sv` header, which is XNOR feedback with a *left* shift – encodes the same polynomials differently. Get it wrong and you lose the full period, and the failure is quiet either way. Sweeping all 15 non-zero seeds on this RTL at `WIDTH=4` with taps `[4,3]`: three seeds (1, 2, 3 – including `4'b0001`, which reaches zero in **one** step) walk to zero and freeze under the `|r_lfsr` guard; three (6, 11, 13) DO revisit their seed, so `lfsr_done` asserts for them; the remaining nine settle into a cycle that never revisits the seed and never assert it. `[4,1]` runs the full period of 15 from every seed.

Width	Polynomial	Tap Positions	Period
3	x^3+x^2+1	[3,1]	7
4	x^4+x^3+1	[4,1]	15
5	x^5+x^3+1	[4,1]	31
8	$x^8+x^6+x^5+x^4+1$	[7,6,5,1]	255
16	$x^{16}+x^{15}+x^{13}+x^4+1$	[16,14,5,1]	65535
24	$x^{24}+x^{23}+x^{22}+x^{17}+1$	[24,23,18,1]	16777215
32	$x^{32}+x^{22}+x^2+x+1$	[23,3,2,1]	4294967295

The polynomials themselves are the standard primitive set; only the tap *encoding* differs from the published one. The full 168-width table in this module's RTL header (`rtl/common/shifter_lfsr_fibonacci.sv`) is already converted – use it directly.

47.7.10 Why the Encoding Differs

This module computes `fb = ^(lfsr & taps)` and shifts it into the MSB, so a tap at position `t` contributes `x^(t-1)` and the characteristic polynomial is:

```
x^WIDTH + SUM over taps of x^(tap-1)
```

For a published polynomial `x^n + x^a + x^b + 1`, the taps here are `[a+1, b+1, 1]`. Two consequences worth remembering:

- **Tap 1 is always present** – it supplies the constant term. A tap set without it cannot be maximal, and typically locks at zero.
- **`n` itself is not a tap** – the register width supplies the leading term.

To convert any row of the standard table: drop the leading `n`, add 1 to every remaining number, then append 1.

47.7.11 Usage Note

The same polynomial works for both Fibonacci and Galois LFSRs, and both produce maximal-length sequences of the same period – but the *tap numbers you pass in* differ, and the state orderings differ. `shifter_lfsr_galois.sv` takes the exponents directly (`[n, a, b]`); this module takes `[a+1, b+1, 1]`.

47.8 Related Modules

- shifter_lfsr_galois — the distributed-XOR Galois form. Same polynomials and period, but the tap numbers are **not** interchangeable with this module's, and the state orderings differ.
- shifter_lfsr — the generic XNOR-feedback LFSR; left-shift, different tap encoding again (see the warning above).

47.9 Testing

`val/common/test_shifter_lfsr_fibonacci.py` exercises this module. It collects 6 parameter cases at the default `REG_LEVEL`.

```
source env_python
pytest val/common/test_shifter_lfsr_fibonacci.py -v
```

47.10 References

- Ward, R., Molteno, T.C.A. “Table of Linear Feedback Shift Registers.” Electronics Technical Report No. 2012-1, University of Otago, 2012 (non-peer technical report) — open maximal-length tap table; note this module’s Fibonacci tap numbering differs from the published Galois/XNOR columns, as the Design Notes explain.
- Moon, T.K. *Error Correction Coding*, 2nd ed., Wiley, 2021, ch. 4, Appendix 4.A “Linear Feedback Shift Registers” (textbook treatment; secondary source).
- Merrick, R. *Getting Started with FPGAs*, No Starch Press, 2024, ch. 6 “Common FPGA Modules”, “The Shift Register” (including “Creating a Linear Feedback Shift Register”) (textbook treatment; secondary source).
- Indexed in `rtl/common/references/README.md` (local PDF: `LfsrTapTable__Ward2012.pdf`).

47.11 Navigation

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48 shifter_lfsr_galois

48.1 Overview

The `shifter_lfsr_galois` module is a Galois Linear Feedback Shift Register: instead of one feedback XOR at the end of the chain, the XOR gates are distributed across the tap positions inside the register. That buys you better timing and higher operating frequencies than the Fibonacci form — which is usually why you're reaching for it.

- Galois (internal XOR) LFSR architecture
- Distributed feedback XOR gates at tap positions
- Right-shift operation with LSB feedback
- Superior timing characteristics for high-speed operation
- Configurable tap positions and polynomial
- Seed loading and cycle detection

48.2 Parameters

Parameter	Type	Default	Description
`WIDTH`	—	8	Width of the LFSR register
`TAP_INDEX_WIDTH`	—	12	Width of each tap index
`TAP_COUNT`	—	4	Number of feedback taps
`TIW`	—	—	Shorthand for TAP_INDEX_WIDTH

48.3 Ports

48.3.1 Inputs

Port	Width	Description
`clk`	1	System clock
`rst_n`	1	Active-low

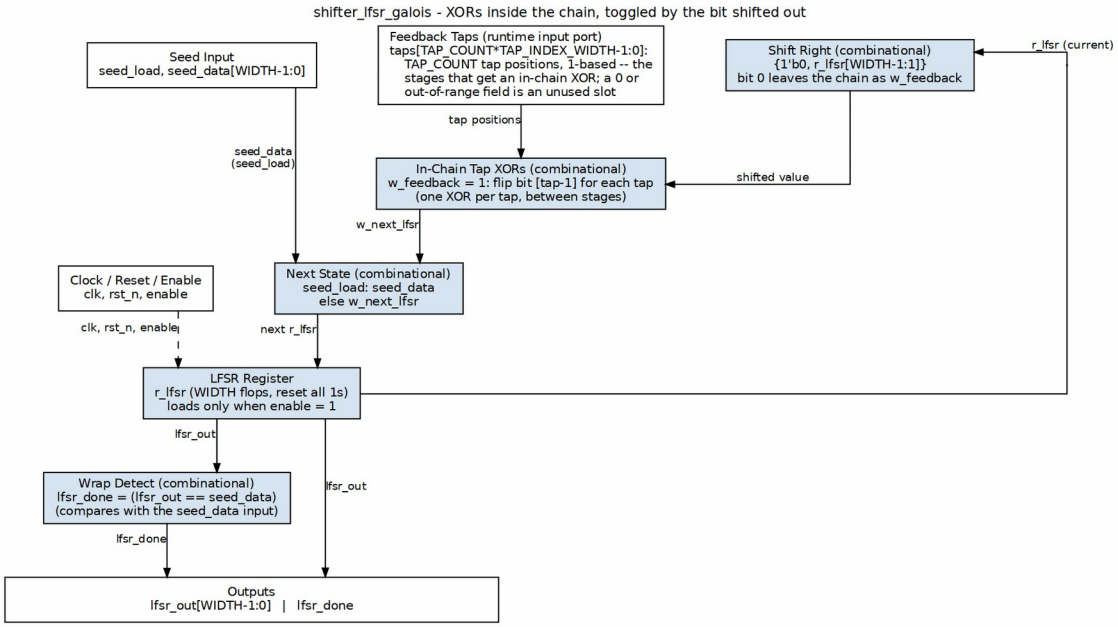
Port	Width	Description
		asynchronous reset
`enable`	1	Enable LFSR operation
`seed_load`	1	Load seed value into LFSR
`seed_data`	WIDTH	Seed value for LFSR initialization
`taps`	TAP_COUNT*TIW	Concatenated tap positions

48.3.2 Outputs

Port	Width	Description
`lfsr_out`	WIDTH	Current LFSR value
`lfsr_done`	1	High when LFSR returns to seed value

48.4 Functional Description

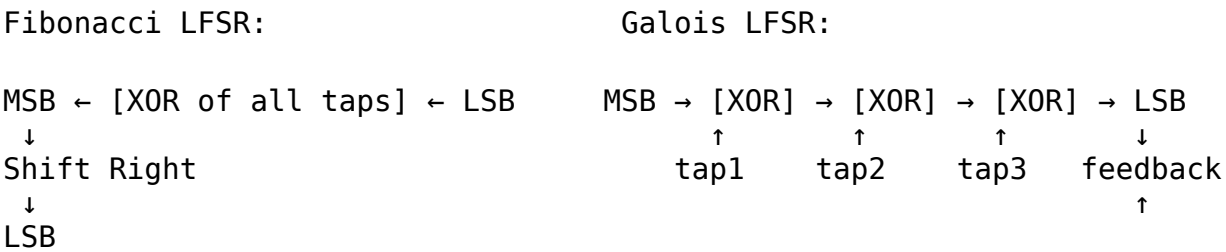
48.4.1 Block diagram



shifter_lfsr_galois block diagram

This is the Galois form: the register shifts right, and the bit leaving at bit 0 is the feedback. When it is 1, the in-chain XORs flip every tap position of the shifted value, one XOR per tap between stages, so there is no wide XOR tree on the critical path. Compare the Fibonacci diagram, whose single XOR tree sits outside the chain. The register resets to all ones.

48.4.2 Key Architectural Differences



48.4.3 Advantages of Galois Architecture

- **Distributed Logic**: XOR gates spread throughout the shift register
- **Better Timing**: Shorter critical paths between flip-flops
- **Higher Frequency**: Can operate at higher clock speeds
- **Localized Routing**: Reduced interconnect complexity

48.4.4 Tap Position Processing

```
// Split concatenated tap positions
always_comb begin
    for (int i = 0; i < TAP_COUNT; i++) begin
        w_tap_positions[i] = taps[i*TIW+:TIW];
    end
end
```

48.4.5 LSB Feedback Generation

```
assign w_feedback = r_lfsr[0];
```

The feedback is just the LSB (bit 0) of the current LFSR state. That's it.

48.4.6 Galois Next State Calculation

```
always_comb begin
    // Start with right shift (include 0 in MSB)
    next_lfsr = {1'b0, r_lfsr[WIDTH-1:1]};

    // Apply Galois feedback taps if LSB is 1
    if (w_feedback) begin
        for (int j = 0; j < TAP_COUNT; j++) begin
            if (w_tap_positions[j] > 0 && w_tap_positions[j] <= WIDTH)
```

```
begin
                // Apply XOR to the tap positions
                next_lfsr[w_tap_positions[j]-1] =
next_lfsr[w_tap_positions[j]-1] ^ 1'b1;
            end
        end
    end
end
```

48.4.7 Sequential Update

```
always_ff @(posedge clk or negedge rst_n) begin
    if (~rst_n) begin
        r_lfsr <= {WIDTH{1'b1}};  // initialization to all 1's
    end else if (enable) begin
        if (seed_load) begin
            r_lfsr <= seed_data;
        end else begin
            r_lfsr <= next_lfsr;
        end
    end
end
```

48.5 Testing

`val/common/test_shifter_lfsr_galois.py` exercises the module, and `val/common/test_shifter_lfsr_galois_sequence.py` checks the generated sequence against a reference model.

```
source env_python
pytest val/common/test_shifter_lfsr_galois.py
val/common/test_shifter_lfsr_galois_sequence.py -v
```

48.6 Timing Characteristics

48.6.1 4-bit Galois LFSR Example

Configuration: - WIDTH = 4 - Polynomial: $x^4 + x^3 + 1$ (taps at positions 4,3) - Seed: 4'b1001

48.6.2 Sequence Generation

Galois right-shift: shift right by 1; if the shifted-out LSB was 1, XOR the tap mask `4'b1100` (bits 3 and 2 — positions 4,3, 1-indexed) into the shifted value.

```
Cycle | LFSR | LSB | after >>1 | XOR 1100? | Next LFSR
------|------|-----|-----------|-----------|----------
0     | 1001 | 1   | 0100      | yes       | 1000
1     | 1000 | 0   | 0100      | no        | 0100
```

```
2     | 0100 | 0   | 0010      | no        | 0010
3     | 0010 | 0   | 0001      | no        | 0001
4     | 0001 | 1   | 0000      | yes       | 1100
...
```

48.7 Usage Examples

48.7.1 High-Speed Systems

```
// Use Galois LFSR for high-frequency applications
// Clock rates > 500MHz where timing is critical
```

48.7.2 Communications

```
// Spread spectrum systems requiring fast sequence generation
// Real-time scrambling/descrambling applications
```

48.7.3 Cryptography

```
// Stream cipher implementations
// Key sequence generation
```

48.7.4 Testing

```
// High-speed BIST pattern generation
// Fast pseudo-random test vectors
```

48.8 Design Notes

48.8.1 1. Reset to All Ones

```
r_lfsr <= {WIDTH{1'b1}};  // initialization to all 1's
```

Where the Fibonacci version resets to all zeros, this one resets to all ones — a valid non-zero state, so it can run immediately.

A zero seed locks this module, and nothing stops you loading one. Unlike `shifter_lfsr_fibonacci.sv`, which shifts only while `|r_lfsr` holds, this module has no zero guard: with `r_lfsr = 0` the feedback bit `r_lfsr[0]` is 0, no taps are applied, and `next_lfsr = {1'b0, r_lfsr[WIDTH-1:1]}` is zero again. The register stays at zero for good — and because `lfsr_done` is the equality `lfsr_out == seed_data`, a zero seed also parks `lfsr_done` high forever instead of pulsing once per period. Drive `seed_data` non-zero on `seed_load`; the reset value is safe, only a loaded seed can do this.

48.8.2 2. Combinational Next State Logic

The next LFSR state is calculated combinationally in the `always_comb` block, then registered in the `always_ff` block. Clean separation: timing concerns stay out of the logic, and the logic stays easy to read.

48.8.33. Conditional XOR Application

```
if (w_feedback) begin
    // Only apply tap XORs when LSB is 1
```

The tap XORs fire only when the LSB feedback bit is 1 — that conditional inversion *is* the Galois behavior.

48.8.44. Bounds Checking

```
if (w_tap_positions[j] > 0 && w_tap_positions[j] <= WIDTH) begin
```

Explicit bounds checking prevents out-of-range array access and handles unused tap positions (value 0).

48.8.55. Right-Shift with Distributed Feedback

The basic operation is a right shift, but with XOR gates inserted at tap positions that conditionally invert bits based on the LSB feedback.

48.8.6 Critical Path Analysis

```
Fibonacci LFSR Critical Path:
All tapped bits → Multi-input XOR → MSB flip-flop

Galois LFSR Critical Path:
Previous bit → XOR gate → Current bit flip-flop
```

48.8.7 Timing Advantages

- **Shorter Logic Depth**: Each bit only sees local XOR gate
- **Parallel Processing**: All XOR operations occur simultaneously
- **Reduced Fan-out**: No single XOR gate with multiple inputs
- **Better Scaling**: Critical path doesn't grow with tap count

48.8.8 Tap Position Mapping

For a polynomial P(x) = x^n + x^a + x^b + ... + 1: - Galois LFSR taps: [n, a, b, ...] – the exponents, used directly - The same polynomial is maximal for both Fibonacci and Galois, but **the tap numbers you pass in are not interchangeable**. This module XORs the mask into the post-shift value, so taps are the exponents; `shifter_lfsr_fibonacci.sv` XORs the tapped bits into the MSB, which shifts the encoding to `[a+1, b+1, 1]`. Passing Galois taps to the Fibonacci module costs the full period, but what you see depends on the seed: measured at `WIDTH=4` with taps `[4,3]` fed to the FIBONACCI module, three of the 15 non-zero seeds walk to zero and freeze (1..3), three revisit their seed so `lfsr_done` still asserts (6, 11, 13), and the remaining nine cycle without ever revisiting. (This module itself with `[4,3]` is maximal: all 15 seeds run the full period – enumerated 2026-08-22.) - Different sequences produced but same mathematical properties

48.8.9 Example Polynomials

Width	Polynomial	Galois Taps	Max Period
3	x^3+x^2+1	[3,2]	7
4	x^4+x^3+1	[4,3]	15
5	x^5+x^3+1	[5,3]	31
8	$x^8+x^6+x^5+x^4+1$	[8,6,5,4]	255
12	$x^{12}+x^6+x^4+x+1$	[12,6,4,1]	4095
16	$x^{16}+x^{15}+x^{13}+x^4+1$	[16,15,13,4]	65535
24	$x^{24}+x^{23}+x^{22}+x^{17}+1$	[24,23,22,17]	16777215
32	$x^{32}+x^{22}+x^2+x+1$	[32,22,2,1]	$2^{32}-1$

Widths 3 through 16 were confirmed by simulation against this RTL; each yields period 2^n - 1. The full 168-width table is in the module's RTL header (`rtl/common/shifter_lfsr_galois.sv`).

48.8.10 Galois LFSR Advantages

- **Speed**: Higher maximum clock frequency
- **Timing**: Better setup/hold characteristics
- **Scalability**: Performance doesn't degrade with tap count
- **Power**: Potentially lower dynamic power

48.8.11 Galois LFSR Disadvantages

- **Complexity**: More complex combinational logic
- **Area**: May require more logic gates for many taps
- **Debug**: Harder to trace intermediate states
- **Understanding**: Less intuitive than classical Fibonacci

48.8.12 When to Choose Galois over Fibonacci

Use Galois LFSR when: - High clock frequency is required (>100MHz) - Timing closure is challenging - Multiple taps are needed (>3 taps) - Power consumption is critical - Synthesis tools favor distributed logic

Use Fibonacci LFSR when: - Educational/demonstration purposes - Legacy system compatibility required - Very few taps (2 taps or less) - Area is more critical than speed - Debugging capabilities are important

48.9 Related Modules

- shifter_lfsr_fibonacci — the single-XOR Fibonacci form. Do **not** pass this page's tap numbers to it; the encoding differs (see Tap Position Mapping above).
- shifter_lfsr — the generic XNOR-feedback LFSR.

48.10 References

- Ward, R., Molteno, T.C.A. "Table of Linear Feedback Shift Registers." Electronics Technical Report No. 2012-1, University of Otago, 2012 (non-peer technical report) — open maximal-length tap table whose tap numbers are the polynomial exponents, as this module uses them.
- Moon, T.K. *Error Correction Coding*, 2nd ed., Wiley, 2021, ch. 4, Appendix 4.A "Linear Feedback Shift Registers" (textbook treatment; secondary source).
- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 4 "Basic Sequential Circuits", Sec. 4.5 Pseudo Noise Sequence Generator (textbook treatment; secondary source).
- Indexed in `rtl/common/references/README.md` (local PDF: `LfsrTapTable__Ward2012.pdf`).

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49 shifter_lfsr

49.1 Overview

The `shifter_lfsr` module is the generic Linear Feedback Shift Register: configurable tap positions for pseudo-random number generation, support for various LFSR polynomials, and cycle completion detection so you know exactly when the deterministic sequence has wrapped.

- Configurable tap positions for different polynomial implementations
- Seed loading capability for sequence initialization
- Cycle completion detection (sequence wrap-around)
- Enable control for conditional operation
- Support for various LFSR widths and tap configurations

49.2 Parameters

Parameter	Type	Default	Description
`WIDTH`	—	8	Width of the LFSR register
`TAP_INDEX_WIDTH`	—	12	Width of each tap index
`TAP_COUNT`	—	4	Number of feedback taps
`TIW`	—	—	Shorthand for TAP_INDEX_WIDTH

49.3 Ports

49.3.1 Inputs

Port	Width	Description
`clk`	1	System clock
`rst_n`	1	Active-low asynchronous reset
`enable`	1	Enable LFSR

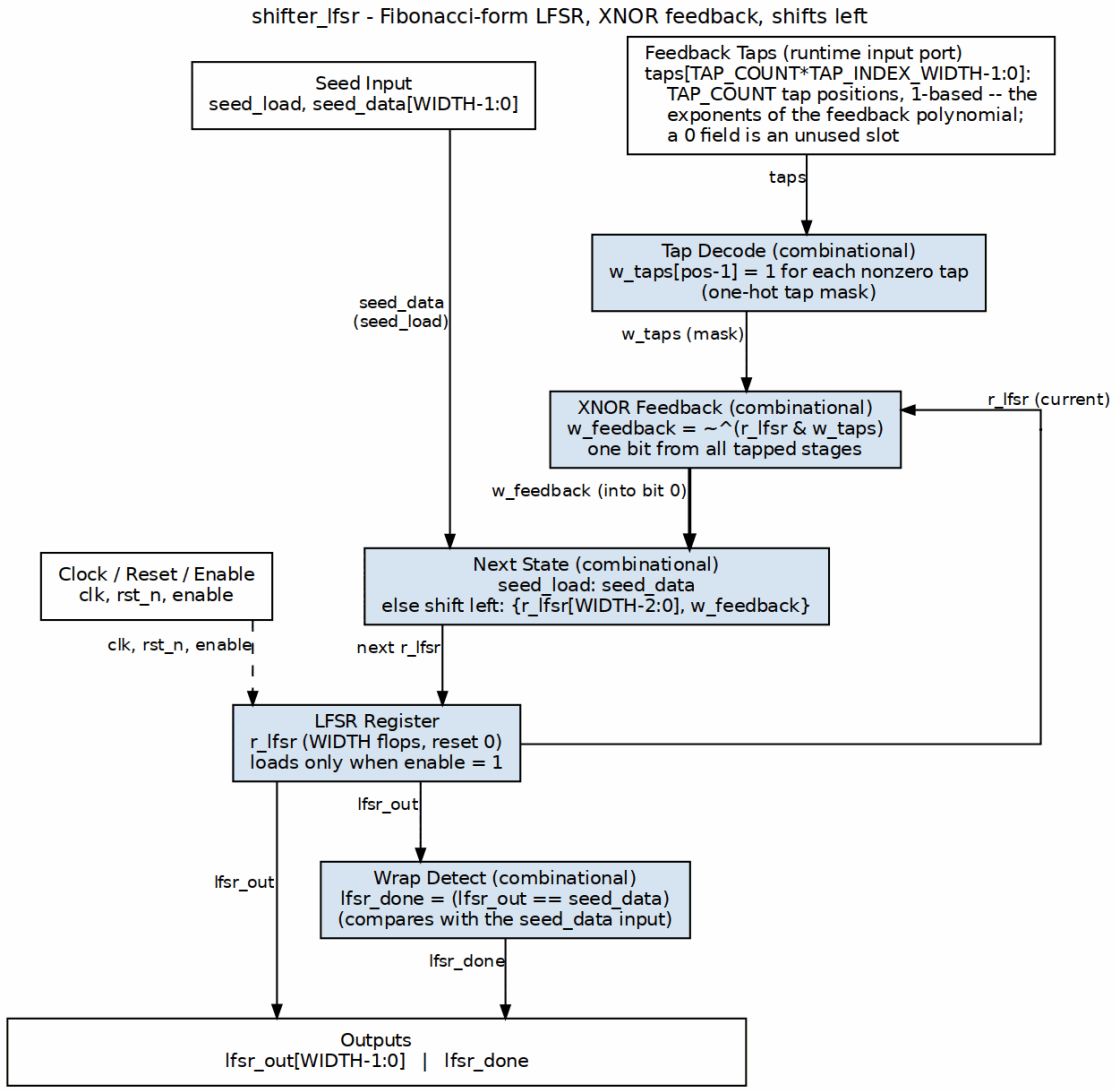
Port	Width	Description
		operation
seed_load	1	Load seed value into LFSR
seed_data	WIDTH	Seed value for LFSR initialization
taps	TAP_COUNT*TIW	Concatenated tap positions

49.3.2 Outputs

Port	Width	Description
lfsr_out	WIDTH	Current LFSR value
lfsr_done	1	High when LFSR returns to seed value

49.4 Functional Description

49.4.1 Block diagram



shifter_lfsr block diagram

The `taps` port (legend at the top right) lists the tap positions; tap decode turns them into a one-hot mask. One XNOR tree outside the shift chain folds every tapped bit of the current register into a single feedback bit, which enters bit 0 as the register shifts left. `seed_load` overrides with `seed_data`, and the wrap detector raises `lfsr_done` when the register comes back to the seed.

49.4.2 Feedback Polynomial

The LFSR implements a linear feedback shift register based on a primitive polynomial. The feedback is calculated by XORing bits at specified tap positions.

49.4.3 Tap Position Processing

```
// Split concatenated tap positions into separate groups
always_comb begin
    for (int i = 0; i < TAP_COUNT; i++)
        w_tap_positions[i] = taps[i*TIW+:TIW];
end

// Convert tap positions to bit mask
always_comb begin
    w_taps = 'b0;
    for (int i = 0; i < TAP_COUNT; i++)
        if (w_tap_positions[i] > 0)
            w_taps[w_tap_positions[i]-1'b1] = 1'b1;
end
```

49.4.4 Feedback Calculation

```
assign w_feedback = ~^(r_lfsr[WIDTH-1:0] & w_taps[WIDTH-1:0]);
```

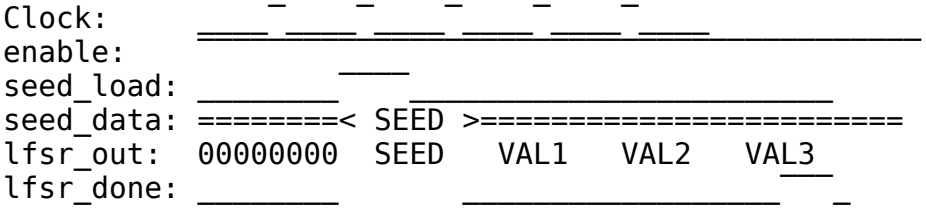
The feedback uses XNOR reduction (inverted XOR) of tapped bits, which is characteristic of maximal-length LFSR sequences.

49.4.5 Shift Register Operation

```
always_ff @(posedge clk or negedge rst_n) begin
    if (~rst_n) begin
        r_lfsr <= 'b0;
    end else begin
        if (enable) begin
            if (seed_load)
                r_lfsr <= seed_data;
            else
                r_lfsr <= {r_lfsr[WIDTH-2:0], w_feedback};
        end
    end
end
```

49.5 Timing Characteristics

49.5.1 Seed Loading and Operation



49.5.2 Sequence Example (3-bit LFSR, taps=[3,2])

Feedback is `~^(lfsr & taps)` (XNOR reduction) with `taps = 3'b110`, and the next state is `{lfsr[1:0], feedback}`. Full period is 7 (all states except the all-ones lockout):

```
Step | LFSR | lfsr & 110 | Feedback | Next
-----|------|------------|----------|------
0    | 001  | 000        | 1        | 011
1    | 011  | 010        | 0        | 110
2    | 110  | 110        | 1        | 101
3    | 101  | 100        | 0        | 010
4    | 010  | 010        | 0        | 100
5    | 100  | 100        | 0        | 000
6    | 000  | 000        | 1        | 001  (wraps to start; period 7)
```

49.6 Usage Examples

49.6.1 Pseudo-Random Number Generation

```
// Initialize with non-zero seed
seed_data = 8'hA5;
seed_load = 1'b1;  // Load seed
// Then enable normal operation
enable = 1'b1;
seed_load = 1'b0;
// lfsr_out provides pseudo-random sequence
```

49.6.2 Built-In Self-Test (BIST)

```
// Use lfsr_done to detect test completion
if (lfsr_done) begin
    // Full test sequence completed
    test_complete = 1'b1;
end
```

49.6.3 Scrambling/Descrambling

```
// XOR data with LFSR output for scrambling
scrambled_data = input_data ^ lfsr_out;
```

49.6.4 Common Use Cases

- Pseudo-random number generation
- Data scrambling and encryption
- Built-in self-test pattern generation
- CRC calculation

- Spread spectrum communications
- Test vector generation
- Noise generation for audio applications

49.7 Design Notes

49.7.1 1. Tap Position Encoding

- Tap positions are 1-indexed (position 1 = bit 0)
- Zero tap position is ignored (allows unused taps)
- Multiple tap positions concatenated into single bus

49.7.2 2. XNOR Feedback

- Uses XNOR (~^) instead of XOR for feedback
- Produces maximal-length sequences for primitive polynomials
- Different from some LFSR implementations that use XOR

49.7.3 3. Cycle Detection

```
assign lfsr_done = (lfsr_out == seed_data) ? 1'b1 : 1'b0;
```

Detects when LFSR returns to its initial seed value, indicating completion of full pseudo-random sequence.

49.7.4 4. Left-Shift Architecture

The implementation uses left-shift with feedback to LSB, which is equivalent to traditional right-shift LFSRs but with different bit ordering.

49.7.5 5. Zero State Handling

- Reset initializes LFSR to all zeros
- All-zeros is a **normal** state for this module, not a lockout. Feedback is XNOR (`~^(r_lfsr & w_taps)`), so an all-zeros register produces `~^0 = 1` and advances on its own – the 3-bit example in this page shows exactly that (`000` follows `100`, then wraps to `001`)
- The lockout state for XNOR feedback is **all-ones**, as stated elsewhere on this page. The all-zeros-avoidance rule belongs to XOR LFSRs such as `shifter_lfsr_fibonacci`, which carries an explicit `|r_lfsr` guard

- Seed loading is therefore optional: it selects where in the sequence you start, but the reset state runs correctly without it

49.7.6 LFSR Polynomial Reference

The module supports various standard LFSR polynomials. Common configurations (XNOR taps):

Width	Tap Positions	Max Period
3	[3,2]	7
4	[4,3]	15
5	[5,3]	31
8	[8,6,5,4]	255
16	[16,15,13,4]	65535
32	[32,22,2,1]	2^{32}-1

49.7.7 Example Tap Configuration for 8-bit LFSR

```
// For WIDTH=8, polynomial x^8 + x^6 + x^5 + x^4 + 1
// Taps at positions 8,6,5,4 (1-indexed)
parameter TAP_COUNT = 4;
parameter TIW = 4;  // 4 bits enough for tap positions up to 8

// Concatenated tap positions: {8,6,5,4}
wire [TAP_COUNT*TIW-1:0] taps = {4'd8, 4'd6, 4'd5, 4'd4};
```

49.8 Related Modules

- shifter_lfsr_fibonacci — Fibonacci (external-XOR) form; carries the `|r_lfsr` zero guard this module doesn't need.
- shifter_lfsr_galois — Galois (distributed-XOR) form for higher clock frequencies.

49.9 Testing

`val/common/test_shifter_lfsr.py` exercises this module. It collects 6 parameter cases at the default `REG_LEVEL`.

```
source env_python
pytest val/common/test_shifter_lfsr.py -v
```

49.10 References

- Ward, R., Molteno, T.C.A. “Table of Linear Feedback Shift Registers.” Electronics Technical Report No. 2012-1, University of Otago, 2012 (non-peer technical report) — open table of maximal-length tap sets (n <= 786) for cross-checking the in-header table; maximal-length LFSR theory itself is standard (Golomb, *Shift Register Sequences*, 1967).
- Moon, T.K. *Error Correction Coding*, 2nd ed., Wiley, 2021, ch. 4, Appendix 4.A “Linear Feedback Shift Registers” (textbook treatment; secondary source).
- Roy, S. *Advanced Digital System Design*, Springer, 2024, ch. 4 “Basic Sequential Circuits”, Sec. 4.5 Pseudo Noise Sequence Generator (textbook treatment; secondary source).
- Merrick, R. *Getting Started with FPGAs*, No Starch Press, 2024, ch. 6 “Common FPGA Modules”, “The Shift Register” (including “Creating a Linear Feedback Shift Register”) (textbook treatment; secondary source).
- Indexed in `rtl/common/references/README.md` (local PDF: `LfsrTapTable__Ward2012.pdf`).

49.11 Navigation

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50 shifter_universal

50.1 Overview

The `shifter_universal` module is the general-purpose shift register: left shift, right shift, parallel load, and hold, with both serial and parallel data in and out. If your design needs bits moved sideways in any direction, this is the part you reach for.

- Four distinct operations: hold, right shift, left shift, parallel load
- Bidirectional serial data input/output
- Parallel data load capability
- Serial data capture during shift operations
- Synchronous operation with asynchronous reset
- Configurable width

50.2 Parameters

Parameter	Type	Default	Description
`WIDTH`	—	4	Width of the shift register. **Must be >= 2** — at WIDTH=1 the shift expressions become illegal part-selects (`o_pdata[WIDTH-1:1]` -> `[0:1]`, `o_pdata[WIDTH-2:0]` -> `[-1:0]`).

50.3 Ports

50.3.1 Inputs

Port	Width	Description
`clk`	1	System clock
`rst_n`	1	Active-low asynchronous reset
`select`	2	Operation select control

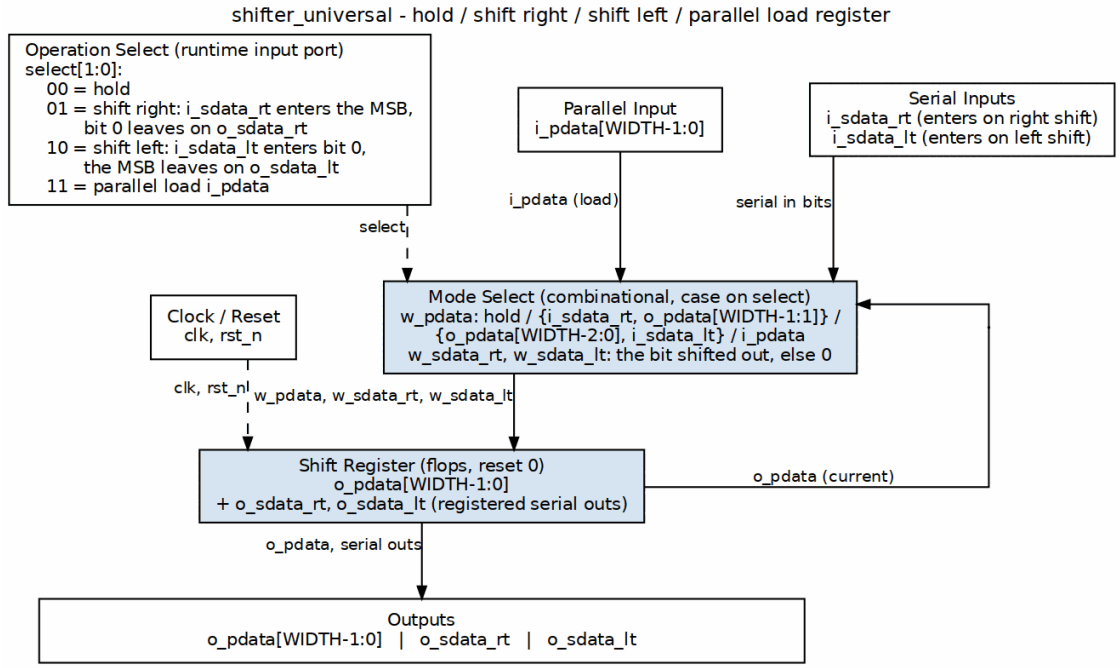
Port	Width	Description
i_pdata	WIDTH	Parallel data input
i_sdata_lt	1	Serial data input (left side)
i_sdata_rt	1	Serial data input (right side)

50.3.2 Outputs

Port	Width	Description
o_pdata	WIDTH	Parallel data output
o_sdata_lt	1	Serial data output (left side)
o_sdata_rt	1	Serial data output (right side)

50.4 Functional Description

50.4.1 Block diagram



shifter_universal block diagram

The parallel and serial inputs enter at the top. The mode select, a plain case on the `select` port (legend at the top left), forms the next word: hold, shift right with `i_sdata_rt` entering the MSB, shift left with `i_sdata_lt` entering bit 0, or a parallel load. The register holds the word and the two serial-out bits, all flopped, and its current word climbs the right edge back into the mode select.

50.4.2 Operation Control

The 2-bit `select` signal controls the shifter operation:

select	Operation	Description
2'b00	Hold	Maintain current state
2'b01	Right Shift	Shift data right; new bit enters from `i_sdata_rt`
2'b10	Left Shift	Shift data left; new bit enters from `i_sdata_lt`
2'b11	Parallel Load	Load parallel input data

50.4.3 Combinational Logic for Next State

```
always_comb begin
    // Default values - hold current state
    w_pdata = o_pdata;
    w_sdata_lt = 1'b0;
    w_sdata_rt = 1'b0;

    casez (select)
        2'b00: begin  // Hold
            w_pdata = o_pdata;
            w_sdata_lt = 1'b0;
            w_sdata_rt = 1'b0;
        end

        2'b01: begin  // Right Shift
            w_pdata = {i_sdata_rt, o_pdata[WIDTH-1:1]};
            w_sdata_lt = 1'b0;
            w_sdata_rt = o_pdata[0];
        end

        2'b10: begin  // Left Shift
            w_pdata = {o_pdata[WIDTH-2:0], i_sdata_lt};
            w_sdata_lt = o_pdata[WIDTH-1];
            w_sdata_rt = 1'b0;
        end
```

```
        2'b11: begin  // Parallel Load
            w_pdata = i_pdata;
            w_sdata_lt = 1'b0;
            w_sdata_rt = 1'b0;
        end
    endcase
end
```

50.4.4 Sequential Logic for State Update

```
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        o_pdata    <= '0;
        o_sdata_lt <= 1'b0;
        o_sdata_rt <= 1'b0;
    end else begin
        o_pdata    <= w_pdata;
        o_sdata_lt <= w_sdata_lt;
        o_sdata_rt <= w_sdata_rt;
    end
end
```

50.4.5 Hold Operation (select = 2'b00)

- **Function**: Maintains current register state
- **Data Flow**: No data movement
- **Serial Outputs**: Both remain low (no shift occurring)
- **Use Case**: Pause operation while maintaining data integrity

50.4.6 Right Shift Operation (select = 2'b01)

- **Function**: Shifts data toward LSB
- **Data Flow**: `i_sdata_rt` → MSB, LSB → `o_sdata_rt`
- **Serial Input**: New data enters from `i_sdata_rt`
- **Serial Output**: Shifted-out data exits via `o_sdata_rt`
- **Use Case**: Serial-to-parallel conversion, delay lines

50.4.7 Left Shift Operation (select = 2'b10)

- **Function**: Shifts data toward MSB
- **Data Flow**: `i_sdata_lt` → LSB, MSB → `o_sdata_lt`

- **Serial Input**: New data enters from `i_sdata_lt`
- **Serial Output**: Shifted-out data exits via `o_sdata_lt`
- **Use Case**: Multiplication by 2, bit-serial processing

50.4.8 Parallel Load Operation (select = 2'b11)

- **Function**: Loads entire parallel word
- **Data Flow**: `i_pdata` → `o_pdata`
- **Serial Outputs**: Both remain low (no shift occurring)
- **Use Case**: Initialization, data injection

50.5 Related Modules

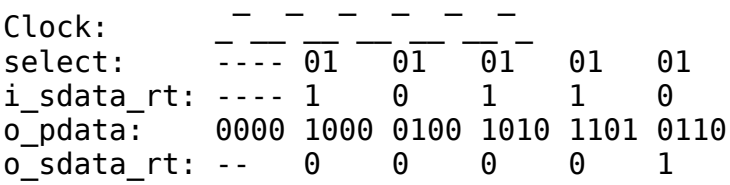
Nothing in the tree instantiates this module and it instantiates nothing: it is a leaf, used directly by whatever design needs it. Its nearest neighbours in `rtl/common/` are:

- `shifter_barrel`
- `shifter_beat_pack`
- `shifter_lfsr`
- `shifter_lfsr_fibonacci`
- `shifter_lfsr_galois`

50.6 Timing Characteristics

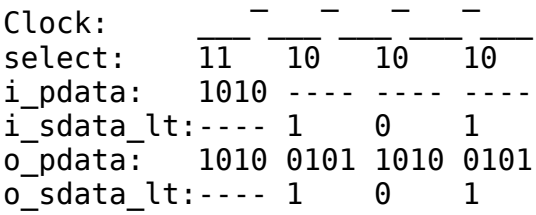
50.6.1 Right Shift Operation Example (WIDTH=4)

The first column is the reset state, before any edge; every later column is the value *after* that cycle's edge. `o_sdata_rt` is registered, so the bit it shows in a column is the LSB that the *previous* column's `o_pdata` ejected.



The single `1` on `o_sdata_rt` is the LSB of `1101`, ejected by the fifth shift and visible in the column that shift produces – not alongside `1101` itself.

50.6.2 Parallel Load followed by Left Shift



50.7 Usage Examples

50.7.1 Serial Communication Interface

```
// Receive shift register for UART
// Right shift to collect incoming serial data
select = 2'b01;
i_sdata_rt = uart_rx;
// When 8 bits received, parallel data available
received_byte = o_pdata;
```

50.7.2 Multiplication/Division by Powers of 2

```
// Multiply by 2: left shift
select = 2'b10;
i_sdata_lt = 1'b0;  // Fill with zeros

// Divide by 2: right shift
select = 2'b01;
i_sdata_rt = 1'b0;  // Fill with zeros (unsigned)
// or i_sdata_rt = o_pdata[WIDTH-1] for arithmetic shift
```

50.7.3 Data Delay Line

```
// Create N-cycle delay using N shifters
// Right shift with data input
select = 2'b01;
i_sdata_rt = data_input;
delayed_data = o_sdata_rt;  // N cycles later
```

50.7.4 Pattern Generator

```
// Load pattern then circulate
// 1. Load initial pattern
select = 2'b11;
i_pdata = initial_pattern;

// 2. Circulate pattern
select = 2'b01;         // Right shift
i_sdata_rt = o_pdata[0];  // Combinational tap -- NOT o_sdata_rt
```

Feed back `o_pdata[0]`, not `o_sdata_rt`. Here's the trap: the serial outputs are registered (`o_sdata_rt <= w_sdata_rt`), so routing `o_sdata_rt` back returns the ejected bit one shift too late and the pattern does not rotate – it runs a `WIDTH+1`-long cycle. With `WIDTH=4` loaded to `1011`, the registered feedback gives `1011 -> 0101 -> 1010 -> 1101 -> 0110 -> 1011` (period 5, and `1110` and `0111` never appear), whereas the combinational tap gives the true rotation `1011 -> 1101 -> 1110 -> 0111 -> 1011`.

50.7.5 Scan Chain for Testing

```
// JTAG-style scan chain
// Load test data
select = 2'b11;
i_pdata = test_vector;

// Shift out for observation
select = 2'b01;
i_sdata_rt = next_scan_in;
scan_out = o_sdata_rt;
```

50.7.6 Common Use Cases

- Serial-to-parallel conversion
- Parallel-to-serial conversion
- Data delay and timing adjustment
- Arithmetic operations (×2, ÷2)
- Pattern generation and circulation
- Scan chain testing
- FIFO buffer implementation
- Digital signal processing pipelines

50.8 Design Notes

50.8.1 1. Bidirectional Serial Interface

The module provides separate serial inputs and outputs for both directions: - **Left side**: `i_sdata_lt` (input), `o_sdata_lt` (output) - **Right side**: `i_sdata_rt` (input), `o_sdata_rt` (output)

That's what lets you chain multiple shifters or hook into bidirectional serial systems.

50.8.2 2. Clean Serial Output Management

Serial outputs are explicitly driven to zero when not actively shifting in that direction — no glitches, no ambiguous states.

50.8.3 3. Asynchronous Reset

Reset immediately clears all outputs to zero, providing deterministic startup conditions.

50.8.4 4. Combinational/Sequential Separation

The design cleanly separates: - **Combinational logic**: Calculates next state - **Sequential logic**: Registers the state on clock edge

That split improves timing predictability and synthesis results.

50.8.5 Chaining Multiple Shifters

```
// Connect shifters in series
shifter1.i_sdata_rt = serial_input;
shifter2.i_sdata_rt = shifter1.o_sdata_rt;
shifter3.i_sdata_rt = shifter2.o_sdata_rt;
```

50.8.6 Bidirectional Operation

```
// Support both shift directions
if (shift_direction == RIGHT) begin
    select = 2'b01;
    i_sdata_rt = serial_data;
end else begin
    select = 2'b10;
    i_sdata_lt = serial_data;
end
```

50.8.7 Control Logic Integration

```
// State machine control
case (state)
    LOAD: select = 2'b11;
    SHIFT_RIGHT: select = 2'b01;
    SHIFT_LEFT: select = 2'b10;
    HOLD: select = 2'b00;
endcase
```

50.9 Testing

`val/common/test_shifter_universal.py` exercises this module. It collects 1 parameter cases at the default `REG_LEVEL`.

```
source env_python
pytest val/common/test_shifter_universal.py -v
```

50.10 References

- Harris, S.L., Harris, D. *Digital Design and Computer Architecture, RISC-V Edition*, Morgan Kaufmann, 2022, Sec. 5.4.2 Shift Registers (textbook treatment; secondary source).
- LaMeres, B.J. *Introduction to Logic Circuits & Logic Design with Verilog*, 2nd ed., Springer, 2019, ch. 7 “Sequential Logic Design”, Sec. 7.3.4 Shift Registers (textbook treatment; secondary source).
- No peer paper: the hold/shift-right/shift-left/parallel-load register is a standard technique. Indexed in `rtl/common/references/README.md`.

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51 sort

51.1 Overview

The Sort module is a pipelined hardware sorting engine built on the odd-even sort algorithm, sorting arrays of configurable size in descending order. It's aimed at high-throughput work: new data accepted every clock cycle, with predictable latency equal to the number of elements being sorted.

51.1.1 Module Declaration

```
module sort #(
    parameter int NUM_VALS = 5,    // Number of values to sort
    parameter int SIZE     = 16    // Size of each value in bits
) (
    input  logic                      clk,       // Clock signal
    input  logic                      rst_n,     // Active-low reset
    input  logic [NUM_VALS*SIZE-1:0] data,       // Packed input data
    input  logic                      valid_in,  // Start sorting when
asserted
    output logic [NUM_VALS*SIZE-1:0] sorted,     // Packed sorted
output
    output logic                      done       // Asserted when
sorting complete
);
```

51.2 Parameters

Parameter	Type	Default	Description
NUM_VALS	int	5	Number of values in the array to be sorted. Determines pipeline depth. Range 2-16 (RTL header documents the supported range as 2 to 16).
SIZE	int	16	Bit width of each individual value. All values must be unsigned integers. Range >= 1 (no upper bound enforced).

51.3 Ports

51.3.1 Inputs

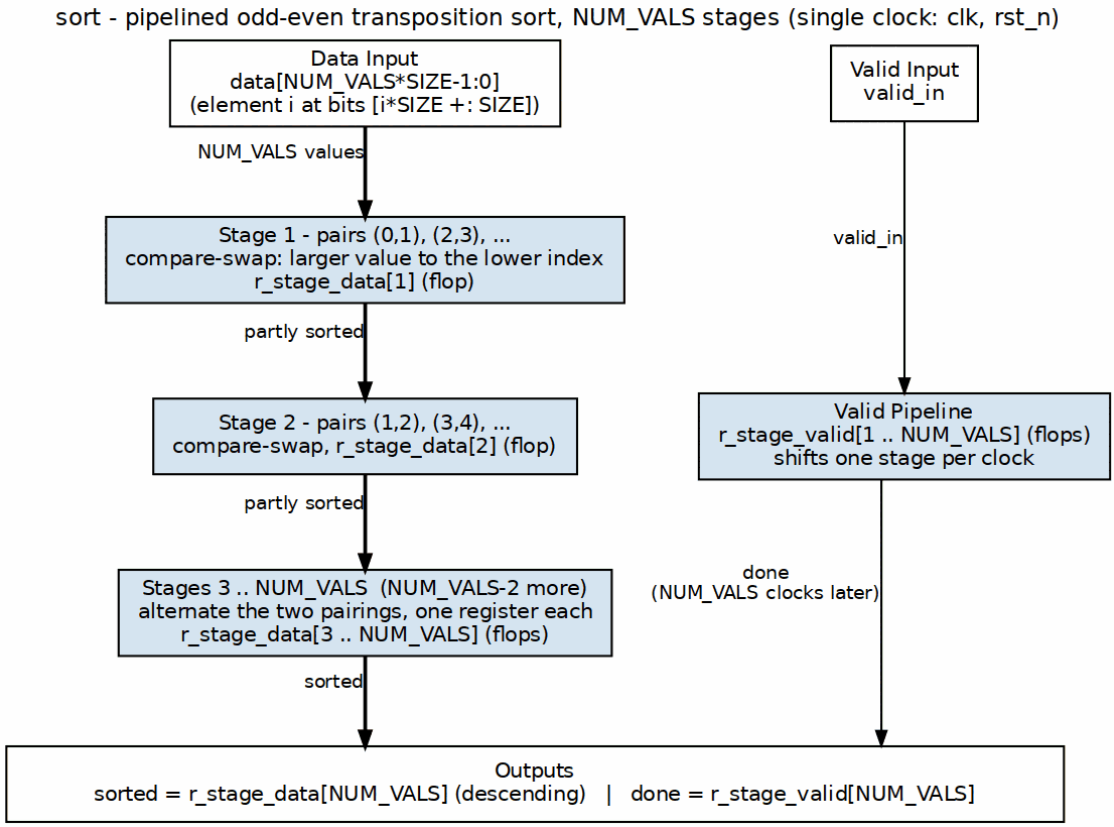
Port	Width	Description
`clk`	1	System clock signal. All operations are synchronous to the rising edge.
`rst_n`	1	Active-low asynchronous reset. Clears all pipeline stages while **asserted (low)**; normal operation is rst_n high.
`data`	`NUM_VALS*SIZE`	Packed input array. Elements are packed as `data[i*SIZE +: SIZE]` for element `i`.
`valid_in`	1	Control signal. When asserted (high) for one cycle, starts the sorting process for the current `data` input.

51.3.2 Outputs

Port	Width	Description
`sorted`	`NUM_VALS*SIZE`	Packed output array sorted in descending order. Same packing format as input.
`done`	1	Status signal. Asserted (high) when the sorting operation is complete and `sorted` output is valid.

51.4 Functional Description

51.4.1 Block diagram



sort block diagram

`data` enters as `NUM_VALS` packed values and passes through `NUM_VALS` registered stages. Odd-numbered stages compare-swap the pairs (0,1), (2,3), and so on; even-numbered stages compare-swap (1,2), (3,4), and so on. Each swap puts the larger value at the lower index, so the output is descending. `valid_in` rides a parallel shift register of the same depth and comes out as `done`, so a result appears `NUM_VALS` clocks after its input. A new input can enter every clock.

51.4.2 Data Format

Input/Output Packing: Arrays are packed into a single wide bus where: - Element 0 occupies bits `[SIZE-1:0]` - Element 1 occupies bits `[2*SIZE-1:SIZE]` - Element i occupies bits `[(i+1)*SIZE-1:i*SIZE]`

Sort Order: Output is sorted in **descending order** (largest to smallest).

51.4.3 Algorithm: Odd-Even Sort

The module implements the **Odd-Even Sort** (also known as **Brick Sort**) algorithm — a particularly good fit for pipelined hardware, as you'll see below.

Odd-Even Sort works by alternating between two types of comparison passes:

1. **Odd Passes**: Compare and swap adjacent pairs at positions (0,1), (2,3), (4,5), etc.
2. **Even Passes**: Compare and swap adjacent pairs at positions (1,2), (3,4), (5,6), etc.

The algorithm requires exactly `NUM_VALS` passes to guarantee complete sorting, regardless of input data.

51.4.4 Example Execution

For input array `[5, 3, 8, 1, 9]` (5 elements):

```
Initial:     [5, 3, 8, 1, 9]

Pass 1 (Odd):  Compare (0,1), (2,3), (4,X)
               [5, 3] → [5, 3] (no swap)
               [8, 1] → [8, 1] (no swap)
               [9, X] → [9, X] (no comparison)
Result:        [5, 3, 8, 1, 9]

Pass 2 (Even): Compare (1,2), (3,4)
               [3, 8] → [8, 3] (swap)
               [1, 9] → [9, 1] (swap)
Result:        [5, 8, 3, 9, 1]

Pass 3 (Odd):  Compare (0,1), (2,3), (4,X)
               [5, 8] → [8, 5] (swap)
               [3, 9] → [9, 3] (swap)
               [1, X] → [1, X] (no comparison)
Result:        [8, 5, 9, 3, 1]

Pass 4 (Even): Compare (1,2), (3,4)
               [5, 9] → [9, 5] (swap)
               [3, 1] → [3, 1] (no swap)
Result:        [8, 9, 5, 3, 1]

Pass 5 (Odd):  Compare (0,1), (2,3), (4,X)
               [8, 9] → [9, 8] (swap)
               [5, 3] → [5, 3] (no swap)
               [1, X] → [1, X] (no comparison)
Final:         [9, 8, 5, 3, 1] -- sorted
```

51.4.5 Algorithm Comparison

The “Space (SW)” column is the classical **in-place software** space bound; “HW Area” is the cost of a fully-pipelined hardware network of that algorithm. For **this** module the network is $O(n^2)$: ~n/2 compare-swaps per stage × n flopped stages (the RTL header: “$O(NUM_VALS^2)$ comparators, $O(NUM_VALS^2 \cdot SIZE)$ registers”). Do not read O(1) as the hardware area.

Algorithm	Time Complexity	Space (SW)	HW Area (this style)	Hardware Efficiency	Pipeline Stages	Best Use Case
Odd-Even Sort	$O(n^2)$	O(1)	**$O(n^2)$**	Excellent	n	Small arrays, predictable timing
Bubble Sort	$O(n^2)$	O(1)	$O(n^2)$	Poor	Variable	Sequential processors only
Bitonic Sort	$O(n \log^2 n)$	O(1)	$O(n \log^2 n)$	Very Good	$\log^2 n$	Power-of-2 sizes, larger arrays
Merge Sort	O(n log n)	O(n)	O(n log n)	Good	log n	Large arrays, general purpose
Insertion Sort	$O(n^2)$	O(1)	$O(n^2)$	Good	Variable	Nearly sorted data

Why Odd-Even Sort for Hardware?

1. **Regular Structure**: Each pipeline stage is identical, making synthesis predictable
2. **Parallel Operations**: All comparisons in each pass happen simultaneously
3. **Fixed Latency**: Always takes exactly `NUM_VALS` cycles regardless of input data

4. **Simple Control**: No complex indexing or variable loop bounds
5. **Scalable**: Pipeline depth scales linearly with array size

51.4.6 Pipeline Structure

The module implements a `NUM_VALS`-stage pipeline where: - **Stage 0**: Input capture (combinational) - **Stages 1 to NUM_VALS**: Pipeline registers with compare-swap logic

The block diagram above shows the stages.

51.4.7 Naming Conventions

Naming is strict, and it pays off when you're reading waveforms:

- **Wire signals** (combinational): Prefixed with `w_`
  - `w_values[stage][element]`: Unpacked arrays for easy manipulation
  - `w_next_values[stage][element]`: Computed next values
  - `w_stage_data_0`: Input stage data wire
  - `w_stage_valid_0`: Input stage valid wire
- **Flopped signals** (sequential): Prefixed with `r_`
  - `r_stage_data[stage]`: Pipeline data registers
  - `r_stage_valid[stage]`: Pipeline valid registers

51.4.8 Compare-Swap Logic

Each pipeline stage implements compare-swap operations based on the stage number:

```
// Determine if this is an odd or even pass
localparam bit IS_ODD_PASS = ((stage-1) % 2) == 0;

always_comb begin
    // Default: pass through unchanged
    for (int i = 0; i < NUM_VALS; i++) begin
        w_next_values[stage][i] = w_values[stage-1][i];
    end

    if (IS_ODD_PASS) begin
        // Odd pass: compare (0,1), (2,3), (4,5), etc.
        for (int i = 0; i < NUM_VALS-1; i += 2) begin
            if (w_values[stage-1][i] < w_values[stage-1][i+1]) begin
                w_next_values[stage][i]   = w_values[stage-1][i+1];
```

```
                w_next_values[stage][i+1] = w_values[stage-1][i];
            end
        end
    end else begin
        // Even pass: compare (1,2), (3,4), (5,6), etc.
        for (int i = 1; i < NUM_VALS-1; i += 2) begin
            if (w_values[stage-1][i] < w_values[stage-1][i+1]) begin
                w_next_values[stage][i]   = w_values[stage-1][i+1];
                w_next_values[stage][i+1] = w_values[stage-1][i];
            end
        end
    end
end
```

51.4.9 Reset Behavior

- **Asynchronous Reset**: All pipeline stages are cleared while `rst_n` is **asserted (low)**
- **Data Reset**: All `r_stage_data` registers are cleared to zero
- **Valid Reset**: All `r_stage_valid` registers are cleared to zero
- **Output**: Both `sorted` and `done` outputs go to zero during reset

51.4.10 Control Flow

The Sort module does **not** use explicit Finite State Machines (FSMs). Control is just the pipeline itself — a **pipeline-based control scheme**:

1. **Input Stage (Combinational)**:
    - Accepts new `data` when `valid_in` is asserted
    - No state storage at input stage
2. **Pipeline Progression (Sequential)**:
    - Valid signal propagates through pipeline automatically
    - Each stage processes data independently
    - No centralized control or state machine needed
3. **Output Indication**:
    - `done` signal is simply the valid signal from the final pipeline stage

- No complex output state management required

51.5 Related Modules

Nothing in the tree instantiates this module and it instantiates nothing: it is a leaf, used directly by whatever design needs it. Its nearest neighbours in `rtl/common/` are:

- (none in this category)

51.6 Timing Characteristics

51.6.1 Timing Characteristics

- **Latency**: `NUM_VALS` clock cycles from `valid_in` assertion to `done` assertion
- **Throughput**: 1 array per clock cycle (once pipeline is full)
- **Critical Path**: One compare-swap operation — a `SIZE`-bit magnitude comparator feeding a 2:1 mux, i.e. several logic levels (consistent with the ~1.5-2.0 ns figures in the synthesis table below, not "1-2 gate delays")
- **Pipeline Depth**: `NUM_VALS` stages

51.6.2 Valid Signal Flow

```
Cycle:    1      2      3      4      5      6
valid_in: 1      0      0      0      0      0
r_stage_valid[1]: -      1      0      0      0      0
r_stage_valid[2]: -      -      1      0      0      0
r_stage_valid[3]: -      -      -      1      0      0
r_stage_valid[4]: -      -      -      -      1      0
r_stage_valid[5]: -      -      -      -      -      1
done:     0      0      0      0      0      1
```

51.7 Usage Examples

51.7.1 Typical Use Cases

1. **Packet Processing**: Sorting priority fields or timestamps
2. **Signal Processing**: Median filtering, statistical operations
3. **Graphics**: Z-buffer sorting, polygon vertex ordering
4. **Control Systems**: Sorting sensor readings or control parameters

51.7.2 Integration Considerations

1. **Clock Domain**: Ensure `clk` is stable and meets timing requirements
2. **Reset Strategy**: Include proper reset sequencing in system design
3. **Data Formatting**: Pack input arrays correctly according to element ordering
4. **Flow Control**: Monitor `done` signal for output validity
5. **Resource Planning**: Consider pipeline depth impact on latency-sensitive applications

51.7.3 Performance Optimization

1. **Pipeline Depth**: Use smallest practical `NUM_VALS` for lowest latency
2. **Data Width**: Use smallest practical `SIZE` for best area efficiency
3. **Clock Frequency**: Critical path is typically one comparison operation
4. **Resource Sharing**: Multiple instances can share input/output buses if properly arbitrated

51.8 Design Notes

51.8.1 1. Parameterizable Design

- **Flexible Array Size**: `NUM_VALS` supported range is 2 to 16 (per the RTL header)
- **Configurable Data Width**: `SIZE` sets the bits per element. The RTL applies no validation to it – any `SIZE >= 1` elaborates; wide values simply cost a wider comparator on the critical path
- **Automatic Pipeline Scaling**: Pipeline depth automatically adjusts to `NUM_VALS`

51.8.2 2. High Throughput Pipeline

- **Back-to-Back Operations**: New data can be accepted every cycle
- **Overlapped Processing**: Multiple arrays can be in the pipeline simultaneously

- **Predictable Performance**: Fixed latency regardless of data patterns

51.8.3 3. Synthesis-Friendly Design

- **Regular Structure**: All pipeline stages are nearly identical
- **Local Connectivity**: No long combinational paths across stages
- **Inference-Friendly**: Uses standard SystemVerilog constructs for reliable synthesis

51.8.4 4. Packed Data Handling

- **Packed Buses**: Efficient use of bus resources
- **Unpacked Internal Arrays**: Easy element-wise manipulation
- **Automatic Packing/Unpacking**: Transparent data format conversion

51.8.5 Synthesis Results

Typical synthesis results for common configurations:

Configuration	Pipeline Stages	Critical Path	Area (LUTs)	Frequency
3×8-bit	3	~1.2ns	~150	>800 MHz
5×16-bit	5	~1.5ns	~400	>650 MHz
8×32-bit	8	~2.0ns	~1200	>500 MHz

Results vary by target FPGA family and synthesis tool settings

51.8.6 Future Enhancements

Potential improvements for future versions:

1. **Bidirectional Sorting**: Parameter to select ascending vs descending order
2. **Early Termination**: Detect when array is already sorted and terminate early
3. **Variable Width Elements**: Support for mixed-width data elements
4. **Priority Encoding**: Include secondary sort keys for equal elements
5. **Power Optimization**: Clock gating for unused pipeline stages

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This documentation describes the Sort module implementation. For the latest updates and additional examples, refer to the source code and testbench files.

51.9 Testing

The module comes with a comprehensive CocoTB testbench that includes:

51.9.1 Test Levels

1. **Basic Tests**: Core functionality verification
2. **Random Tests**: Extensive random data patterns
3. **Boundary Tests**: Edge cases and limit conditions
4. **Pipeline Tests**: Throughput and overlapped operation verification
5. **Reset Tests**: Proper reset behavior validation

51.9.2 Test Coverage

- **Functional**: All possible input combinations for small arrays
- **Parametric**: Multiple `NUM_VALS` and `SIZE` combinations
- **Temporal**: Pipeline timing and valid signal propagation
- **Stress**: Back-to-back operations and maximum throughput

51.10 References

- Habermann, A.N. “Parallel Neighbor-Sort (or the Glory of the Induction Principle).” Technical Report, Carnegie-Mellon University, Computer Science Dept., Aug. 1972 (DTIC AD-759 248). DOI: 10.1184/R1/6608258.v1 — the original odd-even transposition sort: alternating odd/even neighbor compare-exchange passes sort N values in N parallel steps, which is this module’s N-stage pipeline.
- Indexed in `rtl/common/references/README.md` (local PDF: `OddEvenTranspositionSort__Habermann1972.pdf`)

51.11 Navigation

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